

वार्षिक प्रतिवेदन ANNUAL REPORT 2023



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद - 500 030, तेलंगाना, भारत
ICAR-National Academy of Agricultural Research Management
Rajendranagar, Hyderabad-500 030, Telangana, India

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ICAR-National Academy of Agricultural Research Management

(ISO 9001 : 2015 Certified)

Rajendranagar, Hyderabad-500030, Telangana, India

<https://www.naarm.org.in>

Citation:

NAARM, 2024. Annual Report (2023). ICAR-National Academy of Agricultural Research Management, Rajendranagar, Hyderabad-500030, pp 1-240, ISSN: 2349-7548.

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Printed at:

Printography, Ranigunj, Secunderabad - 500 003.

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ABBREVIATIONS

ABI	Agri-Business Incubator
ADG	Assistant Director General
AEO	Agriculture Extension Officer
AFOLU	Agriculture, Forestry and Other Land Use Sector
AgIn	Agrinnovate India Limited
AI	Artificial intelligence
AICRP	All-India Coordinated Research Project
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
ANOVA	Analysis of Variance
APLDA	Andhra Pradesh Livestock Development Agency
APMC	Agricultural Produce Market Committee
APSIM	Agricultural Production Systems sIMulator
ARS	Agricultural Research Station
ASEAN	Association of South East Asian Nations
ASPIRE	Association for Scientific Pursuits for Innovative Research Enterprises
ASRB	Agricultural Scientists Recruitment Board
ATARI	Agricultural Technology Application Research Institute
BESTIU	Bharatiya Engineering Science & Technology Innovation University
BIG	The Biotechnology Ignition Grant
BIRAC	Biotechnology Industry Research Assistance Council
CBC	Capacity Building Commission
CBU	Capacity Building Unit
CCSMC	Catalytic Capital Support Management Committee
CCVEC	Centre for Continuing Veterinary Education and Communication
CDC	Career Development Center
CGIAR	Consultative Group on International Agricultural Research
CHIRPS	Climate Hazards Group Infrared Precipitation with Station data
CII	Confederation of Indian Industry
CIL	Coromandal International Limited
CIPHET	Central Institute of Post Harvest Engineering & Technology, Ludhiana
CLD	Causal Loop Diagram
COLLAgE	Centre for Lifelong Learning in Agricultural Education
CSIR	Council of Scientific & Industrial Research
CSKHPKV	CSK Himachal Pradesh Krishi Vishwavidyalaya
CSSRI	Central Soil Salinity Research Institute
CTO	Chief Technical Officer
DBT	Department of Biotechnology
DPR	Directorate of Poultry Research
DRDL	Defence Research and Development Laboratory
DSSAT	Decision Support System for Agrotechnology Transfer

DST	Department of Science & Technology
DWRP	Developing Winning Research Proposal
EDP	Executive Development Program
ESAF	Evangelical Social Action Forum
ESDTP	Entrepreneurial Skill Development Training Program
ET	Evapotranspiration
F&V	Fruits & Vegetables
FAITT	Foundation for Aquaculture Innovations & Technology Transfer
FET	Field Experience Training
FICCI	Federation of Indian Chamber of Commerce and Industry
FIRST	Farm, Innovations, Resources, Science and Technology
FLAME	Front Line Agri-Marketing Excellence
FOCARS	Foundation Course for Agricultural Research Service
FPC	Farmer Producer Company
FPO	Farmer Producer Organization
GADVASU	Guru Angad Dev Veterinary and Animal Sciences University
GBI	Global Bio-India
GBPUAT	Govind Ballabh Pant University Of Agriculture & Technology
GHG	Greenhouse Gas
GHGPI	Greenhouse Gas Platform India
GoI	Government of India
HWC	Human-Wildlife Conflict
CIAE	Central Institute of Agricultural Engineering
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
ICT	Information and Communication Technology
IFAD	International Fund for Agricultural Development
IFPRI	International Food Policy Research Institute
IFS	Integrated Farming System
IIPM	Indian Institute of Plantation Management-Bengaluru
IISWC	Indian Institute of Soil & Water Conservation, Dehradun
IoT	Internet of Things
IRC	Institute Research Council
IRIC	International Rice Informatics Consortium
ISRO	Indian Space Research Organisation
JDA	Joint Director of Agriculture
KSA	Knowledge, Skills, and Attitudes
KSNUAHS	Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences
KUFOS	Kerala University of Fisheries and Ocean Studies
KVASU	Kerala Veterinary and Animal Sciences University
KVK	Krishi Vigyan Kendra
LFCS	Live Fish Carrier System
LiFE	Mission Lifestyle for Environment
LST	Land Surface Temperature
MABIF	Madurai Agribusiness Incubation Forum
MAFSU	Maharashtra Animal & Fishery Sciences University

MANAGE	National Institute of Agricultural Extension Management
MANIT	Maulana Azad National Institute of Technology
MDP	Management Development Program
MNI	Market Yard of National Importance
MOA	Memorandum of Agreement
MOOC	Massive Open Online Courses
MORE	Marketing Officers' Resurgence & Excellence
MOU	Memorandum of Understanding
MRU	Malla Reddy University
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NABARD	National Bank for Agricultural and Rural Development
NABCONS	NABARD Consultancy Services Private Ltd
NAES	National Agricultural Extension System
NAHEP	National Agricultural Higher Education Project
NAIF	National Agriculture Innovation Fund
NAIF-ABA	National Agricultural Innovation Fund- Agri-business Incubators
NARES	National Agricultural Research and Education Systems
NBA	National Board of Accreditation
NCCT	National Council for Cooperative Training
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NEAT	New Entrants Assimilation Training
NEP	National Education Policy
NGO	Non-Governmental Organization
NIAS	National Institute of Advanced Studies, Hyderabad
NIDHIPRAYAS	NIDHI-Promotion and Acceleration of Young and Aspiring technology entrepreneur
NIRD & PR	National Institute of Rural Development and Panchayat Raj
NSCSTI	National Standards for Civil Service Training Institutions
OUAT	Odisha University of Agriculture & Technology
PGDM-ABM	Post Graduate Diploma in Agribusiness Management
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PMC	PRAYAS Monitoring Committee
PM-KISAN	Pradhan Mantri Kisan Samman Nidhi Programme
PMRC	Project Monitoring and Review Committee
PoP	Package of Practice
PPPs	Public-Private Partnerships
PRT	Primary Response Teams
PSA	Principal Scientific Advisor
PVNRTVU	PV Narasimha Rao Telangana Veterinary University
QGIS	Quantum Geographic Information System
RAC	Research Advisory Committee
RLBCAU	Rani Laxmi Bai Central Agricultural University
RothC	Rothamsted Carbon Model
RPF	Recruitment, Procurement, and Finance
RRT	Rapid Response Teams

RSETI	Rural Self Employment Training Institutes
Ru-TAG	Rural Technology Action Group
RVSKVV	Rajmata Vijayaraje Scindia Krishi Vishwavidyalaya
SAIRD	Sri Aurobindo Institute of Rural Development
SAUs	State Agricultural Universities
SCSP	Scheduled Caste Sub Plan
SDGs	Sustainable Development Goals
SKNAU	Shri Karan Narendra Agriculture University
SMC college	Sheth M. C. College Of Dairy Science
SMS	Subject Matter Specialist
SVVU	Sri Venkateswara Veterinary University
SWOT	Strengths, Weaknesses, Opportunities and Threats
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TBI	Technology Business Incubator
TCE	Teaching Competency Enhancement
TNJFU	Tamil Nadu Dr. Jayalalithaa Fisheries University
ToT	Training of Trainers
UAS	University of Agricultural Sciences
UTs	Union Territories
WRDA	Warehousing Development Regulatory Authority
ZTMC	Zonal Technology Management Centre

FROM THE DIRECTOR'S DESK



Dr Ch. Srinivasa Rao
Director

It is my great pleasure to write preamble for the annual report of NAARM for the year 2023. The Academy has made remarkable contribution in its mandated areas of training and capacity building, research and policy advocacy and academic programmes. It is my proud privilege to inform that during the year 2023, the Academy has organized its flagship programme “Foundation Course for Agricultural Research Service” in two batches (112th FOCARS and 113th FOCARS). The Academy also conducted 5 Leadership Development Programmes, 48 Need-Based Programmes, two workshops/conferences/webinars, 28 off-campus programmes, two MOOC programmes, two FOCFAU programmes, 23 skill development programmes at different KVKs for the benefit of farmers. The Academy continued to play a key role in enhancing the capacities of Individuals and Institutions of NARES in Agricultural Research and Education Management.

I am happy to share that Shri Narendra Singh Tomar Ji, Hon’ble Union Minister of Agriculture & Farmers Welfare has laid the foundation stone for three new facilities of the Academy viz., New Faculty Building, Farmers House and Ladies/Girls Hostel on June 17, 2023. The Academy organised a Policy Workshop on Integrating Public and Private Systems for Effective

Extension Service on 12th January 2023; and a National Conference on “Policy Development and Implementation Strategies for Academia Industry-Government Linkages for Agricultural Higher Education in India” at ICAR-NAARM in March, 2023. The Academy also contributed to G20 events on climate resilient agriculture and one-health at Hyderabad and Bangalore besides organizing India-Brazil meeting under the Chairmanship of Secretary, DARE & DG, ICAR.

The PGDM-ABM students at NAARM represent 16 States and 23 universities providing a truly national character to the Course in tune with the NAARM’s focus as a National Academy in Agribusiness Management. Further, it is to mention that more girl students are taking admission in the Course, which has reached more than 50 per cent in the current Batch of 2023-25. It is heartening to state that the PGDM-ABM Programme got the accreditation by National Board of Accreditation (NBA) for a period of three years till 2025. SANKALP 8.0, the national-level B-fest by the students of Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) was organized during 22–23 September, 2023 and garnered an impressive response with over 9300 registrations from more than 1000 institutions, culminating in 200+ finalists in eight distinct events. Seven students of our PGDM-ABM programme brought laurels to the Academy by winning various case studies/quiz competitions during various B-Fests organized by reputed management institutions.

I am glad to state that the Academy has also facilitated several new policies, institutional mechanisms, and initiatives leading to enhanced capacities in NARES for Research Project Management (RPM), Information and Communication Management (ICM), Human Resources Management (HRM), Intellectual Property Management (IPM), Technology Management (TM), Massive Open Online Courses (MOOCs), and in several other areas. The Academy has very significantly contributed towards research in addition to the capacity building activities. During the year, 23 research projects are in operation at the Academy, of which 14 are in-house and 9 are external projects. The Faculty published 62 Research Papers in peer reviewed journals and brought out 6 Policy papers.

Both the Centres of Excellence of the Academy have contributed significantly in 2023. The Centre of Lifelong Learning in Agricultural Education (COLLAge), one-of-its kind state-of-the-art facility, has developed educational content in digital format and resulted in the development of MOOCs which provided the capacity building through online mode. Another Centre of Excellence, a-IDEA (a Technology Business Incubator) supported 81 startups across India, organized 2 capacity building programs, one sensitization program for students; and 4 immersion programs to network start-ups and FPOs. The a-IDEA also organized

its First Graduation Ceremony of Agri-Startups on 17 Mar, 2023. AGRI UDAAN 6.0 which is the Food & Agribusiness Accelerator Program was launched at Bengaluru on 07 July 2023. Continuing its contribution for the creation of human resources to manage FPOs, a-IDEA conducted 2 Certificate Courses on Agribusiness Management Development programmes for prospective CEOs of FPOs located in the States of Telangana and Andhra Pradesh.

ICAR-NAARM celebrated its 48th Foundation day to commemorate the establishment of the institution and to reflect on its achievements & contributions. On this auspicious occasion, the Academy based on the contribution has given annual awards to different categories presented to the staff, and felicitation done to the innovative farmers, best performing Start-ups, Farmer Producer Organizations (FPOs), Best Collaborating Institute, print & electronic media personnel, who have active engagement with academy. ICAR-NAARM, popularly known as the Green Campus bagged the Silver Garden Certificate Award in the 7th Garden Festival by Govt. of Telangana for maintaining the institute landscape, gardens and lawns in the best presentable manner.

I take this opportunity to thank ICAR, RAC and IMC for their constant support to the Academy to take up the various activities of national importance.



(Ch Srinivasa Rao)

Director

Date: 31-05-2024

Place: Hyderabad

EXECUTIVE SUMMARY

The National Academy of Agricultural Research Management (NAARM), a Premier Capacity Strengthening Institute of Indian Council of Agricultural Research (ICAR), is striving ahead to build the individual and institutional competencies of agricultural professionals serving in agricultural research, education and extension system of the National Agricultural Research and Education System (NARES), apart from extending research and policy advocacy support. The Academy serves as a think tank for NARES to facilitate strategic management of agricultural human capital in the country. The academy strives to fulfil its mandate by working on a strategic framework of six thematic areas, viz. Research Systems Management, Human Resource Management, Information & Communication Management, Education Systems Management, Extension Systems Management and Agribusiness Management.

During the year under report, NAARM has organized a total of 96 programmes reaching about 6558 participants. These encompassed a variety of training programs, such as two Foundation Courses for Agricultural Research Service attended by 186 participants, five Leadership Development Programs engaging 109 participants, 48 Need-Based Programs involving 1508 participants, two Workshops/Symposiums with 270 participants, and two Massive Open Online Courses (MOOC) with 3054 participants. Under SC-Sub Plan programme, the Academy also organized 23 skill development programmes at different KVKs for the benefit of farming fraternity. Academy also

conducted six capability building programmes for 146 newly recruited young agribusiness professionals and marketing officers of Industry. New training modules on gene editing and systems modelling were introduced during 2023. Apart from it, restructuring of FOCARS under the Chairmanship of Dr P.K. Joshi, was also taken up.

As a part of ‘Think-tank Policy’ mandate, the Academy provides advisory services, research policy advocacy in the critical areas of importance in agriculture. Accordingly, NAARM has organized a Policy Workshop on Integrating Public and Private Systems for Effective Extension Service; and a National Conference on “Policy Development and Implementation Strategies for Academia- Industry-Government Linkages for Agricultural Higher Education in India”.

During 2023, NAARM faculty published 58 Research Articles in National as well as International Journals. The Academy also published five Policy papers, one Strategy Paper, three Books, 11 Book Chapters, 12 Training Manuals and one Policy Brief. Currently, 23 research projects are in operation at the Academy of which 14 are in-house and nine are external funded projects. The Academy signed six MoUs with other universities and institutes in areas of mutual interest. To reach maximum end users/beneficiaries, around 20 programmes, events and activities of the Academy were widely covered in print and electronic media during 2023. Besides, around 6965 undergraduate students from 91

different colleges and agricultural universities across India visited the Academy this year as a part of their mandated all India Study tour.

NAARM's Post Graduate Diploma in Management-Agri-Business Management (PGDM-ABM) programme is accredited by National Board of Accreditation (NBA) for three years till 2025. At present, the 14th batch of PGD-ABM (2022-24) strength is 58 students whereas 15th batch of PGD-ABM (2023-25) strength is 61. Seven students of our PGDM-ABM programme brought laurels to the Academy by winning various case studies/quiz competitions during various B-Fests organized by reputed management institutions. SANKALP 8.0, the national-level B-fest by the students of (PGDM-ABM) was organized at NAARM during 22-23 September, 2023 and garnered an impressive response with over 9300 registrations from more than 1000 institutions, culminating in 200+ finalists in eight distinct events.

a-IDEA, Technology Business Incubator of NAARM, launched a National Level Ideation Competition "Aggnite"; celebrated National Startup Day on 16 Jan, 2023, organized State Level Millet Conclave on 10 Mar 2023 to commemorate the International Year of Millets. a-IDEA also organized its First Graduation Ceremony of Agri-Startups on 17 March, 2023 besides AGRI UDAAN 6.0 which is the Food & Agribusiness Accelerator Program at Bengaluru, Karnataka on 07 July 2023. The incubator supported 81 startups across India, conducted two capacity building programs, one sensitization program for students; and four immersion programs in UP, Tamil Nadu, Telangana and

AP. It also conducted two Certificate Courses on Agribusiness Management Development programmes for prospective CEOs of FPOs in Telangana and Andhra Pradesh. Three trainings were conducted by a-IDEA in collaboration with SBI, National Bank Staff College and Bankers Institute of Rural Development. a-IDEA of NAARM executed 31 MoUs with different ICAR institutes and with five other organizations.

The Academy also contributed to G20 events on climate resilient agriculture and one-health at Hyderabad and Bengaluru besides organizing India-Brazil meeting under the Chairmanship of Secretary, DARE & DG, ICAR. a-IDEA and CII organized exposure visit of Bhutanese Start-ups who won the pitching competition during the Bhutan-India Start-up Summit on 26th April, 2023. The Academy also associated in webinar on 'Technology, Policy, Programs and Institutions for Resilient Dryland Systems' as "PRE - COP28" entitled "*Impact of Climate Change on the Dry Areas*" organized by the International Dryland Development Commission (IDDC), Regional Action on Climate Change (RACC), Japan; Nizami Ganjavi International Center (NGIC) the African League of Young Masters (ALYM), hosted by Alexandria University in virtual mode on September 12, 2023.

Shri Narendra Singh Tomar, Hon'ble Union Minister of Agriculture & Farmers' Welfare, has laid the foundation stone for three new infrastructure facilities of the Academy i.e., New Faculty Building, Farmers Training Centre and Ladies/Girls Hostel in the presence of Shri Kailash Choudhary, Hon'ble MOS for Agriculture & Farmers' Welfare and Dr Himanshu Pathak, Secretary & Director General, on June 17, 2023.

The Sports Contingent of the Academy secured six prizes during the ICAR Inter-Zonal Sports Meet-2022 organized at ICAR-NDRI during September, 2023; and also at ICAR South Zone Sports Meet 2023 organized at ICAR-IIHR Bengaluru during December 2023. NAARM organized a blood donation camp on 31st August, 2023 with the help of PVNR Memorial Trust and Blood Centre. ICAR-NAARM, bagged the Silver Garden Certificate Award in the 7th Garden Festival by Govt. of Telangana for maintaining the institute

landscape, gardens and lawns of the Academy in a best presentable manner. Dr Ch. Srinivasa Rao, Director, NAARM, was bestowed with fellowship of Indian National Science Academy (INSA)-2023. Further, the staff and faculty of the Academy won awards and accolades for their academic work and excellence, and many were nominated as members / Chairman for various committees of ICAR and other organizations. NAARM is further targeting to reach new heights in skilling human resources of NARES.

ICAR-NAARM: An Overview

1

The National Academy of Agricultural Research Management (NAARM), located in Hyderabad, was established by the Indian Council of Agricultural Research (ICAR) in 1976. It operates within the National Agricultural Research and Education System (NARES) with the primary goal of enhancing capacity in agricultural research, education, and extension education. The Academy is dedicated to offering policy advocacy and conducts various capacity-building programs for researchers, academics, extension specialists, scholars, students, and other stakeholders in the agricultural research and education system.

NAARM aims to increase both institutional and individual capacity for innovation within NARES, recognizing the crucial role of agricultural

research in the country's food security and economic prosperity. It emphasizes leadership, governance, and innovation as essential elements for transforming NARES into a more pluralistic innovation system. By fostering a scientific work culture, the academy seeks to strengthen the sense of community among agriculture scientists and faculty nationwide. NAARM acts as a catalyst for the intended change within NARES to address current and future challenges effectively.

The Academy utilizes its expertise in managing human resources in agricultural research to advance agribusiness management, technology management, and education technology management. The Academy offers higher education programmes such as the Post



Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) with AICTE approval and accreditation from the National Board of Accreditation (NBA).

The academy's Technology Business Incubator, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture), has gained national recognition for its innovative approach to incubating Agri Startups. Through a comprehensive approach involving capacity building, education, research, consultancy, and policy support, NAARM is committed to strengthening the leadership, governance, and innovation capacities of NARES. It aspires to be an institution of excellence facilitating a culture of dynamic management in agricultural research, education, and frontline extension. Positioned as a think tank for agriculture, NAARM actively contributes to plans aimed at improving the National Agricultural Research and Education System and ensuring the long-term viability of Indian agriculture.

1.1 Strategic Framework

The Academy built a six-tier strategic framework by its Vision, Mission, and Mandate to guide various activities towards the defined goals and targets.

These are:

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and agribusiness management education and research in NARES.
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses.

2. Education System Management for Enriched Learning Quality and Environment

- Enhance capacities for agricultural education policy, planning, and evaluation in institutions of NARES.
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

3. Enhancing Capacities for Leadership and Governance

- Institutionalize a framework for leadership development at all levels of NARES - early, mid-career, senior professional, and manager across all functions (research, education, extension)
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation.

4. Extension Systems Management in a Market-driven Environment

- Enhance the operational, adaptive, and generational capacity of front-line agricultural extension systems to address emerging challenges.
- Enhance capacities for the use of ICTs to provide customized knowledge, skills, and solutions to farmers, farmer groups, rural communities, and social networking.

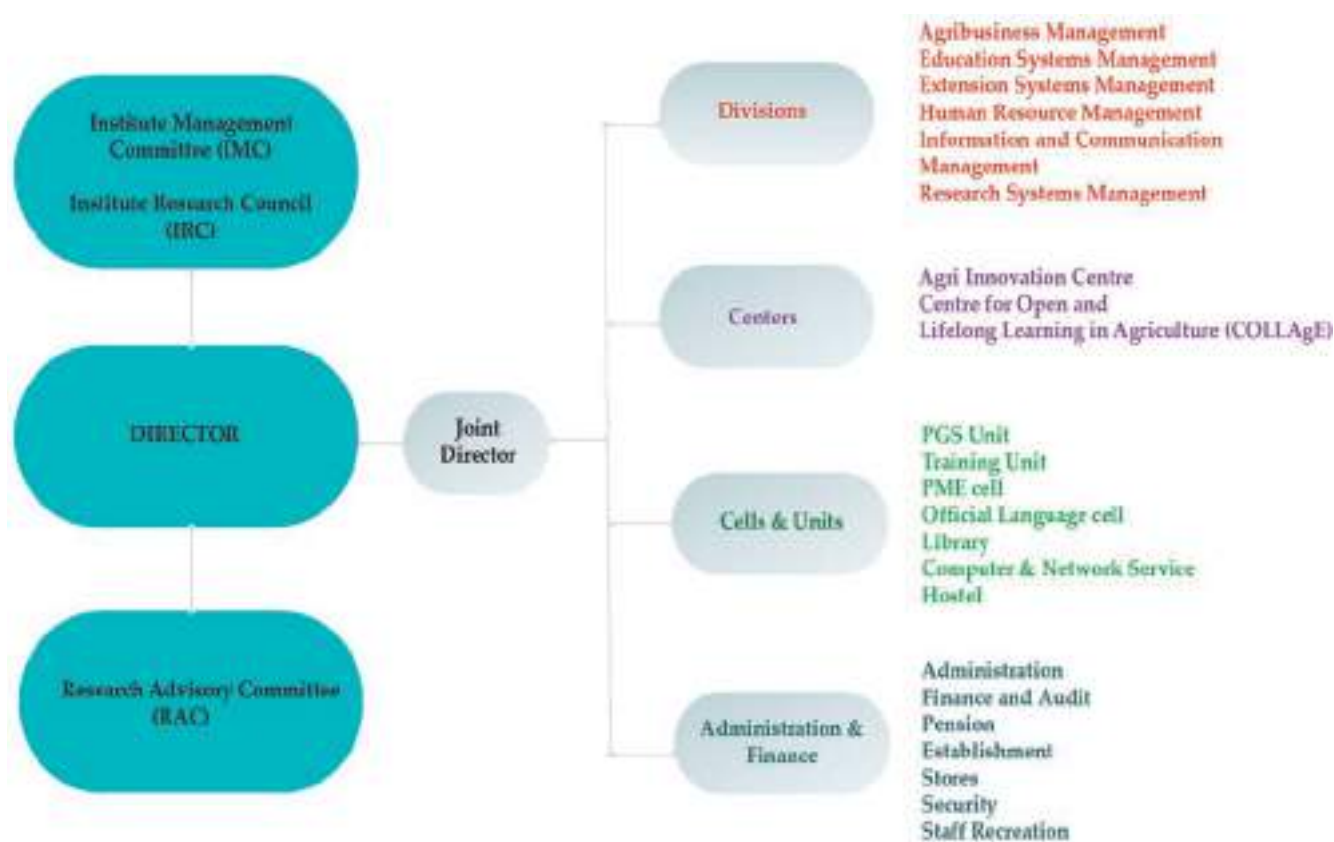
5. Information and Communication Management for Promoting Innovation and Governance

- Enhance capacities for integrating ICTs and the Internet of Things (IoTs) in institutional management and governance.
- Enhance capacities for managing data, information, and knowledge in agriculture, scientific communication, and institutional governance.

6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring, and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change, and globalization.
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies.

1.2 Organization and Management



1.2.1 Research Advisory Committee (RAC)

The Indian Council of Agricultural Research (ICAR) constituted a new committee for IX Research Advisory Committee (RAC) of ICAR-NAARM for a period of three years w.e.f 22.06.2023 as given below.

Dr. Govinda Rajulu Chintala Former Chairman, NABARD	Chairman
Shri Mohan Kumar Mishra Secretary, National Council for Cooperative Training (NCCT), Ministry of Cooperation, Govt. of India, New Delhi	Member
Dr. Gopala Krishna Vice President, Nagarjuna Fertilizers Pvt. Ltd.	Member
Dr. P. K. Joshi Former Director, NAARM, Hyderabad	Member
Dr. Arvind Kumar (Padmasri Awardee) Ex-VC, RLBCAU, Jhansi	Member
Dr. Ch. Srinivasa Rao Director of the Institute	Member
Dr. R.C. Agrawal DDG (Education), Indian Council of Agricultural Research, Krishi Anusandhan Bhawan II, Pusa, New Delhi	Member
Two non-official members representing agricultural/rural interests on the Management Committee of the Institute for the period of their membership of the Institute Management Committee.	Member
Dr. Tavva Srinivas Head, PME and Head, Division of Agribusiness Management, ICAR-NAARM, Hyderabad.	Member Secretary

1.3 Linkages/Networking/ Collaborations

NAARM has a strong strategic and functional network with 113 ICAR Research Institutes and 76 Agricultural Universities (AUs) and plays a key role in developing the capability of these organisations, the agribusiness industry, scientists, and academia. Its network includes numerous institutions at the national and international levels.

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as members of the Institute Management Committee of NAARM for a period of three years w.e.f. 02.08.2023.

Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Chairman
Director of Agriculture, Department of Agriculture, Government of Telangana	Member
Spl Commissioner and Director, Department of Agriculture, Government of Andhra Pradesh	Member
Director (Research), Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	Member
Dr. Sheikh N. Meera, Director, ATARI, ZONE-X, Hyderabad	Member
Dr. G. Byju, Director, CTCRI, Thiruvananthapuram, Kerala	Member
Dr. C.N. Ravishankar, Director, CIFE, Mumbai	Member
Dr. O.P. Yadav, Director, CAZRI, Jodhpur	Member
Dr (Mrs) Seema Jaggi, ADG (HRD), ICAR, KAB-II, New Delhi	Member
AS & FA, DARE/ICAR	Member
CAO (Sr Grade), NAARM	Member Secretary

1.3.1 National Institutes/ Organizations

1. All ICAR institutes and Agricultural Universities.
2. Various departments of the Government of India like NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), Capacity Building Commission, National Science & Technology Entrepreneurship Development Board (NSTEB).

3. The University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.
4. Ministry of Environment, Forest and Climate Change (MoEF & CC), National Board of Accreditation (NBA), Protection of Plant Varieties & Farmers' Rights Authority (PPV&FRA), National Fisheries Development Board (NFDB) and State Biodiversity Boards.
5. Management Institutions located at Hyderabad viz. the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD&PR), and Institute of Public Enterprise (IPE); and across the country such as Indian Institutes of Management (IIMs).
6. State Departments of Agriculture of various States.
7. Private Sector Organizations (Agri Input Companies, Service Companies, etc.).
8. Non-Government Organizations (NGOs).
9. Other organizations viz. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), National Bank for Agricultural and Rural Development (NABARD) Centre for Innovation Incubation and Entrepreneurship (CIIE), etc.
10. Krishi Vigyan Kendras (KVKs).

1.3.2 International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
2. World Bank, FAO, UNDP, and other United Nations Organizations
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka, and other ASEAN countries.
6. NARS in other African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia and the Philippines.
7. Leading Land Grant and State Universities of the USA and other Universities in Europe, Australia, and other developing and developed countries.

The Academy's mission revolves around innovation, governance, and knowledge, guided by its vision documents (2030 and 2050) and those of ICAR. Recognizing the pivotal role of emerging science and technologies in various sectors, the Academy emphasizes their integration into the agricultural value chain. The changing landscape of agri-education, with a shift from 'teaching to learning outcome,' underscores the need for new institutional mechanisms and enhanced system capacities. The Academy also underscores the importance of continuous stakeholder engagement to improve the performance of NARES, fostering transparency and accountability. Consequently, research, capacity building, and education activities align with a six-tier strategic approach. By implementing this strategy, the Academy aims to catalyze the planned transformation of NARES. The subsequent sections detail the annual progress and accomplishments in research, capacity building, and other activities within the six strategic areas for the year 2023.

2.1 Agribusiness Management for Inclusive Growth

The paradigm shift in agricultural development strategies, transitioning from a narrow production-centric approach to a holistic systems outlook, highlights the escalating significance of the agribusiness sector. Recognized as a catalyst for economic advancement, agribusiness, and its

ancillary industries are now central to policies and strategies aimed at fostering investments in agro-enterprises and augmenting agro-based value chains. This evolution necessitates a comprehensive understanding of agribusinesses, agro-industries, and agri-food value chains, underscoring the theme of agribusiness management for inclusive growth. This theme encompasses multifaceted initiatives spanning research, education, capacity building, and consulting, all geared toward cultivating future leaders in agribusiness.

The overarching objective is to bolster capacities in food and agribusiness management education and research within NARES (National Agricultural Research and Education Systems) and to fortify entrepreneurship development and strategic management within agribusinesses. The confluence of rapid urbanization, increasing household disposable incomes, shifting food preferences, and the globalization of the agricultural economy amplifies the imperative for research and capacity building in agribusiness management. To effectively address these imperatives, the Academy prioritizes the following focal areas for research, capacity building, and education:

1. Agribusiness Environment: Emphasizing value chain analysis for agri-commodities and services, understanding consumers' demands, analyzing price behavior, and forecasting trends.

2. **Public Policy and Impact:** Investigating market reforms, especially in the agriculture sector, such as new agriculture policy, Electronic National Agriculture Market (e-NAM), etc., and their ramifications on the agri-startup ecosystem in the country, exploring financial inclusion initiatives and their spillover effects, and evaluating the dissemination and convergence of crop insurance schemes.
3. **Markets and Institutional Innovation:** Examining market integration dynamics of commodities, assessing the role of Farmer Producer Companies (FPCs) and Farmer Producer Organizations (FPOs), optimizing inputs and service delivery systems, exploring diversification and intensification strategies for high-value commodities, and leveraging business analytics for informed decision-making.
4. **Nurturing Future Agribusiness Leaders:** Fostering quality education in agribusiness management to equip aspiring leaders with the requisite knowledge and skills.

2.1.1 Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/Value-Added Products

Among the top ten technologies identified for preparing the techno-economic feasibility report, the process for production of the microbial method for the production of protein isolate/concentrate from oilseed cakes/meals and the Live Fish Carrier System (LFCS) were studied during the reporting period. Collected process details of the technologies viz., Live Fish Carrier System (LFCS) and microbial method for production of protein isolate/concentrate

from oilseed cakes/meals in consultation with Agrinnovate India Ltd. and ICAR-CIPHET, Ludhiana, Punjab. Traditionally the live fish is transported in mini trucks that are modified with polythene lining for holding water. Non-availability of the aeration system and filtration system and due to the continuous discharge of excreta by the fish, the fishers have to face economic loss due to sudden high mortality of fish in such carriers which may go up to 100%. The carriers are prone to accidents and overturns due to faulty design. To sell the fish in live condition and to earn more income, the Live Fish Carrier System (LFCS) was developed by ICAR-CIPHET, Ludhiana, Punjab. It has automatic aeration, filtration, and evaporative cooling systems ensuring continuous availability of good quality water and less fish mortality (<1%). As per the reports, the cost of the vehicle would be approx ₹ 2.0 lakh for the complete system with a discounted payback period of 0.3 years (3.59 months) and an Internal Rate of Return (IRR) of 354.5% indicating a highly profitable venture. The ICAR-CIPHET (Ludhiana) has licensed this technology to E-Magic Electric, Jalandhar through Agrinnovate India Ltd., New Delhi. This technology has also been accepted by the International Fund for Agricultural Development (IFAD) for further promotion.

Microbial method for production of protein isolate/ concentrate from oilseed cakes/meals: The plant protein market of India is currently estimated as Rs 2850/- crores and forecasted to Rs 4300/- crores by 2023 at a CAGR of 8.6%. Plant protein is used in protein supplements, texturized vegetable proteins, imitation dairy products, seafood products, the beverage industry, infant food formulations, weaning food

formulations, bakery products, and meat analogs for various purposes. Globally, alkaline extraction followed by acid precipitation or ultrafiltration method is followed. The strong alkaline and acidic conditions alter the functional properties of the protein, which adversely affects its quality. This unique process developed by ICAR-CIPHET comprises novel bacterial strains isolated from a food sample for producing protein from de-oiled meal/flour and it provides about 5% higher yield of protein as compared to the existing chemical process and the extracted protein is of superior quality in terms of solubility, wettability, water absorption capacity and degree of hydrolysis. The technology was transferred to two industries i.e., M/s Samyog Health Foods, Navi Mumbai, and M/s Trishveda Naturals Pvt. Ltd, Ludhiana, Punjab for commercialization through Agrinnovate India Ltd. (AgIn), New Delhi.

2.1.2 New Market Reforms and E-NAM in India: Effect on Marketing Choices and Price Realization for Smallholders

Over the decades, the country has transitioned from food deficiency to a state of food surplus, challenging the realization of profitable returns on farm investments. In April 2016, e-NAM was introduced to integrate all the physically dispersed agricultural markets, viz. APMC mandis. Currently as on Dec 31, 2023, a total of 1389 mandis across 23 states and 4 Union Territories (UTs) are brought on the e-NAM platform. This was the first step towards the “One Nation, One Market” concept to bring both farmers and different trade functionaries on a single platform, resulting in more transparent and competitive price discovery for agricultural commodities. The new market reforms were

expected to create an ecosystem wherein the farmers and traders enjoy the freedom of choice relating to the sale and purchase of agricultural produce. Such an ecosystem seamlessly connected required new investment in infrastructure catering to the needs of purchasers including traders, processors, exporters, etc. enriching the value chain. In 2023, the Govt of India has also constituted a high-level expert committee to promote inter-mandi and inter-state trade through Market Yard of National Importance (MNI). This platform will provide an opportunity for farmers of the participating states to sell their surplus produce beyond its state boundaries. It would enable to creation of digital ecosystems that leverage the expertise of various segments of the agriculture value chain.

During the reporting period, the capacity development in warehouses in different states during the last 5 years was analyzed and also compared the price received by the farmers for their sale in e-NAM and non-e-NAM mandis. In recent years, the number of warehouses registered with the Warehousing Development Regulatory Authority (WDRA) has significantly increased, however, the average capacity of these warehouses has decreased. The reason for this may be more warehouses with smaller capacities coming up in recent years, while earlier large warehouses under state and central agencies got the registration. Similarly, there is a large variation in warehouse capacity among all the states of India.

The study recommends more incentives for investment in warehouses for agricultural purposes, particularly connected mandis. Several functional agricultural mandis is also required to be expanded. The back-end infrastructure of

most of the mandis requires serious large-scale investment for the functioning of e-NAM, which comprises warehousing, cold storage, assaying

laboratories, grading facilities, and certification mechanisms among others.



Figure 2.1.1a: WDRA registration of warehouses in India

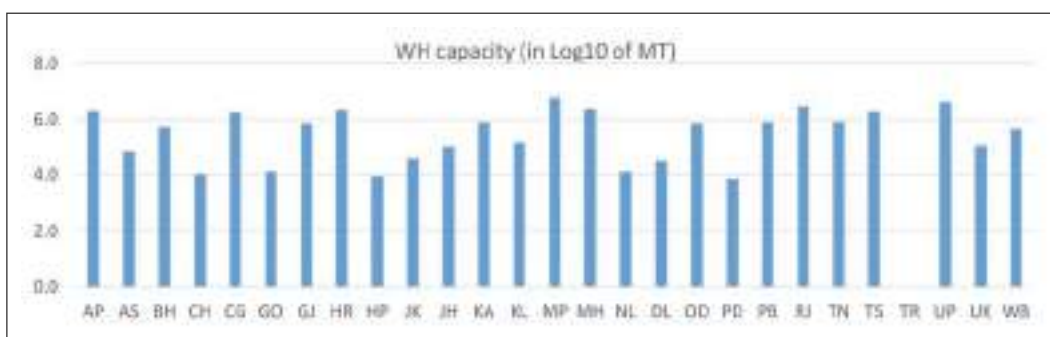


Figure 2.1.1b: State-wise warehouse capacity registered during 2021- 2023 (in Log10 of MT)

2.1.3 Agri Startups in India: Decoding the Determinants and Success Factors

In recent times, the agricultural sector, encompassing crops, livestock, and fisheries, has experienced a surge in the establishment of numerous startups. Concurrently, the government has introduced several initiatives aimed at fostering the growth of these startups. Against this backdrop, this study was initiated with the following objectives: to explore the motivations driving the entry of entrepreneurs into the agricultural startup landscape, to scrutinize the operational processes adopted by agricultural startups, to identify the key factors influencing the success of such startups, and to present case studies exemplifying instances

of successful agricultural startups. While the government recognizes all types of registered entities, all studied startups operate as private limited companies. The sector witnessed a gradual rise in new ventures, peaking in 2019 (29%) and 2017 (25%), but dipped slightly in recent years possibly due to the pandemic. Unlike the average young founder across industries, most agri-startup founders (54%) fall within the 30-40 year age group. Typically, startups have two founders and a majority boast STEM credentials: 56% are engineers, and 48% hold management degrees, with 15% possessing both qualifications (Figure).

Launching an agri-startup was a deliberate choice for most founders, driven by identified business

opportunities (Figure). Analyzing their target clientele, B2B models lead the pack, followed by B2C and B2G initiatives. Notably, a sizeable 78% of startups cater to actors downstream in the agricultural supply chain, highlighting their perceived potential. Farmers themselves directly benefit from only 22% of these ventures.

Technology solutions take center stage, with 44% of startups offering tech-based products and services to their customers. However, early-stage ventures face obstacles like limited capital access, insufficient guidance and planning, difficulties in talent acquisition, and challenges in maintaining customer relationships. As startups mature and progress through the traction and scaling-up phases, their organizational structures solidify, and innovation in products or processes becomes more evident.



Figure 2.1.2a: Founder's Educational Background

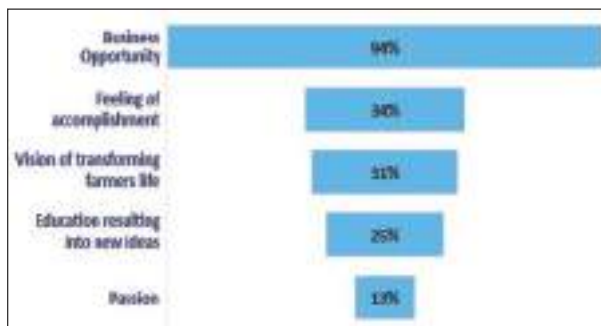


Figure 2.1.2b: Motivation for Entering into Startup Space

2.1.4 Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

The study aimed to enumerate the socio-economic profile of small ruminant farmers and conduct a thorough livelihood analysis. Secondly, it delved into the examination of production and marketing practices adopted by these farmers concerning small ruminants. Thirdly, the study sought to identify the constraints faced by small ruminant farmers within the value chains in the designated study area. Lastly, the research aimed to provide valuable insights by suggesting strategies and policies that could empower small ruminant farmers to enhance their income through the adoption of efficient value chains. This comprehensive investigation underscored the commitment to understanding and addressing the challenges faced by small ruminant farmers in India while working towards sustainable and income-enhancing solutions in the context of value chains.

2.1.4.1 Production practices of sheep farmers

Cent percent of the farmers in the study area were rearing the sheep in an extensive system. The mean grazing time (hrs) and distance (km) in the study area was 9.41 hrs and 5.98 km per day respectively. The majority (82.50%) of sheep farmers didn't follow supplementary feeding while only 17.50 % of the sheep farmers were providing supplementary feeding. Homegrown ewes were the source of ewes for majority (92.50%) of sheep farmers.

2.1.4.2 Marketing pattern of sheep

The majority (70.00%) of the farmers marketed their sheep at 3-6 months of age, followed by 17.50 % at 9-12 months of age and 12.50 % at 7-8 months of age respectively (Figure 2.1.3a).

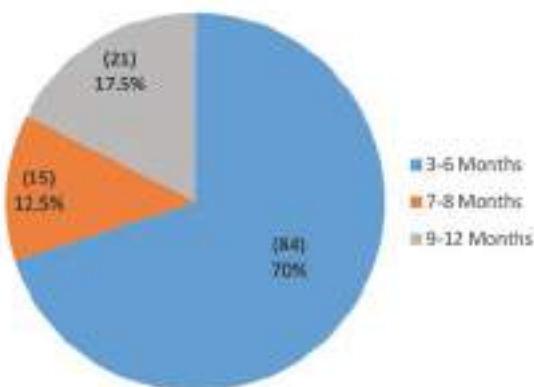


Figure 2.1.3a: Marketing age of sheep in Telangana

Figures in parenthesis indicate the number of sheep farmer
(Source: Primary Survey, 2019)

Marketing pattern revealed that nearly one-third (34.17%) of the farmers sold their sheep to the “local traders + outstation traders”, followed by 29.17%, 15.83%, 5.83%, 5.00%, 4.16% and 3.33% selling their sheep to “local traders”, “local traders + consumers”, “outstation traders”, “local traders + consumers”, “outstation traders + consumers” and “local traders + consumers + outstation traders + consumers” respectively.

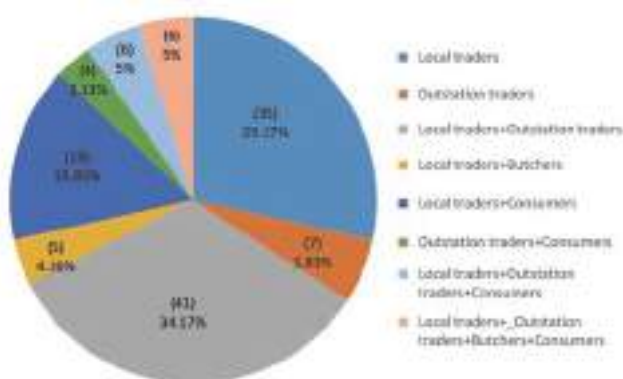


Figure 2.1.3b: Marketing Pattern in the Sale of Sheep in Telangana

Figures in parenthesis indicate the number of sheep farmers
(Source: Primary Survey, 2019)

traders + outstation traders + consumers”, “local traders + butchers” and “outstation traders + consumers” respectively (Figure 2.1.3b). The majority (91.67%) of the shepherds sold their sheep herds at their homes followed by 5.83 % of the farmers in livestock markets and 2.50 % of the farmers in haat. The majority of the farmers (75.00%) expressed that the price for marketable stock was fixed based on physical appearance.

2.1.4.3 Production practices of goat farmers

Cent percent of the farmers in the study area were rearing goats under a semi-intensive system.

2.1.4.4 Marketing pattern of goat

The majority (69%) of the farmers marketed their goat at 9-12 months of age, followed by 17% at less than 9 months of age and 13% at 12-18 months of age, and the remaining 1% at more than 18 months of age, respectively (Fig. 2.1.4a).



Figure 2.1.4a: Marketing age of goat in Rajasthan

Figures in parenthesis indicate the number of goat farmers
(Source: Primary Survey, 2021)

Marketing pattern revealed that about 61% of the farmers sold their goats to the local traders, followed by 20%, 14% and 5% of them found selling their goats to outstation traders, butchers, and consumers, respectively.



Figure 2.1.4b: Marketing Pattern in the Sale of goat in Rajasthan

*Figures in parenthesis indicate the number of goat farmers
(Source: Primary Survey, 2021)*

It is understood that small ruminant farmers are found still far away from modern-efficient markets. The main reason is the absence of any institutional support and innovations in goat farming and markets as in dairy and poultry. Efforts should be made to formalize the relationship/contract between farmers and village traders.

2.1.5 Farmer Producer Companies in India: A study on their management practices and business potential

The study has made notable progress by delving into the operations of 10 Farmer Producer Companies (FPCs) in Tamil Nadu and four FPCs in Telangana. A comprehensive survey involving 250 farmer members was conducted to gain insights into the intricacies of FPCs. The study has focused on highlighting the strengths and weaknesses of these FPCs, offering a nuanced understanding of their management practices. Additionally, the research endeavors to estimate the benefits accrued to the members through their active participation in these FPCs.

2.1.6 Developing and Pilot Testing Specialized Knowledge Products on Human-Wildlife Conflict Mitigation for Agriculture and Veterinary sector in India

In order to accomplish the objectives of the project, various activities were carried out during the reporting period, A Field Mission to Rajaji Landscape was organized. As a part of the project, a Virtual Training was organized for Faculty and Experts of the State Agriculture Universities and KVKs on “Holistic Approach to Human-Wildlife Conflict Mitigation and One Health” from 15-16 May 2023 where 31 faculty members of eight State Agriculture Universities; five KVKs and two ICAR institutions participated. Besides, training sessions were conducted for the Rapid Response Teams (RRT) and Primary Response Teams (PRT), Panchayat and Forest Department staff at Kodagu and Gorumara landscapes. The information collected during the field missions to Kodagu, Gorumara, and Rajaji landscapes and different project activities enabled in bringing out a technical report on “Business Potential, Plan and Value Chain of Alternative Crops in HWC Landscapes”; 20 case studies; Six Crop Fact Sheets; eight Implementers Tool Kits on Agriculture; Participants Handbook on Reducing Vulnerability; Desk Report and Training materials on four Sessions including one health and HWC mitigation in the CBP of the Academy.

2.1.7 Training Program on Soft Skills and Entrepreneurship Development for Students

An off-campus training program on “Soft Skills and Entrepreneurship Development” was organized during 03 – 05 January 2023 at the College of Dairy Science, Amreli, Kamdhenu

University, Gujarat. A total of 120 B.Tech, Dairy Technology students including 19 girls from Dairy Science College, Amreli attended the program. The program covered the broad areas of soft skill development, business idea generation, plan preparation, and business plan pitching wherein participants presented their business ideas. The sessions were taken by NAARM faculty as well as guest faculty. The valedictory session was graced by Dr NH Kelawala, Hon'ble

Vice Chancellor, Kamdhenu University, Gujarat, and Dr. G Venkateshwarlu, Joint Director, ICAR-NAARM. During his valedictory address, Dr Kelawala highlighted the opportunities available as an entrepreneur in the dairy sector. Dr. Venkateshwarlu briefed about a few entrepreneurs in the dairy sector and suggested the students to take cues from them and become job providers rather than job seekers.



Figure 2.1.5: Participants of Training Program on Soft Skills and Entrepreneurship Development

2.1.8 Skills and Entrepreneurship Development for Faculty

An off-campus training on “Soft Skills and Entrepreneurship Development” for the Faculty of Kamadhenu University was organized from 08-10 August 2023. A total of 25 Faculty including 13 women participants in online mode attended the program representing five colleges under Kamadhenu University viz., Polytechnic in

Animal Husbandry, Sardarkrushinagar; Fisheries Science College, Veraval; SMC College of Dairy Science, Anand; Fisheries Research Station, Okha; and College of Dairy Science, Amreli. The program was sponsored by Kamadhenu University. The objective of the program is to promote and nurture entrepreneurship abilities in the Faculty along with equipping them with necessary soft skills.



Figure 2.1.6: Participants of Training Programme on Soft Skills and Entrepreneurship Development

2.1.9 Training on “Entrepreneurial Skill Development” (For B.Tech, Dairy Technology students of NDRI, Karnal)

The training program on “Entrepreneurial Skill Development” for B.Tech Dairy Technology students of NDRI, Karnal was organized from 18th September - 19th October 2023.

The program was aimed to achieve several objectives, including providing insights into Agri-business, particularly in the Dairy Sector, fostering entrepreneurship and innovative thinking skills, and equipping participants with the ability to develop business plans, DPRs, and effective funding pitches. Additionally, the program focused on the holistic development of participants, enhancing their personality, communication, and other soft skills. The program accommodated 25 students, with 2 of them being girls. The program comprised 58 sessions that encompassed both theoretical and practical aspects and covered diverse topics such as Health and lifestyle management, Business Plan Development, Marketing in the Dairy sector, Data analysis software, Business Communication, and Startup Ecosystem, along with case studies, discussions, and Business Plan Development, etc.



Figure 2.1.7: Entrepreneurial Skill Development for Students of NDRI, Karnal

2.1.10 Collaborative Capability Building Programme for Young Agri-Business Professionals, New Entrants Assimilation Training (NEAT) - Phase II for Officials of M/S Coromandel International Ltd.

The Academy organized the training program during 13-17 February 2023. The main aim of the program was to provide a basic understanding

of agriecosystem and to upgrade the knowledge and skill in the sales and marketing domain. A total of 33 sales and marketing officers of CIL representing 10 states participated in the training program. In his valedictory address, Dr G Venkateshwarlu, Joint Director, ICAR-NAARM highlighted the importance of such partnership between academia and the corporate sector for the dissemination of the latest knowledge and tools in the interest of the farming community.



Figure 2.1.8: Participants of NEAT, Phase II, Batch-1

2.1.11 Collaborative Capacity Building Programme for Young Agri-Business Professionals. New Entrants Assimilation Training (NEAT) Phase II for Officials of M/S Coromandel International Ltd.

NEAT phase II Batch 2 – a capacity building program was held at ICAR – NAARM, Hyderabad in collaboration with Coromandel Sales Force Learning Academy from 20-24 February 2023. This 5-day training capsule was designed to cater to the job requirements of these young Agri-business professionals who were either associated with marketing or Agronomic activities conducting demonstrations on farmers' fields to prove the worth of their company product to the dealers & retailers. The training sessions focussed on business and marketing aspects, creativity, problem-solving,

etc. along with some core concepts of soil health-based nutrient application & marketing and opportunities in the fertilizer industry. In all, there were 29 Agri-business professionals out of which 15 were associated with Marketing and 14 were Agronomists. They represented 11 states of our country namely AP, Telangana, Bihar, Karnataka, MP, Odisha, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh & West Bengal.



Figure 2.1.9: Participants of NEAT, Phase II, Batch-2

2.1.12 Collaborative Capacity Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (NEAT) Phase II Batch 3 for Officials of M/s Coromandel International Ltd.

Academy organized this Programme from 13 – 17 March 2023. The objectives of the program were to promote and nurture new entrants to the organization and to equip them with management tools needed for making successful managers.

The training program included sessions such as Time Management, Digital Marketing Techniques, Target Setting, Delegation, Creativity and Experiential Learning, Fertilizer Industry: New Opportunities, Soil health based nutrient application and marketing, Public Speaking-Video recording of individual

performance and review. Creativity and Problem-Solving, Work-Life Balance in Organizations, Farmers' buying Behavior-Nutrients and Agri-inputs; Improving Self Confidence; Group Dynamics and intra-team interactions; MS Excel for day to day business; Personality Quotients for Professional Excellence; Entrepreneurial Skill for Excellence; Health and Life Style Management. Two Sessions viz., Farmer Success Stories & ROI and New Business Initiatives were handled by the M/S Coromandel International Ltd. Practical handouts were shared with the participants during practice sessions. Practical exercises were done during the sessions on Group Dynamics and intra-team interactions; MS Excel for day-to-day business; and Public Speaking Video making.



Figure 2.1.10a: Valedictory Function of the NEAT Phase II Batch 3 Training Programme for Officials of M/s Coromandel International Ltd.



Figure 2.1.10b: Participants of Collaborative Capacity Building Programme for Young Agri-Business Professionals NEAT Phase II Batch 3

Participants were made into different groups to practice the Hands-on exercises. After everyday training, the participants were asked to give answers to the questions from the respective day sessions through Google Forms.

2.1.13 Mentoring and Training program for Agri startups (ABIIGROW 1.0)

Under the NAIF-ABI Scheme of ICAR, a-IDEA organized an ABIIGROW-Mentoring and training program from 15-17 March 2023 for Agri startups supported by ICAR ABIs and a-IDEA. The objective of the program was to enlighten startups on financial and business aspects and to create an ideal networking platform for startups from agriculture and allied sectors. Total 40 participants from 33 startups participated in this training program.



Figure 2.1.11: Participants of Training program Agri startups (ABIIGROW 1.0)

2.1.14 Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs) in India

The National Academy of Agricultural Sciences (NAAS) and ICAR- NAARM, Hyderabad jointly hosted a Brainstorming Session on “Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private

Partnerships (PPPs) in India” at NASC, New Delhi on 25 May 2023, with Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM as convener and by Dr. G. Venkateswarlu, JD and Dr. D. Thammi Raju, Principal Scientist of NAARM as co-conveners. The session, chaired by Dr. Himanshu Pathak, President of NAAS, and Secretary- of DARE and DG-ICAR, aimed to foster collaboration between public and private partners and propose solutions to challenges in utilizing PPPs in agriculture. The session began with Prof K C Bansal, Secretary of NAAS, highlighting NAAS’s role as a think-tank and policy-making academy. Presentations were made by Dr. Ch. Srinivasa Rao, Dr. D. Thammi Raju, and Dr. Tushar Pandey, discussing the need for PPPs in agriculture, successful models in different sectors, and areas for PPP implementation. Valuable insights and recommendations were shared by panelists during the session. Dr. Himanshu Pathak emphasized the importance of strong partnerships with the private sector from problem identification to technology commercialization. He emphasized the significance of the session’s recommendations in finalizing guidelines for successful PPP implementation in the agriculture sector.



Figure 2.1.12: Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs) in India

2.1.15 Collaborative Capacity Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (Phase -1) Batch I for Officials of M/s Coromandel International Ltd.

The Collaborative Capacity Building Programme for Young Agri-Business Professionals, specifically the New Entrants Assimilation Training (Phase -1) Batch I, spanned from 5-9 June 2023, encompassing five days with a total participation of 23 individuals. The principal objective of the program was to provide foundational training to recently appointed officials of CIL. The training approach incorporated a diverse pedagogy involving lectures, case studies, management games, and guest lectures in the domains of crop science and agribusiness management. Additionally, participants engaged in hands-on activities focused on communication and team building. The program also ensured that participants were brought up to speed with the latest developments in agronomy, crop protection, weed science, and fertilizers, fostering a comprehensive understanding of crucial aspects in the agricultural sector.



Figure 2.1.13: Participants of NEAT, Phase-1

2.1.16 Collaborative Capability Building Programme for Young Agri-Business Professionals of CIL MORE (Marketing Officers' Resurgence & Excellence) for Officials of M/s Coromandel International Ltd.

The Academy organized a training program on Collaborative Capability Building Program for Young Agri-Business Professionals of CIL, MORE (Marketing Officers' Resurgence & Excellence) in collaboration with Coromandel International Limited from 4 - 8 September 2023. The main aim of the program was to provide a basic understanding of agroecosystems and to upgrade knowledge and skills in the sales and marketing domain. In all, there were 29 Agri-business professionals, Assistant managers, senior marketing officers, and Agronomists of CIL representing 11 states participated in the training program. In his valedictory address, Dr



Figure 2.1.14: Participants of Collaborative Capability Building Programme for Young Agri-Business Professionals of CIL MORE (Marketing Officers' Resurgence & Excellence)

G Venkateshwarlu, Joint Director, ICAR-NAARM highlighted the importance of such partnership between academia and the corporate sector for the dissemination of the latest knowledge and tools in the interest of the farming community.

2.1.17 Collaborative Capability Building Programme for Young Agri-Business Professionals: New Entrants Assimilation Training (NEAT-II) for Officials of M/s Coromandel International Ltd.

The academy organized the training program on “Collaborative Capability Building Programme for Young Agri-Business Professionals: New Entrants Assimilation Training (NEAT-II)” from 11 – 15 September 2023. The program was designed with key objectives to empower participants in the dynamic field of agri-business. Participants in this program are geared towards learning about essential aspects of work-life balance within organizations, honing effective communication skills crucial for business managers, cultivating awareness of climate change and nutrient management, and understanding the dynamics of group interactions in the workplace. Furthermore, the program emphasizes the development of entrepreneurial skills, excellence in target setting, effective delegation, and fostering creativity through experiential learning activities. Hands-on engagement in public speaking, coupled with constructive reviews, contributes to the practical knowledge and confidence-building



Figure 2.1.15: Participants of Collaborative Capability Building Programme for Young Agri-Business Professionals: New Entrants Assimilation Training (NEAT-II)

of participants, equipping them with valuable skills for success in the agri-business sector. The collaborative nature of this capability-building initiative ensures that young professionals receive a comprehensive and practical education, enhancing their capabilities and preparing them for challenges in the agri-business realm.

2.1.18 Collaborative Capability Building Program for Young Agri-Business Professionals MORE (Marketing Officers’ Resurgence & Excellence for Officials of M/s Coromandel International Ltd.

This Program is a 5-day (09-13 October 2023) capacity-building program (MORE) that has a mix of lectures, case studies, and management and data Visualization practicals. The participants were also updated with the latest developments in field crops, climate change, marketing, and entrepreneurship. Every day, participants shared their learnings on the previous day through assessment tests and also feedback of respective classes. The participants were also trained on the design thinking processes for boosting sales, teamwork, stress and conflict management, decision-making, virtual leadership, data visualization and interpretation, agribusiness, and entrepreneurship.



Figure 2.1.16: Participants of Collaborative Capability Building Program for Young Agri-Business Professionals MORE (Marketing Officers’ Resurgence & Excellence)

2.1.19 Collaborative Capability Building Program for Marketing Officers [NEAT] – Phase II

The academy organized the Collaborative Capability Building Program for Young Agri-Business Professionals from 20 – 24 November 2023 with 28 participants. This training program was meant for the employees of Coromandel International Limited (CIL) who had undergone NEAT Phase 1 training earlier. A total of 28 participants nominated by the CIL working as Officers (Marketing, Logistics, etc), Senior Marketing Officers, Agronomists, and Assistant Managers participated in the training. During the programme, the participants were given insights on improving different aspects like soft skills, domain knowledge, and quantitative techniques. Tests were conducted every day to assess the learnings of the trainees.



Figure 2.1.17: Participants of Collaborative Capability Building Program for Marketing Officers [NEAT] – Phase II

2.1.20 Collaborative Capability Building Program for Young Agri-Business Professionals Front Line Agri-Marketing Excellence (FLAME) for Officials of M/s Coromandel International Ltd.

The academy organized the Collaborative Capability Building Program for Young Agri-Business Professionals Front Line Agri-

Marketing Excellence (FLAME) program from 04 - 08 December 2023. The course was designed for the marketing executives of Coromandel International to identify the problem, develop viable solutions through their products, and increase the sales of Coromandel International Ltd by these young agribusiness professionals. The objective was to enhance the competency and improve marketing proficiency of the newly recruited Marketing Executive Trainees of Coromandel International Ltd.

2.1.21 Training program on Entrepreneurial Skill Development for Students of RVSKVV, Gwalior

The academy organized an Entrepreneurial Skill Development program from 04 - 23 December 2023 to impart knowledge, on entrepreneurial skill development to the B.Sc (Agri/Hort) students of RVSKVV, Gwalior. The program had theoretical as well as demonstrations, where necessary, on the subject including business plan presentations by the participants at the end of the training.



Figure 2.1.18: Participants of the Entrepreneurial Skill Development Program for students of RVSKVV, Gwalior

2.1.22 One-Day Workshop on “Development of Soft Skills for Entrepreneurship among Agricultural Graduates”

A One-day workshop on “Development of Soft Skills for Entrepreneurship among Agricultural

Graduates” was organized at Ch. Chhotu Ram (P.G.) College, Muzaffarnagar, U.P. on 05 December 2023, under World Bank supported National Agricultural Higher Education Project (NAHEP) and ICAR- National Academy of Agricultural Research (NAARM), Hyderabad. The main objective of this workshop is to develop Soft Skills for Entrepreneurship and support agri-students in rural setups in the Hindi-speaking belt of Northern India. A total of 100 agri graduates participated in this one-day workshop. Shri Kapil Dev Aggarwal, Hon’ble Minister of State (Independent Charge), Vocational Education and Skill Development Department, Govt. of U.P. also graced the occasion and addressed the students about the need for soft skills.



Figure 2.1.19: One-day workshop on the Development of Soft Skills for Entrepreneurship among Agricultural Graduates

2.2 Education Systems Management for Enriched Learning Quality and Environment

The Division was extensively involved this year in overall educational research, policy, and technology-driven capacity-building initiatives. Three projects on agricultural education and one project each on Rural Technology Action

Group (RuTAG) and Agri-business incubation were undertaken. Under the Project on Development of Implementation Framework for National Education Policy - 2020 in Agricultural Education, the extent of implementation of NEP, issues, and challenges in implementation of NEP 2020 were analyzed and interpreted. Under the research study on MOOCs in Digital Agricultural Education - Strategies to Enhance MOOCs effectiveness, the Success Rate Analysis of MOOC Participants was studied to distinguish between certified and noncertified learners. As a part of the MOOC in Dairy Education, a total of 26 modules were developed covering three courses. The division also organized both on and off-campus programs for faculty for Agricultural Universities and ICAR Institutions. Seven off-campus programs and four on-campus programs were organized covering 198 and 101 faculty respectively besides two vigilance officers programs covering 69 participants. A workshop was organized to finalize the course content for “AI in Agriculture”. Two MOOC courses were organized covering 3056 participants. The MOOC course on Commercial Dairy Farming was organized in collaboration with ICAR-NDRI, Karnal with 917 participants, and the second MOOCs on Digital Teaching Techniques was organized by ICAR-NAARM with 2139 participants and generated revenue of Rs. 8.5 lakhs.

2.2.1 Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education

The project “Advancements in Leadership for Agricultural Higher Education within the Indian Council of Agricultural Research” has made substantial progress. A total of 152 programs

were organized, benefiting approximately 18,000 students. The 5th External Advisory Panel meeting was conducted with globally renowned experts in the Agricultural Education System on 9 March 2023. Furthermore, a National Conference on Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India took place on 12-13 March 2023, with participation from 15 vice-chancellors of State Agricultural Universities (SAUs) and Principal Secretaries of State Governments. Additionally, 27 One Day Awareness programs were conducted at 27 SAUs, training around 3,000 students on Soft skills and Entrepreneurship, benefiting 61 master trainers and 6,000 students from 51 SAUs. The project also encompassed activities by 5 Career Development Centers and 1 Faculty Development Center, benefitting 15,100 students. Furthermore, data on Technology Commercialization, Career Development Infrastructure, and Inspired Teachers were collected from 52 State Agricultural Universities. The project achieved the establishment of a Smart Board Room Facility at ICAR-NAARM and organized a Training Workshop on Fostering Academic Leadership in Agricultural Education at GBPUAT, Pantnagar. Three one-day workshops on Sustenance of Career Development Centers and NAHEP component 2 were conducted in KVASU, Kerala; AAU, Anand; and CSKHPKV, Palampur. The initiative also involved promoting an Entrepreneurship Culture among Hindi medium students through two workshops in North India. Lastly, the A-I-G Model underwent pilot testing at GBPUAT, Pantnagar, Uttarakhand, in collaboration with Shobhit University, Meerut, UP, and an industry partner.

2.2.2 Development of implementation framework for National Education Policy - 2020 in Agricultural Education

The new project has been conceived to investigate awareness about the NEP 2020 among different actors of AUs and ascertain different actors' perceptions of the preparedness of Agricultural HEIs in implementing NEP – 2020. Based on the findings, it is proposed to develop a framework for the effective implementation of NEP-2020 in the Agricultural Higher Education System with special reference to philosophical, educational, organizational, and leadership perspectives. The data pertaining to the awareness, attitude, and preparedness of the faculty and administrators in the implementation of NEP 2020 were collected from 110 respondents (98 Faculty and 12 Administrators) and analyzed using suitable statistical tools and interpreted suitably. The significant achievements of the project include the development of a survey instrument to capture the awareness, attitude, and preparedness about the NEP 2020 among different actors (Faculty and Administrators) of AUs. The important findings include low awareness of NEP among Assistant Professors, and Associate Professors and comparatively high for HoDs and Administrators. The Assistant Professors and HoDs have a more positive attitude compared to Associate Professors and Professors. However, the administrators have high preparedness for the implementation of NEP. The extent of implementation of NEP, issues, and challenges in the implementation of NEP 2020 were also analyzed and interpreted.

Two workshops on the Development of an implementation framework for National Education Policy - 2020 in Agricultural Education were organized involving the academic leaders of different AUs. The workshop deliberated on key issues viz. 1) Evolution into large multidisciplinary universities with at least 3000 students by 2030, 2) Enhancing gross enrolment ratio >10 percent per year, 3) Education Management, Governance & Academic leadership, 4) Integration of vocational & private education and 5) Digital Education.

2.2.3 MOOCs in Digital Agricultural Education - Strategies to Enhance MOOCs effectiveness, Inclusivity, Reach

The focus of this study revolves around MOOCs in Digital Agricultural Education, specifically targeting strategies to enhance their effectiveness, inclusivity, and reach. The outlined objectives encompass the analysis of factors influencing course completion rates in MOOCs, the development of metrics for gauging MOOC effectiveness, the exploration of learner perceptions and the impact of MOOCs on completion patterns, and the formulation of strategies to boost participation. Noteworthy achievements include the creation of a comprehensive instrument consisting of 19 statements designed to identify factors crucial for measuring MOOC success and completion. By employing established methods for evaluating MOOCs, completion rates were scrutinized, revealing that rates calculated based on Active Learners exceeded those based on Enrolled Learners by 9.5 to 18.3% across all MOOCs offered by the Academy. The study emphasized that completion rates considering Active Learners, defined as those who visited the course

page at least once during the course period, provided a more realistic measure. Additionally, a Success Rate Analysis was conducted to differentiate between certified and noncertified learners. The findings indicated that certified learners accessed resources at almost double the rate of noncertified learners, with total access days for uncertified learners being approximately half or less than their certified counterparts. The study also highlighted the importance of the first 5-8 days of a course as, crucial in determining a learner's ability to complete MOOCs.

2.2.4 Operationalizing Digital Content for MOOCs in Dairy Education

This study delves into the operationalization of digital content for Massive Open Online Courses (MOOCs) in Dairy Education, with a primary focus on two key objectives. The first objective involves the development of suitable protocols and courseware tailored for online education in Dairy Technology. The second objective aims to scrutinize learner behavior and patterns of learning specifically within the context of MOOCs for Dairying. Noteworthy achievements stemming from this project include the successful offering of a MOOC on Commercial Dairy Farming through the NAARM e-learning portal. A total of 917 participants registered for the program, with 175 of them qualifying for certification. Additionally, the project recorded 26 modules for three courses: Technology of Traditional Indian Dairy Products, Cheese Science and Technology, and Food Packaging and Labelling. These achievements will contribute to introducing advancing digital education in the field of Dairy Technology through targeted course development and insightful analyses of learner engagement in MOOCs.

2.2.5 Rural Technology Action Group (RuTAG 2.0)

ICAR-NAARM has received a sanction order from Rural Technology Action Group (RuTAG 2.0) which is an initiative of Principal Scientific Advisor (PSA) to the Government of India since 2004. RuTAG was devised to be a mechanism to achieve rural advancement through Science and Technology (S&T) interventions. This project will be collaborated with CIAE, Regional Centre, Coimbatore, and IIPM, Bangalore with a total budget sanction of Rs.1.51 Crores for a period of 3 years. The project will facilitate the commercialization of Grassroots Innovations in the Agri and Allied sectors. ICAR-NAARM organized an Accelerator Mentorship Program for selected 11 Innovators for a period of 3 days (22-24 November 2023).



Figure 2.2.1: Participants of Rural Technology Action (RuTAG 2.0)

2.2.6 Training Workshop on “Education Management and Academic Leadership” (Under NAHEP Component 2)

The Academy organized a training workshop on Education Management and Academic Leadership from 06-10 February 2023 with an objective to enhance the efficiency of the Indian Agricultural Education System under NAHEP – Component 2. Twenty-two enthusiastic administrators and senior faculty from agricultural universities including 02 lady

participants from 16 states participated in the training workshop. During this program, the participants were provided with theory and practical sessions on various topics related to education management and academic leadership by eminent personalities. The members of the NAHEP External Advisory Panel i.e. Dr. Brajesh Singh, Distinguished Professor, Western Sydney University, Australia also virtually interacted with participants.



Figure 2.2.2: Participants of training workshop on “Education Management and Academic Leadership” (Under NAHEP Component 2)

2.2.7 Establishment of Agri-Business Incubation (ABI) Centers

From 15-17 March 2023, an impactful Mentoring and Training program was meticulously organized, focusing on Agri start-ups. This initiative resulted in the successful incubation of 14 start-ups operating in the Agriculture and allied sectors. The program further facilitated four Immersion sessions tailored for start-ups and Farmer Producer Organizations (FPOs), fostering an environment conducive to growth and innovation. An additional highlight was the execution of a Start-up-Industry Interface Program on May 18, 2023, creating valuable opportunities for collaboration. A new call for applications was opened on January 10, 2023, resulting in an impressive response of 40 startup applications. Following a rigorous selection

process, six startups emerged successful and were chosen for incubation. Notably, an individual startup, Prajwal from Bangalore, garnered distinction by being selected for the BIRAC BIG 22nd call. A testament to the success of these initiatives is the substantial fund generation by the start-ups, accumulating a noteworthy sum of 1.5 Crores over the past year, underscoring the positive impact of the mentoring and training endeavors on the financial sustainability of these emerging enterprises.

2.2.8 Online training program on Digital Teaching Techniques

An online training program on Digital Teaching Techniques was conducted by the Academy from 13-17 February 2023 for 15 Faculty for the Faculty of Sri Konda Laxman Telangana State Horticultural University. The program touched on various contemporary aspects of Overview of E-Learning, Content Authoring Tools, Digital Content creation methods, technologies, and digital resources that every participant needs to be updated about. In-house faculty and three Guest faculty covered all the sessions.



Figure 2.2.3: Participants of the online training program on Digital Teaching Techniques

2.2.9 Enhancement of Teaching Competencies for Faculty of Kamdhenu University Amreli, Gujarat

The Academy organized an off-campus training program on “Enhancement of Teaching

Competencies” for the Faculty of Kamdhenu University Amreli, Gujarat from 21-23 February 2023. The objective of the program was to enhance the effectiveness of teaching capabilities in Agricultural Education. Twenty-seven enthusiastic faculty of various constituent colleges of Kamdhenu University including 02 lady participants participated actively both online and offline in the workshop. During the three-day training program, the participants were provided with theory and practical sessions on various topics related to teaching management and mastering the art of teaching by the faculty of the academy, teaching with technology, National Education Policy, etc. Feedback on various topics covered during the program was analyzed and is very encouraging. The usefulness and scope of the topics were rated very satisfactorily for all.



Figure 2.2.4: Participants of off-campus Training Programme on Enhancement of Teaching Competencies

2.2.10 Digital Teaching Techniques for Kamdhenu University, Amreli, Gujarat

From 14 – 16 March 2023, Kamdhenu University in Amreli, Gujarat, hosted a three-day workshop on “Digital Teaching Techniques” with a total of 25 participants. The main objectives of the program were to sensitize participants to the practices of digital course content development and to explore various open online platforms

for educational purposes. A notable innovation introduced during this workshop involved comprehensive training for participants in creating digital course content and utilizing collaborative tools. The initiative aimed to equip educators with the skills and knowledge needed to effectively leverage digital teaching techniques, fostering a more dynamic and engaging learning environment at Kamdhenu University.



Figure 2.2.5: FDP on Competency Enhancement in ARE

2.2.11 Faculty Development Programme for OUAT, Bhubaneswar

The academy organized the Faculty Development Programme for OUAT, from 24 July – 3 August 2023 with the objectives, to enhance capacity for participants the working competency of the participants in research areas and its current scenario, to equip participants with the latest knowledge, and to build competency in teaching skills & management. The schedule encompassed with National perspective on contemporary and Future Challenges in Agricultural Education & management; Critical thinking, Interdisciplinary Teamwork, and Professional values and Ethics for Effective participation in and Contribution to Agricultural and Rural Development; Personality Enhancement and Communication, Leadership Development, Motivation and Organizational Behaviour and Information and Knowledge Resources for Impact-making. The Core content

and delivery included, Multi multi-instructional strategy consisting of interactive lectures, hands-on practices, structured exercises, observational visits to IIMR, RTP and DPR.



Figure 2.2.6: Faculty development programme for OUAT, Bhubaneswar

2.2.12 Massive Open Online Course (MOOC) on Digital Teaching Techniques

A MOOC on “Digital Teaching Techniques” was offered during 1- 31 October 2023 for teachers and aspiring teachers/faculty in the e-learning portal of NAARM. It was eleventh in the Academy’s series of MOOCs. A total of 2139 participants from all over the country have registered, of whom 781 participants have completed the requirements for certification which accounted for different activities of the participants during the course period. The entire course was offered through online mode using pre-recorded interactive lectures developed in the Centre for Lifelong Learning in Agricultural Education (COLLAge) and learning material which were provided every week. The evaluation of this program was based on the reflections, assignment, course and content access, and final evaluation through a quiz.



Figure 2.2.7: MOOC on Digital Teaching Techniques

2.2.13 Brainstorming workshop on “Finalizing the content for MOOC on AI in Agriculture”

As a part of enhancing the capacities of NARES in the field of “Artificial Intelligence in Agriculture” a MOOC on “AI in Agriculture” was proposed. In this connection, a Brainstorming workshop was organized to finalize the content of the said MOOC, on 08 November 2023, with the following objectives. 1). To sensitize the stakeholders of the agri domain, about various concepts of Artificial Intelligence, Machine Learning, IoT, and Cloud Computing, 2). To apprise the trainees of AI tools and technologies for Agriculture. 3). To demonstrate various ICT applications/platforms developed by Startups in the field of Artificial Intelligence. Experts from ICAR institutes, SAUs, and Industry have participated.



Figure 2.2.8: Brainstorming workshop to finalize the content for MOOC on “AI in Agriculture”

2.2.14 Training program on Enhancement of Teaching Competence for the Faculty of UAS, Raichur

The “Enhancement of Teaching Competence” program, scheduled for 6-8 December 2023, for the Faculty of UAS, Raichur, spanned three days with the primary objective of sensitizing participants to diverse methodologies in teaching and learning. Additionally, the program aims to foster awareness and competence in utilizing

digital technologies for educational purposes. Through this initiative, faculty members are encouraged to enhance their teaching skills and embrace innovative approaches that leverage digital tools to elevate the overall quality of education.

2.2.15 FDP on Competency Enhancement in Agricultural Research and Education for the Faculty at KUFOS, Kochi

The ICAR-NAARM organized a five-day Faculty Development Program on “Competency Enhancement in Agricultural Research and Education” for the Faculty of Agriculture from 20 to 24 November 2023 under SCSP of NAARM. The program was organized at KUFOS, Kochi under the SCSP of ICAR-NAARM. It covered several invigorating and emerging topics. A total of 24 participants representing 5 Universities – Kerala University of Fisheries and Ocean Studies(KUFOS), Kerala Veterinary and Animal Sciences University(KVASU); Tamil Nadu Veterinary and Animal Sciences University(TANUVAS); Kerala Agricultural University(KAU), Professor Jayashankar Telangana State Agricultural University (PJTSAU) participated in the training program. In his, presidential address Dr. T. Pradeep Kumar, Vice-Chancellor, KUFOS emphasized the fact that the role of teachers is very important for the effective education of students. While addressing the gathering Dr. Ch. Srinivasa Rao, Director of NAARM, Hyderabad, stressed the role of teachers in all aspects of education and research activities to improve the performance of the university and improve individual skills. He emphasized how faculty can overcome their weaknesses and convert them into opportunities, so that they can evolve as a better person and make great

contributions. Dr. G. Venkateshwarlu, Joint Director, NAARM, underlined the recent and innovative technology developments in teaching skills and abilities to be developed by the teaching faculty to cope with the developments and demands in the frontier areas of agriculture and allied sectors. He also stressed the importance of scientific communication being a major relevance in the research works.



Figure 2.2.9: Faculty Development Program on Competency Enhancement in Agricultural Research and Education

2.2.16 Improving Non-Cognitive Skills and Soft Skills for Assistant Professors from Affiliated Colleges of Maharashtra Animal & Fishery Sciences University (MAFSU), Nagpur

ICAR-NAARM organized a 5 day training program on “Improving Non-Cognitive and Soft Skills (INC&SS)” for Assistant professors of Maharashtra Animal & Fishery Sciences University (MAFSU), Nagpur from 11-15 December 2023. A total of 22 participants representing affiliated colleges of MAFSU attended the program. The objective of the program was to improve non-cognitive skills and to enhance abilities in writing research articles for prestigious journals. Topics covered in this program included Group Dynamics

and Teamwork; Use of technology in learner evaluation; Business Communication through mainstream media; Interpersonal relationships; How to write research articles in high impact journals; Motivation; Stress Management; Emotional Intelligence; Orientation to One Health; National Education Policy-2020: Avenues and Strategies for Veterinary Education; Challenges and opportunities in Animal and Fisheries Education and research and achieving excellence



Figure 2.2.10: Participants of the training program on “Improving Non-Cognitive and Soft Skills (INC&SS)” for Assistant professors of Maharashtra Animal & Fishery Sciences University (MAFSU)

2.3 Enhancing Capacities for Leadership and Governance

The Human Resource Management Division has adopted a strategic approach to the effective and efficient management of people in the organization so that they contribute to the organization’s ability to obtain a competitive edge. This will help in maximizing employee performance in the service of an employer’s strategic objectives. The division participates in the creation and execution of the ICAR’s human resource policies and procedures. Human Resource Management deals with issues related to Performance Management, Organizational Development, Safety, Wellness, Employee

motivation, Teamwork, and training individuals and groups on different soft skills required for individual and organizational development in ICAR. The HRM Division's primary goal is to maintain better workplace relationships through the application and evaluation of policies, development, procedures, and Programmes related to human resources. Leadership and management development are two important key areas of HRM. The Strategic framework of the thematic area is to enhance capacities for leadership and governance by

1. Institutionalizing a framework for leadership development at all levels of NARS – early, midcareer, senior professional, and manager across all functions (research, education, extension).
2. Institutionalize good governance in NARS by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation.

The Human Resource Management Division places a strong emphasis on management development for the improvement of individual and team leadership abilities. An organization's objectives and goals can be accomplished by enhancing individual and organizational performance, which is typically accomplished by performing a training Programme termed as management development Programme (MDP). The training aims to raise an employee's skill levels and has a longer-term perspective to prepare them for upcoming challenges. The major objectives of these include effectiveness, learning outcome, and performance of participants in their professional careers. There is always a real

need for leadership development Programmes that aim to develop creative solutions for both society and public administrations. The leadership development Programme was started to monitor how senior managers reacted to systemic achievements, failures, and results. One of the most popular ways to improve managerial and leadership abilities inside an organization is through management development Programmes. When it comes to building high-performance organizations, leaders and managers are seen as a very potent group. Management Development is a systematic awareness, ability, attitude, and behavioral process that helps in the development of future leaders.

2.3.1 Executive Development Programme on Leadership Development

The Executive Development Programme (EDP I of 2023) on Leadership Development for Research Managers of ICAR was organized from 22-27 May 2023. While inaugurating the Programme, Dr. MV Gupta, recipient of the World Food Prize & Padma Shri Award, spoke about the emerging challenges in science leadership and the need for convergences of stakeholders and nurturing of multidisciplinary teams for better research outputs. The Programme, with 21 participants including 4 Assistant Director Generals (ADGs) and 17 Directors of ICAR institutes, focused mainly on cross-learning from other scientific organizations such as DRDL, CSIR, ISRO, NIAS, etc., apart from HRM exercises, viz. Collaboration, Leadership attributes, and Team building. The participants were also exposed to administrative and financial management in ICAR, and happiness & well-being with Yoga classes. The Programme also covered a special session on practical aspects of dealing

with parliamentary affairs. The Programme was concluded with a valedictory session on 27th May 2023 wherein Dr. Srinivasa Rao, Director NAARM complimented the participants for successful completion and called upon them to translate the learnings into action.



Figure 2.3.1: Participants of the Executive Development Programme on leadership Development held during 22-27 May, 2023

2.3.2 Executive Development Programme on Leadership Development

The ICAR–NAARM organized another Executive Development Programme on Leadership Development for 25 newly recruited Research Managers comprising five ADGs and 20 Directors of various ICAR institutes representing all Subject matter divisions during 21-26 August 2023. Dr Jacqueline d'Arros Hughes, the Director General of ICRISAT participated in the inaugural Programmeme on 21st August 2023 as



Figure 2.3.2: Participants of the Executive Development Programme on leadership development held during 21-26 August 2023

Chief Guest and addressed the gathering. In her address, she embraced all the facets of leadership from Mahabharata to New Age Leadership and Management and tenets of effective leadership. The valedictory Programme was graced by Padma Sri Dr. M.V. Gupta, World Food Prize Laureate and Former Assistant Director-General, World Fish Center (CGIAR), Penang, Malaysia held on 26th August 2023. In his address, Dr Gupta emphasized the tangible research outputs, outcomes, their socio-economic impacts, quality of research, and linkages with stakeholders.

2.3.3 Workshop on Fostering Academic Leadership in Agriculture

The Academy in collaboration with the Faculty Development Centre of GB Pant University of Agriculture and Technology, Pantnagar, Uttarakhand organized a 5-day workshop (4-8 September, 2023) on Fostering Academic Leadership in Agriculture under NAHEP Component 2. The main purpose of this



Figure 2.3.3: Workshop on Fostering Academic Leadership in Agriculture held during 4-8 September, 2023

workshop was to connect potential leaders of agriculture education and foster collaboration for enhancing the quality and relevance of agricultural education. The workshop involved sharing experiences, networking, developing

action plans, community of practice, group work etc. The workshop also deliberated on the implementation of NEP-2020. Senior Official of Shobhit University, a private University also participated. The important issues such as NEP 2020 implementation, Ranking framework, IT initiatives in agricultural education, impact of NAHEP Component 2, etc. Challenges and opportunities in Agricultural Research, Institution Building and Resolving Conflicts, etc.

2.3.4 Training Programme on Motivation, Positive Thinking, and Communication Skills for ICAR Technical Officers (T-5 and above)

The academy organized the training Programme on Motivation, Positive Thinking, and Communication Skills for ICAR Technical Officers (T-5 and above) from 11-15 September, 2023. The objective of the Programme is to improve skills and efficiency in discharging functions and providing constructive technical support in meeting the organizational goals and targets. The Programme was intended to build the capacity of technical officers through appropriate modes. Both in-house and external faculty were invited to deliver the sessions. The training methodology consisted of a blend of lectures, case analysis, and hands-on exercises. The Programme covered the topics in the following broad areas of motivation, positive thinking, communication skills, trust building, interpersonal relationships, teamwork and team building, health management and stress management, etc. The certificates were awarded by Dr G. Venkateswarlu, who was the chief Guest during the valedictory session. Twenty-one participants from different ICAR institutes attended the training.

2.3.5 Training Programme on Vigilance Perspectives for ICAR Officers

As a part of the Vigilance Awareness Campaign, ICAR NAARM organized a three-day training Programme on 'Vigilance Perspectives for ICAR Officers' from 18-20 October 2023. The Programme was attended by 35 Assistant Administrative Officers and Section Officers, representing 32 ICAR institutions and Head Quarters. The primary objective of this training Programme is to sensitize and develop the perspectives of vigilance among functionaries of ICAR institutions. While inaugurating the training Programme, Shri Rajeev Lal, Joint Secretary (Admn) emphasized the need for capacity building of administrative functionaries at regular intervals and requested to uphold the spirit of the campaign 'Say no to corruption; commit to the Nation'. Dr Ch Srinivasa Rao, Director of NAARM and president of the function categorically indicated that preventive vigilance is of utmost importance in curbing corruption. He also cautioned the participants on three important aspects – RPF (Recruitment, Procurement, and Finance).



Figure 2.3.6: Training Programme on Vigilance Perspectives for ICAR Officers held during 18-20 October 2023

2.3.6 Training Programme on Vigilance Perspectives for ICAR Officers

ICAR NAARM organized another three-day training Programme on 'Vigilance Perspectives for ICAR Officers', the second in a series during 1-3 November, 2023 as a part of the Vigilance Awareness Campaign and Vigilance Awareness Week. The Programme was attended by 34 Assistant Administrative Officers representing 31 ICAR institutions spread over 17 states. The primary objective of this training Programme is to sensitize and develop the perspectives of vigilance among functionaries of ICAR institutions. While inaugurating the training Programme, Shri Rajeev Lal, Joint Secretary (Admn) emphasized that adapting to changes is very critical for the success of organizations; which needs to be adopted at every level,

considering the shortage of staff in ICAR. Dr G Venkateshwarlu, Joint Director of the Academy advised to dispose of different cases duly following the procedures at their level, instead of referring to Council. On this occasion, Dr. D Thammi Raju, Vigilance Officer narrated different activities undertaken during the Vigilance Awareness Campaign. As a part of Vigilance Awareness Week, several competitions were held for school children, PGDM-ABM students, and staff at the Academy. Dr. Ch Srinivasa Rao, Director of the Academy and Chief Guest of the Valedictory Function distributed prizes to all the winners and requested the participants (AAOs) of the training Programme to be proactive and create an enabling environment in their institutions for better research outcomes.



Figure 2.3.7: Training Programme on Vigilance Perspectives for ICAR Officers held during 1-3 November, 2023

2.3.7 Management Development Programme (MDP) on Leadership Development (A Pre-RMP Programme)

This Programme is organized for HODs and other senior Principal Scientists in NARES to help participants understand their personality dimensions and explore leadership challenges and core competencies needed in the present-day context during 12-23 June, 2023. It was

attended by 14 participants from 14 ICAR institutes and 1 from SAU. They comprised 11 males and 4 females. The Programme helped them in developing understanding and skills in research and technology management, information and knowledge management, and administration and finance management and adapting ideas of diverse skills for sharpening capacities to effectively execute various roles in

the organization. The Programme consisted of a blend of lectures, self-exploration instruments, case analysis, experiential learning, and group discussion. The Programme provided excellent opportunities for mutual interaction and information sharing among the participants to develop leadership opportunities in the organization.



Figure 2.3.8: Participants of the Management Development Programme (MDP) on Leadership Development (A Pre-RMP Programme) held during 12-23 June, 2023

2.3.8 Management Development Programme on Leadership Development (A pre-RMP Programme)

The Academy organized a Management Development Programme on Leadership Development from 11-22 December 2023. There were 25 participants in the Programme out of which 3 were women participants. 22 participants were from the ICAR institutes, 2 participants were from SAUs, and one from ICAR KVK. They represented 17 states of India from Kashmir in the north to Kerala in the south and NE states in the East to Rajasthan in the west. This Programme had 30 sessions which consisted of a blend of lectures, self-exploration instruments, case analysis, experiential learning, and group discussion. The Programme provided

opportunities for mutual interaction and information sharing among the participants. And all twelve days, there was a yoga session. The topics covered leadership and personality development, research and technology management, information and knowledge management, administration, finance, and vigilance management. Apart from 18 NAARM faculty, 4 Guest Faculty had taken sessions. This gave insights into the leadership issues in different sectors.



Figure 2.3.9: Participants of Management Development Programme on Leadership Development (A pre-RMP Programme) held during 11-22 December, 2023

1.4 Extension Systems Management in a Market-driven Environment

The agricultural landscape is transforming from subsistence nature to business orientation. Today, farming in India needs to be approached as a business and the farmers need to be considered as entrepreneurs. Building the capacity of farmers as entrepreneurs requires the integration of various service providers such as input dealers, processors, marketers, and exporters apart from information and technology facilitators. In India, the system that is responsible for serving all the tangible and non-tangible inputs needed by the farming community is the National Agricultural Extension System (NAES). This system comprises frontline extension components operated by ICAR Institutes, SAUs, ATARIs, and KVKs,

which cater to the technology assessment and refinement, demonstration, and skill training needs of the farmers. Mainstream extension served by the development departments, indulge themselves in technology advisory, input distribution, and implementation of welfare and development schemes of central and state sectors. The private extension component vigorously promotes the technologies developed by them, while the tertiary sector comprises non-governmental organizations (NGOs), Farmer Producer Organizations (FPOs), etc., who operate to facilitate access of farmers to technology demonstration, training, and other facilities such as inputs, processing, marketing, credit, insurance, etc.

At present, the components of NAES adopt convergence among the various service providers on a limited scale and on certain occasions. Extending farmer-centric extension, by transforming the farming community as business operating entrepreneurs necessitates not only the partnership and convergence among the various extension players but also access of farmers and farmers' organizations towards the various services rendered by these components. Such a scenario needs the use and applications of advanced Information and Communication Technologies (ICTs), application of participatory methodologies, advanced extension management principles and priorities, promotion of para-extension professionalism, and mainstreaming gender through adequate and appropriate capacity building.

To address such contemporary and emerging issues in Agricultural Extension and the related capacity-strengthening needs of all the components of NAES, the Academy established

a Division viz., Extension Systems Management (XSM), with a focus on enhancing sustainable livelihood security and farmers' empowerment. The Division has a two-pronged strategy towards enhancing operational, adaptive, and generative capacity for frontline agricultural extension systems to address emerging challenges and enhancing capacities for the use of ICTs to provide customized knowledge, skills, and solutions for social networking. To cater to the needs of the stakeholders of the Extension System in the country, the Division undertakes research and policy advocacy; and capacity-building programs for the top, middle, and entry-level extension personnel of frontline extension functionaries and outreach activities. The themes on which the Division organizes all the activities include (i) Extension Programme, Policy, and Management; (ii) Institutional Innovations; (iii) Gender Mainstreaming and (iv) ICT applications in Agriculture and knowledge sharing. This particular section explains various activities undertaken by the scientists of the Division of Extension System Management, during 2023.

2.4.1 Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India

The study aims to develop a methodological framework for the assessment of gender attribution in developmental schemes of agriculture across ministries which would result in the formulation of a status report on mainstreaming gender in the agriculture sector and in turn recommend reforms in identified government schemes for gender mainstreaming. Under this study, selection of 53 Ministries (based on programs for women

related to agriculture) was done and then, the number was narrowed down to 8 Ministries viz. Ministry of Agriculture & Farmers' Welfare; Ministry of Rural Development; Ministry of Food Processing Industries; Ministry of Environment, Forest & Climate change; Ministry of Labour & Employment; Ministry of Fisheries, Animal husbandry and Dairying; Ministry of Development of North Eastern Region and Ministry of Cooperation. Since, Ministry of Cooperation had no development program fitting into our criteria, the analysis on other seven Ministries revealed that there were 20 programs with 100% allocation for women and 38 programs had 30% allocation for women in agriculture.

2.4.2 Management and Impact Assessment of Farmer FIRST Project (FFP)

This research project has been implemented since 2017 with financial support from ICAR. ICAR-NAARM is a partner institution in the project management consortium, mandated to support the management of FFP through capacity building, documentation, information showcasing, and impact assessment. Onsite field visits were made to FFP villages of ICAR-Indian Institute of Soil & Water Conservation (IISWC), Dehradun; Chaudhary Sarwan Kumar Himachal Pradesh Krishi Vishvavidyalaya (CSKHPKV); Palampur; and ICAR-Central Soil Salinity Research Institute (CSSRI), Karnal. Focused group discussions were organized with farmers & FFP teams and good practices for scaling up were documented. Three farmers from FFP villages were selected and awarded with the "Innovative Farmers Award" on the occasion of 48th Foundation Day of ICAR-NAARM on September 1, 2023.



Figure 2.4.1: Glimpses of Farmer FIRST Project activities

2.4.3 Agripreneurship for Sustainable Agricultural Development- Technological and Institutional Innovations and Strategies

This research project sponsored by National Agricultural Science Foundation (NASF) has been implemented since May 2023 with the objectives, of mapping the nature of problems in agriculture and assessing the extent of risks in entrepreneurship development; identifying technological and institutional innovations in the study area to overcome the risks/ constraints in entrepreneurial development; exploring

the Institutional innovations for community mobilization and collective action; and to identify the attributes for success of Agripreneurship and to design strategies for sustainable agriculture through Agripreneurship. The first Advisory Committee Meeting for the NASF project was conducted and convened on 2 November 2023, at ICAR-NAARM whereas, the scale construction on “Constraint analysis of agri-preneurship” is in progress.



Figure 2.4.2: Participants of the Advisory Committee Meeting of NASF Project

2.4.4 A Multidimensional Study on Impact of Pradhan Mantri Kisan Samman Nidhi Programme (PM-KISAN) on the Livelihood of Farmers of Southern States of India.

This empirical study, sponsored by Indian Council of Social Science Research (ICSSR), is envisaged on the Impact of the Pradhan Mantri Kisan Samman Nidhi Programme (PM-KISAN) on the Livelihood of Farmers of the Southern States of India. The study was conducted in five southern states of India namely Tamil Nadu, Kerala, Andhra Pradesh, Telangana, and Karnataka with the objective of analyzing the trend of implementation of Pradhan Mantri Kisan Samman Nidhi (PM KISAN) across these states and to quantify and compare the impact of PM KISAN on varied domains of livelihood of farmers using a composite index. Further, this

study intends to predict the determinants of the spending pattern of money received through PM KISAN and to model the Construct of intention to invest the money received through PM-KISAN in Agriculture. In this connection, we have collected data from PM Kisan beneficiaries and non-beneficiaries of Medchal (Less Active), Nalgonda (High Active) and Vikarabad (Moderate Active) districts of Telangana State.

2.4.5 Training of Trainers Workshop for Entrepreneurship Development among Agri-graduates

The Academy organized a Training of Trainers (ToT) Workshop during 23-24 January 2023 for faculty members of SAUs to develop soft skills and entrepreneurship among agri-graduates. The training was organized with the aim to create a pool of master trainers in agricultural universities who would in turn train the agri-graduates in their respective universities. The Workshop was organized as a part of the World Bank-sponsored National Agricultural Higher Education Project Component - 2A. In his address, Dr Ch. Srinivasa Rao, Director, ICAR-NAARM offered to extend handholding support to selected SAUs to create awareness among the agri-graduates for entrepreneurship development throughout the length and breadth of the country. Dr Surya Rathore informed the participants about the impact of the earlier workshop held in the year 2019, stating that it created 34 master trainers who have trained more than 3000 students by organizing 28 programs in 25 Universities. Twenty-seven faculty members representing 27 State Agricultural Universities belonging to 15 states participated in the workshop. Consequent upon the completion of the TOT program, each participant organized a one-day workshop on

“Development of Soft Skills for Entrepreneurship among Agri Graduates” at their respective universities. These workshops were held during February and March 2023, and 2896 students were trained.



Figure 2.4.3: Trainers Workshop for Entrepreneurship Development among Agri-graduates

2.4.6 VIII MDP for Newly Recruited Senior Scientists & Heads of Krishi Vigyan Kendras

The Phase I Training of VIII Management Development Program for Newly Recruited Senior Scientists and Heads of Krishi Vigyan Kendras was organized at the Academy during 31 January to 14 February 2023. This is a collaborative program, planned and implemented by ICAR-NAARM, Hyderabad; Agricultural Extension Division of ICAR; and ICAR-ATARIs exclusively for the newly recruited Senior Scientists and Heads of KVKs. It consisted 3 phases viz., 1st Phase of 15 days at ICAR-

NAARM, 2nd Phase of 10 days at Best KVK in other Zone (ICA-KVK Kalaburagi, ICAR-KVK Ujjain and ICAR-KVK West Godavari) and 3rd Phase of five days at respective ATARIs. Fifteen newly recruited Heads of KVKs representing seven ATARI Zones participated in this program. The program content consisted of modules on the organization and management aspects of KVKs, communication (oral and written) management, human resource management, technological content and media management, and administration and finance management as relevant to KVKs. Dr. Udham Singh Gautam, Deputy Director General (Agricultural Extension), ICAR graced the valedictory session of the program as Chief Guest, and Dr Ch. Srinivasa Rao, Director, ICAR-NAARM presided over. Dr Sheikh N. Meera, Director, ICAR-ATARI, Zone X, Hyderabad was the Guest of Honour.



Figure 2.4.4: VIII MDP for Newly Recruited Senior Scientists & Heads of Krishi Vigyan Kendras

2.4.7 Programmes Organized under SCSP

S.No	Programme	Venue/ Organized by	Dates
01	Skill Development Training Programme on Vermicompost Production Technology	ICAR-NAARM/SAIRD KVK, Gaddipally, Telangana	02-03 February 2023
02	Improved Crop Management Technologies and Value Addition	ICAR-NAARM	13-15 February 2023; 16-18 February 2023
03	Skill development training program on Integrated Farming System	ICAR-NAARM/ KVK, Raichur, Karnataka	23-28 February 2023
04	Skill Development Training Program on Integrated Farming System	ICAR-NAARM/ KVK, Namakkal, Tamil Nadu	13-17 March 2023
05	Skill Development Training Program on Backyard Poultry	ICAR-NAARM/ SAIRD KVK, Gaddipally, Telangana	23-24 March 2023
06	Skill Development Training Program on Scientific Beekeeping	ICAR-NAARM/ SAIRD KVK, Gaddipally, Telangana	24-31 March 2023
07	Skill Development Training Program on Enriched Vermicompost Production	ICAR-NAARM/KVK, Lam, Guntur, Andhra Pradesh	27-29 March 2023
08	Skill Development Training program on Food Quality & Standards in Food Processing for Entrepreneurial Development	ICAR-NAARM/KVK, Lam, Guntur, Andhra Pradesh	27-29 March 2023
09	Skill Development Program on Integrated Farming System	ICAR-NAARM/KVK of TNJFU	08-12 May 2023
10	Skill Development Program on Advanced Aquaculture Techniques	ICAR-NAARM/KVK of TNJFU	08-12 May 2023
11	Skill Development Training on Mushroom cultivation	ICAR-NAARM/KVK, Pathanamthitta, Kerala	11-15 July 2023
12	Training Program on Promotion, Processing, and Value Addition of Millets	ICAR-NAARM	07-09 August 2023
13	Skill Development Training Program on Integrated Farming System-Climate Smart Dryland Technologies	ICAR/NAARM	11-13 August 2023; 16-18, August 2023
14	Skill Development Training Programme on Integrated Crop management in Chilli	ICAR-NAARM/SAIRD KVK, Gaddipally, Telangana	28-30 August 2023
15	Skill Development Training Program on Value added Products with Millets	ICAR-NAARM/SAIRD KVK, Gaddipally, Telangana	07-12 October 2023
16	Training Program on Promotion of Nutrition-Sensitive Agriculture	ICAR-NAARM	01-03 November 2023
17	Skill Development Training Program (SDTP) on “Improved Crop Management Technologies and Value Addition of Millets”	NAAS, Hyderabad Chapter, ICAR-NAARM/KVK, Amdalavalasa, Andhra Pradesh	01 to 03, November 2023

S.No	Programme	Venue/ Organized by	Dates
18	Training Program on Backyard Poultry Management	ICAR-NAARM/ KVK, Yagantipalle, Andhra Pradesh	4-6 December 2023
19	Skill Development Training Program (ESDTP) on Value Addition of Milk and Meat Products	NAAS, Hyderabad Chapter, ICAR-NAARM/SVVU, Tirupati, Andhra Pradesh	11-15 December 2023
20	Skill Development Training Program on Mushroom Cultivation	ICAR-NAARM/KVK, KSNUAHS, Shivamogga	18-22 December 2023

2.4.8 Skill Development Training on Vermicomposting at KVK, Gaddipally, Telangana

ICAR-NAARM organized Skill Development Training Programme on “Vermicompost Production Technology” for scheduled caste farmers under SCSP at SAIRD KVK, Gaddipally, in Suryapet district of Telangana from 02 to 03 February 2023. The topics covered in the training program were vermicompost preparation and application method, management of vermicompost, practical vermi bed preparation, types of sheds used for vermicompost, inoculation of earthworms, sprinkling of water, identification and harvesting of vermicompost, precautions to be taken while harvesting. There were sessions on plant protection measures to be taken during the life cycle of earthworms,



Figure 2.4.5: Skill Development Training on Vermicomposting at KVK, Gaddipally, Telangana

identification of earthworms and materials required for vermicomposting, vermiwash preparation, application of vermiwash and management of vermiwash tank etc. The participants acquired skills on the various approaches to vermicomposting. Fifty-eight farmers from seven mandals of Suryapet district participated in the training programme.

2.4.9 Training program on Improved Crop Management Technologies and Value Addition

The Academy organized a training program on Improved Crop Management Technologies and Value Addition from 13 to 15 February 2023 and 16 to 18 February 2023 with multifaceted objective aimed at empowering farmers with essential knowledge and skills. Firstly, the program focuses on training farmers in judicious use of nutrients across various crops, value-addition techniques in millets, crop diversification, micro-irrigation, improved agricultural practices in cereals and pulses, and entrepreneurial opportunities within the horticultural sector. Secondly, the program aspires to contribute to the development and promotion of sustainable crop management technologies, aligning to enhance agricultural

productivity while minimizing environmental impact. A key emphasis is placed on the third objective, which is to elevate the value-addition processes for crops. By doing so, farmers can access higher-value markets, thereby augmenting their income. Finally, the program is designed to provide comprehensive training and extension services to farmers, facilitating the widespread adoption of improved crop management practices and value-addition techniques. Overall, this training initiative seeks to equip farmers with the necessary tools to optimize their agricultural practices, increase productivity, and create additional avenues for income generation through value addition.



Figure 2.4.6: Participants of the Training Program on Improved Crop Management Technologies and Value Addition

2.4.10 Skill development training program on Integrated Farming System

The Academy organized a skill development training program on “Integrated Farming System” under SCSP in collaboration with Krishi Vigyan Kendra, Raichur for SC farmers from 23 - 28 February 2023. A total of 30 farmers were trained on aspects such as fisheries in integrated farming systems, soil conservation as a part of integrated farming systems, bee-keeping as a subsidiary occupation in integrated farming systems, vermicomposting in integrated farming systems, animal- husbandry, poultry, sheep and

goat- rearing in integrated farming system, processing and value addition of agricultural and horticultural produce, nutrition garden and its management, horticulture nursery management for additional income, bio-fertilizers in integrated farming system, role of integrated farming system in doubling the farmers income, organic farming and quality seed production.



Figure 2.4.7: Participants of Skill Development Training Program on Integrated Farming System

2.4.11 Skill Development Programme on Integrated Farming System at KVK-Namakkal, Tamil Nadu

The Academy organized 5-day Skill Development Training Programme on “Integrated Farming System” under SCSP in collaboration with Krishi Vigyan Kendra (KVK), TANUVAS Namakkal, Tamil Nadu for SC farmers from 13 – 17 March 2023 at KVK Campus. As a part of the training curriculum, trainees were taken for an exposure visit to irrigated upland of an integrated farming system and integration of paddy cum fish culture unit developed at the Regional Research Station, TNAU, Vridhachalam to exposed farming system models and interacted with scientists for effective utilization of land resources and higher income generation from various allied components. During the training,

demonstrations on vermicompost production, Azolla production, compost-making methods, mushroom production, honey bee rearing, fodder seeds processing, etc were organised. In the valedictory program, important critical inputs viz., vermicompost bed with earthworms, Azolla bed with Azolla seeds, one-month-old Aseel bird chicks, vegetable seeds kit for kitchen garden development, fodder seeds for fodder bank establishment, mineral mixture for livestock, waste decomposer, neem oil, and yellow sticky trap were distributed to all the trainees. A total of 40 SC farmers from Mettupatti, Pilippaikuttai, Thusur, Sambamedu, Vadugam, E. Pudupatti, Nadupatti and Valayapatti villages of Namakkal District participated and benefitted.



Figure 2.4.8: Skill Development Programme on Integrated Farming System at KVK- Namakkal, TN

2.4.12 Skill Development Training Programme on Backyard Poultry

The Academy organized a skill development training program on “Backyard Poultry” under SCSP from 23 - 24 March, 2023 in collaboration with Krishi Vigyan Kendra (KVK), Gaddipally, Telangana. The objective of the program was to enhance the livelihood of farmers through income generation by having a self-employment

activity. Around 100 farmers from seven mandals of Suryapet District participated in the program. To start the backyard poultry farming by the trainees, 10 Gramapriya chicks (one month old) were distributed to each trainee. During the program, Dr G. Venkateshwarlu, Joint Director, ICAR-NAARM highlighted that backyard poultry farming can boost up family income as it involves low-cost investment but gives higher economic returns. Dr M Balakrishnan, Principal Scientist & Chairman SCSP, ICAR-NAARM while briefing about the importance of the training program, enlightened the trainees towards adopting the backyard poultry farming successfully as an income generation activity. Sri B. Lava Kumar, Senior Scientist and Head (i/c), Krishi Vigyan Kendra, Gaddipally; Smt. N Sugandhi, Program Assistant (Home Science), Krishi Vigyan Kendra, Gaddipally were also present.



Figure 2.4.9: Participants in Training programme on Backyard Poultry

2.4.13 Skill Development Training on Scientific Beekeeping at KVK, Gaddipally, Telangana

The Academy organized a skill development training program on “Scientific Beekeeping” for SC farmers during 24 - 31 March, 2023. The program was conducted under SCSP in collaboration with Krishi Vigyan Kendra (KVK), Gaddipally, Telangana. The unemployed rural youth, farmers, and students participated actively in the program and got educated on various aspects of scientific beekeeping and its use in pollination of various crops. A total of 25 farmers, rural youth and students from Suryapet and Nalgonda districts participated in the program. In his address, Dr G. Venkateshwarlu, Joint Director, ICAR-NAARM informed the participants about the importance of honeybees in agriculture, getting additional sources of income by rearing honeybees and pollination



Figure 2.4.10: Training on Scientific Beekeeping at KVK, Gaddipally, Telangana

can be improved in crops. He also explained the commercial importance of rearing honeybees. Dr M Balakrishnan, Principal Scientist & Chairman SCSP, ICAR-NAARM briefed about the importance of beekeeping and asked the participants to become entrepreneurs. Sri B. Lava Kumar, Senior Scientist and Head (i/c), Krishi Vigyan Kendra, Gaddipally; Mr. Ch. Naresh, SMS (Horticulture), Krishi Vigyan Kendra, Gaddipally were also present.

2.4.14 Training on Enriched Vermicompost Production for Enhancing the Income of Farmers at KVK, LAM Guntur, Andhra Pradesh

The Academy organized a skill development training program on “Enriched Vermicompost Production” for enhancing the income of SC farmers during 27 – 29 March, 2023. The program was organized under SCSP at KVK, Lam, Guntur district of Andhra Pradesh. A total of 37 farmers from Guntur, Palnadu and Bapatla districts,



Figure 2.4.11: Training on Enriched Vermicompost Production for Enhancing the Income of Farmers at KVK, LAM Guntur, Andhra Pradesh

Andhra Pradesh participated in the program. Chief Guest, Dr G. Venkata Naidu, Director of Extension (DE), SVVU Tirupati, Hyderabad, in his address, elaborated on the vermicompost in farming and insisted that every person has the responsibility of preserving and protecting the soil health for better future. Dr M. Balakrishnan, Principal Scientist & Chairman SCSP, ICAR-NAARM informed the farmers about the use of vermicompost to protect the soil health and to get quality crop yields. He stressed that the farmers to start the vermicompost units without fail for better income and healthier life. Dr KR Subramanyam, Principal Scientist and Head; Dr M. Ganga Devi, Scientist (Agronomy), KVK, Lam, were also present. In the valedictory program, portable vermicompost units were distributed to the participants.

2.4.15 Food Quality & Standards in Food Processing for Entrepreneurial Development at KVK, Lam, Guntur, Andhra Pradesh

The Academy organized a skill development training program on “Food Quality & Standards in Food Processing for Entrepreneurial Development” for SC women farmers under SCSP at KVK, Lam, Guntur district of Andhra Pradesh during 27 – 29 March, 2023. A total 60 SC women farmers from different villages of Lam, Jonnalagadda, Vengalayapalem, Namburu, Ponnekkal, Kattevaram, Nandivelugu, Thumuluru, Revendrappaadu, Peddhakonduru, Tadikonda, Tadepally, Kunchinipally, Venegandla and Pedhapalem of Guntur district, Andhra Pradesh participated in the program. During his address, Dr G. Venkata Naidu, Director of Extension, SVVU, Tirupati, informed the

participants that the Government of India is focusing and taking all necessary steps to boost the production and income of farmers and farm women in rural areas. Dr. M. Balakrishnan, Principal Scientist & Chairman SCSP, ICAR-NAARM, Hyderabad suggested and encouraged women to set up small scale units on millets and other products which are suitable to the local market, as there is a growing demand for millets. He stressed that women should come up with a start-up proposal and supply millet products to all institutions in future. Dr KR Subramanian, Principal Scientist; Dr M. Deepa, Scientist (Home Science), KVK, Lam, were also present.



Figure 2.4.12: Entrepreneurial Development Program on Food Quality & Standards in Food at KVK, Lam, Guntur, Andhra Pradesh

2.4.16 Skill Development Programme on Integrated Farming System

In collaboration with KVK of TNJFU, Nagapatinam, Tamil Nadu, a 5-day skill development program on “Integrated Farming System” (under SCSP) was organized during 08 - 12 May 2023 for the benefit of SC farmers. Dr. Ch. Srinivasa Rao, Director, NAARM, Hyderabad participated in the program as Chief Guest. The program covered integrated farming system models for effective utilization

of farm resources, more net profit from less land areas, rural employment generation, alleviating malnutrition in rural areas, and doubling the farmers' income. While addressing farmers, Dr. Ch. Srinivasa Rao motivated and advised the farmers to adopt suitable IFS components like mushroom production, sericulture, poultry farming, and fish culture by forming FPOs for better marketing of their produce. A total of 40 farmers including farm women, and rural youths from Sikkal, Panimedu, Mozhithidal, Vadakku Poigainallur, Anthakudi and Melakarairuppu villages of Nagapatinam district participated in the program.



Figure 2.4.13: Skill Development Programme on Integrated Farming System

2.4.17 Skill Development Programme on Advanced Aquaculture Techniques

The Academy, in collaboration with KVK-TNJFU Nagapatinam, Tamil Nadu, organized a 5-day skill development program on “Advanced Aquaculture Techniques” (under SCSP) during 08 – 12 May 2023 for the benefit of SC farmers. The program aimed at enhancing economy of rural fish farming community



Figure 2.4.14: Skill Development Programme on Advanced Aquaculture Techniques

through scientific interventions in aquaculture and advanced fish culture technologies. The training program was inaugurated by Prof. Dr. G. Sugumar, Vice Chancellor, TNJFU, Nagapatinam. Dr. Ch. Srinivasa Rao, Director, NAARM, Hyderabad participated as Chief Guest in the program. Dr. M. Rajakumar, Director of Extension Education, TNJFU, Nagapatinam welcomed the participating farmers engaged in aquaculture. Dr. M. Balakrishnan, Principal Scientist, NAARM delivered a keynote address

highlighting the importance of the program and its implementation. A training manual on Advanced Aquaculture Techniques in Tamil was released during the program.

2.4.18 Skill Development Training on Mushroom Cultivation

ICAR-NAARM organized 5-day Skill Development Training on “Mushroom cultivation” exclusively for SC community as part of the SCSP programme during 11 - 15 July 2023 at ICAR -Krishi Vigyan Kendra, Pathanamthitta, Kerala. The training concentrated on different aspects of developing skills in mushroom cultivation to set up an enterprise as an income source. The training focused on mushroom cultivation (Milky and Oyster), mushroom spawn production, pest and disease management in mushroom cultivation,



Figure 2.4.15: Skill Development Training in Mushroom Cultivation

harvesting, packaging and marketing, mushroom processing and value addition, Project formulation, Finance & banking. Technical sessions were handled by Dr. A V Mathew, Prof. Plant Pathology (Retired), Kerala Agricultural University; Mr. Aswin, Assistant Manager, SBI, Kozhenchery, Dr. Shana Harshan, SMS (Home Science), KVK, Pathanamthitta and Sri. Alex John, SMS (Plant Protection), KVK, Pathanamthitta. Forty persons attended the training. As part of the training, an exposure visit to ‘Vijay Mushroom’ Mavelikkara was also

organized. An interaction meet was organized on August 16, 2023 at KVK, where Dr. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad along with Dr. V. Venkatasubramanian, Director, ICAR-ATARI, Bengaluru, Dr. M. Balakrishnan, Principal Scientist, ICAR-NAARM, Hyderabad had face to face interaction with the ex-trainees to evaluate the effectiveness of the training and the problems faced by farmers in initiating the mushroom enterprise.

2.4.19 Training Programme on Promotion, Processing, and Value Addition of Millets

The Academy organized the training program on “Promotion, Processing, and Value Addition of Millets” during 07 – 09 August 2023. The training program was designed with a comprehensive set of objectives aimed at empowering farmers with diverse skills and knowledge. Primarily, the program sought to train farmers on various aspects of crop diversification, emphasizing nutri-sensitive agriculture. It also highlighted the role of animal sector activities in boosting the rural economy and dealt entrepreneurial activities through agro-based enterprises. Additionally, the program covered essential topics such as the planning and maintenance of



Figure 2.4.16: Participants of the Training Programme on Promotion, Processing and Value Addition of Millets

nutrition gardens and incorporation of natural and organic farm practices like vermicompost and bio-fertilizer preparations. The training further included hands-on sessions on preparing fruit squash and practical training on murabbas and fruit rolls. Through this holistic approach, the program aimed to equip farmers with the knowledge and practical skills needed for crop diversification, nutritional awareness, and value addition in millets, contributing to the overall enhancement of agricultural practices and rural economic development.

2.4.20 Skill Development Training Program on Integrated Farming System-Climate Smart Dryland Technologies

The Skill Development Training Programme on “Integrated Farming System-Climate Smart Dryland Technologies”, conducted in two phases from 11 to 13 August 2023, and 16 to 18 August 2023, each spanning three days, garnered the participation of 120 individuals, including 60 female participants. The program aimed to raise awareness about conservation agriculture, livestock production, and, notably, weather-based farming practices, while also enhancing practical knowledge in integrated nutrient and



Figure 2.4.17: Skill Development Programme on IFS

pest management. The dual-phase training initiative focused on equipping participants with the skills necessary for climate-smart dryland technologies, emphasizing sustainable farming practices that consider climatic conditions for optimal agricultural outcomes.

2.4.21 Skill Development Training on Integrated Crop Management in Chilli at KVK, Gaddipally, Telangana

ICAR-NAARM organized a 3-day Skill Development Training Programme on “Integrated Crop management in Chilli” for SC farmers under SCSP at SAIRD KVK, Gaddipally, in Suryapet district of Telangana from 28 to 30 August 2023. Twenty-five farmers from



Figure 2.4.18: Skill Development Training on Integrated Crop Management in Chilli at KVK, Gaddipally, Telangana

nine villages like Gajulamalkapuram, Chetla mukundapuram, Kotapahad, Isthapuram, Nassempet, Apoor, Mohammadapuram and Dupahad of Suryapet district participated in the training program. Dr. M. Balakrishnan, Principal Scientist & Chairman SCSP ICAR-NAARM,

Hyderabad addressed the farmers and informed about the export opportunities for quality chillies. Sri B. Lavakumar, Senior Scientist & Head (I/c), KVK Gaddipally explained the various programs undertaken by the KVK for the development of SC. As part of the program, critical inputs namely chili seedlings, azadirachtin, solar light traps, sticky traps, bio-inoculants (biofertilizers and biopesticides), macro and micronutrients were distributed.

2.4.22 Skill Development Training Programme on Value-added Products with Millets

In connection with the International Year of Millets-2023, The NAAS Hyderabad chapter, in association with ICAR-NAARM, organized a 6-day Skill Development Training Program on “Value added Products with Millets” for SC women under SCSP at SAIRD KVK, Gaddipally, in Suryapet District of Telangana during 07 - 12 October 2023. A total of 25 SC women from Gaddipally, Ponugodu, Keethavarigudem,



Figure 2.4.19: Skill Development Training Programme on Value Added Products with Millets

Kalmalcheruvu and Nereducherla villages of Suryapet district participated in the program. The main aim of the program was addressed and the participants were encouraged to be an entrepreneur by establishing small-scale value-added millet units. They were suggested to add millet recipes to their daily diet for improving their health. B. Lavakumar, Senior Scientist & Head (I/c), KVK Gaddipally explained the health benefits of millets and the present market demand for millet products. An exposure visit to millet processing units was also arranged for trainees.

2.4.23 Skilled-based training programme on Promotion of Nutrition-Sensitive Agriculture

The Academy organized a training program on “Promotion of Nutrition-Sensitive Agriculture,” during 01 – 03 November 2023. The program aimed at promoting a comprehensive approach to addressing malnutrition and improving public health. Firstly, the program aimed to increase both the production and consumption of diverse, nutrient-rich crops, recognizing them as crucial elements in combating malnutrition. Secondly, the program focused on educating farmers and communities about the significance of nutrition-sensitive agriculture, emphasizing its profound impact on overall health and well-being. Lastly, a pivotal objective of the program is to establish linkages between the agriculture, health, and nutrition sectors. This strategic connection aims to ensure a holistic approach to addressing malnutrition and food insecurity, recognizing the interconnectedness of these sectors in fostering sustainable solutions. By equipping participants with practical skills and knowledge, this training program strived to contribute to the promotion

of nutrition-sensitive agriculture, thereby fostering improved public health outcomes and addressing the multifaceted challenges associated with malnutrition.



Figure 2.4.20: Participants of the Skilled-based training program on “Promotion of Nutrition-Sensitive Agriculture”

2.4.24 Training program on Improved Crop Management Technologies and Value Addition of Millets

The NAAS Hyderabad chapter, in association with ICAR-NAARM, conducted a 3-day skill development training program (SDTP) on “Improved Crop Management Technologies and Value Addition of Millets” for the SC farmers of four villages (Santha Kothavalasa, Kondavalasa, Kondraguda and Isakapalem) of Srikakulam District during 01 - 03 November 2023 under



Figure 2.4.21: Participants of the training program on Improved Crop Management Technologies and Value Addition of Millets

SCSP at KVK, Amadalavalasa, Andhra Pradesh. A total 75 farmers including 28 women participated in the training. The farmers were trained on the aspects related to improved package of practices of millets, nutritional significance and health benefits of millets, importance of bio-fertilizers, preparation of diversified value-added products with millets, packing, branding, FSSAI registration procedures. An exposure visit to millet processing unit at Agricultural Research Station (ARS), Vizianagaram was also arranged.

2.4.25 Skill Development Training for Rural Women on Backyard Poultry Management

ICAR-NAARM organized a three-day training program on “Backyard Poultry Management” under SCSP in collaboration with Krishi Vigyan Kendra (KVK), Yagantipalle, AP for SC women during 4 – 6 December 2023. A total of 40 SC farmers from Veepanagandla and Devenuru villages of Miduturu Mandal, Nandikotkur Division of Nandyal district, Andhra Pradesh were trained. The participants gained both theoretical and practical experience in poultry management in the backyards, diseases commonly observed in poultry, and their management. To start the backyard poultry farming each trainee was distributed with 30 Rajashri chicks, tarpaulin cover and net for Azolla cultivation, and a training manual. Dr. M. Balakrishnan, Principal Scientist and Chairman, SCSP Committee, ICAR-NAARM, briefed the participants about the SCSP program and its implementation and enlightened the trainees towards adopting Backyard Poultry Farming as an income generation activity. Smt. Dhanalakshmi, Senior Scientist & Head KVK, in her address, highlighted that backyard poultry farming can boost family income as it involves

low-cost investment but gives higher economic returns. Earlier, the program was welcomed by Shri Krishnamurthy, SMS (Animal Husbandry).



Figure 2.4.22: Skill Development Training for Rural Women on Backyard Poultry Management

2.4.26 Entrepreneurial Skill Development Training Programme on Value Addition of Milk and Meat Products at SVVU, Tirupati, Andhra Pradesh

The NAAS Hyderabad Chapter, in association with ICAR-NAARM, organized the Entrepreneurial Skill Development Training Program (ESDTP) on “Value Addition of Milk and Meat Products” for women farmers under SCSP program at Centre for Continuing Veterinary Education and Communication (CCVEC), SVVU, Tirupati during 11 – 15 December 2023. A total of 20 women farmers from the SC community of Akkurthi, Yellampalli, and Vagaveedu villages of Erpedu Mandal, Tirupati participated in the program. Dr. M. Balakrishnan, Principal

Scientist & Chairman SCSP ICAR-NAARM, Hyderabad explained the aim of the program and encouraged the participants to be entrepreneurs by establishing small-scale units of value-added milk and meat products. As part of the program trainees also took exposure visits to milk and meat processing units. Dr. G Venkata Naidu, DE, SVVU explained the importance of value addition of animal products to improve farmers’ income from livestock farming. The significance and potential of producing organic animal products



Figure 2.4.23: Training Programme on Value Addition of Milk and Meat Products at SVVU, Tirupati, Andhra Pradesh

can greatly contribute to improving their livelihoods. Dr. Y. Ravindra Reddy, Associate Dean of Dairy Technology College briefed about the significance of value addition in mitigating post-harvest losses, promoting industrialization, and generating employment opportunities. Dr. Ch. Srinivasa Rao, Director of ICAR-NAARM, participated in the valedictory program as the Chief Guest. He addressed the farmers, emphasizing the importance of adding value to milk and meat products. Dr. Rao stressed the need for farmers to gain awareness about value addition to improve their income from livestock farming. He encouraged the trainees to consider starting a business based on value-added milk and meat products. During the event, the Chief Guest released a training manual on Value Addition of Milk and Meat Products. This manual serves as a self-explanatory guide for preparing value-added products. In the valedictory session, trainees displayed their milk and meat products.

2.4.27 Skill Development Training Programme on Mushroom Cultivation

ICAR-NAARM organized a 5-day Skill Development Training Program on “Mushroom Cultivation” for the farmers under SCSP program at ICAR-Krishi Vignan Kendra, KSNUAHS, Shivamogga, during 18 – 22 December 2023. A total of 40 participants including 8 women participants from Shivamogga, Davanagere, and Raichur districts of Karnataka participated in the program. Dr. Ch. Srinivasa Rao, Director of ICAR-NAARM participated in the inaugural session as the Chief Guest and he enlightened the participants about the importance of mushrooms in gaining sustainable income. Dr. Rao stressed the need for farmers to gain

awareness about mushroom cultivation to improve their income from farming. During the event, the Chief Guest released a training manual on Mushroom cultivation. Dr. Hemla Naik, Director of Education, KSNUAHS, Shivamogga expressed that farmers can get better marketing linkages through group approaches like Farmers Producers Organization (FPO). He also expressed that due to the nutritive value of mushrooms and one of the important sources of vitamin B12 for vegetarians, the mushroom has demand in the market this could be exploited through FPOs. Dr. K.T. Gurumurthy, Director of the extension, KSNUAHS, Shivamogga in his presidential address encouraged the farmers to go for value addition of the mushroom to get more profit and enlightened the audience about the importance of mushrooms as a part of the IFS component for effective and efficient utilization of resources. Dr. M. Balakrishnan, Principal Scientist & Chairman SCSP ICAR-NAARM, Hyderabad addressed the trainees and he expressed that mushroom is one of the most important sources of regular income in 25-30 days. As part of the program, trainees also took an exposure visit to the mushroom cultivation unit at Kerehalli, Uluvi hobli of Soraba Taluk. Trainees visited Farm Fresh Organics run by Mr. Manasvi Hegde.



Figure 2.4.24: Inauguration of Training Programme on Mushroom Cultivation

2.5 Mobilizing Science and Technology for Innovation and Sustainable Development

A major yardstick of research output in agricultural and allied sectors is identification of useful technologies, both tangible as well as intangible, that aid in enhancing the relevance of the national agricultural research system. Thus prioritization, monitoring, and evaluation of research projects gains importance in assessing these technologies both via suitable metrics and in institutionalisation of the process. Moreover, a proper mechanism towards intellectual property aspect in research management not only helps in the filing of patents, plant variety rights, copyrights, trademarks, trade secrets etc., once technologies or research outputs have been finalized, but also in transfer and/ or commercialization of agricultural technologies. In addition, technology management can be done via foresight exercises to set priorities for research on the basis of future technological possibilities and societal challenges, re-orient research and innovation systems to cater these needs and to create awareness about alternative possibilities including consideration of unconventional yet plausible ways of research, technology and innovation policy. In order to achieve the objectives under this theme, the Division of Research Systems Management has earmarked the following areas in which the mandated activities of systematic studies, education and training programs are carried out: (1) Research Management and Priority Setting; (2) Intellectual Property, Technology and Innovation Management; (3) Research in the context of Food Security, Globalization and Sustainability.

2.5.1 Indian Foodgrain Production Towards 2050: Implications for Self Sufficiency, Food Security and SDG Achievements

The study intends to project demand for and supply of key food commodities by the years 2025, 2030, 2040 and 2050 and assess the resultant demand-supply gap aiding policy makers to make Indian agriculture self-sufficient, and food secure and in addition achieve Sustainable Development Goals (SDGs). For this, data collection from secondary sources is being done.

2.5.2 Commercialization of Agricultural Technology Scheme

This work is carried out under National Agriculture Innovation Fund (NAIF) (Component-I: Innovation Fund of Intellectual Property & Technology Management (IP&TM) Unit of ICAR coordinated via Zonal Technology Management Committee (ZTMC). The ZTM Unit at NAARM (NAARM-ZTMU) actively engages in enhancing expertise within the realms of IP management and technology transfer/commercialization. This is achieved through the provision of comprehensive training programmes and acting as a dedicated “help desk” for agribusiness incubators, Startups, and State Agricultural Universities (SAUs). Additionally, NAARM-ZTMU supports other ICAR institutes by conducting patent searches as a valuable resource for prior art in patent filings. Functioning as an institutional mechanism, NAARM-ZTMU plays a pivotal role in safeguarding and overseeing innovations and IPs originating from the institute. During the calendar year 2023, the unit has successfully filed seven trademarks and three copyright applications for software and literary works developed at NAARM. Furthermore,

ZTMU at NAARM is deeply involved in research activities aimed at comprehending the evolving trends in agricultural technologies.

2.5.3 Dynamics of Agriculture and Food Systems in Different Agroecosystems of India

During the period under report, the agricultural food system for one district has been analyzed to conduct the same in other districts within one agroecological zone. In addition, a simple causal loop diagram (CLD) for problem scoping (Food availability, accessibility, and affordability) has been developed.

2.5.4 Holistic Approach to Human-Wildlife Conflict Mitigation and One Health for the faculty and experts of the State Agricultural Universities and Krishi Vigyan Kendras (KVKs)

The two-day virtual training during 15 – 16 May 2023 was facilitated by the ICAR-National Academy of Agricultural Research Management (ICAR-NAARM) under the Indo-German



Figure 2.5.1: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM addressing the participants of the two day Training

Collaborative project ‘Human-Wildlife Conflict Mitigation in India’. It was organized in two stages viz., May 15 on HWC mitigation by the faculty members and experts from GIZ and ICAR-NAARM and May 16 on Trainer’s laboratory where participants join as co-trainers to create awareness among the farmers of the study area. This training program aimed to train faculty members and experts from the state agriculture universities and KVKs to gain knowledge on HWC mitigation and for the effective implementation of various training programs, particularly on HWC management, for the target audiences.

2.5.5 Gene Editing and Technology Management in Agriculture

The academy organized the training programme on “Gene Editing and Technology Management in Agriculture” during 10-14 July 2023. The training programme was structured into three main modules viz., theoretical aspects of gene editing, the theoretical foundation of technology management in agriculture, and practical hands-on sessions focused on gene editing. The training covered crucial topics such as Biotechnological Tools for Crop Improvement, Genome Editing



Figure 2.5.2: Participants of Training on Gene Editing and Technology Management in Agriculture

for High Grain Yield in Rice, Bioinformatics Tools for Genome Editing, Genome Editing Crops Commercialization: Industry Perspective, Therapeutic Genome Editing, Guide RNA Cloning and Construct Preparation, Primer Editing, and Base Editing, Biosafety Regulations of Genome-Edited Crops, Genetic Transformation and Plant Development, Technology Valuation and Commercialization, IPR and Protection of Biotechnological Inventions, Editing Characterization and Phenotyping and Genome Editing for Abiotic Stress Tolerance. The program was designed in an innovative and collaborative mode with the Indian Institute of Rice Research, Hyderabad.

2.5.6 Agricultural Production System Modelling

The academy organized the training program on “Agricultural Production System Modelling” during 25- 29 September 2023. The program offered a comprehensive understanding of recent modeling approaches and strategies in agriculture. The primary objectives included providing participants with insights into the application of various simulation models such as APSIM, DSSAT, and RothC. The program equipped participants with hands-on training, enabling effective utilization of these simulation models. Participants learnt to evaluate different agricultural management practices and assess their impact on crop yield, soil health, and plant processes, particularly under potential future climatic scenarios. Additionally, the program included practical training on simulation models using R software, enhancing participants’ skills in time series analysis and modelling. By focusing on practical applications and hands-on experiences, the program empowered

participants with the knowledge and tools necessary for informed decision-making in the dynamic field of agriculture.



Figure 2.5.3: Participants of Agricultural Production System Modelling

2.5.7 Online Training Programme on Design Thinking in Agricultural Research and Education

The Online Training Programme on “Design Thinking in Agricultural Research and Education”, conducted at the academy during 9 – 13 October 2023, engaged a total of 34 participants, with 14 female participants. The primary objectives of the program were to instill design thinking perspectives among researchers and educators, implement design thinking approaches in research and education management through specially designed web-based open-access



Figure 2.5.4: Participants of the Online Training Programme on Design Thinking in Agricultural Research and Education

software, and foster competency for the implementation of research ideas and the delivery of education with empathy, involving clients and other stakeholders. The program incorporated innovations to enhance the learning experience, offering participants the opportunity to improve their design-thinking mindset. The majority of exercises were conducted online, allowing participants to interact with specifically developed online software such as AHP Analyser, Research Concept Writer, and Project Logframe Writer.

2.5.8 Training Programme on Enhancing Teaching Competency, Academic Management, and Developing Winning Research Proposals

Training on “Teaching Competency Enhancement (TCE), Academic Management (AM) and Developing Winning Research Proposal (DWRP)” for the Faculty of Maharashtra Animal & Fishery Sciences University, Nagpur was held at ICAR-NAARM during 18-22 December 2023. This Program was offered to improve the teaching competency of the faculty, in line with New Education Policy 2020. 15 Assistant Professors/ Associate Professors/Professors, representing 8 affiliated colleges of MAFSU attended the Training Program. There were a total of 18 sessions, which were spread over a five-day schedule. In-house faculty covered all the sessions. The sessions covered topics such as National Education Policy 2020: Challenges and Opportunities for Agricultural Education; Teaching Management, and how to write Research Papers in High-

impact Journals. Technology-enabled teaching; e-learning principles for teaching management; Learner-Centric Teaching Management; Learning-Centric Teaching Management; AI and Cloud Education Management; How to propose Winning Research; Design Thinking in Research Management; Emotional Intelligence; Stress Management while handling teaching modules; Challenges and opportunities in Animal Husbandry and fisheries Sciences Education; Positive Thinking and Achieving Excellence; Student-Centered Learning; Communication of Education Information through Mainstream Media; IP issues and technology management and a local visit to ICAR - Directorate of Poultry Research (PDP).



Figure 2.5.5: Training Programme on “Enhancing Teaching Competency, Academic Management and Developing Winning Research Proposals

2.6 Information and Communication Management for Promoting Innovation & Governance

In the era of Artificial Intelligence, there is a need for managing data generated from research organizations to effectively utilize them for scientific advancements and decision making. Emerging technologies are revolutionizing the traditional way of doing farming. Drones, sensors, IoT, smart farms, etc. are commonly heard within the NARES ecosystem. The Academy has taken a proactive lead to enable various stakeholders to understand, use, and implement these in their respective disciplines. The Division of Information and Communication Management is promoting them through capacity-building programmes, workshops, and research. The division has started during this year on equipping the scientists working in bioinformatics with various computing tools, particularly for analyzing next-generation sequences. The other important areas of the division include data analytics, geospatial analysis, artificial intelligence etc.

2.6.1 Mega Environment Analysis Tools for AICRP Trials

The project is aimed at incorporating additional resources such as weather in AICRP Data to develop prediction models for genotype models using statistical and other approaches. These models are expected to be useful in predicting the performance of genotypes in untested locations using the information available on tested location performances and weather variables in both tested and untested locations. During the year, 3 datasets, each containing 7000 data points, were selected from the

AICRP Maize. Each set contained about 50-55 genotypes performances at 15-20 locations. The trait considered for the prediction is Cob Weight. The prediction problem was also modified into a classification problem by considering cob weight > 3.5 kg/plot as the desired outcome. GDD and average temperature were considered. Random Forest Regression Model for prediction, as well as classification, was tried for the dataset along with traditional statistical models such as Generalized Linear Mixed Model with Genotype as Random and Locations as Fixed or Random. Based on the preliminary results, the average prediction errors range around 10-15% in the majority of the cases and in some cases, it was more than 20%. No single model performed consistently well in all situations. Performances of the prediction models developed will be compiled and submitted in the final report.

2.6.2 Comprehensive Geospatial Analysis of Environmental Change in Farming

Remote sensing technologies with high-resolution satellite imagery provide the opportunity to monitor and estimate greenhouse gas (GHG) emissions, particularly from sources such as agricultural activities. These tools enable the creation of detailed maps of land use and land cover, the assessment of vegetation health through indices like NDVI, and the monitoring of agricultural fields to estimate emissions associated with different practices. Additionally, infrared imagery aids in identifying heat emissions from various sources, contributing to the overall understanding of GHG dynamics. Integrating these data with GIS and employing advanced processing techniques such as machine learning enhances the accuracy of emission estimates, providing valuable insights for climate

change mitigation and adaptation strategies. The objectives of this project are to assess GHG emission (CO_2 , NO_2 , CH_4), and develop a suitable methodology for the estimation of methane emission and environmental pollution distribution patterns. The main focus is on Methane emission from the agriculture sector, such as paddy cultivation. The parameters used to estimate the methane emission such as Land Surface Temperature (LST) data, Precipitation data, Evapotranspiration (ET) data, Area under rice cultivation, and Methane flux value for rice.

High-resolution satellite images are processed and downloaded from Google Earth Engine (GEE) Platform. LST data was obtained from the Landsat -8 Collection 1 Tier 1 composites scenes in each 8 days with 30-meter spatial resolution, Evapotranspiration (ET) data were obtained

from MOD16A2 Version 6 it is an 8-day interval with a resolution of 500-meter, Precipitation data obtained from CHIRPS (Climate Hazards Group Infrared Precipitation with Station data) which incorporates 0.05° resolution satellite imagery approximately 5 km spatial resolution and rice sowing area data were obtained from the Directorate of Economics and Statistics, MoAFW, GoI. The methane flux value taken from the research papers. to estimate methane. Based on this methodology, the complete spatial distribution of CH_4 emission analysis has been done, the projected methane emission from rice field were analyzed and validated with GHG Platform India (AFOLU by GHGPI) (<https://www.ghgplatform-india.org/>) from 2014-2018 and year wise methane emission from rice fields for Andhra Pradesh given in Fig .

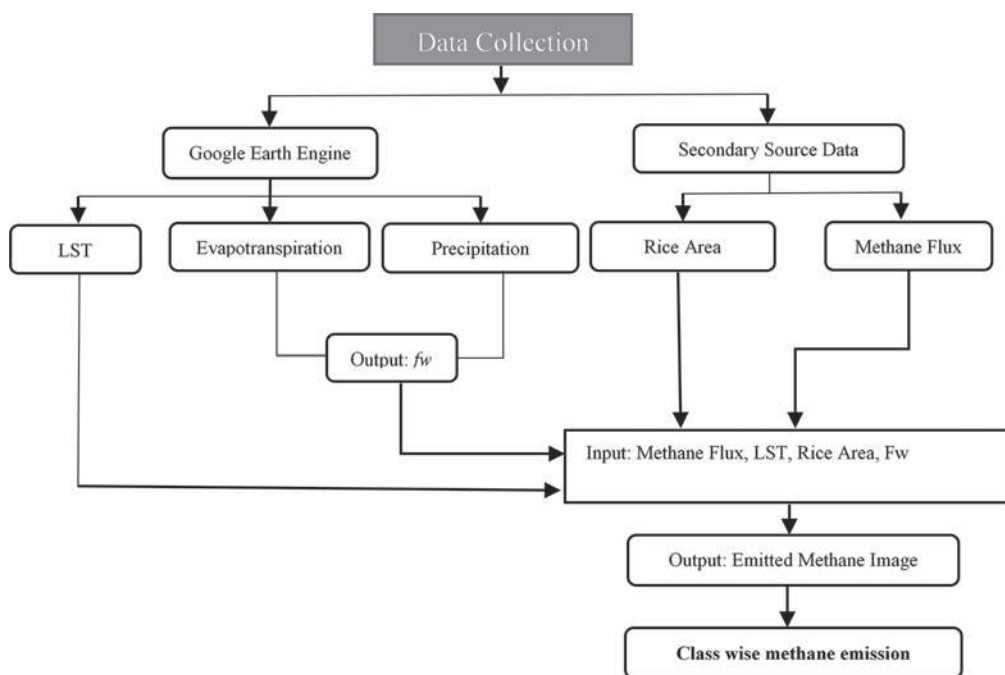


Figure 2.6.1: Data flow diagram



Figure 2.6.2: Model estimation of methane emission from rice fields compared with GHG platform-India

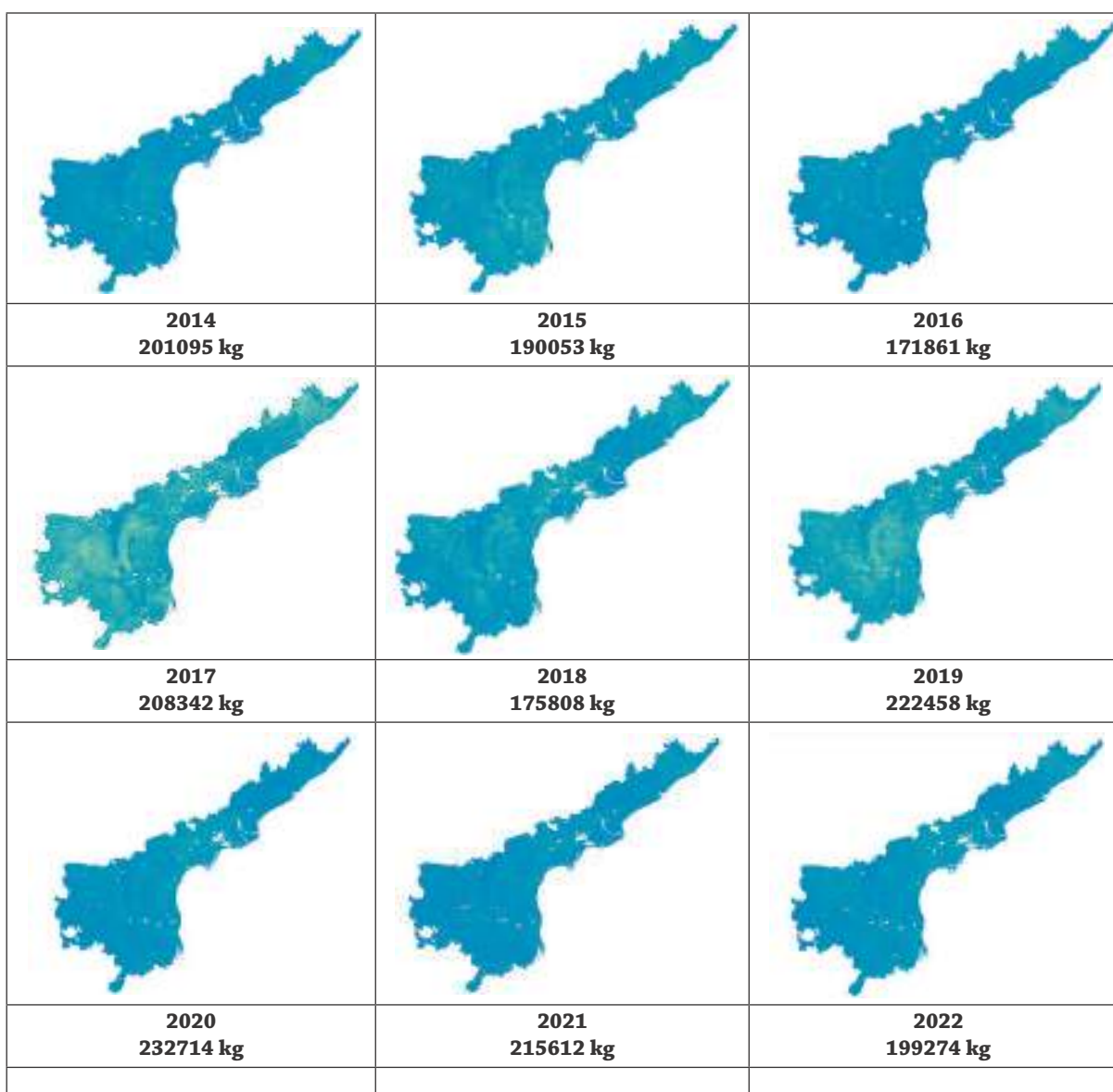


Figure 2.6.3: Yearwise methane (CH₄) emission map for Andhra Pradesh.

2.6.3 Impact Assessment of SCSP implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries

As a constituent of ICAR, NAARM is actively implementing various activities and interventions under the Scheduled Caste Sub Plan (SCSP) to support SC farmers, including youth and women, in selected districts such as Nandyal and Kurnool in Andhra Pradesh state. The proposed project aims to study the social and economic backwardness of the Scheduled Caste (SC) population in our country, in comparison to other population categories in the adopted villages under ICAR-NAARM. It has been observed that approximately 71% of SC farmers work as agricultural laborers in these villages. These districts have been chosen based on their SC population, which is at least 50% or higher. Since 2019, NAARM has been adopting villages

in these districts to provide agricultural inputs and offer skill development programs, with the aim of improving their income and livelihoods. It is expected that these interventions will bring about the desired impact over a significant period of time. More than 24 Skill development training programs for the farmers and farm women such as Integrated farming system, Backyard Poultry, Integrated Crop Management in Chilli, Value addition to Millets, Value addition to Milk and meat products, mushroom cultivation, Advanced Aquaculture Techniques, Vermicomposting, Food Processing for Entrepreneurial Development etc. were organised. A total of 1191 farmers have been trained under the skill development and capacity building program in agriculture and allied sectors. These farmers represented from Telangana, Andhra Pradesh, Karnataka, Kerala, Uttar Pradesh, and Tamil Nadu states.



Figure 2.6.4: Impact Assessment of SCSP implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries

In this project, the objective-wise interview schedule and questionnaires were prepared. Primary data collection on socio-economic status of beneficiaries is completed based on the selected parameters. Critical inputs were demonstrated among the sub-plan beneficiaries. Demonstration and Critical Input Supply under SCSP (2022-23) viz., Cherukucherla of Midthur Mandal of Nandhyal District on 27th June 2023 and Peddakadubur of Peddakadubur villages of Kurnool district in Kharif 2022. A total number of 672 farmers were benefited. As part of the project, policy guidelines have been prepared for the successful implementation of the SCSP Scheme at ICAR institutes.

2.6.4 KRISHI- ICAR Research Data Repository for Knowledge Management

Under ICAR-Research Data Repository for Knowledge Management, the major activities involve maintaining the AICRP Information System for Sorghum, Small Millets, and Maize and support for AICRP on Castor, Sunflower, Safflower, and Vegetable Crops. During the year, all three AICRPS, Sorghum, Small Millets, and Maize created their trials as well as analyzed their data using the automation system. Based on the experiences gained through analysis modules of the AICRP Information system, a new analytical workbench for analyzing Multi-Location Trials is being developed. Multi-Location Trials are the same experiments conducted over different locations and possibly over years also. These kinds of trials are common in many agricultural experiments as they are used not only for varietal improvements but also for technology development. The online tools provide only a single option for analyzing the

data. The analytical workbench provides a more personalized analytical tool for the researchers and at the same time hides the statistical complexities and enables them to concentrate their efforts on interpreting the results. The workbench is being created using the R-Shiny App. This app can be launched either through a web server or through a personal computer having the R and R-Shiny packages loaded. The interface is similar to any web application but can run on a desktop. The app can be started by using the interface to choose the folder where the excel files of the experiments are located. Multiple excel files or a single excel file can be selected from the folder. Once the files are read successfully, the user can initiate the individual location-wise analysis and pooled analysis across locations. For the individual locations, two files, one in the form of excel files consisting of detailed summarized values such as Means, CV, SEM, P-Values etc, and a CSV file in the form of reports, are generated. For the pooled analysis, a single excel file consisting of complete summary measures are provided. The workbench has been tested for these two steps. The bi-plot generation step is being incorporated in the workbench.

2.6.5 Identification and characterization of superior haplotypes for agronomically important traits in Rice (*Oryza sativa* L.)

Rice (*Oryza sativa* L.) is the major staple food for over and above half of the world's population. The sequencing data created a "digital library" of the 3,000 accessions, allowing natural genetic differences in rice to be identified. The project is aimed to characterize superior haplotypes for heat stress and low P tolerance using a 3K rice genome panel and to develop haplotype-specific

molecular markers. Key genes associated with heat tolerance in rice from the available genomic information were collected. Single nucleotide polymorphism variation was obtained across the selected target genes from the International Rice Informatics Consortium (IRIC) SNP Seek database.

2.6.6 Identification of similar research studies using machine learning techniques

This research project aims to develop a methodology for identifying similar research studies. Subsequently, the methodology will be employed to identify similar research studies conducted in NARES. Initially, the model is intended to be built and tried for rice-related studies. The data pertaining to research articles on rice are being collected from journal databases like Web of Science, Scopus, or J-Gate. After the collection of data, different language models and similarity indices will be used to identify similar studies. The output of this project will be useful in streamlining the R & D investments.

2.6.7 Online Training Programme on Impactful ICT Applications and Technologies in Agriculture

The Academy organized an online training program on “Impactful ICT Applications and Technologies in Agriculture” during 06-10 February 2023. The program was designed into 3 sections which covered (i) Concepts of ICTs, Artificial Intelligence, Machine Learning, Deep Learning & IoT; (ii) Python programming and Hands-on Python programming. (iii) Presentation by various start-ups explaining their AI-powered digital platforms that are currently implemented for various stakeholders.

All the resources required for the program were shared in online mode, which includes an ecosystem of digital resources required for hands-on practical sessions as well. The program was attended by 33 participants from 17 ICAR institutes & 16 State Agricultural Universities who belonged to 16 agricultural disciplines. The program covered the importance of ICT/AI methodologies in the current scenario, where more than 100 billion devices are connected to the web and generate a lot of data. The program also covered the importance of capacity building on ICT/AI tools in the NARES and detailed about various initiatives by start-ups of a-IDEA.



Figure 2.6.5: Participants of the online training program on impactful ICT applications and technologies in agriculture

2.6.8 Application of Bioinformatics in Accelerating Agricultural Research

The Application of Bioinformatics in Accelerating Agricultural Research, conducted over five days from February 13 to February 17, 2023, witnessed the active participation of 24 individuals, including 8 female participants. The primary objective of the program was to provide researchers with a comprehensive orientation on a diverse array of bioinformatics tools essential for advancing agricultural research. Participants were immersed in practical experiences, gaining exposure to various bioinformatics software

and tools. The hands-on training focused on the analysis of biological data using potent open-source software like R and Galaxy, enabling researchers to enhance their proficiency in leveraging bioinformatics for agricultural research applications.



Figure 2.6.6: Training programme on Application of Bioinformatics in accelerating Agricultural research

2.6.9 Data Visualization using R

The program was organized during 1-8 March 2023 with 108 participants. This program was meant to expose the participants to R & R Studio and equip them to create impactful data visualization using various R packages. The content of the workshop included hands-on sessions on the Basics of R, Data Management/wrangling, and Visualization of different types of data viz., longitudinal data, spatial data, and text data. Several R packages such as ggplot2, dplyr, tidyr, tmap etc were employed and the applicability of various functions available in these packages was taught to the participants.

2.6.10 Use of information and communication technology in teaching

The Academy organized an off-campus training program focused on harnessing information and communication technology (ICT) in teaching, conducted during 4 - 6 July 2023. The program was designed to enhance teachers' knowledge

of ICT tools and resources, develop their skills in using ICT to create and deliver engaging and interactive lessons, help them integrate ICT into their teaching practice in a meaningful way, and promote the use of ICT to improve student learning. The training program covered an extensive array of topics to provide a well-rounded understanding of ICT integration. Participants delved into the exploration of various online platforms suitable for courses, delving into course management systems, and examining diverse online data collection methods. Additionally, the program focused on the nuances of digital content creation and value addition, acquainting teachers with a range of open education resources. In-depth discussions were held on the latest tools and technologies pertinent to teaching, with a particular emphasis on digital content creation and management. This multifaceted approach aimed to equip educators with comprehensive insights and practical skills, empowering them to leverage ICT effectively in their teaching endeavors. This off-campus training program was a concerted effort to advance teachers' proficiency in ICT, with the ultimate goal of elevating the quality of education through strategic and meaningful integration of technology in the teaching and learning process.



Figure 2.6.7: Training program on the use of information and communication technology in teaching

2.6.11 Advances in mobile application development

The ICAR-National Academy of Agricultural Research and Management in Hyderabad conducted a five-day Capacity Building Training program on “Advances in mobile application development” during 7-11 August 2023, in a virtual format. The program aimed to increase awareness and impart essential skills in mobile app development. Participants engaged in theory and hands-on practical sessions covering various aspects, including an overview of the Android platform, Java programming, Android Studio installation, development of Android applications, and database connectivity. The training also delved into no-coding technologies for Android app development. Similarly, a training program on “Advances in Mobile Application Development” took place from February 20-24, 2023, with a focus on enhancing awareness of the latest developments in the field. The sessions covered theoretical aspects of the Android platform, emphasizing foundational knowledge for effective mobile app development. Practical sessions centered around Java programming, Android Studio utilization,



Figure 2.6.8: Participants of the training program on advances in mobile application development

and database connectivity, while also introducing participants to no-coding technologies for innovative Android app development. Both programs aimed not only to raise awareness of advances in mobile application development but also to equip participants with diverse skills for navigating the dynamic landscape of mobile technology confidently.

2.6.12 Training Programme on Analysis of Experimental Data Using R

The academy organized a “Training program on analysis of experimental data using R” during 21-25 August 2023. The training program focused on experimental Data. An introductory lecture on experiments in Agriculture was delivered initially to refresh the issues to be considered before planning as well as going for statistical analysis. The R was introduced to the participants initially. Basic data management such as filtering, creating categorical variables, merging, subsets using R was demonstrated using different types of datasets. Simple and typical graphs using the ggplot2 library were also covered. The analysis of experimental data from simple designs such as Completely Randomized Design, and Randomized Block Design were covered including different types of multiple comparisons and visual representation of multiple comparison results. Analysis of factorial experiments, split plots, and strip plot designs were covered. Analysis of experiments where observations are collected over time on the same experimental units using repeated measurement ANOVA was also discussed with an example. The various options available for pooling data both from varietal improvement trials as well as from technology validation were discussed. The use of orthogonal polynomials, the need for

transformation of data, creation of randomized layouts, analysis of augmented data, and generation and analysis of Alpha Lattice designs when the number of treatments is more were also covered. Multiple Linear Regression and Regression Diagnostics, Principal Component Analysis, and Cluster Analysis were some of the multivariate techniques covered during the training program.

2.6.13 Training Programme on Impactful ICT Applications and Technologies in Agriculture

The three-day Training Programme on “Impactful ICT Applications and Technologies in Agriculture”, scheduled for 12-14 September 2023, aims to familiarize faculty members with key concepts such as Artificial Intelligence, Machine Learning, IoT, Cloud Computing, and existing ICT Applications developed by Startups, focusing on tools and technologies applicable to agriculture. The program provided hands-on demonstrations of diverse ICT technologies, enabling participants to gain proficiency in rapid application development. As part of the training, emphasis was placed on introducing participants to innovative tools and technologies that are emerging in the field of agriculture.



Figure 2.6.9: Training Programme on Impactful ICT Applications and Technologies in Agriculture

2.6.14 Online Training Programme on Next-Generation Sequencing and Data Analysis

The online Training Programme on “Next Generation Sequencing and Data Analysis” organized by ICAR-National Academy of Agricultural Research Management, Hyderabad during 16-20 October 2023 concluded today. A staggering 97 participants including 40 women participants representing 27 research disciplines from 36 ICAR Institutions, 14 Agricultural Universities, and 3 private organizations participated in the programme. The program provided the participants an insight into the enormous potential and application of Next Generation Sequencing and its various pipelines.



Figure 2.6.10: Participants of Online Training Programme on Next Generation Sequencing and Data Analysis

During the virtual training, the participants were provided with theory and hands-on practical sessions on various bioinformatics software and tools such as Linux, Python, and R related to the emerging requirements of NGS. This program laid the foundation and equipped the participants with the necessary skills to analyze NGS data independently. In his inaugural Address Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad highlighted the importance of bioinformatics for undertaking frontier

research in Next Generation Sequencing such as transcriptomic, proteomics, genomics, and metabolomics called upon the participants to undertake innovative and collaborative research to improve agricultural productivity.

2.6.15 Training Programme on Analysis of Experimental Data Using R

The training program was conducted from 20-24 November 2023 and a total of 37 participants participated in the program. The training program focused mainly on the analysis of experimental data. An introductory lecture on experiments in agriculture was delivered to refresh and highlight the issues to be considered before planning as well as going for statistical analysis. A few experimental situations in the form of problems and different possible layouts were shown and the trainees were asked to choose any of the layouts and discuss the pros and cons of the layout. The R software was introduced to the participants initially. Basic data management such as filtering, creating categorical variables, merging, and subsets using R was demonstrated using different types of datasets. Simple and typical graphs using the ggplot2 library were also covered. The analysis of experimental data from simple designs such as Completely Randomized Design, and Randomized Block Design were covered including different types of multiple comparisons and visual representation of multiple comparison results. Analysis of factorial experiments, split plots, and strip plot designs were covered. The various options available for pooling data both from varietal improvement trials as well as from technology validation were discussed. Use of Orthogonal Polynomials, the need for transformation of data, creation of randomized layouts, analysis of augmented data,

generation, and analysis of Alpha Lattice designs when the number of treatments is more were also covered. Multiple Linear Regression and Regression Diagnostics, Principal Component Analysis, and Cluster Analysis were some of the multivariate techniques covered during the training program.

2.6.16 Training program on Geospatial Analysis using QGIS & R

The Geospatial Analysis using QGIS & R training program, spanning from 20 to 24 November 2023, over five days, saw the participation of 13 individuals, including 3 female participants. Specifically designed for Scientists, Faculty, and Technical Officers engaged in Natural Resource Management at ICAR Institutes, State Agricultural Universities, and KVKs, the training covered a comprehensive curriculum. Topics ranged from an introduction to geospatial technologies in agriculture, relational database and geodatabase concepts for GIS, to the fundamentals of GIS, GPS, and remote sensing. The program delved into exploring data from open-source platforms for geodata, spatial database design, and implementation



Figure 2.6.11: Participants of Training program on Geospatial Analysis using QGIS & R

using Quantum GIS open-source software, and spatial analysis techniques such as clipping, joining, buffering, interpolation, and krigging. Furthermore, participants gained insights into image processing and land use land cover using R Studio.

2.6.17 10- days Training-cum-Certificate Course on Drone Technology in Agriculture

ICAR- National Academy of Agricultural Research Management (NAARM), Hyderabad in collaboration with G B Pant University of Agriculture and Technology, Pantnagar, and Shobhit Institute of Engineering & Technology (Deemed-to-be- University), Meerut organized 10- days Training-cum-Certificate Course on Drone Technology in Agriculture during 04-13 December 2023 at School of Agricultural Sciences and Technology, Shobhit University, Meerut. This program was initiated as part of NAHEP Component 2 objective to assist agricultural universities in strengthening their linkages with industry – Development of a model for Academia-Industry-Government linkages. A total of 36 students from GBPUAT, Pantnagar; SVPUAT, Meerut; and Shobhit Institute of Engineering & Technology (Deemed to-be- University), Meerut participated.



Figure 2.6.12: Participants of the 10 days Training-cum-Certificate Course on Drone Technology in Agriculture

2.7 Foundation Courses

2.7.1 The 112th Foundation Course for Agricultural Research Service (FOCARS)

The 112th Foundation Course started on 11 April 2023. It was inaugurated by Dr Himanshu Pathak, Secretary DARE and DG, ICAR. The Guest of Honor was Dr. MV Gupta, Padmasree Awardee and World Food Prize Winner. There were 117 trainees who joined in this course including 42 female scientists. All the trainees took the oath of allegiance to serve the country and constitution of India administered by Dr. Ch Srinivasa Rao, Director, ICAR-NAARM.

The training had 3 phases: The first phase had orientation and capacity building, second phase had Field Experiential Training (FET) and the third phase covered a multidisciplinary perspective. There were interaction sessions every Saturday by inviting different groups of



Figure 2.7.1a: Inauguration speech by Dr Himanshu Pathak, Secretary DARE and DG, ICAR

speakers namely Women Directors, Directors of ICAR institutes, Award Winning Young Scientists, Industry Representatives, Startup founders, and many more. Regular yoga sessions and sports activities to reduce stress are also part of the training curriculum.

Industry visit was also organized to Godfrey Philips, Ongole; ITC tobacco processing unit, Perala; Kadiyam Nursery; Central Tobacco Research Institute, Rajahmundry; Indian Institute of Oil Palm Research, Pedavegi; Directorate of Floriculture Research regional station, Rajahmundry. The participants also visited the International Crops Research Institute for the Semi-Arid Tropics, the Indian Institute of Millet Research, the Indian Institute of Oilseeds

Research, and the National Meat Research Institute in Hyderabad. The participants were sent to 19 different KVKs across the country as part of their FET module.

The valedictory program was organized on July 10, 2023. Dr. Padma Raju, Former Vice Chancellor, of ANGRAU, Andhra Pradesh was the Chief Guest during the valedictory programme.



Figure 2.7.1b: Participants of 112th FOCARS



Figure 2. 7.1c: Special Interactive Sessions of 112 FOCARS



Figure 2. 7.1d: Industrial visit for 112th FOCARS



Figure 2. 7.1e: Best 112th FOCARS Trainee Award received by Dr Harshit Kumar

2.7.2 The 113th Foundation Course for Agricultural Research Service (FOCARS)

Hon'ble Chairman, NABARD Shri. Shaji, K.V., officially inaugurated the 113th Foundation Course for Agricultural Research Service on 18 July 2023 at the ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad. The foundation course witnessed the enrolment of 69 scientist probationers (27 Females + 42 male) representing 32 ARS disciplines and hailing from 21 states / UTs of India. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, led the scientist's trainees through an affirmation of allegiance to serve our country and its constitution.

The first phase was to provide orientation and capacity building, the second phase was in the form of Field Experiential Training (FET), and the third phase covered multidisciplinary perspective agricultural research. During the first phase of the program, major emphasis was laid on configurable approaches around themes viz., “National and International Agricultural Research Systems”, “Agricultural Price Policy Perspectives”, “Intellectual Property Rights and Technology Management”, “Human Resource Management”, “Scientific Writing, Documentation, Presentation and Communication Management”, “Participatory Technology Development and Dissemination”, “Research Project Planning and Budgeting”, “Introduction to Administration and Financial Management including Procurement and Purchase Procedures”, etc. The Second Phase of Field Experience Training (FET) provided a first-hand opportunity for the scientist trainees to focus research efforts on field realities and needs of stakeholders by employing participatory rural appraisal methodology. Finally, the third phase was organized with more interactive sessions with leading researchers to expose the scientist probationers to the challenges and opportunities in agricultural research

Interaction sessions have been conducted every Saturday since the first phase, inviting diverse groups of speakers including Women Directors, Directors of ICAR Institutes, Young Scientist awardees, Industry Representatives, Progressive farmers, and Startup founders.

Industrial visit for Probationary Scientists of 113 FOCARS batch was planned and executed to enable scientist probationers to gain practical hands-on exposure to important agri-industries

and marketing systems in Andhra Pradesh during 18-19 Aug, 2023. The visits included Tobacco Board's e-Auction Floor at Kandukur, ICAR- CTRI-RS and the KVK, Kandukur, Sri Dhanalakshimi Cotton & Rice Mills (P) Ltd, Vallabha Feeds (P) Ltd. Narsaraopet and Cattle Breeding Station at Chadalawada.

The participants also visited the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), Indian Institute of Rice Research (IIRR), Indian Institute of Millet Research (IIMR), Indian Institute of Oilseeds Research (IIOR), Directorate of Poultry Research (DPR), National Meat Research Institute (NMRI) and Rural Technology Park (RTP) of NIRD, Hyderabad.

The 113th Foundation Course was completed on 17.10.2023. Dr. BN Mathur, Former Director, of ICAR- NDRI & NAARM, was the Chief Guest and Dr. RC Agrawal, DDG (Agriculture Education) was the Guest of Honor during the valedictory program.



Figure 2.7.2a: Inaugural address by Shri. Shaji, Chairman, NABARD



Figure 2.7.2b: Participants of 113th FOCARS



Figure 2.7.2c: Special Interactive Sessions of 113 FOCARS



Figure 2.7.2d: Industrial Visit for Probationary Scientists of 113 FOCARS



Figure 2.7.2e: Valedictory Program of 113 FOCARS

2.7.3 Foundation Course for New Faculties of OUAT, Bhubaneswar & PVNRTVU, Hyderabad

ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad organized 21 days Foundation Course for the Faculty of Orissa University of Agriculture & Technology (OUAT), Bhubaneshwar, and PV Narasimha Rao Telangana Veterinary University (PVNRTVU), Hyderabad from January 24 to February 13, 2023. A total of 31 participants (18 male & 13 female) participants from agriculture, veterinary and horticulture disciplines attended the foundation

course. The objectives of the foundation course were to improve the participant's Knowledge, Skills, and Attitudes (KSA) change in Facets of Education, the Process of Teaching & Learning, Writing Winning Research Proposals, National Education Policy (NEP) - 2020, Project Management Techniques, Competencies & Group Dynamics, Entrepreneurship & Startups, etc. Participating trainees were also taught about handling various RTIs, Statutes, and Regulatory authorities present in the education and research in the university system. A total of 69 sessions including lectures and practicals



Figure 2.7.3a: Foundation Course for New Faculties of OUAT, Bhubaneswar & PVNRTVU, Hyderabad

were completed by eminent experts on various aspects of teaching and learning. Educational trips were also arranged for the participants to make better off-farm experiences.



Figure 2.7.3b: Participants of the Foundation Course for Faculty of Agricultural Universities (FoCAU)

2.7.4 IX Foundation Course for Faculty of Agricultural Universities (FoCAU)

The academy organized the IX Foundation Course on Faculty Advancement in Agriculture Universities (FoCAU) from September 7 to September 27, 2023, with four distinct objectives. Firstly, the course aimed to orient participants to both national and international agricultural scenarios. Secondly, it sought to enrich the knowledge, skills, and attitude (KSA)

of participants in various aspects of education, teaching-learning processes, and education technology. Thirdly, the course aimed to expose participants to research and technology transfer processes. Lastly, it aimed to sensitize participants to organizational capacities related to human capital, administration, and financial management. The participating faculty, representing four universities—Rani Laxmi Bai Central Agricultural University (RLBCAU) in Jhansi, Sri Venkateswara Veterinary University (SVVU) in Tirupati, Tamil Nadu Dr. Jayalalithaa Fisheries University (TNJFU) in Nagapattinam, and Guru Angad Dev Veterinary and Animal Sciences University (GADVASU) in Ludhiana—contributed to a diverse and collaborative learning environment. The course comprised of 43 lecture modules, covered a spectrum of topics including agricultural education, research, extension, advanced teaching technologies, career enhancement strategies, human resource management, and administrative cum financial rules and regulations. Additionally, educational and exposure visits were organized to further enrich the experience and knowledge of the participating young faculty from agricultural universities.



Figure 2.7.4: Participants of the Foundation Course for Faculty of Agricultural Universities (FoCAU)

2.8 Research Projects (Completed & On-going)

2.8.1 Completed Research Projects during 2023

S No	Project Code	Project Title	Team	Duration	
				From	To
In-House					
1	AGED/NAARM/SIL/2020/004/00170	Identification of Best Practices Developed by the Leadership in Scientific and General Administration in NARES	RVS Rao, P Ramesh, Alok Kumar, K Kareemulla, S Senthil Vinayagam, S Ravichandran, D Thammiraju, Ch Srinivasa Rao, G Venkateshwarlu	June 2020	February 2023
2	AGED/NAARM/SIL/2018/009/00148	Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India	B Ganesh Kumar, Sanjiv Kumar, D Thammi Raju, PC Meena, Tavva Srinivas, A Dhandapani	October 2018	September 2023
3	AGED/NAARM/SIL/2019/012/00166	Agri Startups in India: Decoding the Determinants and Success Factors	Sanjiv Kumar, K Srinivas, Ranjit Kumar, Umesh Hudedamani, KH Rao, Senthil Vinayagam, Surya Rathore	June 2019	May 2023
4	AGED/NAARM/SIL/2021/002/00178	Strategy for Strengthening National Water Policy	K Kareemulla, S Ravichandran, KV Rao, PS Brahmanad	January 2021	March 2023
Extra-mural Projects					
1	AGED/NAARM/CCL/2021/004/00180	Developing and Pilot Testing Specialized Knowledge Products on Human Wildlife Conflict Mitigation for Agriculture and Veterinary sector in India	Ch Srinivasa Rao, P Krishnan, Tavva Srinivas, Bharat S Sontakki, Ranjit Kumar, Sanjiv Kumar, G Venkateshwarlu	December 2021	May 2023
2	AGED/NAARM/SOL/2019/005/00159	Establishment of Skill Development Bioinformatics Infrastructure Facility (Skill-BIF) for Agricultural Biotechnology (DBT)	M Balakrishnan, SK Soam, P Supriya, Satendra Kumar Mangrauthia, Manish Pandey	December 2018	March 2023
3	AGED/NAARM/SOL/2018/013/00152	Profiling Resources devoted to R&D in Agriculture Sector	M Balakrishnan, P Krishnan, M Ramesh Naik, H Umesh, Tavva Srinivas, SK Soam	August 2020	June 2023

2.8.2 Ongoing Research Projects

2.8.2.1 In-House Projects

S No	Project Code	Project Title	Team	Duration	
				From	To
1	AGED/NAARM/SIL/2022/004/00184	Motivational Factors Driving Scientific Personnel: Identification of Strategies for Scientific Excellence	P Ramesh, RVS Rao, BS Yashavanth, G Venkateshwarlu	April 2022	April 2024
2	AGED/NAARM/SIL/2020/001/00167	New Market Reforms and E-NAM in India: Effect on Marketing Choices and Participation and Price Realization for Small holders	Ranjit Kumar, Sanjiv Kumar, A Dhandapani, K Srinivas, PC Meena	June 2020	March 2024
3	AGED/NAARM/SIL/2020/002/00168	Techno-Economic Feasibility Studies for ICAR Agro- Processing Technologies/Value Added Products	Srinivas Tavva, B Ganesh Kumar, GRK Murthy, K Kareemulla, Sudha Mysore	June 2020	September 2024
4	AGED/NAARM/SIL/2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N Sivaramane, Ranjit Kumar, PC Meena, Bharat S Sontakki, Vijay Avinashilingam, P Venkatesan, Tavva Srinivas	October 2018	September 2024
5	AGED/NAARM/SIL/2023/002/00190	Disaggregated food product demand: Insights and predictive analysis	N Sivaramane, P Venkatesan, Tavva Srinivas,	September 2023	August 2025
6	AGED/NAARM/SIL/2020/010/00176	Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India	Surya Rathore, P Krishnan, Lipi Das, Veenita Kumari, B S Yashavanth	January 2021	December 2024
7	AGED/NAARM/SIL/2022/001/00181	MOOCs in Digital Agricultural Education-Strategies to Enhance Effectiveness, Inclusivity and Reach	GRK Murthy, VV Sumanth Kumar, S Senthil Vinayagam, P Supriya	April 2022	March 2025
8	AGED/NAARM/SIL/2022/005/00185	Development of implementation framework for National Education Policy -2020 in Agricultural Education	D Thammi Raju, GRRK Murthy, P Ramesh, Bharat S Sontakki P Venkatesan, NAVijay Avinashilingam, N Sivaramane, G Venkateshwarlu, Ch Srinivasa Rao, Senthil S Vinayagam	October 2022	September 2025

S No	Project Code	Project Title	Team	Duration	
				From	To
9	AGED/NAARM/SIL/2020/007/00173	Mega Environment Analysis Tools for AICRP Trials	A Dhandapani, Yashavanth BS, Bal SK	June 2020	May 2024
10	AGED/NAARM/SIL/2021/003/00179	Impact Assessment of SCSP implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries	M Balakrsihnan, B Ganesh Kumar, P Venkatesan, Ramesh Naik P Supriya, B SYashavanth, B S Sontakki, D Thammi Raju, N Sivaramane, PC Meena, Vijay Avinashilingam, G Venkateshwarlu	July 2021	June 2024
11	AGED/NAARM/SIL/2022/002/00182	Comprehensive Geo-Spatial Analysis of Environmental Change in Farming	PD Sreekanth, <u>D Thammi Raju</u> , P Venkatesan, Ramesh Naik	February 2022	February 2024
12	AGED/NAARM/SIL/2022/007/00187	Identification of similarity in research studies using machine learning	BS Yashavanth, Dhandapani A, Ramasubramanian V	February 2022	October 2025
13	AGED/NAARM/SIL/2020/009/00175	Dynamics of Agriculture and Food Systems in Different Agroecosystem of India	Mude Ramesh Naik, Umesh Hudedamani, N Sivaramane, BS Yashavanth, N Subash, Monoranjana Mohanty, Nishant K Sinha, Ch Srinivasa Rao, Ramasubramanian V, Damodar Reddy, SJ Balaji	January 2022	December 2024
14	AGED/NAARM/SIL/2023/003/00191	Indian Food Grain Production Towards 2050: Implications for Self Sufficiency, Food Security, SDG's achievements	MB Dastagiri, Ramasubramanian V, Damodar Reddy, Umesh Hudedamani, M Ramesh Naik	October 2023	March 2025

2.8.2.2 Externally Funded Research Project (Sponsored/Collaborative/Grant-in-Aid)

S No	PME Code	Project Title	Team	Budget (In Lakhs)	Sponsored by	Duration	
						From	To
1	AGED/NAARM/COP/ 2017/001/00121	Management and Impact Assessment of Farmer FIRST Project	P Venkatesan, Bharat S Sontakki, N Sivaramane	114.05	ICAR, New Delhi	March 2017	September 2023
2	AGED/NAARM/SOL/ 2022/008/ 00188	Distance Education Courses on Extension Management for Agricultural Professionals	Sandhya Shenoy	21.00	ICAR	January 2022	January 2025
3	AGED/NAARM/COL/2023/001/00189	Agripreneurship for Sustainable Agricultural Development: Technological and Institutional Innovations and Strategies	P Venkatesan, BS Sontakki, N Sivaramane, S Senthil Vinayagam	150.63	NASF, ICAR	June 2023	May 2026
4	AGED/NAARM/SOL/2016/002/00112	Establishment of Agri-Business Incubation (ABI) Centers under XII Plan Scheme for NAIF	SS Vinayagam, Umesh Hudedamani, Vijay Avinashilingam, BS Yashavanth,	79.15	ICAR	January 2016	December 2023
5	AGED/NAARM/COL/2023/004/00192	Rural Technology Action Group (RuTAG) 20	S Senthil Vinayagam, NA Vijay Avinashilingam, Sanjiv Kumar, Umesh Hudedamani	151.45	O/o PSA to GOI, New Delhi	October 2023	September 2026
6	AGED/NA ARM/COP/ 2019/001/00155	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education (NAHEP)	SK Soam, D Thammi Raju, N Srinivasa Rao Alok Kumar, VV Sumanth Kumar, Surya Rathore, Sanjiv Kumar, M Balakrishnan, SS Vinayagam, P Krishnan, BS Yashavanth	1704.17	ICAR-NAHEP	February 2019	March 2024
7	AGED/NA ARM/SOL/ 2015/007/00110	KRISHI: ICAR Research Data Repository for Knowledge Management	A Dhandapani, SK Soam, Umesh H, Supriya P	76.00	Data Management, ICAR HQ	November 2015	March 2026

S No	PME Code	Project Title	Team	Budget (In Lakhs)	Sponsored by	Duration	
						From	To
8	AGED/NAARM/COL/2023/005/00193	Identification and characterization of superior haplotypes for agronomically important traits in Rice (<i>Oryza sativa</i> L)	P Supriya, Satendra Kumar Mangrauthia, M Balakrishnan, RM Sundaram,	27.20	DST	September 2023	August 2026
9	AGED/NA ARM/SOL/ 2008/003/0067	IP Management and Transfer/ Commercialization of Agricultural Technologies Scheme	Umesh Hudedamani, N Sivaramane, Ramesh Naik, Sanjiv Kumar	19.15	ICAR, New Delhi	April 2014	March 2024

2.9 Educational Programme

2.9.1 Admission of 15th Batch Students of PGDM-ABM

In the admissions process for the Post Graduate Diploma in Management - Agri-Business Management (PGDM-ABM), a total of 1023 applicants applied for the program. From this pool, 168 candidates were selected for interview. The interviews were conducted at various centres, including the main centre at ICAR-NAARM where 65 applicants attended, and satellite centres in Kochi, Kolkata, Nagpur, and New Delhi, with 24, 9, 17, and 53 applicants respectively. Ultimately, 61 candidates were successfully admitted into the PGDM-ABM programme, reflecting a rigorous selection process aimed at identifying the most promising candidates for advanced studies in agri-business management.

2.9.2 NBA Accreditation and AIU Equivalence

PGDM-ABM programme got the accreditation by National Board of Accreditation (NBA) during March 2023. The period of accreditation is valid for 3 years till 2025. The Expert Committee visited the academy during 16-18 December,



Figure 2.9.1: NBA Accreditation and AIU Recognition

2022. Later, the Academy applied for Association of Indian Universities (AIU) recognition. The programme got recognized for the same period as NBA accreditation. Now, PGDM-ABM students can apply for higher studies in various universities across the country.



Figure 2.9.2:NBA Expert Team at ICAR-NAARM

2.9.3 Sankalp 8.0

ICAR-NAARM marked the splendid occasion of SANKALP 8.0, the national-level B-fest by the students of Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM), during 22 – 23 September, 2023. SANKALP 8.0 garnered an impressive response with over 9300 registrations from more than 1000 institutions, culminating in 200+ finalists in eight distinct events. The programme was inaugurated by Mr. Siva Sankar Bathala, Head Human Resources, M/s Nuziveedu Seeds Ltd. Several competitions among students in the

field of entrepreneurship development such as business plan, idea and branding were conducted. Sankalp 8.0 concluded with a valedictory event, awarding the winners of all competitions. The event received an overwhelming support from many agri-based industries.



Figure 2.9.3: Dr Ch. Srinivasa Rao addressing the participants of SANKALP 8.0



Figure 2.9.4: Participants of SANKALP 8.0

2.9.4 Educational Visit of PGDM-ABM Students

Two educational visits were organized for PGDM-ABM 2023-25 students as part of their study. The first visit was organized on 17 December 2023 to the Gudimalkapur flower market. The students interacted with various stakeholders in the market viz. flower, vegetable, fish traders, and wholesale buyers of these produce. They were sensitized about the trading process in the market along with the value chain of the agricultural produce specifically flowers.

The second visit was organized to agricultural input dealers at Chottupal on Dec 26, 2023. The students got an opportunity to meet seed, fertilizer, pesticide, and farm machinery dealers. The students learned about the supply-demand of all the inputs and their supply chain practices. They also met bank officials to get information about the agricultural loans in the region.



Figure 2.9.5a: Collection of information from various stakeholders in the market by PGDM-ABM students



Figure 2.9.5b: Collection of information from seed, fertilizer, pesticide, farm machinery dealers by PGDM-ABM students



Figure 2.9.5c: Group Photo of PGDM-ABM students

2.9.5 Campus Placement of PGDM-ABM 2021-23 Batch Students

All the students of PGDM-ABM 2021-23 batch were placed in various agri-business companies after completion of their course. They were placed in different sectors, viz. Banking & Finance (32%), Agri-Input (28%), Agritech & Startup (12%), FMCG (14%), Consultancy (7%), Supply Chain (5%), Commodity & Warehousing (2%). The average package of this batch was Rs. 10.63 lakh per annum and the maximum being Rs. 13.70 per annum secured by 4 students.

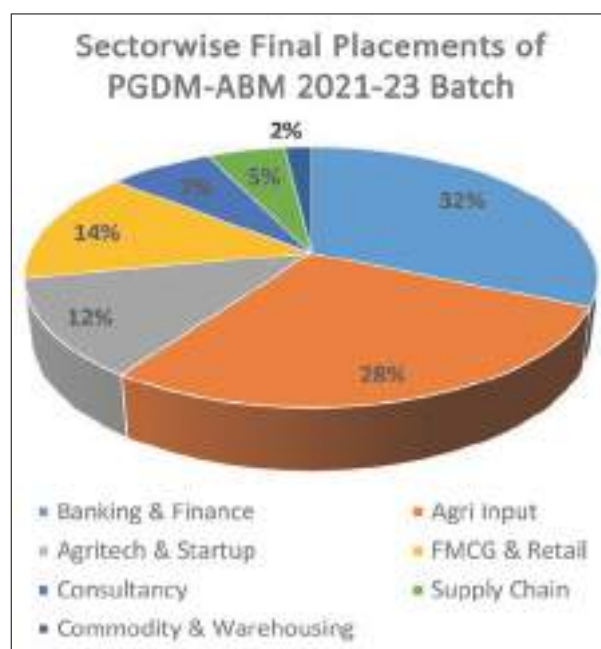


Figure 2.9.6: Campus Placement and Average Package of PGDM-ABM 2021-23 Batch Students

2.9.6 Awards won by PGDM-ABM 2022-24 batch

The PGDM-ABM students participated in various business competitions organized by other management colleges across the country. The students won many competitions and won prizes in these business competitions.

Name of Students	Event and Organizer	Position
Shivaraj Kumar Chavan	‘Vivechan-The Case Study Challenge’ in Agrovon 2023 at C.C.S National Institute of Agricultural Marketing (NIAM), Jaipur	Winner
Apoorva Chourasia and Kanchan Dubey	Khoj - The Social Idea, Case Study Competition in Krishi Chanakya 2023 in MANAGE, Hyderabad	1 st Runner-up
Kajal Khobragade, Sourabh Singh Pokhriyal, and Mrunal Kharat	“Akanksha: B-Plan Krishi Chanakya 2023” in Krishi Chanakya 2023 in MANAGE, Hyderabad	1 st Runner-up
Kajal Khobragade, Sourabh Singh Pokhriyal, and Mrunal Kharat	Business Plan Competition, Spardha in VAMNICOM, Pune	Winner
Adarsh Kumar and Shivaraj Kumar	Online quiz competition, AGQUEST organized by Agzistence	Winner
Yerrabelli Ramya	Olympus’23 badminton tournament in MANAGE, Hyderabad	1 st Runner-up



Figure 2.9.7: Awards won by PGDM-ABM 2022-24 batch

2.9.7 Project Reports Submitted by 13th Batch (2021-23) PGDM-ABM Students

Sl No	Roll No	Student Name	Title of the Project	Guide Name
1	ABM-2101	Abhishek R	A Study of m-Commerce Applications in Agriculture	Dr Umesh Hudedamani
2	ABM-2102	Abinav E Basi	Study on Challenges and Opportunities of Big Grantees	Dr S Senthil Vinayagam
3	ABM-2103	Alana Raphy	Study of Opportunities in Micro-Housing Loan and Competitor Analysis	Dr PC Meena
4	ABM-2104	Amrut Basavaraj Salimath	Understanding Tech Enabled Integrated Procurement Model Through Digital Saathi Application in Maize Crop	Dr Ch Srinivasa Rao
5	ABM-2105	Ankireddy Mahathi	Finding Efficient Strategies to Improve Sourcing of Pongamia Beans and Selling Karanja Cake while Maximizing Social Impact	Dr Mude Ramesh Naik
6	ABM-2106	Anmol Mahajan	Implications of Technological and Policy Changes on Use of Nitrogenous Fertilizers in India	Dr Ranjit Kumar
7	ABM-2107	Anshika Sharma	Assessment of Microfinance Sector in Punjab	Dr P D Sreekanth
8	ABM-2108	Apoorva	Understanding e-Rupi and its Implication on Agriculture	Dr A Dhandapani
9	ABM-2109	Arem Bhanu Prakash	Analyze and Identify the Potential Geographical Markets for Seaweed- Based Products in Telangana	Dr VV Sumanth Kumar
10	ABM-2110	Ashutosh Upadhyay	Consumer Behaviour and Market Potential for Value-Added Millet Products	Dr Alok Kumar
11	ABM-2111	Asifpasha F	Growing Opportunities: A Study of Agritech Startup Funding and Market Trends	Dr Sanjiv Kumar
12	ABM-2112	Attravanam Venkatesh	Managing Packaged Food Wastage in India	Dr Ranjit Kumar
13	ABM-2113	Balram Patidar	Study on Various Crop Produce Procurement Models	Dr Yashavanth BS
14	ABM-2114	Banala Pavan Reddy	Dragon Fruit Cultivation in India: Scope, Constraints and Value Chain Analysis	Dr Surya Rathore
15	ABM-2115	Bavisetti Sai Satya Lakshmi Sruthi	Frozen Food Marketing: SWOT Analysis	Dr D Thammi Raju
16	ABM-2116	B Pranay Kumar	Study on Backward Integration of Millet Start-Ups	Dr P Ramesh
17	ABM-2117	Bijender	Farmers' Buying Behavior for Phosphatic Fertilizers in Western Uttar Pradesh	Dr Tavva Srinivas
18	ABM-2118	Sneha Bute	Analysis of Export Potential of Millets Businesses of Millets- Based Start-Ups	Dr Ranjit Kumar

Sl No	Roll No	Student Name	Title of the Project	Guide Name
19	ABM-2119	C Anand	Gender-Inclusion in Agribusiness Management Education: A Case of PGDM-ABM at ICAR-NAARM	Dr Bharat S Sontakki
20	ABM-2120	Chethan CS	Solar Sector Analysis (Roof Top Solar) – from Debt Finance Perspective	Dr SK Soam
21	ABM-2121	Chintalapudi Ramana Sairam	Supply Chain Analysis and Market Compatibility Model in B2C Segment for Fish and Shrimp in Hyderabad	Dr P Venkatesan
22	ABM-2122	Chirra Naveen Kumar	Study on Livelihood Dimension of Tribals in Bhadrachalam District of Telangana	Dr PC Meena
23	ABM-2123	Devashish Bhimrao Dhadwe	Market Analysis and Strategy Development for Wheat Seed Sales Growth in the State of Maharashtra	Dr B Ganesh Kumar
24	ABM-2124	Gaurav	Apple Value Chain Analysis for Shimla District Himachal Pradesh	Dr GR Ramakrishna Murthy
25	ABM-2125	Gourineni Prathima	Estimation of Market Potential of Natural Sweeteners in Rare Sugars Market	Dr NA Vijay Avinashilingam
26	ABM-2126	Gupala Mounika	Greendelights Backend Operations (Jeevika Farm Retail Business Model)	Dr K Kareemulla
27	ABM-2127	Illamparithi S	Credit Market Analysis in Tamil Nadu and Applicability of Imperfect Collateral Loans	Dr G Venkateshwarlu
28	ABM-2128	Jammula Sravani	Technological Interventions to Improve the Efficiency of the Cold Supply Chain for Aquaculture	Dr M Balakrishnan
29	ABM-2129	Shweta Jaware	An Analysis of Variations in the Maize Production System in Bihar	Dr Sivaramane N
30	ABM-2130	Jayasurya M	A Study on Post Harvest Losses in Mango and its Management in Tamil Nadu	Dr B Ganesh Kumar
31	ABM-2131	Jyoti Singh	Front-End Operation for Better Management of Green Delights E-Commerce Platform	Dr S Ravichandran
32	ABM 2132	Kalithi Alekhya	International Trade Scenario of Mango	Dr MB Dastagiri
33	ABM-2133	Kantamureddy Devi Pravallika	A Study on Consumption Pattern of Dryfruits in Pre- and Post-Covid Period in Andhra Pradesh	Dr Umesh Hudedamani
34	ABM-2134	Karthik AS	Study on Impact of Immersion Program of a-idea	Dr S Senthil Vinayagam
35	ABM-2135	Krishnasree S	Spatial Market Integration and Price Transmission of Tomato in Karnataka	Dr A Dhandapani
36	ABM-2136	Mallikarjun Kamanalli	Awareness of E-Commerce for Agri-Input (Specialty Plant Nutrient) in Karnataka	Dr RVS Rao
37	ABM-2137	Srishti Ankush Mhatre	A Study on the Wine Consumption Pattern in India	Dr PD Sreekanth

Sl No	Roll No	Student Name	Title of the Project	Guide Name
38	ABM-2138	Nancy Gupta	Vegetable FPO Mapping and Suitable Business Model	Dr B Ganesh Kumar
39	ABM-2139	Nehal Ahuja	Study of Agro-Chemical Market of Chilli in Maharashtra and Madhya Pradesh	Dr VV Sumanth Kumar
40	ABM 2140	Nikhil Mund	Comparative Study on Economic Feasibility of Subsidized and Non-Subsidized Custom Hiring Centres	Dr Ch Srinivasa Rao
41	ABM-2141	Nithya Vinod	Consumption Behaviour of Food Prepared Away from Home in Hyderabad	Dr Sivaramane N
42	ABM-2142	P Sreevidya	Behavioural Analysis of the Consumption Pattern of Basmati Rice	Dr Yashavanth BS
43	ABM-2143	Polepaka Parimala Paul	Status of Farmer Producer Companies in Telangana	Dr Surya Rathore
44	ABM-2144	Pragya Srivastava	Fertilizer Industry Outlook and Government Intervention in India	Dr Sanjiv Kumar
45	ABM-2145	Pulugujju Swetha	Export Performance of Potato in India	Dr Tavva Srinivas
46	ABM-2146	R Manognya	Perceptions of Women Professionals on Gender Issues in Agri Business Industry	Dr Bharat S Sontakki
47	ABM-2147	Rohini G	Study on Consumer Behaviour Towards Organic Food Products	Dr P Venkatesan
48	ABM 2148	Shaik Bepari Mohammed Junaid	Comparative Analysis of Consumer Buying Behaviour with reference to Perishable Agricultural Products	Dr DThammi Raju
49	ABM-2149	Shaik Sameer	Study on Fresh Food E-Commerce in South India	Dr GR Ramakrishna Murthy & Dr P Supriya
50	ABM-2150	Shwetha P	Customer Engagement Analysis of Royye Raju – an Aquaculture App	Dr P Supriya
51	ABM-2151	Simrandeep Kaur	Analysis of Consumer Perception And Growth of Plant-Based Meat and Milk Products	Dr NAVijay Avinashilingam
52	ABM-2152	Sourabha Kumar Naik	Business Models for Urban Agriculture (Terrace Gardening)	Dr K Kareemulla
53	ABM-2153	Sourav Kumar Dubey	Automated Irrigation System: Farmers' Awareness and Challenges	Dr N Srinivasa Rao
54	ABM 2154	Sreelakshmi Murali Nair	Study of Consumer Behaviour on App-Based Meat Purchase in Kerala	Dr M Balakrishnan
55	ABM-2155	Sreenitha S	Analysis of Credit Utilization By FPOs in Telangana	Dr Sivaramane N
56	ABM-2156	Sukhpreet Singh	Exotic Fruits and Vegetables in Punjab: Farmers' and Consumers' Perspectives	Dr S Ravichandran
57	ABM-2157	Umang Kumar Shyam	Ramban Brand Awareness and Recognition Among Farmers	Dr R V S Rao

2.9.8 Summer Internship of 14th batch (2022-24) PGDM-ABM Students

Sl No	Roll No	Student Name	Title of the Project	Guide Name
1	ABM-22101	Adarsh Kumar	Assessment of Selective Herbicides- "Triskele/Trishuk" in Uttar Pradesh	Dr Sivaramane N
2	ABM-22102	Aishwarya GS	Retail Sector Analysis, Finding indicators for US apparel industry, Standardizing the farmer database for India and making it database ready	Dr SK Soam
3	ABM-22103	Akhila	Building the Brand: On-Ground Activation, Brand Awareness, and harnessing market Intelligence for Inera by Absolute	Dr K Srinivas
4	ABM-22104	Apoorva Chourasia	To evaluate the annual impact assessment of the Sustainable Guar Initiative	Dr G Venkateshwarlu
5	ABM-22106	Avinash Kumar Jha	Galaxy® NXT Launch: Segmentation and positioning in Soybean markets of MP	Dr MB Dastagiri
6	ABM-22107	Baloju Rahul	Analysis of new business opportunities and the scope of Hybrid Paddy in Western Odisha	Dr PC Meena
7	ABM-22108	Banda Abhinaya	Study on the scope of sustainable practices in paddy for regenerative agriculture and mapping the impact of project operations	Dr M BalaKrishnan
8	ABM-22109	Bhor Swapnil Machindra	Data-driven study of agriculture in Pune cluster	Dr Surya Rathore
9	ABM-22110	Chappidi Paul Joseph	Study on the scope of Sustainable practices in Paddy for Regenerative Agriculture and Mapping the impact of Project Operations	Dr P Supriya
10	ABM-22111	D Pramod Kumar	Overview of Groundnut and Sunflower Market	Dr Ranjit Kumar
11	ABM-22112	Dilbagh Singh Patel	Market mapping in context of opportunities for Fullpage enabled DSR	Dr N Srinivasa Rao
12	ABM-22113	Guguloth Gunavardhan	Study on the Scope of Sustainable practices in Paddy for Regenerative Agriculture and Mapping the impact of Project Operations	Dr P Ramesh
13	ABM-22114	Hari Prasad Vasantula	Study on the Scope of Sustainable practices in Paddy for Regenerative Agriculture and Mapping the impact of Project Operations	Dr B Ganesh Kumar
14	ABM-22115	Himanshu Rajpoot	Post Harvest Technology for agriculture commodities	Dr S Senthil Vinayagam
15	ABM-22116	Indudhar SM	Preparation of business proposal with assigned sugar mill (NSL Sugars Ltd) under UPL's shashwat mithaas initiative in Karnataka	Dr Sreekanth PD
16	ABM-22117	Ishita Chattopadhyay	Client Engagement & CSR Activities by Financial Industry Players	Dr Sivaramane N

Sl No	Roll No	Student Name	Title of the Project	Guide Name
17	ABM-22118	Jeevitha N	Assessing Digital solutions for Farmers in Groundnut and Sunflower output Marketing	Dr VV Sumanth Kumar
18	ABM-22119	Jithin Babu	Study of Pulses market in Bihar with special focus on a lentil processing unit in Nalanda and nearby districts	Dr Sanjiv Kumar
19	ABM-22120	Jupally Kavya Sri	Analysing Oilseed Cropping Pattern Evolution: A Historical Perspective on Shifting trends	Dr SK Soam
20	ABM-22121	Kanchan Dubey	Climate financing and exploring the scope for developing financial products to address it	Dr Mude Ramesh Naik
21	ABM-22122	Kartik	Study on wheat flour market, the industry and its prospects in Bihar	Dr Tavva Srinivas
22	ABM-22123	Keerti Raje	Study on the scope of sustainable practices in paddy for regenerative agriculture and mapping the impact of project operations	Dr Surya Rathore
23	ABM-22124	Kharat Mrunal Pandurang	To access onion production clusters and streamline it with respect to existing clusters of S4S	Dr Umesh Hudedamani
24	ABM-22125	Khobragade Kajal Hiwaraj	To evaluate the annual impact assessment of the Sustainable Guar Initiative	Dr V Ramasubramanian
25	ABM-22126	Kritika Subhashini	Optimization of Pongamia Beans Supply Chain for Terviva, Identifying Challenges and Opportunities along with Quality Assurance and Quality Control	Dr Mude RameshS Naik
26	ABM-22127	MR Gourish	Analysing Oilseed Cropping Pattern Evolution: A Historical Perspective on Shifting Trends	Dr Ch Srinivasa Rao
27	ABM-22128	Madhusudan M Sakare	Preparing road map for developing Kocide 3000 in Chilli, Cotton and Pomegranate crops	Dr M BalaKrishnan
28	ABM-22129	Musale Dhamaraka Kishan Rao	Scope of Financing Cattle Feed in Rajasthan	Dr Tavva Srinivas
29	ABM-22130	Muthina Poojitha	Overview of Groundnut and Sunflower Market	Dr Yashavanth BS
30	ABM-22131	Nalla Pruthwi Teja	Market Place for fruits	Dr NA Vijay Avinashilingam
31	ABM-22132	Nitesh Dhakad	To assess the existing suppliers of fresh Raw material and the existing FPO/FPC-led grains procurement centers	Dr PC Meena
32	ABM-22133	Patil Saurav Ramesh	Data Driven study of agriculture in sangli cluster	Dr P Venkatesan

Sl No	Roll No	Student Name	Title of the Project	Guide Name
34	ABM-22134	Pentakota Kavya	Evaluate the granular segment market trends against stem borer in transplanted rice	Dr NA Vijay Avinashilingam
35	ABM-22135	Pooja DN	Data Driven study of agriculture in Shivamogga cluster	Dr P Ramesh
36	ABM-22136	Poonam	Analysing the market dynamics of Onion with focus on production, export, and import trends	Dr Yashavanth BS
37	ABM-22137	Relangi Gayathri	Supply and Demand of Rice in India	Dr PC Meena
38	ABM-22138	Rokalla Lalitha Bhavani	Assessment of structure and satisfaction level of Farmer influence programs conducted by Agri-input companies	Dr GR Ramakrishna Murthy
39	ABM-22139	Saineni Abhigna	Market research to strengthen the use and engagement of REACH ADAMA India Kisan app	Dr B Ganesh Kumar
40	ABM-22140	Samhitha Padhi	Understanding various nursery practices for Verimark Positioning in Label Crops	Dr Sanjiv Kumar
41	ABM-22141	Satyam Purohit	Streaming supply chain of pomogrant from farm to market	Dr D Thammi Raju
42	ABM-22142	Saurabh Dixit	Mapping of vegetable-growing villages	Dr G Venkateshwarlu
43	ABM-22143	Saurabh Singh Pokhriyal	Building the Brand: On-Ground Activation, Brand Awareness, and Harnessing Market Intelligence for Inera by Absolute	Dr D Thammi Raju
44	ABM-22144	Seeram Supriya	Business Excellence Assessment of FPOs and Coffee pricing dynamics in Andhra Pradesh	Dr Sanjiv Kumar
45	ABM-22145	Sheik Karishma Alias Nafish	A Comprehensive Study on NBFCs and Fintech Industry and Securitization Structure	Dr N Srinivasa Rao
46	ABM-22146	Shikha Arora	Arrear Analysis and Find out the Reasons for non-payment of IEL Clients	Dr A Dhandapani
47	ABM-22148	Shivaraj Kumar Chavan	Profiling of Mango Farmers towards Establishing new Mandi	Dr P Supriya
48	ABM-22149	Sumit Sharma	To evaluate the annual impact assessment of the Sustainable Guar Initiative	Dr A Dhandapani
49	ABM-22150	Surakanti Nandini	Kharif market size study on corn	Dr S Senthil Vinayagam
50	ABM-22151	Thombare Viraj Shantaram	Supply Chain analysis of Pomegranate Crop in Solapur Region of Maharashtra	Dr GR Ramakrishna Murthy
51	ABM-22152	Thukuntla Nithya Sri	An Overview of Ground and Sunflower	Dr Sreekanth PD
52	ABM-22153	Tumati Bhardwaj	Generating Leads and sourcing of Mangoes	Dr V Ramasubramanian
53	ABM-22154	Umesh Chandra Ravirala	Assessment of Toki potential in Ap and Telangana	Dr K Srinivas

Sl No	Roll No	Student Name	Title of the Project	Guide Name
54	ABM-22155	Vaishnavi Bajpai	To Understand the market trend and consumer behaviour towards Cuprofix among cotton growers in Ganganagar Zo	Dr P Venkatesan
55	ABM-22156	Vajjala Tejaswini	To evaluate the impact of the market access program on Black Pepper in AP	Dr VV Sumanth Kumar
56	ABM-22157	Vinay Kumar	Supply chain management of mango in ramanagara district	Dr Umesh Hudedamani
57	ABM-22158	Yashwanth Kumar Gusidi	To understand the Resistance issues of fop chemistry and new trends in graminicides in Soybean: A study in Madhya Pradesh	Dr Ch Srinivasa Rao
58	ABM-22159	Yerrabelli Ramya	Study crop agronomy of Cotton, focussing on Pest pattern and understanding the future needs of the Cotton growers	Dr Ranjit Kumar

2.9.9 Students (MSc & Ph. D) Guided by NAARM Faculty as Major Guide/ Supervisor

Sl. No.	Name of the faculty	Student Name	Title of the thesis	Programme Name (M.Sc./Ph.D)	University / Institute Name
1	Dr. Ramasubramanian V.	Samir Barman	Bootstrap Resampling Based Prediction Intervals for Time Series Forecasting	Ph.D. (Agricultural Statistics)	ICAR-Indian Agricultural Research Institute, New Delhi
2	P Venkatesan	Tadepalli Yamini	Social Network Analysis (SNA) of Stakeholders in Agriculture for Identification of Key Communicators in Transfer of Technologies in Andhra Pradesh	Ph. D. (Agricultural Extension)	Acharya N. G. Ranga Agricultural University, Agricultural College, Bapatla, Andhra Pradesh
3	P Venkatesan	Pola Anuhya	An Impact Analysis of Farmer FIRST Programme in ATARI Zone – X	Ph. D. (Agricultural Extension)	Acharya N. G. Ranga Agricultural University, SV Agricultural College, Tirupati, Andhra Pradesh
4	BS Yashavanth	Rakesh Jammugani	Spatio-temporal Study of Agricultural Labour Wages in Telangana and Neighbouring States	MSc (Agricultural Statistics)	PJTSAU Hyderabad

3.1 Policy Research and Advocacy

In tune with the recommendations of several Review Committees, ICAR has recognized NAARM as a 'think-tank' body of the Council, and has taken steps to strengthen the faculty positions. The cadre positions have been revamped providing non-ARS positions suitable to support its role as a 'think tank'.

The Academy also has taken several initiatives to leverage its strengths for providing research-based management inputs and advice to Agricultural Policy Makers, Planners, Administrators, and others. Based on the progress of institute and externally funded research projects and realizing the importance of challenges and issues faced by the agriculture, the Academy has developed six policy/strategy documents and organized three policy workshops in 2023 and their essence is enumerated in the following sections:

3.1.1 Self-Sufficiency in Edible Oilseeds in India: Strategies and Policies

Global demand for oilseed production has continued to increase. In the Indian economy of agriculture, oilseed crops occupy the second important role after food grains in terms of area and production. Still, there is no self-sufficiency of oilseeds in India. The study analyzes the production-consumption pattern, export-imports scenario, growth rates, elasticities, TOT, instability, constraints, strategies, policies, and national missions for self-sufficiency. During the year 2017-18, regarding trade, Indian oilseeds

shared 0.79% of world exports and 9.72% of world imports. The study showed that the percentage share of oilseeds imports was higher than the exports. Sunflower, palm, mustard, and cotton oil have shown increasing trends in consumption over the years. India has a comparative advantage in groundnut, nigerseed, safflower, and sunflower. The instability in export prices of oilseed crops was found to be higher than imports. The export price elasticity of soybean, mustard, safflower, sesamum, and sunflower was found to be marginally greater than that of imports. The major export destinations of India for groundnut, soybean, niger seeds, and sesamum were Indonesia, the USA, South Korea and Philippines, respectively. The major import destinations of India for groundnut, soybean, sesamum, and sunflower were Germany, USA, Nigeria, and Ukraine, respectively. TMO, NMOOP, and domestic and foreign trade policies are the major existing oilseed policies.

Environmental, technological, socio-economic, organizational infrastructural, and marketing constraints are the major constraints faced by the oilseed producers. Decomposition analysis of past growth in oilseeds indicates 32% has come from area expansion and the remaining 68 percent from yield increase. By taking oilseed areas to irrigated cultivation, yields can be raised to 45% (Groundnut), 23-60% (Sunflower), 26-63% (Castor), 17-53% (Safflower) and 16-42% (Rapeseed & Mustard). Revolutionary frontier technologies for productivity enhancement and

repeat of successes achieved in TM in Oilseeds initiated in the late 1980s and 1990s and increase in MSP to oilseed crops are the important policy measures for enhanced production. The study concludes that the lack of self-sufficiency in edible oils is a serious problem. The continuous deficit in edible oils is to be addressed with integrated strategies. Area expansion in non-traditional regions is to be specifically focused. Revolutionary technologies for productivity enhancement, promotion of satellite farming and Customs hiring of micro irrigation system in oilseeds grown rain-fed areas, a repeat of successes achieved in TM in oilseeds initiated in the late 1980s and 1990s, MSP increase and procure at MSP, FPO's, e-NAM, digital Marketing role are the major strategies for oilseeds self-sufficiency.

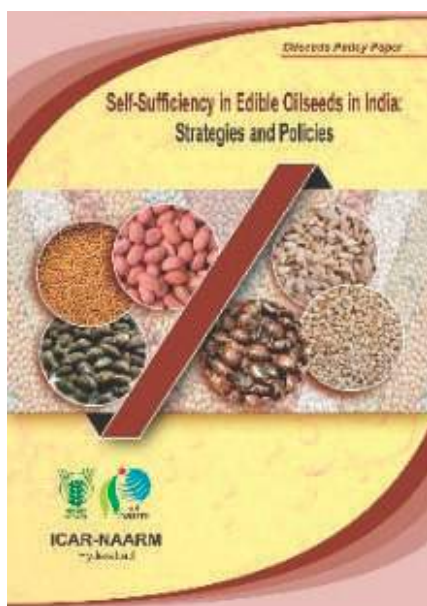


Figure 3.1.1: Self-Sufficiency in Edible Oilseeds in India: Strategies and Policies

3.1.2 Road Map for Rehabilitation of 26 Million Hectares Degraded Lands in India

This policy document provides a thorough analysis of approaches and suggestions for restoring 26 million hectares (Mha) of degraded land in India. The road map spotlights various issues brought about by land degradation in the nation, with an emphasis on sustainable development and environmental preservation. It also highlights the seriousness of land degradation in India and highlights the important 26 Mha area that needs immediate care. Numerous factors, including overgrazing, industrial activity, unsustainable agricultural methods, and deforestation, have contributed to this degradation. Wide-ranging effects of this degradation include reduced soil fertility, decreased biodiversity, depletion of water supplies, and decreased ecosystem health overall. The authors provide a strategic framework based on sustainable land management techniques to address these issues.

The understanding of India's various agro-climatic zones and ecological zones, each of which requires specialized interventions catered to its own features, is fundamental to its methodology. It stresses that in order to achieve ecological balance and enhance effectiveness, adopting region-specific techniques is crucial. The promotion of integrated watershed management, which attempts to improve water availability, replenish groundwater, and stop soil erosion, is a crucial component of the suggested roadmap. It emphasizes the necessity of stakeholders and community involvement in these projects, highlighting the significance of local ownership and knowledge for long-term results. The study also addresses agroforestry's

function in land rehabilitation, noting its ability to boost biodiversity, improve soil health, and give farmers alternative sources of income. The paper recommends the inclusion of land rehabilitation initiatives in federal and state laws. It also emphasises to provide farmers and landowners with incentives and support systems so that they may embrace sustainable methods. The dissemination of cutting-edge technology and best practices depends on the strengthening of research and extension services. Conclusively, the study offers a thorough and workable plan for the restoration of 26 million hectares of degraded land in India. It hopes to direct policymakers, stakeholders, and communities toward a more resilient and sustainable future for India's land resources by fusing scientific ideas with doable tactics. This all-encompassing strategy, which is based on the ideas of sustainable development, provides a way to improve the lives of millions of people who depend on these degraded areas and restore natural equilibrium.

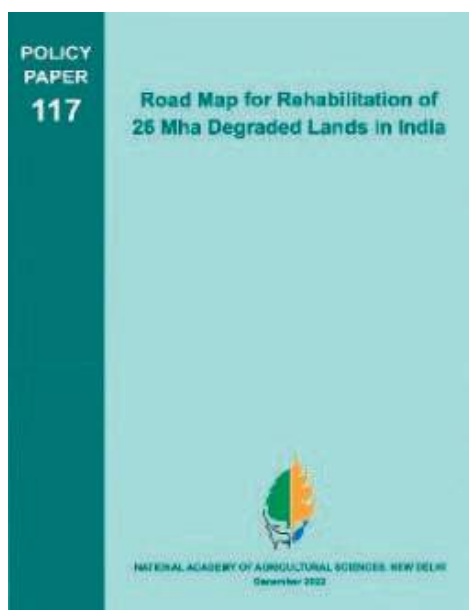


Figure 3.1.2: Road Map for Rehabilitation of 26 Mha Degraded Lands in India

3.1.3 Fostering Entrepreneurial Ecosystem in India: The Agriversity Way

Creating and promoting entrepreneurship could be the tipping point in aligning Indian agriculture with the philosophy of ATMA NIRBHAR KRISHI and ATMA NIRBHAR BHARAT. Towards this end there has been a lot of policy push from both central and state governments. Agriversities, the core institutions in developing human resources at secondary and tertiary levels, have very high prospects of taking up this significant responsibility. Though various institutions and agencies are currently promoting entrepreneurship in agriculture, real impetus and growth would come from an enabling 'ecosystem'. Apart from making remarkable progress since independence, agriculture remains India's primary driver of economic growth and employment.

Agriculture accounts for 20.2% of the total Gross Domestic Product (GDP), provides jobs for about 65% workforce, and contributes to 14.1% of total exports in the year 2020-21. About 70% of the Indian population resides in rural areas and depends on agriculture directly or indirectly. Indian agriculture is plagued with challenges like small and fragmented landholdings, lack of good quality seeds, timely credit, inadequate storage, marketing infrastructure, lack of value addition of produce, climate change risks, etc. This calls for a radical change in our approach towards solving the problems of agriculture.

Agripreneurship or entrepreneurship in agriculture can contribute to modernizing agriculture, improving production, creating job opportunities, and contributing to the overall growth of the farm sector in the country.

Agripreneurship can be a crucial driver for agricultural transformation and the socio-economic empowerment of rural youth and farmers. It enables them to operate successfully in the increasingly competitive global economy. Thus, agriculture should be promoted as an enterprise and farmers as entrepreneurs. When India aspires to become a 5 trillion GDP economy by 2025, it is essential to achieve excellence in the education system by making it globally relevant, competitive, and industry-oriented, while simultaneously promoting self-employment through entrepreneurship development and innovation among the students as well as the faculty members. The agripreneurs will significantly contribute to employment generation and economic growth in the farm sector. However, there is a considerable gap between the Agri-graduates passing out every year and the job opportunities available in the market. Thus, there is a low placement of agricultural graduates after the completion of their education.

In this background, the Academy has developed a policy paper “Fostering the Entrepreneurial Ecosystem in India: The Agriversity way” highlighting the importance of agripreneurship. Efforts are needed to replace the job seeker tag with job providers among agricultural students as agripreneurship is not just for employment; it has other dimensions like innovation, new products, and modernizing agriculture for the benefit of our agricultural graduates as well as for the upliftment of the country's economy. Agriculture graduates need to be sensitized about entrepreneurial skills and motivated to venture into agripreneurship to solve the problem of unemployment. This cannot happen by disjointed

efforts and strategies. The best option is to create and foster an excellent enabling ecosystem with multipronged and connected strategies and operational guidelines for awareness creation, entrepreneurship education, financial support, market knowledge, institutional linkages, etc. Local eminent professionals can be considered as "Master Instructors" in educational institutions in various subjects, such as traditional local arts, vocational crafts, entrepreneurship, agriculture, or any other topic where local expertise exists. Given the various policies set forth, there is a need for simultaneous growth of the agripreneurial ecosystem, attracting the agri-graduates to take this bold step to create a sustainable entrepreneurial ecosystem.

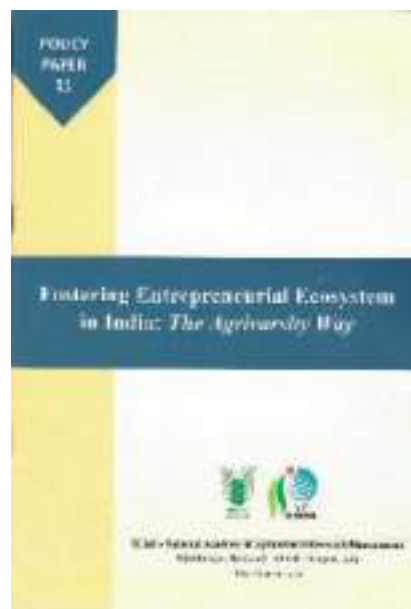


Figure 3.1.3: Fostering Entrepreneurial Ecosystem in India: The Agriversity Way

3.1.4 Organic Farming in India: Production Constraints and Policy Recommendations

Agriculture is a way of life for more than 67% of the population in India and it is necessary to ensure that it is productive, sustainable, profitable and

able to bring prosperity among farm families. This can be achieved through sustainable agriculture by efficient use of natural resources such as soil, water, energy and biodiversity. The production of food grains increased by six folds from 51.0 to 315.7 mt between 1950-51 and 2021-22 (MoA & FW, 2021-22). This increase can be attributed to large-scale adoption of high yielding varieties (HYVs) combined with other green revolution technologies (GRTs) in cereal crops. Nevertheless, excessive and indiscriminate use of synthetic fertilizers and pesticides during post-green revolution had negative effects on soil health and sustainability besides having repercussions on social, ecological and economic issues.

Many alternative farming systems have been emerging, which are claimed to be sustainable and environment friendly. One among them is organic agriculture. The main aim of organic agriculture is to optimize the use of on-farm resources and minimise off-farm resources. So, the philosophy is to “feed the soil and not the plants” which means giving back to the nature of what has been taken from it. Considering the significance of organic farming the Academy has brought out the document “Organic Farming in India: Production Constraints and Policy Recommendations”

Policy suggestions and Recommendations include, identification of niche areas and commodities for promoting organic farming; production practices for organic farming; R&D interventions for promoting organic farming; role of infrastructure, policy and institution for promoting organic farming; certification and marketing of organic produce; dissemination of knowledge

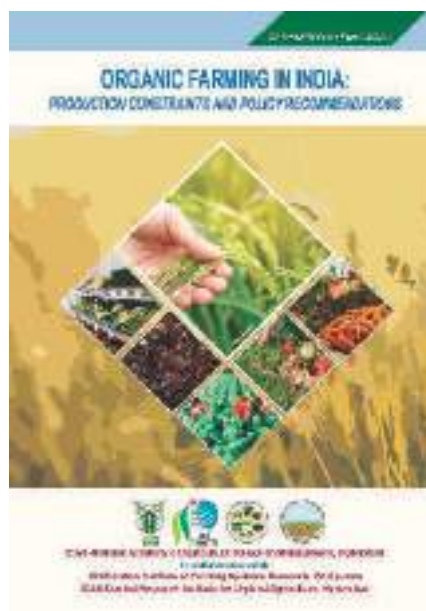


Figure 3.1.4: Organic Farming in India: Production Constraints and Policy Recommendations

3.1.5 Strategies for sustainability of Vegetable Street Vendors in India

Vegetable street vending plays a crucial role in India's urban food supply chain, providing affordable fresh produce to millions of consumers, while offering livelihood opportunities for a significant segment of the population. However, the sector faces numerous challenges related to infrastructure, regulation, market access, and socio-economic factors. Focusing on the vital, yet often overlooked sector of vegetable street vending, the Academy has developed a policy paper which sheds light on the challenges faced by these vendors and explores strategies for enhancing their sustainability and livelihoods. It outlines the socio-economic background of vegetable street vendors, highlighting their diverse profiles and the integral role they play in the informal economy.

Many vendors come from marginalized communities and rely on vending as their primary source of income, making their sustainability

essential for poverty alleviation and urban food security. One of the central themes explored in the paper is the infrastructure and working conditions of vegetable street vendors. It discusses the lack of proper vending zones, storage facilities, and sanitation, which not only affect the vendors' quality of life but also impact food safety and hygiene for consumers. These challenges are compounded by the inconsistent enforcement of regulations and the threat of eviction from public spaces. In addition to infrastructure issues, the study delves into the market dynamics and challenges faced by vendors in accessing markets and establishing fair pricing mechanisms. Competition from organized retailers and middlemen often squeezes the margins for street vendors, affecting their profitability and economic sustainability.

To address the challenges, the paper proposes a series of recommendations aimed at enhancing the sustainability of vegetable street vendors. These include improving infrastructure through the development of dedicated vending zones with basic amenities such as water and sanitation facilities. Strengthening linkages with wholesale markets and introducing price stabilization mechanisms are also suggested to improve market access and ensure fair returns for vendors. Furthermore, the authors advocate for policy interventions that recognize the contribution of street vendors to urban food security and economic growth. Formalizing the sector through registration and providing social security benefits can improve the livelihoods and resilience of vendors, empowering them as valuable contributors to the economy.

In conclusion, "Sustainability of Vegetable Street Vendors in India" provides a comprehensive analysis of the challenges and opportunities faced by this vital sector. By highlighting the need for infrastructure development, market access improvements, and policy support, the paper offers a roadmap towards enhancing the sustainability and livelihoods of vegetable street vendors. This research not only contributes to academic understanding but also serves as a valuable resource for policymakers and stakeholders seeking to promote inclusive and sustainable urban food systems in India.



Figure 3.1.5: Strategies for sustainability of Vegetable Street Vendors in India

3.1.6 Promotion of Inter-Disciplinary Research in Indian Agriculture: Strategies and Opportunities

Agricultural research has an impressive record in India, having contributed to the Green Revolution. It has gone through different phases since then, and is now in the public eye. This new attention stems from the country's ambitious agricultural growth target of four percent per

annum and Governments intention of doubling farmers income. It also reflects the need to respond to the emerging challenges. Some of the emerging trends in Indian agriculture include, plateauing of productivity levels of key food crops like rice, new opportunities in science and technology and the organization of research, changes in the agricultural economy and the demand for food, as well as challenges linked to natural resource management and climate change. In order to make significant impact in food production systems, there is a need to have a deep understanding of all the disciplines that impact agriculture. This understanding can be enhanced by continued focus on basic and strategic research under controlled environments and operational research connecting with the stakeholders. This calls for an interdisciplinary approach in our research. Inter-disciplinary research in agriculture does not rise from lab or green house alone but continued accumulation of new knowledge leading to specialization and interaction with the stakeholders. Academicians, researchers and policymakers have advocated inter-disciplinary approach for solutions to the new and complex challenges. Inter-disciplinary research (IDR) is becoming increasingly important in agricultural research for meeting societal needs and addressing real-world issues that demand collaborative approaches.

Inter-disciplinary research may involve diverse disciplines within a single scientific ecology, and also cross-cultural boundaries as in the study of humans and their environment. Leading research organizations in the world claim that complex problems demand innovative solutions that can be achieved by IDR. IDR leads researchers in different disciplines to meet at

the interfaces and frontiers of these disciplines and even to cross frontiers to form new disciplines. Many researchers argued that IDR science has a positive influence on knowledge generation and innovation. IDR in Agriculture has involved multiple stakeholders in areas such as agricultural science, environmental science, community development entrepreneurship and social justice etc. Traditionally, the relationship between these stakeholders has been one of proximity, but not integration. Now, there are calls for greater inter-disciplinary focus and collaboration to meet the economic and societal needs of people and develop strategies that are environmentally more sustainable. There needs to be a greater recognition among the different disciplines involved, that the link between the economy, society, and nature is intrinsically inter-disciplinary. An inter-disciplinary approach requires stakeholders to move beyond traditional ways of thinking and problem-solving and embrace.

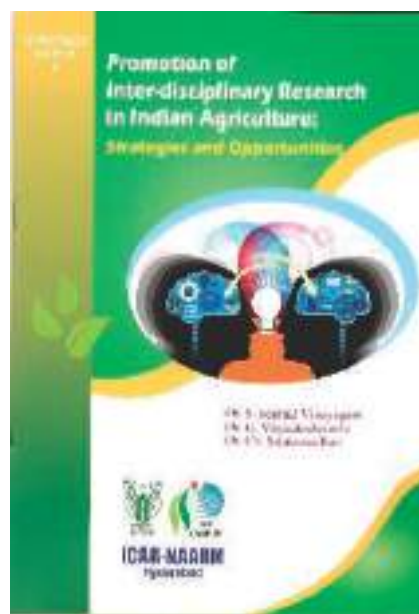


Figure 3.1.6: Promotion of Inter-Disciplinary Research in Indian Agriculture: Strategies and Opportunities

3.2 Policy Workshops

3.2.1 National Policy Workshop on Integrating Public and Private Systems for Effective Extension Service

The Academy organized a National Policy Workshop on Integrating Public and Private Systems for Effective Extension Service in hybrid mode on 12 January 2023 as a part of the Academy's 'Think-tank Policy' mandate. The main objective of the workshop was to critically analyze the Strengths, Weaknesses, Opportunities, and Threats (SWOT) of public and private extension systems and to suggest a framework for integrating them with implementable guidelines in terms of institutional, structural, functional, and legal requirements. Dr AK Singh, Vice Chancellor of Rani Laxmi Bai Central Agricultural University, Jhansi was the Chief Guest for the Inaugural session, which was presided over by Dr Ch.Srinivasa Rao, Director, ICAR-NAARM. Dr Sharan Angadi, Director, Foundation for Advanced Training in Plant Breeding, & COO, Indian Biotech MSME & Startups iBioM, Bengaluru; and Dr Ved Prakash Chahal, ADG (Agricultural Extension), ICAR, New Delhi; Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM and Dr S. Prabhukumar, Former Director, ATARI, Ludhiana and Bangalore and currently Adjunct Faculty with ICAR, NAARM graced the occasion and presented their perspectives representing private and public extension systems.

During the workshop, three technical sessions were organized in the form of panel discussions on the themes of SWOT of the Public Extension System, SWOT of the Private Extension System, and Framework for Integrating Public and Private Systems for Effective Extension Service.

The plenary and concluding session captured the essence of the deliberations and outlined the draft proceedings and recommendations. About 60 delegates representing the agricultural extension ecosystem of the country (Farmers, Entrepreneurs/Start-ups, ATARIs, SAUs, KVKs, MANAGE, NGOs, Private sector, and Faculty of ICAR, NAARM) participated in the workshop and contributed to the deliberations.



Figure 3.2.1: Policy Workshop on Integrating Public and Private Systems for Effective Extension Service

3.2.2 Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India

National Conference on "Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India" was organized at

ICAR-National Academy of Agricultural Research Management, Hyderabad during 12-13 March 2023. The Chief Guest, Dr. R C Agrawal, National Director, NAHEP & DDG (Agril. Education) highlighted the salient achievements of the NAHEP during the preceding years. He also shared the importance of the project in building the institutional mechanism for knowledge sharing, executing the actionable MoUs, and restructuring of agricultural curriculum.

Dr. Ch. Srinivasa Rao, Director, NAARM expressed his happiness for the active involvement of stakeholders from diverse areas and departments and assured them to bring out the actionable quality policy document. Dr. P L Gautam, Former Chairman of PPVFRA, NBA, DDG (Crop Sciences) and presently NAHEP-External Advisory Panel member flagged the issues relating to MoUs, agreements, and reciprocal sabbatical leaves to pursue higher education. Dr. N S Rathore, Former DDG (Education) also spoke on the occasion. Dr. Rajendra Prasad, Director, IASRI, New Delhi highlighted the role of his team in executing the project objectives. More than 65 delegates including 16 Vice Chancellors from Agricultural Universities and 19 other senior representatives from industry, innovative farmers, FPOs, Start-ups participated in this conference. Dr Anuradha Agrawal, National Coordinator (NAHEP component 2), Dr Sudeep, PI of NAHEP Component 2 also shared their views.

The important recommendations of this conference include, every University shall have its own IPR Policy with clear-cut terms and conditions to be a win-win situation for Academia and Industries; options for having

‘Open Source Product Licensing’ may be explored as most private industries are MSMEs, and 51% are from rural areas; industry and Government may nominate their members to different decision-making bodies (RAC/IRC/GB/BoM/



Figure 3.2.2a: National Conference on “Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India”

BoS/ Syndicate) of the Universities / Institutions and vice-versa; develop a consortium of Indian agro-based industries to attract a 2 2% CSR fund for quality education and research; Policies should curb inbreeding or at least ensure that the students get the opportunity to work with other institutions or companies in the form of live projects, summer internships or project work; bringing industry officials on board of academia and in Government bodies to ensure better collaboration; the course curriculum may



Figure 3.2.2b: Participants of Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India



Figure 3.2.2c: Dr Brajesh Singh, Distinguished Professor, Western Sydney University, Australia interacting with participants

be framed in consultation with industry and Government to incorporate the most recent advancements; Universities should adopt standard procedures for providing lab testing, lab reports, certification and other facilities so that the Industry can avail of these facilities hassle-free.

3.2.3 A Policy Dialogue on “Agricultural Education System in the USA and Potential Collaboration with India”

A policy dialogue on “Agricultural Education System in the USA and Potential Collaboration with India” was organized on 07-09-2023 at ICAR-NAARM in collaboration with NAHEP Component 2, and NAAS Hyderabad Chapter. Dr. Kumar Venkitanarayanan, Professor & Associate Dean of Research, College of Agriculture, Health and Natural Resources, University of Connecticut, USA, and Dr Anup Kollanoor Johny, Associate Professor, Department of Animal Science, University of Minnesota, USA addressed the participants on Agricultural Education System in USA. In his presidential address, Dr. Ch. Srinivasa Rao, Director and convener of the NAAS Regional Chapter-Hyderabad appreciated the speaker for an elaborate coverage of the topic and stressed the need for more collaborations in

terms of different activities viz student-ships, senior faculty training, etc as per the need. He also called for One Health, which could be very inclusive comprising components of Human, crop, animal, and environmental health. The program was organized in hybrid mode and a total of 70 participants offline and more than 100 participants online from various ICAR institutions, SAUs, NAAS, Regional Chapter-Hyderabad fellows, and associates actively participated.



Figure 3.2.3. Dr. Kumar Venkitanarayanan addressing the participants about the Agricultural Education System in the USA

3.2.4 Profiling Resources Devoted to R&D in the Agriculture Sector

In a study on “Strengthening & Integrating Agriculture Science and Technology” in collaboration with International Food Policy Research Institute (IFPRI), Washington and the Department of Science and Technology (DST), the ICAR-NAARM was involved in the fourth round of survey being conducted in India to gather research performance data for the period 2018-19 to 2020-21. Nationwide, these data sets are used by the DST. As a part of this collaborative research the data pertaining to (i) human resources deployed (ii) financial resources used (iii) research focus and (iv) research outputs

were collected from all ICAR institutions and Agricultural Universities for the period from 2018-19 till 2020-21. In this project, successfully collected 80% of the required data and uploaded it to the DST and NAARM Data Drive. As a result, an interactive Dashboard that allows us to draw meaningful insights from the datasets was developed. The Government of India (GoI) has recently released the S&T Indicators tables and the R&D Statistics for the year 2020-21. These tables integrate the data collected by NAARM specifically for the agricultural sector. The participation in this project has made ICAR-NAARM as an integral part of a national

initiative aimed at collecting S&T data for the farm sector. This, in turn, will greatly contribute to evidence-based policymaking in India. The Academy has also developed a prototype for the Dashboard jointly with one of the IT firms empaneled with NAARM, to aid in data-driven decision making in the domain of agricultural research and education. Overall, this project not only highlights the successful data collection efforts of the Academy but also highlights its commitment to advancing the agricultural sector through innovative technology and evidence-based decision-making.

Centres of Excellence

4

4.1 Centre for Agri-Innovation (CAI)

The Centre for Agri-Innovation of the Academy functions through the technology business incubators from ICAR-NAIF (National Agriculture Innovation Fund), Department of Science & Technology (DST), National Bank for Agriculture and Rural Development (NABARD), Government of India (GoI) supported Business incubator a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) which also supports activities of BioNest an Agri-Bio-incubator for Agri-Biotechnology Startups. BioNest has received financial support from BIRAC, DBT, GoI.

a-IDEA nurtures agricultural innovations and entrepreneurship in the fields of agriculture and allied sectors in India. Since its inception in the year 2014, a-IDEA, has backed startups covering as many as 14 domains of agriculture

and allied sectors such as dairy, fisheries and animal husbandry. Through its initiatives and programmes, a-IDEA focuses on sensitizing, ideating, incubating and accelerating the innovative early-stage startups/ entrepreneurs that are scalable to become fiercely competitive food and agri-business ventures through capacity building, one on one mentoring, networking and advisory support. The startups backed by a-IDEA have been addressing pressing issues concerning the farming community and those that have a large impact on the lives of farmers, rural and urban stakeholders besides being disruptive in nature.

At present, the following are the key thrust areas of agriculture innovations across the value chain in agriculture on which a-IDEA is emphasizing on:



Figure 4.1.1: a-IDEA focus areas

4.1.1 Agri Udaan 5.0

a-IDEA, Technology Business Incubator (TBI) of ICAR-NAARM, organized Demo Day under AgriUdaan 5.0 on 28 April 2023. This is the fifth Food and Agribusiness Accelerator program in a sequence of a-IDEA amply supported by NABARD, SBI, SIDBI & Caspian Impact Investments, Hyderabad. The objective of this program was to catalyze food and agribusiness startups and provide them a unique platform to pitch their ideas before investors for scaling up to the next level by raising funds along with providing an opportunity to network with other businesses and investors. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and President, a-IDEA, Smt. Susheela Chintala, CGM, NABARD, TSRO, Mr. Shantaram Pai, GM, Agri Business Unit, SBI, Mumbai and Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM and Vice-President, a-IDEA address the gathering.



Figure 4.1.2: Agri Udaan 5.0

The event was attended by 14 startups selected out of 149 applicants under AgriUdaan 5.0. About 110 delegates representing the incubation and agri-startup ecosystem attended the event along with 28 investors from different parts of the country. There was a pre-demo day organized during 26 – 27 April, 2023 before AgriUdaan 5.0, where these startups were groomed to effectively present their ideas/products before investors.



Figure 4.1.3: Agri Udaan 5.0

Table 4.1.1 Agri Udaan 5.0

S. No	Startup Name	No. of Investors Showed Interest
1.	Marut Dronetech Pvt. Ltd., Hyderabad, TS	12
2.	Organikness Pvt. Ltd., Hyderabad, TS	08
3.	Inav Agro Mech Pvt. Ltd, Hyderabad, TS	07
4.	Cisgen Biotech Discoveries Pvt. Ltd., Chennai, Tamil Nadu	07
5.	Auspice Social Networks OPC Pvt. Ltd., Ghaziabad, Uttar Pradesh	06
6.	Renewagri Om Ecommerce Pvt. Ltd., Pune, Maharashtra	05
7.	Univia Pvt. Ltd., Ahmedabad, Gujarat	05
8.	Bee Basket Enterprises Pvt. Ltd., Pune, Maharashtra	04
9.	Attaware Biodegradable Pvt. Ltd., Delhi	05
10.	Anitra Tech Pvt. Ltd., Hyderabad, TS	05
11.	Agriotics Technologies Pvt. Ltd, Pune, Maharashtra	03
12.	Contrivation Labs Pvt. Ltd., Bengaluru, Karnataka	02
13.	Jivanah Platform Services Pvt. Ltd., Hubli, Karnataka	02
14.	Bhojpatta Agripreneur Pvt. Ltd., Saran, Bihar	01

4.1.2 Agri Udaan 6.0

Agri Udaan, Food and Agribusiness Accelerator 6.0 is the 6th edition of the flagship accelerator program of a-IDEA, ICAR-NAARM was launched at Bengaluru, Karnataka on 7 July 2023. It brings on board a range of diverse partners to

create an inclusive and collaborative ecosystem focusing on catalyzing scale-up stage Food and Agribusiness startups through rigorous mentoring, industry networking and investor pitching. It is a unique platform for scale-up stage innovators, entrepreneurs and startups in Food and Agribusiness sectors to showcase their products/services and to receive valuable inputs from Mentors, Incubators, R&D Institutions, Agribusiness Industry, and Investors.



Figure 4.1.4: Agri Udaan 6.0

Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM, emphasized Bengaluru's significance as a startup hub in his welcome address. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, highlighted the growing importance of agricultural startups and their role in addressing farmers' challenges through innovation. Sh. Shantaram Pai, General Manager, Agri Business Unit, SBI, stressed the untapped potential in addressing issues like productivity and efficiency.



Figure 4.1.5: Agri Udaan 6.0

Shri. K. S Mahesh, General Manager, NABARD, Bangalore, commended a-IDEA's efforts in supporting startups by connecting them with mentors, investors, and providing technical, business, and market assistance. Dr. Raghavendra Bhatta, Director, ICAR-NIAP, praised AGRI UDAAN's consistent impact on the startup ecosystem. Angel Investor Dr. Ravi Shankar highlighted its successful Public Private Partnership model. Previous cohorts' positive feedback and experiences were shared by startups like FarmOR Agri-Solutions and Dhisathi Robotics. The event drew around 100 participants from various startup-related sectors.

The application for Agri Udaan 6.0 was opened on the 7 July 2023 and the date for submission of the application was given on the 04 September 2023. Following the launch of Agri Udaan 6.0, Roadshows were conducted in 4 locations to promote and create awareness about Agri Udaan.

1. ABIS, TBI, TNAU, Coimbatore on 28 July 2023
2. NGI-TBI & NIT, Coimbatore on 28 July 2023
3. NABARD Regional Office, Pune on 04 August 2023
4. ICAR-IISR, Lucknow on 08 August 2023

Nearly 80-100 participants attended each of the roadshow. The top 10 startups cohort of Agri Udaan 6.0, was selected from a pool of 156 applications.

Table 4.1.2 List of startups selected for Agri Udaan 6.0

S.No	Startup Name	Domain	Product/Services
1.	Bullwork Mobility Private Limited, Karnataka	Agri-EV	Product portfolio includes cutting-edge electric vehicles that reimagine the way farming and utility operations are carried out.
2.	Country Chicken Co, Telangana	Poultry	Retail- world's first odourless meat store concept; India's first naturally raised healthy chicken & eggs brand; Input Marketplace- first of its kind country chicken industry aggregator
3.	Saswat Financial Technologies Pvt Ltd, Maharashtra	Cattle-Fintech	With a focus on cattle finance, fintech startup serving the needs of 140 million small and marginal farmers who have limited access to lending and agri-inputs. By providing a tech-driven, hi-touch, low-cost platform that is easily accessible to smallholder farmers.
4.	Verdant Impact Private Limited, Rajasthan	Cattle Advisory	Full stack animal husbandry platform. Providing farmers with backward and forward linkage for purchase and sales of livestock. With our state of the art product - Animal ICU, we provide digital livestock healthcare services to the farmers for the wellbeing of their animals.
5.	Samparkbindhu Solutions Pvt Ltd, Karnataka	E-Marketplace	An AgriTech company developing platforms and products to add Value in the post-harvest space. Mission is to ensure a bigger share of consumer's rupee to farmer community in the post-harvest space.
6.	Hindusthaan Feedcare Pvt Ltd, Tamil Nadu	Animal Feed	Startup aims to revolutionize the sports animal nutrition industry by offering a comprehensive range of natural and specialized feeds tailored to the unique dietary requirements of various athletic animals.
7.	Foo Foods India Private Limited, Kerala	Food	Extended shelf life for non veg foods upto 365 days without refrigeration and zero preservatives, value for money foods which is affordable for mass segment consumers, cost effective in terms of transportation, storage, distribution and display
8.	Manjha Technologies, Haryana	Aquaculture	Manjha provides full stack solution for aquaculture sector which includes marketplace for fish/ shrimp seed supplier, supply of other qualitative aqua input like feed/ health supplements from top brands, farm consultancy and forward market linkage to ensure better price for final harvest.
9.	A-Thon Allterrain Private Limited, Karnataka	Agri-Machine	A-THON's vision—to revolutionize Indian agriculture with Assisted Autonomous All terrain BOTs and small-scale AgriBOT's. A-THON's mission is rekindling the agricultural sector's allure among the younger generation.
10.	Sengathali biofiber Pvt Ltd, Tamil Nadu	Waste2Wealth	Banana waste contains valuable silk grade fiber. It is useful for making 100% banana fiber based cloth and the banana sap contains rich pottasisium and plant growth materials are present.

4.1.3 Accelerator Mentorship Program (AMP)

AMP for the startups selected under AGRI UDAAN 6.0, Food and Agribusiness Accelerator during 22-24 November 2023. The objective of the program was to provide networking opportunities, gain insights into industry trends, and find inspiration, empowering them to integrate learning into strategic decision-making for a successful entrepreneurial journey.



Figure 4.1.6: The objective of the Accelerator Mentorship Program

Dr. S. Senthil Vinayagam, CEO, a-IDEA warmly welcomed the esteemed chief guest Mr. Hemendra Mathur, Chair, FICCI Task Force on Agri Start-ups, New Delhi; Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM; Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM; Scientists; a-IDEA Team and participants of the AGRI UDAAN Accelerator Mentoring Program.



Figure 4.1.7: Accelerator Mentorship Program (AMP)

4.1.4 Aggnite 3.0

National Level Ideation Competition named “Aggnite” was conducted by a-IDEA to motivate the students towards entrepreneurship in agriculture and to educate them on Startup ecosystems and entrepreneurship skillsets. The application for AGGNITE 3.0 were received during 16 January - 16th February, 2023. The total number of applications received was 201. The first level of screening was initial eligibility in which 42 applicants were selected for the second round of evaluation. In second level of



Figure 4.1.8: Aggnite 3.0

screening, out of 42, 11 applicants were selected for third round (pitching round). The applicants were evaluated and marks were given on the basis of various factors like problems being addressed, innovative idea, prototyping plan, competency of team & presentation skills. The whole evaluation process was conducted in virtual mode. In the third round, 4 applicants were selected as winners of AGGNITE 3.0.

- **1st Awardee** - Ms. Rafiya Mushtaq - Sher-E-Kashmir University of Agricultural Sciences and Technology of Kashmir - **(Cash Prize -25,000/-)**
- **2nd Awardee Mr. Rohit Sharma** - Lala Lajpat Rai University of Veterinary and Animal Sciences, Hisar, Haryana - **(Cash Prize -15,000/-)**
- **3rd Awardee Ms. Deepika A** - Nehru Institute of Technology, Coimbatore, Anna University, Chennai - **(Cash Prize -10,000/-)**
- **Ms. Sameena Lone** - SKUAST-Kashmir, Shalimar, Srinagar, J&K was awarded with 'Best Innovative Idea' award.

4.1.5 Memorandum of Agreement

a-IDEA signed a Memorandum of Agreement with 35 ICAR Institutes with the aim to support startups by organizing joint events, extending lab facilities, co-networking etc. The details are given below.



Figure 4.1.9: Memorandum of Agreement Signing

Table 4.1.3: List of Institutes that inked MOA with a-IDEA

S. No	Name of the Institution	Date of Execution
1.	ICAR- CIFT COCHIN	26 May 2023
2.	ICAR-CIFA, Bhubaneswar	
3.	ICAR-IIPR, Pedavegi	
4.	ICAR-CIRCOT, Mumbai	
5.	ICAR-NIASM, Baramati, Pune	
6.	ICAR-CITH, Srinagar, J&K	
7.	ICAR- NRCO Pakyong, Sikkim	
8.	ICAR-IISR Lucknow	
9.	ICAR – NRCM Nagaland	
10.	ICAR-CIAH, Bikaner	
11.	ICAR – NISA, Namkum	
12.	ICAR-CSSRI, Karnal	
13.	ICAR-IIWBR, Karnal	
14.	ICAR-NRRI, Cuttack	
15.	ICAR-IIWM, Bhubaneswar	
16.	ICAR-CIWA, Bhubaneswar	23 August 2023
17.	ICAR-IIHR, Bengaluru	
18.	ICAR VPKAS Almora Uttarakhand	
19.	ICAR-IISWC, Dehradun	
20.	ICAR DMAPR, Boriavi, Anand Gujarat	
21.	ICAR ATARI, CRIDA Campus, Hyd	
22.	ICAR-NBSS&LUP, Nagpur	
23.	ICAR- ATARI, Zone IV, Patna	
24.	ICAR-DCR, Puttur	
25.	ICAR-DOGR, Pune	
26.	ICAR-IIMR, Ludhiana	
27.	ICAR-IISR, Kozhikode, Kerala	
28.	ICAR -CIRB, Sirsa Road, Hisar	
29.	ICAR-CTRI, Rajahmundry	
30.	ICAR NRC on Litchi, Muzaffarpur	
31.	ICAR-CTCRI, Thiruvananthapuram	
32.	ICAR-DRMR, Bharatpur	
33.	ICAR-NRC on Equines, Hisar	
34.	ICAR-NRC on Seed Spices, Ajmer	
35.	ICAR Research Complex for Eastern Region Parisar	

4.1.6 Cohort Networking & Capacity Building Program

The Cohort Networking & Capacity Building Program was organized for incubatees under Krishiboot and ABI Incubation Programs and grantees of a-IDEA (Beneficiaries of BIRAC BIG Grant and NIDHI PRAYAS Grant) from October 30-1 November 2023. A total of 15 startups across 6 states (Karnataka, Tamil Nadu, Telangana, Maharashtra, Madhya Pradesh, and Kolkata) comprising of 20 people participated in the 3-day program. The startups were from the ideation stage to PoC and some were even at the MVP stage.



Figure 4.1.10: Cohort Networking & Capacity Building Program

4.1.7 Immersion Programmes

a-IDEA, Technology Business Incubator of ICAR-NAARM, Hyderabad organized four FPOs/FPCs, Farmers, Startups Immersion Programme to connect the FPOs/ FPCs and the farmers with the startups supported by a-IDEA. The details are given below.

Table 4.1.4: The Startups Immersion Programmes

S. No	Location	Date	Startups attended	No. of FPOs attended
1.	MABIF, Madurai, Tamil Nadu	04 March 2023	22	80
2.	Nalgonda, Telangana	26 June 2023	15	53
3.	ICAR-IISR, Lucknow	08 Aug 2023	20	50
4.	ICAR-CTRI, Rajahmundry	17 Nov 2023	18	50



Figure 4.1.11: Inauguration of Immersion Programme

4.1.8 Agri Entrepreneurship Development Programme for Students of SKNAU, Jobner, Rajasthan

The a-IDEA, a Technology Business Incubator of ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad organized a sensitization program on “Entrepreneurship Development in Agriculture and Allied Sectors” for the students of Shri Karan Narendra Agriculture University (SKNAU), Jobner to impart knowledge on entrepreneurship on 23 October 2023.

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM in his presidential remarks stressed the importance of converting waste to value. He highlighted the food losses in India are equal to UK food consumption for a year. He further

underlined the need to work on various post-harvest operations viz., processing, grading, storage, and efficient marketing mechanisms for reaping major monetary benefits to the ultimate stakeholders. Dr. A.K.Gupta, Dean and Faculty Chairman, SKNAU, Jobner, who was the chief guest during the program asked students to inculcate entrepreneurial qualities like planning, passion, business savvy, confidence, and a never-give-up attitude required for the entrepreneur to reach the pinnacle in their profession. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM, who was also present during the occasion highlighted the importance & opportunities in startup India. He also explained the need to reorient the agriculture course curriculum along the lines of entrepreneurship. He also mentioned about the various events viz., Aggnite, Krishiboot, and the immersion program organized by a-IDEA.

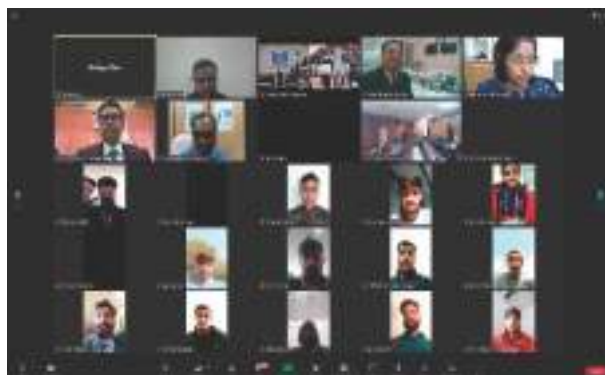


Figure 4.1.12: Agri Entrepreneurship Development Programme for Students of SKNAU, Jobner, Rajasthan

Dr. N.A. Vijay Avinashilingam, Addtl. CEO, a-IDEA, ICAR-NAARM, Hyderabad reiterated the need and highlighted the importance of startups and the process of incubation in entrepreneurship. Dr. B.Ganeshkumar, Principal Scientist stressed the necessity of skillsets required for the students and also said that they should be a risk taker and visionary to become a successful entrepreneurs. Mr. Venkata Sai Teja Duggempudi, Founder & CEO, of FarmOR Agri -Solutions Pvt Ltd & an entrepreneur working on supply chain explained his successful entrepreneurial journey. About 500 students and faculty members virtually participated in the program.

4.1.9 Agribusiness Management Development Program (ABMDP) for prospective CEOs of FPOs

4.1.9.1 ABMDP for prospective CEOs of FPOs – Telangana State

a-IDEA, ICAR-NAARM organized 50 days training program on Agribusiness Management Development Program on from 17 July - 01 September 2023 for 31 participants. This unique course, designed to nurture prospective CEOs of FPOs, is poised to revolutionize the leadership landscape within the agricultural sector with the funding support from NABARD TSRO. In the lieu of success, 100% of the students got placed at various POPIs such as Access Development Services, BASICS, CSA, Dhan Foundation, Ekalavya Foundation, Evangelical Social Action Forum (ESAF), Grant Throtan, IRDS, NABCONS, NESTHAM, PEACE and Astittva Welfare Foundation.



Figure 4.1.13: Participants of the ABMDP for prospective CEOs of FPOs – Telangana State

Followed by 50 days training program, a two days' refresher program for "COEs of FPOs" was organized on 09 - 10 January 2024 with objective of understanding the challenges facing by the CEOs after taken the charge in FPO and provide the solution for challenges and filling the knowledge gap. Around 22 CEOs are attending the program, during the program CEOs are interacted with CGM, NABARD TS RO and other NABARD officials at NABARD office and also interacted with experts.

4.1.9.2 ABMDP for prospective CEOs of FPOs – Andhra Pradesh State

a-IDEA, ICAR-NAARM organized 50 days training program on Agribusiness Management Development Program from 15 November 2023 - 03 January 2024 for 30 participants. This unique course, designed to nurture prospective CEOs of FPOs, is poised to revolutionize the leadership landscape within the agricultural sector with the funding support from NABARD APRO.

In the lieu of success, among the selected candidates 22 are selected as CEOs of FPOs in Andhra Pradesh and remaining 08 are selected for the state of Telangana.



Figure 4.1.14: Participants of the ABMDP for prospective CEOs of FPOs – Andhra Pradesh State

4.1.9.3 NABARD Resource Support Agency

NABARD APRO has empanelled a-IDEA, ICAR-NAARM as a resource supporting agency for supporting FPOs situated in the North coastal regions of Andhra Pradesh state, under which the institute has been mandated with a series of training programs to train the POPIs, CEOs & BoDs of FPOs on different aspect of FPOs business activities.

Table 4.1.5: List of training programs with details

S. No	Name of training program	Participants	No. of Participant's attended
1.	Interface Meet with POPIs "Transformative Interventions in Value Chain & Business Development of FPOs" zone A (North coastal Districts), Andhra Pradesh on 07-08 Sep 23 at a-IDEA, ICAR-NAARM, Hyderabad	POPIs	16
2.	Interface Meet with POPIs "Transformative Interventions in Value Chain & Business Development of FPOs" zone A (North coastal Districts), Andhra Pradesh on 09 October 23 at a-IDEA, ICAR-CTRI, Rajahmundry	POPIs	26
3.	Training Program on "Transformative Interventions in Value Chain & Business Development and Sustainability for CEOs of FPOs" (Training program for CEOs of Vizianagaram and Srikakulam FPO) on 10 - 14 Oct 2023 at Union RSETI, Srikakulam, Andhra Pradesh	CEOs	32
4.	Capacity building program on "Transformative Interventions in Value Chain & Business Development for Sustainability of FPOs" (Training program for CEOs of Srikakulam, Vizianagaram and Vishakhapatnam FPOs) on 07 - 11 Dec 2023 at BCT-KVK, Vishakhapatnam, AP.	CEOs	24

4.1.10 Meetings

4.1.10.1 Governing Board Meeting

Eleventh Governing Board Meeting of a-IDEA was held on 17 March 2023 under the Chairmanship of Dr. Ch. Srinivasa Rao, President, a-IDEA. Dr. Ch Srinivasa Rao, President, a-IDEA & Director, ICAR-NAARM, Dr.G Venkateshwarlu, Vice-President, a-IDEA and Joint Director, ICAR-NAARM; Mr. Debashis Padhi, NABARD; Dr. Renuka Rani, NIRD & PR; Heads of various divisions of NAARM and a-IDEA staff attended the meeting. During the meeting, Dr. S. Senthil Vinayagam, CEO & Principal Scientist of a-IDEA presented the progress of a-IDEA since 13 March 2022 to 16 March 2023.



Figure 4.1.15: Governing Board Meeting

4.1.10.2 Seed Support Management Committee (SSMC) Meeting

The 13th Seed Support Management Committee (SSMC) Meeting was held on 11 July 2023. This crucial meeting was conducted in both physical and virtual modes, reflecting the flexibility of the modern business environment. During this event, a total of 10 promising startups had the opportunity to present their pitch decks, showcasing their innovative ideas and business plans. To ensure a comprehensive assessment, nine esteemed jury members thoroughly reviewed these presentations. As a result of their

deliberations, three outstanding startups were chosen to receive investments from the BIRAC SEED Fund, with each receiving Rs. 20,15,25 lakhs to fuel their entrepreneurial endeavors.

4.1.10.3 Catalytic Capital Support Management Committee (CCSMC) Meeting

Catalytic Capital Support Management Committee (CCSMC) Meeting was held on 14 July 2023 marked the third and final round of evaluations for the NABARD CCF 2.0. This crucial meeting was conducted in both physical and virtual modes, reflecting the flexibility of the modern business environment. During this event, a total of 11 promising startups had the opportunity to present their pitch decks, showcasing their innovative ideas and business plans.

4.1.10.4 PRAYAS Monitoring Committee (PMC) Meeting

Ninth meeting of PRAYAS Monitoring Committee (PMC) was held during August 24-25, 2023 under the chairmanship of Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM, Hyderabad. This committee comprised of representatives of the incubator, technical/domain experts, financial/investment experts, representative of industry/industry association and successful entrepreneurs. The PMC decided to select 10 applicants in total for the NIDHI PRAYAS Grant under 4th call.

4.1.10.5 Project Monitoring and Review Committee (PMRC) Meeting

PMRC meeting was organized to review the progress of activities conducted by a-IDEA with the support of NABARD.

Table No. 4.1.6: Project Monitoring and Review Committee (PMRC) Meeting details

S.No	PMRC	Date
1.	10 th PMRC	04 January 2023
2.	11 th PMRC	10 May 2023
3.	12 th PMRC	13 July 2023
4.	13 th PMRC	17 October 2023

4.1.11 Other Activities

4.1.11.1 Agri-Startups Graduation Ceremony

a-IDEA, TBI of ICAR - NAARM, Hyderabad organized its First Graduation Ceremony of Agri-Startups on 17 March 2023 at ICAR-NAARM, Hyderabad. To commemorate the success of the Agri Startups and to support and encourage the startup ecosystem in agriculture and allied sector. Twenty Agri-startups who are incubated under a-IDEA's incubation program titled "KRISHIBOOT 1.0" were received certificates and mementos and four winners of Aggnite 3.0 also awarded certificate and cash prize.



Figure 4.1.16: Agri-Startups Graduation Ceremony

4.1.11.2 Startups - Scientist Interface Meeting

Startups and 112th FOCARS interface meeting have been organized by ICAR-NAARM on 29 April 2023 at ICAR-NAARM campus, four

a-IDEA, startups namely Agri Vijay, Marut Dronetech Pvt Ltd, Farmor Agri Solutions Private Limited, and Dhi Sathi Robotics Pvt Ltd (Farm Sathi) are interacted with scientist and share their startup stories and innovation with the young scientist. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM and Vice-President of a-IDEA preside over the program and Dr. N. A. Vijay Avinashilingam, Principal Scientist, ICAR-NAARM and Additional CEO of a-IDEA moderate the program.



Figure 4.1.17: Startups - Scientist Interface Meeting

4.1.11.3 Startup-Industry Interface Program

a-IDEA, TBI, ICAR-NAARM organised “Startup-Industry Interface Program” at ICAR-NAARM, Hyderabad on 18 May 2023 in collaboration with ASSOCHAM & FICCI with an objective to share knowledge, brand building, economic growth, networking, financial support, collaboration



Figure 4.1.18: Startup-Industry Interface Program

possibilities, joint licensing & market traction between agri industries & a-IDEA supported startups. Thirty start-ups, more than 17 industries from PAN India participated in the programme.

4.1.11.4 Startups-Bankers Interface

a-IDEA organised Startups – Bankers Interface at ICAR - NAARM, Hyderabad on 19 December 2023 with an objective to connect the Bankers of Telangana and Andhra Pradesh. About 10 startups supported by a-IDEA, ICAR-NAARM from across agriculture and allied sectors presented their products, services and demonstrated their advantages in front of 20 bankers from 13 banks (NABARD, CBI, ICICI, HDFC, UCO, AXIS, BOI, SBI, IDFC, IndusInd, Kotak, SBI, UBI, Grameen Bank etc.,) from different places across Andhra Pradesh and Telangana. The event acted as an ultimate platform for networking between bankers & startups for successful networking and for further collaboration.



Figure 4.1.19: Startups-Bankers Interface

4.1.12: Visitors

1. Confederation of Indian Industry (CII) organized Exposure-Cum-Orientation Visit to a-IDEA, ICAR-NAARM on 26 April 2023 for Bhutanese Startups who won the pitching competition during the Bhutan – India Startup Summit.

2. Ms. Guha Poonam Tapas Kumar, IAS, Additional Secretary to Govt. & CEO, ORMAS, Government of Odisha visited a-IDEA, ICAR-NAARM on 11 May 2023 to understand about the concept of TBI and its role in strengthening startups. A team of five officials along with Additional Secretary interacted with scientists and a-IDEA team and appreciated the activities. They expressed willingness to organize one Immersion Program of Startups, FPOs at Bhubaneswar in collaboration with a-IDEA.
3. Shri C S R Murthy, CGM, FSDD, HO, Mumbai visited a-IDEA on 26 July 2023 and interacted with Prospective CEOs of FPOs under NABARD TSRO Project. He also interacted with a-IDEA officials and reviewed the performance.
4. Dr. Anup Kollanoor Johny, an Associate Professor in Animal Science at the University of Minnesota, Saint Paul, USA, and Dr. Kumar Venkitanarayanan, Professor and Associate Dean of Research at the College of Agriculture, Health, and Natural Resources, University of Connecticut, USA are visited a-IDEA on 7 Sep 2023 to know about startup ecosystem.

4.1.13 Participation

4.1.13.1 India international Science Festival (IISF), Bhopal

a-IDEA team participated in the four-day event, India international Science Festival (IISF), organized by the Union Ministry of Science and Technology at Bhopal during 21-24 January, 2023. The festival was held at Maulana Azad National Institute of Technology (MANIT), Bhopal and aimed at showcasing the country's

strengths and achievements in science, technology and innovation. The event witnessed the participation of Startups, Industrialists, students from various universities. a-IDEA supported startups viz., Quantrello Digital Pvt. Ltd, Water N Spices Pvt Ltd. And Fermentech Labs Pvt. Ltd also showcased their products in the festival.

4.1.13.2 ASPIRE BioNEST Conference, Hyderabad

Team a-IDEA participated in the two-day National Conference on “Building Entrepreneurial Ecosystem in Universities: Opportunities & Challenges” organized at the School of Life Sciences, University of Hyderabad during 3-4 February 2023. The conference was organised by the Association for Scientific Pursuits for Innovative Research Enterprises, University of Hyderabad (ASPIRE).

4.1.13.3 Kisan Agri Show, Hyderabad

a-IDEA team participated in the Kisan Agri Show organised at Hitex, Hyderabad during 3-5 March 2023. The event witnessed huge attendance of various stakeholders in which seven startups of a-IDEA viz., Dhi Sathi Robotics Pvt Ltd (Farm Sathi), FarmOR Agri Solutions Pvt Ltd, Nyasta Gramojvala Solutions PVT LTD, Pluss Avanced Technologies, Isaayu Farms Pvt.Ltd, Aumsat Technologies, Marut Drones Pvt Ltd exhibited their product/service & received the good response from the attendees.

4.1.13.4 Millet Mela, Mahabubnagar

The Ministry of Food Processing Industries (MoFPI) has launched the PM-FME scheme as part of the *Aatmanirbhar Bharat* Initiative to support micro food processing enterprises. To celebrate IYoM 2023, TSFPS organized *Millet*

Utsavam in Mahabubnagar district during 25-26 March 2023, in collaboration with MoFPI. The *Utsavam* was conducted to raise awareness about millet foods and healthy diets through various activities, such as consultation workshops, food stalls, and buyer-seller interactions. The event acted as a platform for stakeholders in the millet industry, including government agencies, food businesses, machinery manufacturers, and professional bodies, to showcase their work. This initiative is a significant step towards promoting local food processing industries and improving the country's food and nutritional security.

4.1.13.5 The Food Conclave-2023, Hyderabad

Team a-IDEA participated in the “The Food Conclave-2023” which is the first edition of the Food Conclave hosted by The Government of Telangana on 29 April 2023 at Hyderabad. a-IDEA supported startups namely Isaayu Farms Pvt Ltd. founded by Laxmi Preeti, FarmOR Agri Solutions Pvt. Ltd. founded by Venkata Sai Teja, Organikness Pvt Ltd. founded by Sandeep Thata, Farm Sathi (Dhi Sathi Robotics Private Limited) founded by Sushanth Masana Fisntails Agri Tech Pvt Ltd founded by Rakesh and Urban Kisan founded by Vihari Kanukollu participated in the conclave.

4.1.13.6 TS-AgriTech, Warangal

a-IDEA team represented at the TS-AgriTech event organized by ASSOCHAM at RARS, Warangal from 9-11th June 2023. The event witnessed huge attendance of various stakeholders wherein ten startups of a-IDEA viz., Dhi Sathi Robotics Pvt Ltd (Farm Sathi), FarmOR Agri Solutions Pvt Ltd, Nyasta Gramojvala Solutions PVT LTD, Khethari AgriTech Pvt Ltd,

Renewagri Om E-commerce Pvt Ltd (AgriVijay), Chimertech Pvt Ltd, Bhaumya Innovations, Pasidi Panta, exhibited their product/service & received the good response from the attendees.

4.1.13.7 Millet Conclave 2023, Hyderabad

Team a-IDEA, ICAR-NAARM attended the Millet Conclave 2023, a significant event organized jointly by NABARD (National Bank for Agriculture and Rural Development) and The Hindu Group. This conclave was held at the prestigious Taj Deccan Hotel in Hyderabad on 15 September 2023. The event likely focused on millets, which are a group of highly nutritious, drought-resistant crops.

4.1.13.8 GeoSmart India 2023: Agriculture & Irrigation, Hyderabad

Team a-IDEA participated as delegates, chaired, moderated, and was a panelist in two sessions (Credit Linkages and Funding Support in Agriculture and Way Forward and Recommendations) for GeoSmart India 2023: Agriculture & Irrigation held during 17-19 October 2023 in Hyderabad. Around 3 startups supported by aIDEA exhibited their products and services.

4.1.13.9 Global Bio India 2023, New Delhi

The BIRAC BIG team from aIDEA actively participated in the Global Bio India 2023 event held during 4-6, December 2023, organized by the Department of Biotechnology (DBT) and BIRAC, with a focus on “Transforming Lives-Biosciences to Bioeconomy.” With support from industry associations and Invest India, this three-day event served as a platform to showcase the vibrant biotechnology ecosystem. It featured engaging activities such as panel discussions,

exhibitions, and bio partnering meetings. The event also highlighted the “India-Germany Startup Connect” and offered a valuable startup masterclass led by experts from diverse fields. During the event, one of our BIRAC-BIG Grantee, Chimertech Private Limited, received the Startup Award in the Agriculture category. The event further encompassed insights from entrepreneurs, a public session on science and society, leaving a lasting impact on knowledge, connections, and recognition within the startup ecosystem.

4.1.13.10 AgTech Exhibition, Guntur

a-IDEA participated in the AgTech exhibition and conference held at Lam Farm, Guntur organised by ASSOCHAM in association with ANGRAU on the Latest Agriculture Technologies during 3–5 December 2023. Around 10 startups supported by aIDEA exhibited their products or services which created a platform for good networking with different stakeholders enabling their market traction.

4.1.14 Call for Applications

4.1.14.1 KRISHIBOOT 4.0 Call for Incubation

a-IDEA, TBI invited applications for Incubation to support individuals and early stage start-ups with an innovative idea/PoC/Prototype/MVP/in the field of Agri & Allied sectors. This programme is to help individuals or start-ups with an innovative idea by availing technical, business, laboratory & advisory support.

The application for KRISHIBOOT 4.0 was opened on 27 March 2023 and the last date for application was 10 May 2023. The total number of applications received was 67. The first level of screening was done by the internal a-IDEA team members based on the eligibility criteria. Based on these 38 Startups were selected for the second round of evaluation. The final round of evaluation was done through virtual mode by the external jury members for two days. In the final round of evaluation, 06 out of 38 startups are selected for the Incubation program.

Table 4.1.7: KRISHIBOOT 4.0 Call for Incubation

S.No	Name of the Startup	State	One liner
1.	Uday Aqua Connects Pvt Ltd	Telangana	Training Academy for young entrepreneurs- Aquaculture
2.	Anitra Tech Private Limited	Telangana	Livestock Traceability Platform
3.	Foundation for Aquaculture innovations & Technology Transfer (FAITT)	Chennai	Technology Transfer Foundation to serve Aquaculture farmers
4.	Garudalytics Private Limited	Telangana	Geospatial AI-based platform for harnessing crop data
5.	Inav Agro Mech Pvt. Ltd.	Telangana	Battery-operated ride-type multipurpose agriculture toolbar.
6.	Attaware Biodegradable Pvt. Ltd.	Delhi	Edible Cutlery and Tableware from Jaggery & Grains

4.1.14.2 KRISHIBOOT 5.0 Call for Incubation

The application for KRISHIBOOT 5.0 was opened on 23 June 2023 and the last date for application was 22 July 2023. The total number of applications received were 22. The first level of screening was done by the internal a-IDEA team members based on the eligibility criteria. A total

of 12 Startups were selected for the second round of evaluation. The final round of evaluation was done through virtual mode by the external jury members. In the final round of evaluation, 6 out of 12 startups were selected for the Incubation program and the remaining 9 startups were on boarded from the grants funded by DST and DBT as it was a pre-requisite for availing the grants.

Table 4.1.8: KRISHIBOOT 5.0 Call for Incubation

S.No	Name of the Startup	State	One liner
1.	Brown Reed Agri Waste Innovations	Karnataka	Coconut leaf drinking straws
2.	Krishigati Private Limited	Maharashtra	Self-Propelled Electric Agri. tool Bar
3.	Mr. Venkatesh K (Individual)	Telangana	Modular aeroponics technology, for commercial scale urban farming
4.	Sahaj Agrotech Private Limited	Maharashtra	Digital platform that streamlines the labor hiring process-Naya Kisan App
5.	Mangalam Kalpataru Industries, LLP	Madhya Pradesh	Banana Pulp into Sustainable Packaging
6.	Trisama Vacuura Agri Engineering and Foods Pvt. Ltd	Telangana	The company is engaged in Automotive & Special Products, Branded Products, Branded Producted & Retail and Industrail Products, Projects & Experts.
7.	Dr. Anirban Sur (Individual)	Maharashtra	Smart and Eco-Friendly Milk Chilling Unit for Rural Areas
8.	Svastha Samriddhi Private Limited	Karnataka	Fortified meat protein and multi nutrient products which are ready to consume- hygienic-easily available and logistics friendly
9.	Famnutra Millet Foods Private Limited	Telangana	Involved In Processing of Millets
10.	Nutricape Private Limited	Tamil Nadu	Bio-encapsulation services to boost aquafeed productivity, and enable disease prevention
11.	Arvind Patidar (Individual)	Madhya Pradesh	Transforming Farmers Life in India Using IoT based Onion Spoilage Detection
12.	KiSeva Kiosks Pvt Ltd	Karnataka	Kiosk for Kisan
13.	Pequrel Microelectronics Pvt. Ltd.	Karnataka	Advanced Adaptable Agricultural System A3S
14.	Prishigo Foods & Beverages Pvt. Ltd.,	Tamil Nadu	Edible Cutlery from Banana Bio-waste (Floral Stem Fibre)
15.	Chandralekha Ayekpam (Individual)	Manipur	Natural anti-ageing solution from blue green algae extract.

Table 4.1.9: Graduated Startups under KRISHIBOOT

S. No	Startup Name	Year of Incubation	Domain	One liner
1.	OneAgri Innovations Private Limited, Maharashtra	March 2022	ICT & Advisory	Democratizing knowledge in agriculture using online courses
2.	Mr. P. Sai hemanth, Telangana	March 2022	Vertical Farming	Agroforest is a terrace farming based company leveraging technology of hydroponics to create a whole new farming experience to customer.
3.	Hrishikesh Ray, West Bengal	March 2022	ICT & Advisory services	ICT enabled storage management system, linkage of Farmers-FPO's-SHGs, Agripreneurs, Cold Storage-Warehouses
4.	Mr. Anik Kundu (KrishiHut), West Bengal	March 2022	Agri- Inputs, Precision/ Smart Agriculture/ IoT/AI	One Stop Solution for All the Problems in Agri-input sector
5.	Pluss Advanced Technologies Private Limited, New Delhi	March 2022	Agritech	Manufacturing of Phase Change Materials (PCMs) for thermal energy storage and Polymeric Additives through R&D to develop people-centric and climate-positive innovative clean technologies
6.	Mars Inno India Private Limited, Telangana	March 2022	Agritech	Innovative Automated Mechanization Solutions for Preparation of Natural Manure
7.	Nimble Growth Consultancy Services Private Limited, Karnataka	March 2022	Agritech	Nimble Growth is into production & supply of organic Fruits & Vegetables (F&V) with its proprietary Package of Practice (PoP) along with Production Traceability
8.	AgriFeeder Agricultural services Pvt. Ltd., Bihar	March 2022	Supply Chain Technology	An initiative to felicitate the farmers
9.	Solos Aqua Global Private Limited, Telangana	March 2022	Agri Tech	Solos Aqua Global Private Limited is involved in vertical farming of mud crabs
10.	Droaerody Private Limited, New Delhi	March 2022	Agri Tech	Drone Manufacturing Company based on the principles of "Make in India"
11.	Mr. Jeyavishnu Kumaragurubaran, Tamil Nadu	March 2022	Agri Tech	low-cost alternative fish feeds using biofloc effluents as the main nutrient source.

S. No	Startup Name	Year of Incubation	Domain	One liner
12.	Krishivan Technologies Private Limited, Karnataka	March 2022	Agri Tech	Democratizing The Agri value chain by empowering Farmers
13.	Farmsaathi, Andhra Pradesh	June 2022	Agri Tech	Our products revolutionize traditional ways of farming. Makes you self-reliable with fully Autonomous-Solar Powered Farm Bots.
14.	InstaCoCo, Tamil Nadu	June 2022	Agri Tech	Fresh coconut water vending machine
15.	Increeks Private Limited, Telangana	June 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Supply / Value Chain	AgriTech Partners for providing technology solutions to make agri-value chain operate at higher equilibrium.
16.	Naturex Technologies LLP, Telangana	June 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Fisheries	Precision farming, hydroponics, climate smart agriculture by leveraging the power of technology and data.
17.	VertoX Labs Private Limited, Odisha	June 2022	Soil, Water & Weather, Fisheries, Supply / Value Chain ,Others	Plashbot is one of a kind aquatic weed harvester that removes aquatic weeds
18.	Sattvika Agri Lab Pvt Ltd, West Bengal	Aug 2022	Animal Husbandry/ Dairy	Probiotics along with plant extract which have properties for immune system stimulant
19.	Chimertech Private Limited, Tamil Nadu	June 2022	Animal Husbandry/ Dairy	One Stop IoT integrated mobile application based cloud platform for Disease management and breeding solutions in dairy industry
20.	Aumsat Technologies LLP, Maharashtra	June 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Soil, Water & Weather	Precision-driven, satellite-based, AI-enabled hydrological analysis for locating, predicting, and forecasting groundwater resources
21.	Organikness Pvt Ltd, Telangana	June 2022	Supply / Value Chain-E commerce platform	Organikness is an online marketplace to buy and sell organic, natural and sustainable products.

S. No	Startup Name	Year of Incubation	Domain	One liner
22.	Crop Intellix Private Limited, Andhra Pradesh	June 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Agri-Inputs, Agritech, ICT & Advisory services	Provision of end to end solution for Agri stakeholders
23.	109 Nano Composite Technologies Private Limited, Andhra Pradesh	June 2022	Agritech	Nano fertilizer
24.	One Place Velocious Private Limited, Maharashtra	June 2022	Food Technology, Supply / Value Chain Agritech, Precision/ Smart Agriculture/ IoT/AI/Sensor Technology	Processing, cold storage freezing supply chain aided by robotics and data science, automation
25.	Agro Origin, Assam	June 2022	Food Technology	Healthier substitute of snacks through omni channels for marketing products viz., Retail and e-commerce
26.	My Millets, Telangana	June 2022	Food Technology	Nutritious snack for all age groups, especially children
27.	Farmigai Agro Industries Pvt. Ltd., West Bengal	June 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology	AI app-based monitoring and services provider system for small and marginal rice farmers, and reduces production costs through an integrated approach, and produce healthier farm produce.
28.	GreenGrahi Solutions Pvt. Ltd., Uttarakhand	June 2022	Food Technology, Animal Genomics & Biotechnology, Fisheries	Transforming food waste into animal protein & oil
29.	Renewagri Om Ecommerce Pvt Ltd, Maharashtra	June 2022	Post Harvest Technology, Farm Machinery and Equipment	AgriVijay is India's first Marketplace of Renewable Energy products for farmers & rural households bringing all products in solar, biogas, thermal, wind & electric under one roof
30.	CheckAG Pvt Ltd, Karnataka	Jan 2023	Dairy technology	Milk quality analyser

S. No	Startup Name	Year of Incubation	Domain	One liner
31.	Farmoid Robotech Pvt Ltd, Karnataka	Dec 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology	Transforming agriculture with our extraordinary set of technologies. We are industry first manufacturer of automated modular robots that performs precision farming, Water Management, Soil & Crop Quality/ Health, Help Overcome Labour Shortage and Data Analysis
32.	Khadyam speciality foods private limited, Telangana	Dec 2022	Food Technology	Khadyam is committed to providing the healthy innovative organic food in India
33.	Foo Foods India Private Limited, Kerala	Jan 2023	Food Technology	Selling ready to serve traditional meat dishes
34.	X BOSON AI PVT LTD, Kerala	Sep 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology	AI-Controlled Aeroponics Indoor Vegetable Garden
35.	Applied Genomics Private Limited, Tamil Nadu	Jan 2023	Agritech, Agri biotechnology	Development and validation of indigenous cost effective genotyping technologies to support seed industries
36.	Udvaban Technologies Private Limited, Odisha	Dec 2022	Api culture, ICT/ IoT	IOT enabled beehive monitoring platform
37.	Nyasta Gramojvala Solutions PVT LTD, Telangana	Dec 2022	Precision/ Smart Agriculture/ IoT/AI/Sensor Technology	Leader in Revolutionizing Farming using Technology
38.	TechAdvent Mobility Private Limited, Maharashtra	Dec 2022	Agritech	Confidence in motion
39.	Khethari Agri Tech Private Limited, Telangana	Jan 2023	Agri- Inputs, Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Supply / Value Chain, Agritech	Digital agri platform for farmers to buy quality inputs
40.	Rosys Virtual Solutions Pvt Ltd, Andhra Pradesh	Dec 2022	Agri- Inputs, Precision/ Smart Agriculture/ IoT/AI/Sensor Technology, Supply / Value Chain , Agritech	Engineering Innovation for a Better Future

S. No	Startup Name	Year of Incubation	Domain	One liner
41.	Physis.living, Uttar Pradesh	Dec 2022	Agri-Biotech, Rural innovation, Post-Harvest Technology	Creating natural, sustainable and safe products
42.	Genflow AI, Delhi	Jan 2023	Animal husbandry/ Dairy technology	Tele medicinal services for dairy farmers
43.	Fisntails Agri Tech Pvt Ltd, Telangana	Jan 2023	Aquaculture	Economically Sustainable Aquaculture Ecologically

4.1.14.3 BIRAC BIG

The Biotechnology Ignition Grant (BIG) is a flagship scheme of BIRAC which recognizes and invests in innovative ideas in the biotech sector. Under BIG scheme, BIRAC provides grant-in-aid of upto INR 50 lakhs for a period of 18 months.

The 21st call for applications for BIRAC-BIG Grant was announced on 1 July 2022. In this

Call, a-IDEA received 84 innovative proposals exclusively from agriculture and allied sector. In light of opening of the new call, a series of Nine awareness webinars were organized in collaboration with various institutes like ICAR-CPCRI, AIC NITTE etc. Four startups are selected for 21st Call of BIRAC BIG in the month of Feb 2023.

Table 4.1.10: Startups selected for BIRAC BIG Grant 21st call

S.No	Name of the Startup	State	One liner
1.	Aicheng Innovation and Research Pvt. Ltd	Assam	A reeling Apparatus and method thereof solar powered integrated multipurpose reeling cum re-reeling machine
2.	Chimertech Private Limited	Tamil Nadu	A cloud integrated near infra-red spectroscopy based 24-in-24-in-1 milk parameters adopter for milking machine
3.	Famnutra millet foods Pvt. Ltd	Telangana	Development and commercialization of nutritional and healthy, ready to mix and beverage based on foxtail millet enriched with probiotics for diabetes that acts as functional food
4.	Dr. Parineeti Juvvi (Individual)	Karnataka	Design and development of continuous vacuum frying system for the production of healthy snacks in the food industries

The applications for **BIRAC-BIG Grant 22nd call** was announced on 1 July 2023. In this Call, a-IDEA received 105 innovative proposals exclusively from Agriculture and Allied sector. In light of opening of the new call, a series of

Nine awareness webinars were organized in collaboration with various institutes like ICAR-CPCRI, AIC NITTE etc. **Five startups** were selected during the BIRAC BIG 22nd Call.

Table 4.1.11: Startups selected for BIRAC BIG Grant 22nd call

S.No	Name of the Startup	State	One liner
1.	Mr. Madhumohan R (Individual)	Karnataka	Arecanut Dehusking Machine
2.	Dr. Kalay Khan (Individual)	J&K	Mechanization Intervention in Saffron Cultivation: Flower Harvesting, Corm Digging, and Sorting
3.	Fins and Tails Agritech Pvt. Ltd.	Telangana	Sustainable and High-Density Aquaculture Systems using IoT Devices to achieve predictable and repeatable yield at large scale
4.	Nutricape Private Limited	Tamil Nadu	Bio-encapsulation services to boost aquafeed productivity, and enable disease prevention
5.	Pequrel Microelectronics Pvt. Ltd.	Karnataka	Advanced Adaptable Agricultural System A3S

4.1.14.4 NIDHI PRAYAS

NIDHI-Promotion and Acceleration of Young and Aspiring technology entrepreneur (NIDHIPRAYAS) - Support from Idea to Prototype, is the scheme under NIDHI, aimed at addressing the gap in the very early stage of an idea/proof of concept funding. Rs. 1.2 Cr funding is given for one cycle valid for 30 months and 10 startups could be supported with Rs. 10 Lakhs each.

The **fourth call for NIDHI PRAYAS** was announced on 22 May 2023 and the last date for closure of applications was 21 June 2023.

Out of 278 applications received pan India for innovations across agriculture and allied sectors, 151 applications were shortlisted for second level of screening. The second level of screening was done by external experts on 28 June 2023 and shortlisted 72 applications for third level of screening. The third level of screening was done by two technical experts and one business expert from 19 July 2023 to 06 August 2023. The final selection was done by the PMC members on 24 Aug 2023. The PMC selected **10 applicants** in total for the NIDHI PRAYAS Grant under 4th call.

Table 4.1.12: Startups selected for NIDHI PRAYAS

S.No	Name of the Startup	State	One liner
1.	Mr.Arvind Patidar (Individual)	Madhya Pradesh	Transforming Farmers Life in India Using IoT based Onion Spoilage Detection
2.	Ms. Kousalya Indravasan (Individual)	Karnataka	Kiosk for Kisan
3.	Mangalam Kalpataru Industries LLP	Madhya Pradesh	Banana Pulp into Sustainable Packaging
4.	Physis Living Private Limited	Uttar Pradesh	Solution for No Pollution
5.	Dr. Ayekpam Chandralekha Devi (Individual)	Manipur	Natural anti-ageing solution from blue green algae extract

6.	Prishigo Foods & Beverages Pvt Ltd	Tamil Nadu	Edible Cutlery from Banana Bio-waste (Floral Stem Fibre)
7.	Svastha Samriddhi Pvt Ltd	Karnataka	Fortified meat protein and multi nutrient products which are ready to consume- hygienic- easily available and logistics friendly
8.	Dr. Anirban Sur (Individual)	Maharashtra	Smart and Eco-Friendly Milk Chilling Unit for Rural Areas
9.	Dhi Sathi Robotics Private Limited	Telangana	Autonomous Navigation Kits for Tractors and Farm Robots
10.	Mr. Somasundaram J	Tamil Nadu	Waste to Wealth – Sugarcane Bagasse to Edible Fiber

4.1.14.5 NABARD SEED Fund 2nd Call

a-IDEA, ICAR-NAARM announced the **2nd call for applications** under NABARD CCF from eligible incubatees/ startups with the last date of applications to receive on 18 May, 2023. In response we have received and reviewed 84 applications. Based on the preliminary screening, 41 applications are shortlisted for pitching

through online. The second level of screening was done by Inter team members, where the startups presented online and the committee shortlisted 11 applications out of 41 applications. In the final selection round **04 startups** were selected by the CCSMC Committee Meeting Members on 14 July, 2023 to make investment.

Table 4.1.13: NABARD SEED Fund 2nd Call

S.No	Name of Start-up Company	State	Amount Sanctioned (in INR Lakh)
1.	Attaware Biodegradable Pvt., Ltd, Delhi	New Delhi	25
2.	Marut Drone Tech Pvt., Ltd., Telangana	Telangana	25
3.	Inav Agro Mech Pvt., Ltd., Telangana	Telangana	25
4.	Drakon Innovation Pvt., Ltd.	Kerala	25

4.1.14.6 BIRAC SEED Fund 3rd Call

a-IDEA, ICAR-NAARM announced the 3rd call for applications under BIRAC SEED FUND (Bio Seed) from eligible incubatees/startups with the last date of applications to receive on 25 March 2023. In response we have received and reviewed 62 applications. Based on the preliminary screening, 49 applications are shortlisted for

pitching through online. The second level of screening was done by Inter team members, where the startups presented online and the committee shortlisted 10 applications out of 49 applications. In the final selection round 02 startups were selected by the 13th SSMC Committee Meeting Members on 11 July, 2023 to make investment.

Table 4.1.14: BIRAC SEED Fund 3rd Call

S. No	Name of Start-up Company	State	Amount Sanctioned (in INR Lakh)
1.	Jivanah Platform Service Pvt., Ltd	Karnataka	15
2.	Chimertech Pvt., Ltd	Tamil Nadu	25

4.1.15 Accomplishments of Startups

AgriVijay Team participated in the farmer event at Mahaboobnagar with support of a-IDEA ICAR NAARM. They also witnessed the visit by Honorable State Minister Mr. Goud who appreciated their vision and enforced more propagation at village level in terms of awareness of renewable & green energy products.

Renewagri Om Ecommerce Pvt Ltd (AgriVijay) exhibited in Krishi Mahotsav at Kota, Rajasthan during 24-25 January 2023 organised by Ministry of Agriculture, GoI where Honourable Speaker Loksabha Shri Om Birla ji & Honourable Minister of Agriculture Shri Narendra Tomar ji graced with their presence at stall.

Analytica Chimica Private Limited has cleared the Clean Tech Innovation Challenge initiative by NCL-IIT BHU for solving the real-world problems faced by mining industries inculcating a culture of product innovation, harnessing the creativity & problem-solving mindset agripreneurs.

Isaayu Farms Pvt Ltd participated at International Conference on Innovation to Transform Dry lands organized at ICRISAT, Hyderabad held during 21-23 February 2023. In their stall, they demonstrated automation in hydroponics or terrace farming. In which the startup has gained so many connects to promote urban farming.

Navshali Innovations Pvt Ltd, Karnal, which is working on Pyotam: Healing India with Water Based Ayurvedic Beverages has been selected in Greenr Sustainability Accelerator, a Techno Serve initiative.

Sparrotronics Pvt Ltd, Bengaluru participated in TS – Agritech 2023, a conference and exhibition on Latest Agriculture Technologies from 9-11 June 2023 at Regional Agricultural Research Station (RARS), Warangal, Telangana.

Nyasta Gramojvala Solutions, Hyderabad reached another milestone of “Model Irrigation Village Project” funded by MARS PET CARE and guided by Hon Minister Harish Rao. Nyasta successfully handed over solutions to farmers who are part of the project.

Shree Kamdhenu Panchagavya Udyog, Himachal Pradesh came up with a unique and novel idea looking at Consumers Preference for China based Rakhis on the festive occasion of Rakhsha Bandhan. Instead of opting for Chinese Rakhis, they have developed eco-friendly and herbal rakhis from Cow-Dung. This rakhi contains tulsi seed as well and helps in anti-radiation as a part of their vocal for local India, vocal for handmade initiatives.

Fishy Farmers Private Limited, Telangana who is working on Low Profile Auto Pneumatic Bead Filter has been selected for the “Akamai Cohort V - Water Accelerator Program” with the grant of INR 42 lakhs and also worked on the construction of an R.A.S in Centurion University Orissa, showcased their products to the Central Fisheries minister on the occasion of Fish Farmers Day.

4.1.16 Awards

On the occasion of NAARM 48th Foundation Day on 01 Sep 2023 the following startups received best Startup Award-2023.

- **Verdant Impact** based at Rajasthan founded by Mr. Manish Kumar, the beneficiary of NABARD CCF working in the domain of animal husbandry.
- **Aumsat Technologies LLP** based at Maharashtra founded by Mr. Riddhish Soni, working in the domain of Water Management

Verdant Impact, Rajasthan working in the domain of animal husbandry, was adjudged the winner of the Jury Award in Tech30 – which comprised of 30 young and most promising Indian startups with high potential to define the spirit of entrepreneurship in The Great Indian Techade—the largest congregation of innovators at startup - tech summit, TechSparks 2023 held in Bangalore. Also, by choice of audience, Verdant Impact[®] was the second runner up at the YourStoryTechade TechFest event held in Bangalore.

Vattam Agro & Dairy Industries Private Limited, Tamil Nadu grabbed first prize in Eco hackathon conducted in sync with 13th International Conference on Sustainable Waste Management and Circular Economy, in Somiya Vidhyavihar University, Mumbai.

Aumsat Technologies LLP, Maharashtra has won GCA LLA Award in the Business Adaptation Solutions category. GCA will award with cash prize of fifteen thousand euros (EUR15,000) at COP28 in Dubai, UAE in December 2023.

Chimertech Private Limited, Tamil Nadu has been awarded the Best Startup award - (Agriculture) at Global Bioo-India (GBI) -2023 organized by BIRAC.

Brown Reed Agri Waste Innovations (Sunbird Straws), Karnataka have won Mass Challenge Switzerland 2023 Silver Prize Winners – CHF 10,000. They make drinking straws from dried fallen coconut leaves which helps prevent the burning of leaves, reduces the use of single-use plastic straws, and provides employment to women in rural areas.

Bhaumya Innovations Pvt Ltd, Sangareddy, Telangana working on Multipurpose Harvester has been honored with the prestigious Best Performing Startup Award by Govt. of Telangana.

Aumsat Technologies LLP, Maharashtra was selected for the Private Water Changemaker Innovation Awards 2023 from among the top 30 finalists selected from a pool of exceptionally innovative and impactful projects dedicated to addressing water-related challenges worldwide.

Glimpses of the Events at a-IDEA



Agri Udaan 5.0 Demo Day



Graduation Ceremony of Agri-Startups



MoA with NGI, TBI, Coimbatore



Sankalp 8.0



Agri Udaan 6.0 Launch



Visit of International delegates



Share Holder Agreement



Accelerator Mentorship Program

Glimpses of the call for applications



CALL FOR BIG GRANT

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22nd CALL



Grant-In-Aid upto **50 Lakhs INR**



Apply Before

15th Feb
2023
(5:30 PM)

Who can apply :

Entrepreneurs, Start-ups, Researchers, Students, Individuals having innovative ideas/ PoC/ Prototype

Application link : www.birac.nic.in

Focus Sectors

- Agri Tech and Food & Agribusiness - Food Technology - Healthcare - Agri E-Infra - Agri Supply & Support Systems - Logistics - Vertical Agri - Agri Extension Technology and Value Addition	- Biotechnology - AI & ML - Smart Farming - Agri Infrastructure - Agri Mechanization - Agri Input Supply - Agri Finance - Agri Extension
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AGRI - TECH / FOOD - TECH & ALLIED SECTOR

BIRAC SEED FUND

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(Under Equity and Equity Linked Instruments)

<https://aideansarm.accubate.app/ext/form/848/1/apply>

Eligibility Conditions for start-ups to apply for seed support:

- Startups should be an incorporated enterprise in India
- Startups registered under the (Sanskriti) Act, 2018 and having at least 11 members in its Indian Citizens (Sanskriti) Act include DDI or PDI as the essential requirement after duly putting in place a selection mechanism based on the work requirements.
- Startups working towards innovation, development, deployment, or commercialization of new products, processes, or services in any of the tech stack or intellectual property.

Our Supported Startups

eSence AI, Geeflow AI, GIG FARM, Aghave, COOL, VATTAM, and many more

Deadline : 24th March 2023

Reach us at: coo.aidea@naarm.in | 7704810418



Call for Incubation

KRISHIBOUT

Pitch for Incubation...

Focus Areas

Agriculture and Agribusiness

IoT in Agriculture

Food Biotechnology

Virtual Reality

E-commerce

Precision Farming

IoT, Water & AI & Automation

Smart Technology

Plant Breeding

Agri-Tech Technology

Agri-robotics

Irrigation

Smart Infrastructure

Supply Chain Optimization

Quality Value Chain

Our Services

Mentoring

Networking

Digital Tools

Access to Markets

Funding & Investment Opportunities

Marketing & Sales Support

Technology Transfer

Social Services

Competitive Advantages

NABARD-ICET Incubators
Supports rural entrepreneurs

BIRAC-ICDS Incubators
Supports rural entrepreneurs

Who can apply?

- Innovative business proposition
- Scalable business model
- Strong management team
- Clear market need
- Strong financials
- High growth potential
- Strong network

Phase 4: Portfolio of Incubators

- 14 incubators
- 100+ startups
- 100+ employees

SCAN ME



Last Date to Apply : 28 April 2023

Contact us

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Focus Areas:

- ICT & IIT in Agriculture
- Innovation/Smart Agriculture
- Innovative Food Technology
- Supply Chain Technology
- Fisheries
- Soil Water & Weather Tech
- Farm Mechanization
- Sustainable Inputs
- Agri E-Content
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- Green/Vertical Farming
- Agri-Biotech
- Food-Related Technology
- Farm Fresh Retail

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8303420990
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4.2: Centre for Open and Lifelong Learning in Agricultural Education (COLLAge)

Centre for Open Life Long Learning (COLLAge) has been spearheading the protocols and process standardization for online course development and management, in line with the National Education Policy 2020. The center serves to develop, host, and manage online courses for the institute and other organizations as well. Besides this, the center is engaged in capacity building in digital learning for the entire NARES. Its model for offering MOOCs stands out as a technology that can be adapted by other educational organizations. The testimony for the center came in the form of recognition as one of the top 5 technologies from the Education Division of ICAR during the year 2023.

Digital learning content development and management uses the state-of-the-art facility for digital content production that involves hardware for pedagogy, content production in the form of video, and content refinement through high-end multimedia software.

The center is also contributing to institutional activities like supporting distance education programs, conducting practical sessions on pedagogy, and evaluating of training participants and students. The Centre contributed to conducting internal and final evaluations in the Distance education program, conducting sessions of microteaching and video production involving the trainees of the institute and FOCARS capacity building programs. The final evaluation through the MOOC portal was conducted for 186 FOCARS participants of 112 and 113 batches which involved randomised

question papers for all participants and instant evaluation.

During the year, the center offered 2 MOOCs, and 1 online course and produced 73 educational videos which were converted into digital modules.

4.2.1 Online Course on E-Agribusiness Management

Keeping in view the future demand for online courses as a part of NEP 2020, the center has offered a course on E-commerce Applications in Agribusiness Management for the students of the PG Diploma in Agribusiness Management. The resource material was shared with 18 students of the course and all activities like assignments and final evaluation were done through the MOOC portal of the Academy. Some of the resource material was shared in video format for the students to revisit the lesson as and when needed. The portal facilitated evaluation in descriptive mode also which was done online, thereby eliminating any possibility of human error/bias in evaluation. The portal helped students to learn from anywhere and anytime during the course period, thereby the convenience of learning is enhanced.



Figure 4.2.1: Video production by FOCARS participants



Figure 4.2.2: Online production and editing of Oral Communication of FOCARS participants

4.2.2 MOOCs

In 2023, two MOOCs were offered with a total of 3055 registered participants where 1897 participants were active. Out of this, the MOOC on Commercial Dairy farming was offered in collaboration with ICAR- National Dairy Research Institute, Karnal while the second one was offered on Digital Teaching Techniques. Total certifications from the TWO MOOCs is 956 with a total completion rate of 31.3 %.



Figure 4.2.3: Course development in progress for Bioinformatics

4.2.2.1 Massive Open Online Course (MOOC) on Commercial Dairy Farming

From 2 January to 31 March 2023, a comprehensive Massive Open Online Course (MOOC) on Commercial Dairy Farming was

conducted for 60 days, attracting a substantial total of 917 participants. The primary objectives of this course were to familiarize dairy entrepreneurs and students with the principles, planning, and technical approaches necessary for establishing successful commercial dairy farms. Additionally, participants were guided to comprehend improved and sustainable dairy farm practices, alongside acquiring knowledge in effective business management practices within the dairy sector. An innovative aspect of this MOOC was the introduction of video lessons hosted through a dedicated video streaming service. This pioneering approach ensured that only authorized learners accessing the MOOC portal were granted permission to view the lessons on the designated e-learning website, enhancing the security and exclusivity of the course content.



Figure 4.2.4: MOOC on Commercial Dairy Farming

4.2.3 Digital Content Creation and Management

An off-campus training program on “Digital Content Creation and Management” for faculty of Uttar Banga Krishi Vishwavidyalaya, Cooch Behar, West Bengal was conducted by the Academy during 28-30 March 2023. In today’s digital world, creating and managing content effectively is crucial for organizations

to succeed. A well-designed training program that emphasizes digital content creation and management can help individuals and teams develop the skills and knowledge necessary to meet the demands of the modern digital landscape. With this objective, the training program was organized covering the essential elements of digital content creation, including the development of effective content strategies, the creation of high-quality multimedia content, and the distribution of content across various digital platforms. The training also included an

emphasis on the latest tools and technologies used for digital content creation and management. The training included E-learning & digital teaching methods, communication skills for effective teaching, content capturing in digital teaching, opportunities and challenges in agriculture education, digital education policies, Accessible content creation, digital interactive tools for teachers, Open source web tools for digital content management, teaching management etc.



Figure 4.2.5: Training program on Digital Content Creation and Management

Training and Capacity building

5

5.1 Training Programmes Attended by ICAR-NAARM Employees

Name	Programme	Duration	Place and Organizers
Dr B Ganesh Kumar	Executive Education Programme on Advanced Human Resource Management	December 4-9, 2023	Indian Institute of Management (IIM), Ahmedabad
Dr BS Yashavanth	Holistic Approach to Human-Wildlife Conflict Mitigation and One Health	May 15-16, 2023	Hyderabad, ICAR-NAARM & GIZ
Dr M Balakrishnan	Training programme on Programing using R and Python	November 6-10, 2023	School of Applied Sciences, Department of Biotechnology, Reva University, Bengaluru, Karnataka
Dr M Ramesh Naik	Holistic Approach to Human-Wildlife Conflict Mitigation: Taking a One Health Approach	May 15-16, 2023	ICAR-NAARM and GIZ
Dr NA Vijay Avinashilingam	Introduction to Emerging Technologies	March 31, 2023	Capacity Building Commission (CBC) - Online, NewDelhi
	MDP on Leadership Development (A pre-RMP Program)	Dec 11-22, 2023	ICAR-NAARM
Dr P Supriya	ICAR Sponsored Winter School on Development, Evaluation, and Biosafety Assessment of Genome Edited Crops: Hands-on Training	January 20 to February 09 2023	ICAR-IIRR
	Training programme on Programing using R and Python	November 6-10, 2023	School of Applied Sciences, Department of Biotechnology, Reva University, Bengaluru, Karnataka
Dr S Senthil Vinayagam	Two days sensitization workshop of ICAR – Agri-business Incubation Centres (ABIs)	Sep 21-22, 2023	IPTM unit, ICAR, New Delhi
Dr Umesh Hudedamani	Patent Analytics Course	February 28, 2023 to March 09, 2023	Turnip Innovations, Kolkata (Online)
	Holistic Approach to Human-Wildlife Conflict Mitigation and One Health	May 15-16, 2023	Jointly organized by ICAR-NAARM and GIZ (Online)

5.2 iGot Karmayogi

The "iGot Karmayogi" platform stands as a pioneering initiative by the Government of India, aimed at revolutionizing skill development and capacity building within the public sector workforce. Through this innovative platform, government employees gain access to a diverse range of training modules, resources, and interactive learning experiences tailored to enhance their professional competencies. By fostering a culture of lifelong learning and collaboration, this initiative not only elevates individual capabilities but also strengthens the overall effectiveness and responsiveness of government institutions in serving the nation. NAARM employees successfully completed the course "Introduction to Emerging Technologies" from iGot Karmayogi" platform.

5.3 Capacity Building Programmes Organized by ICAR-NAARM

In 2023, ICAR-NAARM hosted a total of 96 training programs, engaging 6558 participants. These encompassed a variety of training programs, such as two Foundation Courses for Agriculture Research Service attended by 186 participants, five Leadership Development Programs engaging 109 participants, 48 Need-Based Programs involving 1508 participants, 37 Off-campus programs with 1375 participants, two Workshops/Symposiums with 270 participants, and two Massive Open Online Courses (MOOC) with 3054 participants.

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
1	Training Programme on Soft Skill and Entrepreneurship Development for students of Dairy Science College, Amreli, Gujarat	January 3-5, 2023	Dr Sanjiv Kumar	120	Off-Campus
2	Skill Development Programme on Horticulture Crops for the Benefit of Farmers of Macherial District of Telangana State at COE Jeedimetla and COE Mulugu- Hyd	January 05, 2023	Dr P Vijender Reddy	47	Off-Campus
3	National Policy Workshop on Integrating Public and Private Extension Service (Hybrid Mode)	January 12, 2023	Dr Bharat S Sontakki Dr NA Vijay Avinashilingam	53	Workshop
4	Soft Skills for Entrepreneurship Development among Agri-Graduates under NAHEP	January 23-24, 2023	Dr Surya Rathore	27	Workshop

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
5	Foundation Course for Newly Recruited Faculty of OUAT, Bhubaneswar & PVNRTVU, Hyderabad	January 24 to February 13, 2023	Dr NA Vijay Avinashilingam, Dr Alok Kumar	31	FOCFAU
6	VIII th Management Development Programme for Newly Recruited Senior Scientists and Heads of Krishi Vigyan Kendras	January 31 to February 14, 2023	Dr Bharat S Sontakki Dr Surya Rathore	15	PC KVK
7	Commercial Dairy Farming (Online)	February 1 to March 31, 2023	Dr GRK Murthy	929	MOOC
8	Skill development Training Programme on Vermicompost Production Technology for scheduled caste farmers under SC Sub Plan at SAIRDKVK, Gaddipally, in Suryapet district of Telangana	February 2-3, 2023	Dr M Balakrishnan	58	Off-Campus
9	Training Workshop on Education Management and Academic Leadership (under NAHEP)	February 6-10, 2023	Dr D Thammi Raju	22	Need-Based
10	Training Programme on Impactful ICT Applications and Technologies in Agriculture	February 6-10, 2023	Dr VV Sumanth Kumar	33	Need Based
11	Training program on Improved Crop Management Technologies and Value Addition for the benefits of Farmers at Jagannadhapuram, Rajahmundry, AP under SCSP Sub Plan	February 13-15, 2023	Dr M Ramesh Naik	75	Off-Campus
12	Training Programme on Application of Bioinformatics in Accelerating Agricultural Research	February 13-17, 2023	Dr P Supriya	24	Need Based
13	Training Programme on Geospatial Analysis using QGIS & R	February 13-17, 2023	Dr PD Sreekanth	9	Need Based
14	Online Training Program on “Digital Teaching Techniques” for the Faculty of Sri Konda Laxman Telangana State Horticultural University, Hyderabad, Telangana (Online)	February 13-17, 2023	Dr S Senthil Vinayagam	15	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
15	ICAR-NAARM & Coromandel Sales Force Learning Academy Collaborative Capability Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (NEAT)- Phase 2, Batch 1	February 13-17, 2023	Dr M Balakrishnan	33	Need Based
16	Training program on Improved Crop Management Technologies and Value Addition for the benefits of Farmers at Kadiyam, AP under SCSP Sub Plan	February 16-18, 2023	Dr M Ramesh Naik	75	Off-Campus
17	Training Programme on Advances in Mobile Application Development (Online)	February 20-24, 2023	Dr N Srinivasa Rao	29	Need Based
18	ICAR-NAARM & Coromandel Sales Force Learning Academy Collaborative Capability Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (NEAT)- Phase 2 Batch 2	February 20-24, 2023	Dr Surya Rathore	29	Need Based
19	Training Programme on Enhancement of Teaching Competencies for Faculty of Kamdhenu University, Amreli	February 21-23, 2023	Dr D Thammi Raju	30	Off-Campus
20	Skill Development Training Programme on Integrated Farming Systems for the benefits of Farmers at KVK, Raichur, Karnataka	February 23-28, 2023	Dr M Balakrishnan Dr P Supriya	30	Off-Campus
21	Competency Enhancement Programme for Effective Implementation of Training Functions by HRD Nodal Officers of ICAR	February 27 to March 1, 2023	Dr Alok Kumar	16	Need Based
22	Training Programme on Data Visualization using R (Online)	March 1-8, 2023	Dr BS Yashavanth	109	Need Based
23	National Training Workshop on Big Data Analytics (Online)	March 9-10, 2023	Dr S Ravichandran Dr A Dhandapani	97	Workshop

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
24	National Conference on Policy Development and Implementation Strategies for Academia-Industry-Government (AIG) Linkages in Agricultural Higher Education in India under NAHEP	March 12-13, 2023	Dr SK Soam	65	Conference
25	ICAR-NAARM & Coromandel Sales Force Learning Academy Collaborative Capability Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (NEAT)- Phase 2 Batch 3	March 13-17, 2023	Dr Tavva Srinivas	16	Need Based
26	Skill Development Training on Integrated Farming System for SC Formers at KVK, TANUVAS Namakkal, Tamilnadu	March 13-17, 2023	Dr M Balakrishnan	40	Off-Campus
27	Training Programme on Digital Teaching Techniques for the Faculty of Kamadhenu University at Amreli, Gujarat	March 14-16, 2023	Dr GRK Murthy	30	Off-Campus
28	ABIIGROW (Mentoring and Training programme for Agri start-ups)	March 15-17, 2023	Dr S Senthil Vinayagam	43	Need Based
29	ICAR-NAARM & Coromandel Sales Force Learning Academy Collaborative Capability Building Programme for Young Agri-Business Professionals Marketing Officers' Resurgence & Excellence (MORE) Phase 1, Batch 1	March 20-24, 2023	Dr B Ganesh Kumar	19	Need Based
30	Training Programme on Multivariate Data Analysis (Online)	March 20-27, 2023	Dr A Dhandapani	150	Need Based
31	Skill Development training programme on Backyard Poultry – A Sustainable Rural Livelihood for Farmers at KVK, Gaddipalli, Telanagna	March 23-24, 2023	Dr M Balakrishnan	100	Off-Campus
32	Skill Development training programme on Scientific Bee Keeping for farmers at KVK, Gaddipalli, Telanagna	March 24-31, 2023	Dr M Balakrishnan	25	Off-Campus

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
33	Skill Development training programme on Vermicomposting Technology for Farmers at KVK, LAM Farm, Guntur, Andhra Pradesh	March 27-29, 2023	Dr M Balakrishnan	37	Off-Campus
34	Skill Development training programme on Food Processing Technology for farmers at KVK, LAM Farm, Guntur, Andhra Pradesh	March 27-29, 2023	Dr M Balakrishnan	60	Off-Campus
35	Training Programme on Digital Content Creation and Management for Faculty of Uttar Banga Krishi Vishwavidyalaya (UBKV), Kooch Behar, West Bengal	March 28-30, 2023	Dr N Srinivasa Rao	30	Off-Campus
36	112th Foundation course for Agriculture Research Service	April 11 to July 10, 2023	Dr A Dhandapani DSanjiv Kumar	117	FOCARS
37	MDP on Business Plan Development and Accelerating FPOs/FPCs (Online)	April 8 to May 12, 2023	Dr A Dhandapani Dr Sanjiv Kumar	23	Need Based
38	Skill Development on Integrated Farming System under SCSP in Collaborative with Krishi Vigyan Kendra (KVK), TNJFU, Nagapattinam	May 8 -12, 2023	Dr M Balakrishnan	40	Off-Campus
39	Skill Development Training Programme on Advanced Aquaculture Technique for SC Farmers at KVK Nagapattinam, TN	May 8- 12, 2023	Dr M Balakrishnan	40	Off-Campus
40	Holistic Approach to Human WildLife Conflict Mitigation. Taking a one Health approach	May 15-16, 2023	Dr Tavva Srinivas GIZ Project Team	36	Need Based
41	Executive Development Programme on Leadership Development	May 22 - 27, 2023	Dr G Venkateshwarlu Dr D Thammi Raju	21	EDP
42	Training Cum Exposure Visit of NABARD Official in Collaboration with NABARD and a-IDEA of NAARM- Collaborative Programme	May 31 to June 2, 2023	Dr S Senthil Vinayagam	35	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
43	Collaborative Capability Building Programme for young AGRI-Business Professionals New Entrances Assimilation Training (NEAT)- PHASE I	June 5-9, 2023	Dr P Supriya	23.	Need Based
44	Collaborative Capability Building Programme for young AGRI-Business Professionals New Entrances Assimilation Training (NEAT)- PHASE I Batch II	June 12 - 16, 2023	Dr N Sivaramane	14	Need Based
45	MDP on Leadership Development (a pre-RMP Programme)	June 12 -23, 2023	Dr P Ramesh Dr Alok Kumar	15	MDP Pre-RMP
46	Training Programme on use of Information and Communication Technology in Teaching for the Faculty of kamdhenu University, Gujarat	July 4 - 6, 2023	Dr NS Rao	25	Off-Campus
47	Training Programme on Technical Service Rules	July 10-11, 2023	Dr Vivek Purwar	34	Need Based
48	Training Programme on Gene Editing and Technology Mangement in Agriculture	July 10-14, 2023	Dr Umesh Hudedamani Dr P Supriya Dr M Balakrishnan Dr Satendra K Mangrauthia	22	Need Based
49	Certificate Course on Agribusiness Management Development Programme (ABMDP) for Prospective CEOs of FPOs	July 13 to September 1, 2023	Dr S Senthil Vinayagam	31	Need Based
50	Training Programme on Agri Rural Startup (BIRD Mangalore)	July 17-19, 2023	Dr S Senthil Vinayagam	36	Need Based
51	113th Foundation course for Agriculture Research Service	July 18 to October 17, 2023	Dr NA Vijay Avinashilingam Dr BS Yashavanth	69	FOCARS
52	Training Programme on Agri Rural Startup (BIRD Mangalore)	July 20-22, 2023	Dr S Senthil Vinayagam	35	Need Based
53	Faculty Development Programme for Faculty of Odisha University of Agriculture and Technology, Odisha	July 24 to August 3, 2023	Dr P Venkatesan Dr PD sreekanth	19	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
54	Training Programme on promotion, Processing and Value Addition of millites and Promotion of Nutrition-Sensitive Agriculture for Women Farmer Under SCPS at KVK at sitapur, Uttar Pradesh	August 7-9, 2023	Dr M Ramesh Naik	75	Off-Campus
55	Training Programme on Advanced in Mobile Application Development (Online)	August 7-10, 2023	Dr N Srinivasa Rao	45	Off-Campus
56	Training Programme on Soft Skill and Enterpresureship Development for the Faculty of Kamdhenu University, Gujarat	August 8-10, 2023	Dr Tava sriniavas	25	Off-Campus
57	Taining Programme on promotions, Processing and Value Addition Of Millites and Promotions of Nutritions-sensitive Agriculture for Women Farmer Under SCPS at KVK Sitapur	August 10-12, 2023	Dr M Ramesh Naik	75	Off-Campus
58	Skill Development Training on IFS-Climate Smart Agriculture Technologies for the Farmer of Ananthapur District AP with ARS Ananthapur, AP	August 11-13, 2023	Dr Supriya	60	Off-Campus
59	Skill Development Training on IFS-Climate Smart Agriculture Technologies for the Farmer of Ananthapur District AP With ARS Ananthapur, AP	August 16-18, 2023	Dr Supriya	60	Off-Campus
60	Interface Meet with POPIs of FPOs	August 22-23, 2023	Dr S Senthil Vinayagam	30	Need Based
61	Training Programme On Analysis of Experimental Data Using R	August 21-25, 2023	Dr A Dhandapani	27	Need Based
62	Executive Development Programme on Leadership Development for Newly Recruited RMPS in ICAR	August 21-26, 2023	Dr G Venkateshwarlu Dr D Thammi Raju Dr GRK Murthy	25	EDP

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
63	Skill Development Training Programme on Integrated Crop Management in Chilli at a KVK-Gaddipally	August 28-30, 2023	Dr M Balakrishnan	25	Off-Campus
64	Fostering Academic Leadership in Agriculture in Collaborative with Faculty Development Centre, GBPUAT, Pantnagar under NAHEP - off- Campus	September 4-8, 2023	Dr D Thammi Raju	25	Workshop
65	Workshop Cum interface meet with POPIs Transformative Interventions in the value chain and Business Development of FPOs of zone A (North costal AP), AP State	September 7-8, 2023	Dr S Senthil Vinayagam	60	Workshop
66	Collabrative capability Building Programme for Young Agri-Business Professionals of CIL MORE (Marketing Officers Resurgence & Excellence)	September 4-8, 2023	Dr M Balakrishnan	32	Need Based
67	Foundation Course for Faculty of Agricultural Universities (FOCFAU)	September 7-27, 2023	Dr D Damodar Reddy Dr M Balakrishnan	25	FOCFAU
68	Training Programme on Motivation, Positive Thinking and Communication Skill for ICAR Technical Officers (T-5 and above)	September 11-15, 2023	Dr Aneeja	21	Need Based
69	Collaborative Capability Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (NEAT) – Phase II	September 11-15, 2023	Dr M Ramesh Naik	30	Need Based
70	Programme on Impactful ICT Applications and Technologies in Agriculture for the Faculty of Kamdhenu University, Gujarat	September 11-15, 2023	Dr VV Sumanth Kumar	30	Off-Campus
71	Training Enterpreneurial Skill Development Student for B Tech (Dairy Technology) Students of NDRI, Karnal	September 19 to October 18, 2023	Dr K Srinivas Dr Surya Rathore	25	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
72	Training Program on Agricultural Production System Modelling	September 25-29, 2023	Dr M Ramesh Naik Dr D Damodar Reddy	29	Need Based
73	Digital Teaching (Online)	October 1-30, 2023	Dr GRK Murthy	2125	MOOC
74	Skill Development Training Programme on Value Added Products with Millets for women Farmers under SC Sub plan at SAIRD KVK, Gaddipally, Telangana	October 7-12, 2023	Dr M Balakrishnan	25	Off-Campus
75	Training Programme on Design Thinking in Agriculture Research and Education (Online)	October 9-13, 2023	Dr SK Soam	35	Need Based
76	Collaborative Capability Building Programme for Young Agri-Business Professionals (MORE) For Coromandel International LTD.	October 9-13, 2023	Dr B Ganesh Kumar	29	Need Based
77	Training Programme on Next Generation Sequencing and Data Analysis (Online)	October 16-20, 2023	Dr M Balakrishnan	98	Need Based
78	Training Programme on Vigilance Perspectives for ICAR Officers	October 18-20, 2023	Dr D Thammi Raju Vivek Purwar Mukul Rajsingh	35	Need Based
79	Training Programme on Vigilance Perspectives for ICAR Officers	November 1-3, 2023	Dr D Thammi Raju Vivek Purwar Mukul Rajsingh	34	Need Based
80	Improved Crop Management Technologies and Value Addition of Millets to Farmers at Kvk Amudalavalasa, Srikakulam, AP	November 1-3, 2023	Dr Ramesh Naik	75	Off-Campus
81	Training Programme on Analysis of Experimental Data Using R (Online)	November 20-24, 2023	Dr A Dhandapani	37	Need Based
82	Collaborative Capacity Building Programme for Marketing Officers of MORE (for Coromandel International Ltd. Neat Phase-II Batch)	November 20-24, 2023	Dr BS Yashavanth	28	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
83	FDP ON Competency Enhancement in Agricultural Research and Education for the Faculty of Kerala (KUFOs) AT Cochin under SCSP	November 20-24, 2023	Dr M Balakrishnan	24	Need Based
84	Training Programme on Geospatial Analysis using QGIS and R	November 20-24, 2023	Dr PD Sreekanth	10	Need Based
85	Certificate Course on Agri-Business Management Development Programme (ABMDP) for Prospective CEOs of FPOs Andhra Pradesh	November 15, 2023 to January 3, 2024	Dr S Senthil Vinayagam	31	Need Based
86	Sustainable Livelihoods for Rural Women through Backyard Polutry Farming at KVK, Yagantipalle, Nandtal District, AP	December 4-6, 2023	Dr M Balakrishnan	40	Off-Campus
87	Collaborative Capacity Building Programme for Young Agri-Business Professionals-Front Line Agri Marketing Excellence (FLAME) for Coromandel Officials	December 4-8, 2023	Dr NA Vijay Avinashilingam	27	Need Based
88	National Training on Entrepreneurial Skill Development for B.Sc (Agri/ Hort) students of RVSKVV, Gwalior, Madhya Pradesh	December 4-23, 2023	Dr Ramasubramhanian V Dr Umesh Hudedamani	26	Need Based
89	Training Programme on Enhancement of Teaching Competence of Faculty Members of UAS, Raichur, Karnataka	December 6-8, 2023	Dr GRK Murthy	36	Off-Campus
90	MDP on Leadership Development (a Pre-RMP Programme)	December 11-22, 2023	Dr K Srinivas Dr P Ramesh	25	Need Based
91	Training Programme on Improving Non-Cognitive and Soft Skills for the Assistant Professors from Maharashtra Animal & Fishery Sciences University (MAFSU), Nagpur, Maharashtra	December 11-15, 2023	Dr Tavva Srinivas	22	Need Based

S. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Pro-gramme Type
92	Entrepreneurial Skill Development Training Programme on Value Addition Of Milk and Meat Products at SVVU, Tirupati, Andhra Pradesh (In-Person Mode)	December 11-15, 2023	Dr M Balakrishnan	20	Off-Campus
93	Training Programme on Teaching Competency Enhancement, Academic Management and Developing Winning Research Proposal for the Faculty of Maharashtra Animal & Fishery Sciences University (MAFSU), Nagpur, Maharashtra	December 18-22, 2023	Dr R Venkattakumar	15	Need Based
94	Entrepreneurship Development Through Spawn and Mushroom Production for Unemployed Rural Youths at KVK-Shivamogga, Karnataka	December 18-22, 2023	Dr M Balakrishnan	40	Off-Campus
95	Competency Enhancement Programme -Cum-Exposure Visit for Technical and Administrative Staff of YS Parmar University of Horticulture and Forestry, Solan, Himachal Pradesh	December 26-28, 2023	Dr P Vijender Reddy	23	Need Based

5.4 Participation in Seminars/Symposium/Conferences/Workshops

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Inaugurated the ICAR Short Course on Recent developments in the organic production of oilseeds for ushering productivity, soil health and export	ICAR-IIOR, Hyderabad	February 21, 2023
	Managing soil health for sustainable agriculture in dry land areas in the International Conference on Innovations to Transform Dry lands	ICRISAT, Patancheru, Telangana	February 22, 2023
	Annual Conference of Vice-Chancellors and ICAR Directors	NASC Complex, New Delhi	March 4-5, 2023
	Inaugural Session of ICAR-Industry Stakeholder Consultation Meet	NASC Complex, New Delhi	March 06, 2023
	Inaugural Session of Global Millets (Shree Anna) Conference	Virtual mode/ICAR	March 18, 2023
	International Conference on Blended Learning Ecosystems for Higher Education in Agriculture	National Agricultural Science Complex (NASC), New Delhi	March 21-23, 2023
	Chaired the Panel discussion on Accelerating Digital Development Through Effective Capacity Building and Change Management in the International Conference on Blended Learning Ecosystems for Higher Education in Agriculture (ICBLE 2023)	National Agricultural Science Complex (NASC), New Delhi	March 22, 2023
	Special Screening of 100 th Episode of Mann Ki Baat	Raj Bhavan, Hyderabad	April 30, 2023
	NAAS Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs) in India	NASC Complex, New Delhi.	May 25, 2023
	International Innovation and Creativity Conclave (I2C2) Version – 2.0	Vishakhapatnam, Andhra Pradesh	May 28-29, 2023
	National Training Conclave 2023	Pragati Maidan, New Delhi	June 11, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	95th Foundation Day/Technology Day of the ICAR Society	NASC Complex, New Delhi	July 16, 2023
	12th Convocation of UAS, Raichur, Karnataka as Member of Board of Management	UAS, Raichur	July 28, 2023
	Brainstorming Session on the Implementation of Green Credits Programme to discuss on draft notifications on Green Credit Programme and Carbon Credit Trading Scheme	NAAS, New Delhi	August 24, 2023
	G20 Technical Workshop on One Health-Opportunities and Challenges hosted by the Indian Council of Agricultural Research under the Ministry of Agriculture and Farmers Welfare, Government of India	Bengaluru	August 29-31, 2023
	Expert in the Expert Consultation on Ethics and Current State of Research Publications	Virtual mode/ NAAS&NIAB	August 31, 2023
	Inaugural Session of One Day Dialogue on Identification and utilization of Novel genes of value and relevance for improvement of crop varieties suiting future challenges to Indian agriculture	PJTSAU, Hyderabad	September 02, 2023
	Inaugural programme of G20 Technical Workshop on Climate Resilient Agriculture during its G20 Presidency	Hotel Novotel, Hyderabad	September 04, 2023
	Webinar on: PRE - COP28 entitled: Impact of Climate Change on the Dry Areas organized by the International Dryland Development Commission (IDDC); Regional Action on Climate Change (RACC), Japan; Nizami Ganjavi International Center (NGIC) the African League of Young Masters (ALYM)	Hybrid mode/ IDDC	September 11& 12, 2023
	Chaired NGP Rao Memorial Lecture & NGP Rao Lifetime Achievement Award Ceremony	ICAR-IIRR, Hyderabad	September 13, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Brainstorming Session on Green Carbon Credits	Virtual mode/ NAAS	September 24, 2023
	Expert Consultation on COP 28: Preparedness for Indian Agriculture	Virtual mode/ NAAS	September 25, 2023
	Chief Guest in the Inaugural Programme of National Symposium on Strategies for Promotion of Incubatee in Agriculture and Allied Sector in the NE Region of India and Chaired the Technical Session 1: Fostering the Research Institution-Industry Interface Collaboration in the Symposium	ICAR Research Complex for NEH Region, Umiam, Meghalaya	October 4-5, 2023
	International Conference on From Research to Impact: Towards Just and Resilient Agri-Food Systems and Chaired the Session on Adaptation and innovation under a changing climate in the Conference	ICAR and the CGIAR Gender Impact Platform, NASC Complex, New Delhi	October 9-10, 2023
	Expert in the Brainstorming Session on Restoration and Improvement of Soil Health	Hybrid mode/ NAAS	October 28, 2023
	Chief Guest in the Inaugural Programme of Faculty Development Programme (FDP) on Competency Enhancement in Agricultural Research and Education under SCSP	Kerala University of Fisheries & Ocean Studies (KUFOS), Kochi, Kerala	November 20, 2023
	93 rd Annual Session, NASI and Azadi ka Amrit Mahotsav (Atma Nirbhar Bharat) and Symposium on India Secure @75	DAE Convention Centre, Anushaktinagar, Mumbai	December 3-5, 2023
	59th Annual Seminar of Fertilizer Association of India on the Theme Innovations in Fertilizer and Agriculture Sector	Fertilizer Association of India, Delhi	December 7-8, 2023
	Chief Guest in the Valedictory programme of Off campus programme Entrepreneurial Skill Development Training Program on Value Addition of Milk and Meat Products scheduled	SVVU Tirupati	December 15, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Chief Guest in the Inaugural programme of Off campus programme on Entrepreneurship development through spawn and mushroom production for unemployed rural youth	KVK Shivamogga,	December 15, 2023
Dr G Venkateshwarlu	Annual Conference of Vice-Chancellors and ICAR Directors	NASC Complex, New Delhi	March 4-5, 2023
	Inaugural Session of ICAR-Industry Stakeholder Consultation Meet	NASC Complex, New Delhi	March 06, 2023
	National Conference on Policy development & Implementation Strategies for Academia – Industry Government Linkages for Agricultural Higher Education in India	ICAR – NAARM, Hyderabad	March 12-13, 2023
	Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs)	National Academy of Agricultural Sciences (NAAS), New Delhi	May 25, 2023
	National Training Conclave 2023 in the august presence of Shri Narendra Modi, Honble Prime Minister of India	ITPO, Pragati Maidan, New Delhi	June 11, 2023
	XVI Agricultural Science Congress	National Academy of Agricultural Sciences (NAAS), ICAR-CMFRI, Kochi, Kerala.	October 10-13, 2023
Dr Tavva Srinivas	Developing and Pilot Testing Specialized Knowledge Products on Human Wildlife Conflict Mitigation for Agriculture and Veterinary Sector in India	Virtual Mode	February 22, 2023
	Panelist in the project closing workshop on Reflections and Looking Ahead Holistic and Harmonious Jun 16, 2023 Coexistence Approach to Human-Wildlife Conflict Mitigation in India under the theme Way Forward for the effective and participatory implementation of HWC-NAP at national, state and local level Resources, networks and dialogue	Hotel Claridges, New Delhi by GIZ	June 16, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ranjit Kumar	Chaired the Technical Session (7.2): Enabling policy and institutional environments that protect and promote inclusive and sustainable value chains during ICRISAT@50	International Conference on Innovations to Transform Drylands at Patencheru.	February 23, 2023
	Institute Technology Management Committee (ITMC)	ICAR- IIOR, Hyderabad	May 24, 2023
	International Conference on Blended Learning Ecosystem for Higher Education in Agriculture	ICAR-NAHEP	March 21-23, 2023
Dr S Senthil Vinayagam	ICAR Industry Stakeholders Consultation Meet	ICAR, New Delhi	March 06, 2023
	Workshop on Content Preparation for Annual Training Calendar for the year 2023-24	Extension Education Institute, Hyderabad	March 27, 2023
	Startup20X	Atal Incubation Centre-Sri Krishnadevaraya University, Ananthapur	April 13, 2023
	Global Food Basket 2023 Conclave	MSMECC & WASEE, Vijayawada, Andhra Pradesh	April 29, 2023
	Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs)	National Academy of Agricultural Sciences (NAAS), New Delhi	May 25, 2023
	Review-cum-workshop of ITMUs/ ZTMCs/ ABIs	Online ICAR	June 28, 2023
	National Seminar on Agripreneurship: A Key Driver in Agriculture 4.0	VIT, Vellore	July 29, 2023
	Telangana Farmer Producer Organisations (FPOs) Conclave	TSIRD, Rajendranagar	August 22, 2023
	Selection of awards for 9th Foundation Day celebrations	PJTSAU, Rajendranagar	August 26, 2023
	Millet Conclave 2023	NABARD, Hyderabad	September 15, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr S Senthil Vinayagam	Online Brainstorming Session on Strengthening National Agricultural Education System	Trust for Advancement of Agricultural Sciences (TAAS) jointly with the Indian Agricultural Universities Association (IAUA)	October 30, 2023
	National Workshop on Farmer Producer Companies in India: Profile and Potential under the ICAR-NAHEP-CAAST	Kerala Agricultural University, Thrissur, Kerala	December 01, 2023
Dr D Thammi Raju	National Policy Workshop on Integrating Public and Private Systems for Effective Extension Service	NAARM, Hyderabad	January 12, 2023
	National Conference on Academia-Industry-Government Linkages for Higher Education	NAARM, Hyderabad	March 12-13, 2023
	Brainstorming Session on Enhancing Agri-Infrastructure and Agri-Business Development through Public-Private Partnerships (PPPs) in India	NAAS, New Delhi	May 25, 2023
	95 th ICAR Foundation Day and Technology Day Celebrations	NASC, New Delhi	July 16-18, 2023
	Sustenance of CDCs	CSKHPKV, Palampur	October 4, 2023
	Expert Committee Member for selection committee in the discipline of Veterinary and Animal Husbandry Extension	OUAT, Bhubaneswar	November 20, 2023
	Expert Committee Member for the promotion under CAS in the discipline of Veterinary and Animal Husbandry Extension	KVAFSU, Bidar	December 20, 2023
	International Conference for Implementation of NEP 2020 in Agricultural Universities of Maharashtra	Dr BSKKV, Dapoli, Maharashtra	December 28, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr VV Sumanth Kumar	National Policy Workshop on Integrating Public and Private Systems for Effective Extension Service	ICAR-NAARM	January 12, 2023
	Workshop on Harnessing the Power of Digital Agriculture in India	ICRISAT, Hyderabad	March 10, 2023
	National Conference on Policy Development and Implementation Strategies for Academia- Industry-Government Linkages for Agricultural Higher Education in India	ICAR-NAARM	March 13, 2023
Dr B Ganesh Kumar	G20 Technical Workshop on Climate Resilient Agriculture	Hotel Novotel, Airport, Hyderabad organized by ICAR-CRIDA, Hyderabad	September 4-5, 2023
	AGRI UDAAN – Food & Agribusiness Accelerator 6.0 & RuTAG 2.0 Accelerator Mentorship Program	ICAR-NAARM, Hyderabad	November 22, 2023
	International Year of Millets-2023: Millet Promotion by Hyderabad International Hub	ICAR-IIMR, Hyderabad	November 24, 2023
	International Nutri-Cereals Convention 2023 (INCC 5.0)	Hotel Novotel, HICC, Hitech City, Hyderabad organized by ICAR-IIMR, Hyderabad	November 27-28, 2023
Dr K Srinivas	Integrating Sciences & Institutions for Resilient Production System	SKUAST-K, Shalimar Campus	March 17-18, 2023
	XVI Agricultural Science Congress	National Academy of Agricultural Sciences (NAAS), ICAR-CMFRI, Kochi, Kerala.	October 10-13, 2023
	Agricultural Innovation & Entrepreneurship conclave	Navasari Agricultural University, Navsari, Gujarat	December 04, 2023
	National workshop on Sensitizing the Faculties to Inspire the Youths in Bio-entrepreneurship	University of Agricultural Sciences, Dharwad	December 19, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr R Venkattakumar	National Seminar on Evolving Extension Science towards Secondary Agriculture for Sustainable Development	ISEE, New Delhi and UAS, GKVK, Bengaluru, Karnataka	June 22-24, 2023
	National Conference on Next Gen Extension for Evolving Resilient Agri-Ecosystems (NEERAE, 2023)	EEI, Hyderabad, Telangana	September 25-27, 2023
	International Seminar on Exotic and Underutilized Horticultural Crops: Priorities and Emerging Trends (ISEUHC 2023)	ICAR-IIHR, Bengaluru, Karnataka	October 17-19, 2023
	International Technical Workshop on Climate Resilient Agriculture	ICAR-CRIDA, Hyderabad	September 4-6, 2023
	Kisan Diwas	ICAR-NAARM at Bopally Village of Telkapally mandal, of Nagarkarnool district	December 23, 2023
	National Seminar on Evolving Extension Science towards Secondary Agriculture for Sustainable Development	ISEE, New Delhi and UAS, GKVK, Bengaluru, Karnataka	June 22-24, 2023
Dr Surya Rathore	WIPO – IPO PCT Roving Seminar	WIPO & IP, India at ICAR, NAARM, Hyderabad	March 06, 2023
	National Conference on Policy development & Implementation Strategies for Academia – Industry Government Linkages for Agricultural Higher Education in India	ICAR – NAARM, Hyderabad	March 12-13, 2023
	International Conference on Blended Learning Ecosystems in Higher Education in Agriculture	ICAR, New Delhi	March 21 - 23, 2023
	International Conference of CGIAR Gender Impact Platform From Research to Impact: Towards Just and Resilient Agri-food Systems	New Delhi	October 9- 12, 2023
	International Womens Day organized	ICAR – Central Institute of Brackish water Aquaculture, Chennai (Online)	March 08, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr P Venkatesan	National Consultation on KVK and ATMA in Extension Delivery	MANAGE, Hyderabad	April 24 & 25, 2023
Dr NA Vijay Avinashilingam	State Level Millet Conclave	NABARD & ICAR-NAARM	March 10, 2023
	ABIIGROW – (Agri-Business Incubator of ICARs- A Growth Optimization & Retention Workshop) (Mentoring & Training programme for Agri start-ups)	ICAR-NAARM	March 15-17, 2023
	Introduction to Emerging Technologies	Capacity Building Commission CBC, New Delhi	March 31, 2023
	Interaction on Strengthening KVK and ATMA, MP	ATARI, Jabalpur, M.P.	May 12, 2023
	Regional Advisory Group on Millet Value Chain for FPOs	NABARD Andhra Pradesh	September 14, 2023
	Innovators Meet under RuTAG 2.0 Project	Faculty Building a-IDEA	November 09, 2023
	Intellectual Property, Trade Mark Filling and importance (Weekly Seminar)	ICAR-RCER, Patna	November 10, 2023
Dr Ramasubramanian V	Blended Learning Platform (BLP) Workshop	Professor Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	September 11, 2023
	Blended Learning Platform (BLP) Workshop	PV Narsimha Rao Telangana Veterinary University, Rajendranagar, Hyderabad	September 12, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ramasubramanian V	Blended Learning Platform (BLP) Workshop	College of Horticulture, Rajendranagar, Hyderabad, a constituent college of Shri Konda Laxman Telangana State Horticultural University, Siddipet.	September 13, 2023
Dr MB Dastagiri	Webinar for the 2023 VinFuture Prize Nominators	VinFuture Foundation, VinUniversity Campus, Gia Lam, Hanoi, Vietnam.	February 16, 2023
	Nobel Prize Summit 2023 Truth, Trust and Hope	Virtual Mode	May24-26, 2023
Dr Umesh Hudedamani	From Research to Impact: Towards Just and Resilient Agri-Food Systems	CGIAR and ICAR at New Delhi	October 09-12, 2023
Dr M Ramesh Naik	Telangana State Biodiversity Board UNDP Biodiversity Finance Initiative (BIOFIN)– Project launching programme	Aranya Bhavan, Hyderabad, Telangana State.	June 23, 2023
	Climate Smart Agronomy for Resilient Production System and Livelihood Security	Indian Society of Agronomy at Goa	November 22-24, 2023
Dr SK Soam	Member of the committee for recommendation of procedures and eligible officers for foreign training/ visit under NAHEP	Virtual mode	June 12, 2023
	Consultative Group of Geographical Indications, Government of India as an expert member	Lucknow	November 11, 2023
Dr M Balakrishnan	National Conference on Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India	ICAR-NAARM.	March 12-13, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr N Srinivasa Rao	National Conference on Policy Development and Implementation Strategies for Academia-Industry-Government Linkages for Agricultural Higher Education in India	ICAR - NAARM	March 12-13, 2023
	National Conference on Future of Agriculture and Agriculture for Future: Indian perspective.	SKUAST-K & Society for Plant Research Virtual mode	September 4-6, 2023
	Workshop on Fostering Academic Leadership in Agriculture	GBPUAT, Pantnagar	September 4-8, 2023
	A policy dialogue on Agricultural Education System in USA and Potential Collaboration with India	ICAR - NAARM	September 07, 2023
	Workshop on Sustainability of Career Development Centres	CSK Himachal Pradesh Krishi Vishwavidyalaya (CSKHPKV), Palampur	October 04, 2023
	Workshop on Sustainability of Career Development Centres	Anand Agriculture University, Anand, Gujarat	October 10, 2023
	Workshops on Sustainability of Career Development Centres	Kerala Veterinary & Animal Sciences University, Wayanad, Kerala	November 07, 2023
Dr BS Yashavanth	Holistic Approach to Human-Wildlife Conflict Mitigation and One Health	Hyderabad, ICAR-NAARM & GIZ	May 15-16, 2023
Dr P Supriya	National Conference on Future of Agriculture and Agriculture for Future: Indian perspective.	SKUAST-K & Society for Plant Research Virtual mode	September 4-6, 2023
	National Conference on Generative AI in Practice for Empowering Agricultural Research Productivity	Pune Chapter of National Academy of Agricultural Sciences & ICAR-National Research Centre for Grapes, Pune Virtual mode	September 11-12, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr P Supriya	International Conference From Research to Impact: Towards Just and Resilient Agri-food Systems	CGIAR and ICAR at New Delhi	October 09-12, 2023
G Aneeja	International Day of Biological Diversity	NAAS, Hyderabad Chapter	May 22, 2023
	Participated in the online lecture on Global Food Security: Role of Milk and Milk Products	NAAS Hyderabad Chapter	June 01, 2023
Dr Laxman M Ahire	International Conference on Decarbonizing Agriculture	Krishi tantra at TMA Pai International Convention Center, Mangalore, Karnataka	November 25-27, 2023

5.4.1 Participation in Meetings (other institutes)

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	33 rd Board of Management Meeting of PJTSAU	PJTSAU	January 17, 2023
	Meeting of ICAR National Computing Facilities Management Committee	ICAR New Delhi	January 18, 2023
	Review Meeting of Capacity Building Commission	Krishi Bhawan, New Delhi	January 18, 2023
	Sectional Committee Meeting of Telangana Academy of Sciences	IICT, Hyderabad	January 21, 2023
	Chaired the Online Meeting of Capacity Building Unit of DARE/ICAR	Virtual mode/ ICAR	March 01, 2023
	First Meeting of The National Academy of Sciences, India (NASI) Council 2023	Virtual mode/ NASI	March 01, 2023
	Meeting of the Advisory Committee of International Conference on Blended Learning Ecosystems for Higher Education in Agriculture 2023 (ICBLE)	NASC Complex, New Delhi	March 05, 2023
	Expert Group Meeting to develop a Road Map of the Clean Plant Programme (CPP) in India (under Atmanirbhar Bharat Clean Plant Programme of GoI)	Virtual mode/ NAAS	April 03, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	General Body Meeting and Award Presentation Ceremony of Indian National Science Academy (INSA)	Virtual mode/ INSA	May 09, 2023
	Meeting of The National Academy of Sciences, India (NASI) on the occasion of National Technology Day	NASI, Prayagraj	May 11, 2023
	Meeting of the National Academy of Sciences, India (NASI) Council	NASI, Prayagraj	May 13, 2023
	Meeting of Conveners of Regional Chapters, 30 th General Body Meeting and Foundation Day programme of NAAS	NAAS, NASC Complex, New Delhi	June 3-4, 2023
	Bilateral Meeting with Brazil held between Dr. Himanshu Pathak, Secretary & DG, ICAR and the Honorable Vice Minister of Agriculture and Livestock of Brazil during G-20 AMM	HICC, Novotel, Hyderabad	June 16, 2023
	First Advisory Committee Meeting of Agrinnovate India Limited	Hybrid mode	July 20, 2023
	54th Meeting of the Board of Management, University of Agricultural Sciences (UAS), Raichur, Karnataka	UAS, Raichur	July 27, 2023
	Indian Agriculture Advancement Group (IAAG) International Inc. USA. Zoom Meeting with Hon. Secretary, MoA&FW on NMSA and its Role in Making Indian Agriculture Climate Resilient	Virtual mode/ MoA&FW	August 01, 2023
	35th Meeting of the Board of Management of PJTSAU	Hybrid mode	August 11, 2023
	Meeting on the Draft Reports of Annual Capacity Building Plan (ACBP) and Training Need Analysis (TNA) developed for DARE	Virtual mode/ DARE/ICAR	August 18, 2023
	First Meeting of Regional Advisory Group (RAG) – Stakeholders Meeting of NABARD for the year 2023-24	NABARD Regional Office, Hyderabad	August 18, 2023
	Meeting appraise about the Draft reports received from Capacity Building Commission (CBC) on Annual Capacity Building Plan (ACBP) and Training Need Analysis (TNA) developed for DARE/ICAR	Virtual mode/ DARE/ ICAR	August 24, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Meetings of Student-Science Meet, Fellowship Scrutiny Committee Meeting and Joint Meeting of the SAC and SFC of the National Academy of Sciences, India (NASI)	NASI, Prayagraj	August 25, 2023
	10th Meeting of Executive Council of National Council for Cooperative Training (NCCT)	NCCT, New Delhi	October 31, 2023
	51st Management Committee Meeting of Extension Education Institute (EEI)	Virtual mode	December 13, 2023
Dr G Venkateshwarlu	Review Meeting of Capacity Building Commission	Krishi Bhawan, New Delhi	January 18, 2023
	FOCARS Restructuring Committee Meeting	NAARM, Hyderabad	January 23-24, 2023
	Meeting of the Advisory Committee of International Conference on Blended Learning Ecosystems for Higher Education in Agriculture 2023 (ICBLE)	NASC Complex, New Delhi	March 05, 2023
	Bilateral Meeting with Brazil held between Dr. Himanshu Pathak, Secretary & DG, ICAR and the Honorable Vice Minister of Agriculture and Livestock of Brazil during G-20 AMM	HICC, Novotel, Hyderabad	June 16, 2023
	Meeting on the Draft Reports of Annual Capacity Building Plan (ACBP) and Training Need Analysis (TNA) developed for DARE	Virtual mode/ DARE/ICAR	August 18, 2023
	Meeting to appraise about the Draft reports received from Capacity Building Commission (CBC) on Annual Capacity Building Plan (ACBP) and Training Need Analysis (TNA) developed for DARE/ICAR	Virtual mode/ DARE/ ICAR	August 24, 2023
Dr Ranjit Kumar	QRT Committee meeting of ICAR	ICAR- IIOPR	June 14, 2023
	G-20 AWG Side Event (Agriculture Ministers Meeting) on Managing Agribusiness for Profit, People, and Planet	HICC Novotel, Hyderabad	June 15, 2023
	Board of Studies (BoS) meeting of School of Agri-Business & Rural Management, Dr. RPCAU, Pusa	Virtual mode	October 16, 2023
	Institute Management Committee(IMC) meeting of ICAR-CIWA, Bhubaneswar	Virtual mode	December 01, 2023

Name of the staff	Name of the Programme	Venue / Organized by	Date
Dr S Senthil Vinayagam	HR-Restructuring Committee Meeting for DDU-GKY	NIRDPR, Hyderabad	June 08, 2023
	Board of Governors meeting	SINED-TBI, ICAR-NDRI, Karnal	December 16, 2023
Dr D Thammi Raju	FOCARS Restructuring Committee Meeting	NAARM, Hyderabad	January 23-24, 2023
Dr B Ganesh Kumar	23 rd RAC Meeting of ICAR-CIFT, Cochin	ICAR-CIFT, Cochin	February 20-21, 2023
	G20 Agriculture Ministers Meeting	Hyderabad, MoA&FW	June 15-17, 2023
	Bilateral Meeting between Secretary-DARE & DG-ICAR and Vice Minister of Agriculture and Livestock of Brazil on the sidelines of G20 Agriculture Ministers Meeting	Hyderabad, MoA&FW Govt. of India	June 16, 2023
Dr SK Soam	Institute Technology Management Council Meeting as an expert member	ICAR-IIMR	December 13, 2023

6.1 Activities with Capacity Building Commission

6.1.1 Review of Capacity Building Commission Unit (CBCU) Activities of DARE/ICAR by Hon'ble Minister of State for Agriculture & Farmer Welfare

Shri Kailash Choudhary, Hon'ble Minister of State for Agriculture & Farmers' Welfare, reviewed the capacity building activities of DARE/ICAR and the functioning of its Capacity Building Unit in the presence of Capacity Building Commission (CBC) on 18 January 2023 in Krishi Bhavan, New Delhi. The meeting was attended by Dr Himanshu Pathak, Secretary, DARE & DG, ICAR; Dr. Ch Srinivasa Rao, Director, NAARM & Chairperson, Capacity Building Unit for DARE/ICAR; Ms Alka Nangia Arora, Addl Secretary, DARE & FA, ICAR; Dr RC Agrawal, DDG (Agri Edn); Shri Shaleen Agrawal, Director, DARE; Dr G.Venkateshwarlu, Joint Director, NAARM; Shri Rajeev Lal, Joint Secretary (Admin) & Member, CBU and Mr. Hemang Jani, Secretary, CBC.

Dr Himanshu Pathak, Secretary, DARE & DG, ICAR, while giving opening remarks, enlightened the importance of training and capacity building in enhancing the competency and capacity of the individuals working in an organization. He expressed his satisfaction on the significant role played by the NAARM in imparting the training and wished that this review meeting will catalyze the association between DARE/ICAR and CBC

to work closely in the area of capacity building and training. Dr Ch Srinivasa Rao, Director, ICAR-NAARM welcomed earlier the dignitaries and gave an overview of the capacity-building activities of the entire National Agricultural Research and Education System. He also informed about the constitution of the Capacity Building Unit (CBU), a new initiative taken by the Council as per the advisory of CBC. Ms Alka Nangia Arora, Addl Secretary, DARE & FA, ICAR, Dr RC Agrawal, DDG (Agri Edn), and Shri Shaleen Agrawal, Director, DARE graced the event and provided their inputs for the successful implementation of the Annual Capacity Building Plans for the benefit of personnel under DARE/ICAR.



Figure 6.1.1: A snapshot of the event on Capacity Building Commission Unit (CBCU) Activities of DARE/ICAR

6.1.2 CBU of DARE-ICAR participated in National Training Conclave-2023

As a part of Mission Karmayogi, the first-ever National Training Conclave under the auspices of the Capacity Building Commission (CBC) was held at the Convention Centre in Pragati Maidan, New Delhi on 11 June 2023 to foster collaboration among training institutes and strengthen the training infrastructure across the country. Hon'ble Prime Minister, Sri Narendra Modi inaugurated the conclave and highlighted the importance of capacity building, ending silos, and enhancing service delivery. The DARE-ICAR Capacity Building Unit (CBU) comprising Dr. Ch. Srinivasa Rao, Director and Dr. G. Venkateshwarlu, Joint Director from ICAR-NAARM, Dr S.K. Sharma, ADG (HRM) and Sri Rajeev Lal, Joint Secretary (Admin), ICAR and others participated in the Conclave along with HRD Nodal Officers of various ICAR Institutes. During the conclave, eight breakout sessions were organized with panel discussions on important elements of capacity building about the identification of training needs, promotion of knowledge sharing, effective assessment of conducted training, strategic resource planning, faculty selection, onboarding, and development, transforming to a phygital world of capacity building.

6.2 National Campaign and Celebrations

6.2.1 Agricultural Education Awareness Programme

The experiential field visit was organized for girl students of 7th, 8th and 9th class of Progress School, Chandrayanagutta, Hyderabad on 10 January 2023. The students were oriented about the various programs of the Academy through video presentation and were taken to campus tour comprising of herbal garden, and nursery. Total 75 students participated in the event.

6.2.2 National Science Day Celebrations

National Science Day celebrations-2023 were organized at NAARM on 28 February 2023, to develop the scientific temperament among youngsters. Dr Ch. Srinivasa Rao, Director, NAARM, and Dr. G. Venkateswarlu, Joint Director, ICAR-NAARM addressed the students on the occasion. While addressing, ZPHS, Shamshabad, encouraged the students to work hard and be role models for others and conveyed her best wishes to them. Around 210 students of the 8th & 9th Classes of Zilla Parishad Girls High School, Shamshabad took part in the celebrations. The students were taken for a





Figure 6.2.1: National Science Day celebrations at ICAR-NAARM

campus tour in two batches and were appraised of the various facilities like the Agro-technology park, Biodiversity park, Herbal garden, nursery, orchards and floriculture fields on the campus.

6.2.3 Celebration of International Year of Millets

The year 2023 has been declared by the United Nations as the International Year of the Millets. Towards this endeavour, through a collaborative approach, the Department of Agriculture &

Farmers Welfare, Government of India urges everyone including International Organizations, Academia, Hotels, Media, Indian Diaspora, Start-up Communities, Civil Society, and all others in the millet value chain to come forward and join hands to revive the forgotten glory of 'Miracle Millets' through the grand celebration of International Year of Millets - 2023. On 15 March 2023, NAHEP at ICAR-NAARM celebrated the International Year of the Millets - 2023 to promote the forgotten glory of 'Miracle Millets'.



Figure 6.2.2: Celebration of International Year of the Millets at ICAR-NAARM

6.2.4 International Day for Biological Diversity

The NAAS Regional Chapter-Hyderabad in collaboration with the Academy celebrated the International Day for Biological Diversity with the theme 'From Agreement to Action: Building Back Biodiversity' on 22 May, 2023 at ICAR-NAARM. On this occasion, an online special lecture session chaired by Dr. A.K. Singh, Vice President, NAAS and Former DDG (NRM), and co-chaired by Dr. B. Venkateshwarlu, Former VC, VNMKVP, Parbhani was organized. Dr. N. K. Krishna Kumar, Former DDG (Hort.), ICAR delivered a special lecture on 'Silk Diversity Conservation for Socioeconomic Sustainability of Poor Farmers'. A total of 70 participants from various ICAR institutions, SAUs, NAAS, and Regional Chapter-Hyderabad (fellows and associates) participated in the program.



Figure 6.2.3: International Day for Biological Diversity

6.2.5 Celebration of World Milk Day

NAAS Regional Chapter-Hyderabad in collaboration with ICAR-NAARM celebrated World Milk Day (WMD) on 1 June 2023. On this occasion, an online special lecture session chaired by Dr. Ch. Srinivasa Rao, Convener, NAAS- Hyderabad Chapter and Director, ICAR-NAARM was organized. Dr. D. Damodar Reddy, FNAAS welcomed the dignitaries and all participating guests and briefed them about the



Figure 6.2.4: Snapshot of the events on the occasion of World Milk Day

significance of World Milk Day. Dr. Dheer Singh, Director & Vice Chancellor, ICAR-National Dairy Research Institute, Karnal delivered a special lecture on 'Global Food Security: Role of Milk and Milk Products'. The chairman of the session, Dr. Ch. Srinivasa Rao stressed the importance of livestock in rural livelihood and discussed various opportunities in the dairy sector including improvement in production efficiency in the dairy sector. About 70 participants from SAUs, ICAR institutions, including NAAS Regional Chapter-Hyderabad fellows, and associates joined the program.

6.2.6 Celebration of International Yoga Day

ICAR-NAARM Celebrated International Yoga Day on the theme “Yoga for Vasudhaiva Kutumbakam” on 21 June 2023.



Figure 6.2.5: Active participation of members on the occasion of world yoga day

6.2.7 18th Parthenium Awareness Week

As communication received from ICAR-Director, Directorate of Weed Research, Jabalpur and as appealed by Hon'ble Director General, ICAR & Secretary DARE, Govt. of India to keep institute campus Parthenium free under the national major activity of Swachh Bharat Abhiyan, the

18th Parthenium Awareness Week was observed by the Academy during 16-22 August 2023. During the awareness campaign, awareness was created among the campus residents, farm workers, farm staff, and regular employees of the Academy about the Parthenium weed, its health hazards and yield loss in various crops.



Figure 6.2.6: Parthenium Awareness Week at the Academy

6.2.8 Foundation Day Celebrations

ICAR-National Academy of Agricultural Research Management celebrated its 48th foundation day to commemorate the establishment of the institution and to reflect on its achievements and contributions to the field of agricultural research and management. Dr. J C Katyal, Former DDG (Education) and Former Director, of ICAR-NAARM was the Chief guest. He highlighted the importance of ICAR-NAARM in human resource management and to enhance effectiveness of the national agricultural research and education system. He emphasized the need for multidisciplinary, multi-institutional, and multi-agency collaborations to overcome challenges and transform research outcomes into innovations. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM, highlighted the academy's achievements and innovations. He acknowledged

the efforts of the NAARM staff' commitment to research, innovation, and resource generation. He highlighted expanded capacity building programs, off-campus trainings, think-tank research policy papers in emerging aspects of Indian agriculture, achieving 100% placements for PGDM-ABM students, and high-impact achievements in a-IDEA and Technology Business Incubator. ICAR-NAARM has continued as an intellectual hub of ICAR and released three impactful policy papers on this occasion. Awards were given to best innovative farmer, emerging FPO, best start-up, best print and media, and best PG student. Scientists, admin, technical, finance, project, and contractual staff of ICAR-NAARM, PGDM-ABM students, and 113 FOCARS scientist probationers participated in the foundation day program.



Fig 6.2.7: Celebrations of ICAR-NAARM Foundation day on 1st September, 2023

6.2.9 Vigilance Awareness Campaign

The ICAR–National Academy organized Vigilance Awareness Campaign during 16 August 2023 to 15 November 2023. Various activities were organized during the period viz. i) Awareness was created among the trainees, visitors including farmers, staff and students through display of posters in about 25 locations of the Academy

about PIDPI ii) trained 69 Administrative Officers on Vigilance perspectives in two batches of 3 days duration iii) Oath taking by employees of the Academy iv) Painting Competition for school students on the theme - Say no to corruption; commit to the Nation v) Essay Competition on the identified theme for faculty/ officers/ staff in English, Hindi, Telugu vi) Elocution Competition

among PGDMA students on theme vii) Session on “Preventive Vigilance by (Common Session – For Faculty/ Admn / Finance Staff and Trainees) viii) Social Media Campaign on Vigilance Awareness ix) Electronic Display of Theme of Vigilance 2023 in prominent places of the Academy etc. Dr Ch Srinivasa Rao, Director distributed prizes to the winners of various competitions organized during Vigilance Awareness Week, celebrated during 30-10-2023 to 5-11-2023.



Figure 6.2.8a: Integrity Pledge taking by employees of the Academy



Figure 6.2.8b: Painting Competition for children during Vigilance Awareness Week

6.2.10 Communal Harmony Campaign week

Communal Harmony Campaign week was observed in the Academy during 19-25 November 2023. As part of the same, an Essay Competition on the topic entitled “Ways of Fostering Communal Harmony and National Integration” was conducted on 23 November 2023 for the staff of the institute was organized.



Figure 6.2.9a: Essay Competition held during Communal Harmony Campaign week

A Debate Competition was also conducted on November 23, 2023 for PGDM-ABM students on the topic "Can Communal Harmony and National Integration be promoted in the present era?"



Figure 6.2.9b: Debate Competition held during Communal Harmony Campaign week

In addition, banners on Communal Harmony were placed at prime locations of the institute

during the week. Moreover, Flag Day was observed on 24 November 2023 and the monetary contribution made by the staff of the institute towards the distribution of Flags was sent to National Foundation for Communal Harmony, New Delhi.

6.2.11 Kisan Diwas celebrations

ICAR-NAARM celebrated Kisan Diwas at Bopalle village of Telkapally Mandal, Nagarkurnool District, Telangana on 23 December 2023. The purpose was to observe Kisan Diwas on the theme “Cultivating the Future: Innovation and Sustainability in Agriculture” as a part of Swachhta Pakhwada being observed by ICAR from 16-31 December 2023. An interface with the farmers (77) of the village was organized. During the interface, discussion was held on the theme, and various issues on sustainability of agriculture and food security were addressed.



Figure 6.2.10: Kisan Diwas celebrations

The innovative farmers namely Jagathpally Sudhakar Reddy, Aluvala Balchandraiah, Belle Ramulu and Kesumula Narsaiah were felicitated on the occasion of Kisan Diwas. Nutrition along with food security were discussed and saplings of fruit plants were distributed among farmers.

6.2.12 Implementation of Mission Lifestyle for Environment (LiFE) at ICAR-National Academy of Agricultural Research Management

As per the guidelines provided by the Ministry of Environment, Forest and Climate Change, ICAR-NAARM has organized Mission Lifestyle for Environment (LiFE) activities from 25 May 2023 to 5 June, 2023. The activities were organized involving 122th FOCARS trainees, farm staff, contractual workers, housekeeping staff, other staff members of the academy and faculty members. “Plastic Free Campaign and Pond Cleaning at NAARM & Quiz Competition for 112 FOCARS” was organized at the academy. In this regard, collection of plastic waste was taken up on the campus and total of 50 members took part in this activity. Also, a quiz was organized for 117 scientist probationers of the 112th Foundation Course for Agriculture Research Scientist (FOCARS) batch to encourage learning and help the trainees to gain a broad understanding of the importance of environment and its impact in lifestyle. “Awareness in mess and cafeteria to minimize food wastage” was organized. “Best out of Waste” activity was organized where the collection of leaf litter, grass clippings and road sweepings was taken up and the collected waste was shifted to the vermicompost unit for recycling and subsequent use in various farm operations. Total 45 outsourcing personnel were educated on the importance of good waste

management in order to maintain a clean and healthy workplace. Cleaning activities were taken up behind the Scientist home and from Nursery road to the IGH II area, for removing unwanted plants. A total of 50 members participated in this activity. Outsourcing workers, commuters and others at the academy officials/staff were educated on the necessity of keeping the campus clean and refraining from using single-use plastic. “Sensitization of campus residents on cleanliness” a hackathon walk was conducted to sensitize campus residents on awareness on the importance of cleanliness in order to maintain a clean and healthy workplace was carried out. A sensitization programme to around 80 contractual women workers working in various units of NAARM was done. “Essay competition about mission lifestyle awareness” was organized at the academy in three different

languages viz., English, Hindi & Telugu for 25 members including the faculty members and staff members of ICAR-NAARM, on topic “Effect of Urbanization on Human and Environment Health”.

On 5 June, 2023, World Environment Day, a plantation drive was taken at the Agro-biodiversity park of the Academy in which more than 100 staff members participated. During the plantation drive several species such as *Mimusops elengi*, *Eranthimum*, *Areliabalfour*, *Areliavariegated* and *Plumeriapudica* were planted. A total of 500 plants were planted in this area to create awareness on the conservation of important tree/plant species among the residents, students, staff of the academy, trainee participants, commuters of the campus and visitors.



Figure 6.2.11a: Mission Lifestyle for Environment (LiFE) activities organized by ICAR-NAARM



Figure 6.2.11b: Participants of Mission Lifestyle for Environment (LiFE) activities



Figure 6.2.11c: Pledge on Mission Lifestyle for Environment (LiFE)

6.3 Bilateral Meetings and MOU's

6.3.1 ICAR-NAARM coordinated the Bilateral Meeting between Secretary-DARE & DG-ICAR and Vice Minister of Agriculture and Livestock of Brazil

ICAR-NAARM coordinated the Bilateral Meeting between Secretary-DARE & DG-ICAR and Vice Minister of Agriculture and Livestock of Brazil on the sidelines of the G20 Agriculture Ministers' meeting on 16 June 2023. Dr. Himanshu Pathak (Secretary-DARE & DG-ICAR), Dr. Ch. Srinivasa Rao (Director, ICAR-NAARM), Dr. G. Venkateshwarlu (Joint Director, ICAR-NAARM), Dr (Mrs). C. Tara Satyavati (Director, ICAR-IIMR), Dr. B. Ganesh Kumar (Head, HRM Division, ICAR-NAARM) from the Indian side and Mr. Roberto Serroni Perosa, Hon'ble Vice Minister of Agriculture and Livestock of Brazil and Secretary of Commerce and International Relations, Govt. of Brazil and Mr. Angelo de Queiroz Mauricio, Agricultural Attaché, Embassy of Brazil in India from Brazilian side participated in the bilateral meeting. The two sides discussed the possible collaboration in agricultural research in the areas of climate-resilient agriculture, millets, pulses, and sugarcane.



Figure 6.3.1a: Bilateral Meeting between Secretary-DARE & DG-ICAR and Vice Minister of Agriculture and Livestock of Brazil



Figure 6.3.1b: Participants of the Bilateral Meeting

6.3.2 ICAR-NAARM signed MoU with other Universities

To strengthen the Public-Private Partnership, ICAR-National Academy of Agricultural Research Management, Hyderabad signed MoUs with the Malla Reddy University (MRU), Hyderabad, Bharatiya Engineering Science & Technology Innovation University (BESTIU), Gorantla, Andhra Pradesh, IES University, Bhopal, Madhya



Figure 6.3.2a: Signing MoU with Malla Reddy University, Hyderabad



Figure 6.3.2b: Signing MoU with BESTIU, Gorantla, Andhra Pradesh

Pradesh, Sanskriti University, Mathura, UP and Vellore Institute of Technology, Vellore on 27th April, 17th July, 31st July, 21st August and 26th September 2023 respectively. The Academy also signed an MoU with Kamadhenu University, Gujarat on 18th October 2023. The objective of these MoUs is to have a long-term collaboration for the promotion of students' training and

quality postgraduate research in cutting-edge areas. Dr. V S K Reddy, Vice Chancellor, Malla Reddy University, Dr. Roopa Vasudevan, Chancellor, BESTIU, Dr. M. B. Chetti, Vice Chancellor, Sanskriti University and Er. B. S. Yadav, Chancellor, of IES University, participated during the brief signing ceremony organized at the Academy.



Figure 6.3.2c: Signing MoU with IES University, Bhopal, Madhya Pradesh



Figure 6.3.2d: Signing MoU with Sanskriti University, Mathura, Uttar Pradesh

6.4 Review Meetings

6.4.1 First Meeting of IX Research Advisory Committee

The first meeting of IX RAC was held at the academy on 26-27 October 2023. The meeting was chaired by Shri. Chintala Govinda Rajulu, Former chairman, NABARD along with the members Dr Aravind Kumar (Padmasri awardee) Ex-VC, RLBCAU, Jhansi, Dr Gopala Krishna, Vice President, Nagarjuna Fertilizers Pvt. Ltd., Shri Mohan Kumar Mishra, Secretary, National Council for Cooperative Training (NCCT), Dr R.C. Agrawal DDG (Agril. Education), ICAR, KAB-II, Pusa, New Delhi.

Dr Ch Srinivasa Rao, Director, ICAR-NAARM introduced all the members of RAC and welcomed them to the meeting. Dr Tavva Srinivas, Head, ABM division and Member Secretary, IX RAC of NAARM, presented the action taken report on the recommendations given during the 3rd meeting of the VIII RAC held in September 2022. The Director presented a comprehensive overview of the NAARM activities related to research and think tank/policy advocacy including new grants won, number of research papers published, technologies developed, copyrights registered,

policy papers published, MoUs & MoAs signed and the networks established during the preceding year. Dr G Venkateshwarlu, Joint Director, ICAR-NAARM presented the overview of major activities relating to capacity building and academic programs of the Academy. All the Heads of Divisions presented division-wise progress reports of activities and way forward.

The Committee complimented the Academy, Director, and all the Faculty for the progress despite having a low number of faculty. The members expressed satisfaction regarding the action taken based on the recommendations. The committee recommended keeping more focus on the core mandate of the academy, to enhance carbon credit, Integrated Farming Systems (IFS), precision agriculture, Artificial Intelligence in Agriculture, urban farming, etc. Also recommended to have research projects on nutritional security, financial security, and climate vagaries as collaborative efforts. It was also suggested to strengthen the MOOC platform through social media and websites. The committee visited various facilities of the academy such as the Biodiversity park, herbal garden, and TELAge Lab.



Figure 6.4.1a: First meeting of the IX Research Advisory Committee (RAC) in Progress



Figure 6.4.1b: First meeting of the IX Research Advisory Committee (RAC) in progress



Figure 6.4.1c: IX RAC Members with the faculty of the Academy

6.4.2 28th Institute Research Council (IRC) meeting

The 28th Institute Research Council (IRC) meeting was conducted during 25- 26 September 2023. Dr Ch. Srinivasa Rao, Director and Dr G. Venkateshwarlu, Joint Director, ICAR-NAARM, in their opening remarks, addressed the members of the IRC and appreciated the contribution and

hard work of the faculty in conducting various capacity building programmes, academic and research activities as per the mandate of the academy in spite of their limited number.

During the IRC meeting, a total of 18 in-house, 9 extra-mural and 3 new in-house projects were discussed in seven technical sessions including



Figure 6.4.2: 28th Institute Research Committee (IRC)

the plenary, each headed by the Chair and Co-Chair with rapporteur support. Decisions taken or suggestions made during each technical session were prepared by the respective Session Co-Chair. Three extra-mural projects and four In-house projects were completed, and two new in-house projects were approved during this IRC. At present, the total number of projects stands at 23 of which 14 are In-house and 9 are externally funded.

6.4.3 Review meeting NAHEP Component 2

A Virtual Review meeting of NAHEP Component 2 titled “Investments in ICAR Leadership in Agricultural Higher Education” was held on 13 February 2023. This review meeting was conducted to discuss about the completed activities, foreign exposure visits and forthcoming A-I-G Policy workshop organized during 12-13 March 2023. Dr Ch Srinivasa Rao, Director, ICAR-NAARM; Dr Rajender Parsad, Director, IASRI, New Delhi; Dr Anuradha Agrawal, National Coordinator, NAHEP Component 2 New Delhi; Dr Sudeep, PI, NAHEP Component 2, IASRI, New Delhi and Dr S K Soam, CCPI, NAHEP Component 2, ICAR-NAARM, Project Team from NAARM & IASRI attended this review meeting.

6.4.4 5th Meeting of External Advisory Panel

ICAR – NAARM organized 5th meeting of ‘External Advisory Panel’, constituted by ICAR on 9 March 2023. Main purpose of EAP was to provide ICAR/NAHEP with agenda to appraise the progress and formulate the way forward for the achievement of the overall objective of NAHEP i.e. enhancing the quality and relevance of agricultural education. The panel comprises of renowned educationists and agricultural experts

– six from India, three from the USA and one from Australia. The online meeting was chaired by Dr R C Agrawal, National Director, NAHEP and DDG (Education). Dr. Agrawal requested the External Advisory Panel to guide the ICAR pertaining to issues viz. how to bring transformative changes in agricultural education in tune with National Education Policy, interaction of global universities for globalization of agricultural education, collaboration with national, international and government departments, private sector, and sustainability of NAHEP efforts.



Figure 6.4.3: Meeting of External Advisory Panel

During the meeting the External Advisory Panel members viz. Dr David Kraybill, Dr Glen C Shinn, Dr K V Raman, Dr P L Gautam, Dr J C Katyal, Dr P K Joshi, Dr B Venkateswarlu participated and gave valuable suggestions and guidance in the improving the agricultural education in the country. Dr Ch Srinivasa Rao, Director, NAARM and Dr Rajender Parsad, Director, IASRI also participated.

6.4.5 Review Meeting of Career Development Centers (CDCs) and Faculty Development Center (FDC)

The review meeting Career Development Centers (CDCs) and Faculty Development Center (FDC) of the project NAHEP Component 2 titled “Investments in Indian Council of Agricultural

Research Leadership in Agricultural Higher Education was held on 17 August 2023 at Career Development Centre, Sri Venkateswara Veterinary University, Tirupati. Nodal officers and HR Executive officers of 05 Career Development Centres and Faculty Development Centre were attended this review meeting. All the CDCs and FDC presented their progress, success stories of FY 2022-23 and action plan for FY 2023-24.



Figure 6.4.4: Participants of the review Meeting on Career Development Centers (CDCs) and Faculty Development Center (FDC)

6.4.6 Virtual Review Meeting of Career Development Centres

A virtual meeting with nodal officers/coordinators of CDCs under National Agricultural Higher Education Project (NAHEP) Component 2 to discuss and make action plan for conducting



Figure 6.4.5: Review Meeting of Career Development Centres

career fairs for students at CDC centers was held on 17 February 2023. Dr Anuradha Agrawal, National Coordinator, NAHEP Component 2 New Delhi; Dr Sudeep, PI, NAHEP Component 2, IASRI, New Delhi; Dr S K Soam, CCPI, NAHEP Component 2, ICAR-NAARM nodal officers of Career Development Centres and Project Team from NAARM & IASRI attended this review meeting.

6.4.7 Committee Meeting held on FOCARS Restructuring

The meeting of the committee constituted for restructuring of FOCARS met on 23-24 January 2023 at ICAR National Academy of Agricultural Research Management, Hyderabad. The committee thoroughly reviewed the FOCARS program in its entirety starting from objectives, structure, duration, course content and pedagogy, structure and duration of the FET module, institute orientation, professional attachment, and related issues. Key issues like changing role of scientists, changing internal and external ecosystem of ICAR, and resource management were the primary consideration for suggesting changes/recommendations in the objectives, duration, structure, content, pedagogy, and evaluation of FOCARS in the wake of revised qualifications for the ARS. The Committee ascertained the opinions of the faculty of the Academy while making the recommendations.

The committee was chaired by Dr. P.K. Joshi, Former Director of ICAR-NAARM. The members of the committee are Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Dr. C.N. Ravishankar, Director, ICAR-CIFE, Dr. Raman Meenakshi Sundaram, Director, ICAR-



Figure 6.4.6: Committee Meeting on FOCARS Restructuring on 23-24 January, 2023

IIRR, Dr. G.Venkateshwarlu, Joint Director, ICAR-NAARM, Dr. Bharat S. Sontakki, Principal Scientist, ICAR-NAARM and Dr. D. Thammi Raju, Principal Scientist, ICAR-NAARM acted as Member Secretary. Dr B Venkateswarlu, Former Vice Chancellor of VNMKV has attended as Special Invitee.

6.4.8 An Interaction Meeting with Students for Feedback on Career Development Centers

ICAR-NAARM under NAHEP Component 2 established Career Development Centers at 05

Agricultural Universities. NAARM also preparing the sustenance, and institutionalization guidelines for CDCs. A meeting was conducted to collect feedback from students who benefitted through CDCs and was organized at Sri Karan Narendra Agriculture University, Jobner on 16-12-2023. Dr Balraj Singh, Vice-Chancellor, SKNAU, Jobner and Senior Officials, CDC Committee members, Dr S K Soam, CCPI, NAHEP Component 2 and students participated in this meeting.



Figure 6.4.7: Interaction Meeting with Students for Feedback on Career Development Centers

6.5 Visits and Interactions of Dignitaries

6.5.1 DG (ICAR) - Institutes Interaction Meet at ICAR-NAARM

Dr. Himanshu Pathak, Secretary (DARE) & DG (ICAR) had interactions with heads of the Institutes of ICAR located at Hyderabad on 17 June 2023 at ICAR-NAARM. Heads of the ICAR Institutes: IIRR, CRIDA, IIOR, IIMR, PDP, ATARI, NRC Meat, and NAARM along with Heads of the regional station of IIMR participated in the review meeting. Dr. Himanshu Pathak in his address stressed the research focus on emerging challenges in agriculture, technology development, revenue generation, inter-institutional projects among ICAR Institutes and with CG Institutes, CSR funds, and State Government projects. He has highlighted the need for effective budget utilization and sharing research facilities. Directors of Institutes and Heads of regional stations presented the priority-based new initiatives taken by the respective institutes and progress thereon.



Figure 6.5.1: DG (ICAR) - Institutes Interaction Meet

6.5.2 Secretary, ICAR interacts with ARS Probationers

Sh. Sanjay Garg, Additional Secretary (DARE) and Secretary (ICAR) visited ICAR-NAARM and interacted with the 112th Foundation Course

for Agricultural Research Service (FOCARS) on 1 May 2023. Dr. Ch. Srinivasa Rao, Director of ICAR-NAARM welcomed Sh. Sanjay Garg and appraised him about the Foundation Course and innovations brought in for the 112th FOCARS. Sh. Sanjay Garg, in his address, Shri Garg has apprised the scientists about the latest reforms that have been implemented such as online posting, e-HRMS, etc. While emphasizing the importance of Agricultural Research, he advised the scientist probationers to focus their research career on three things, namely, Changing National Priorities, the needs of the Farmers, and orienting toward global trends. Sh. Garg has mentioned that they should work towards Smart Agriculture as it is the need of the hour.



Figure 6.5.2: Interaction of Sh. Sanjay Garg, Additional Secretary (DARE) and Secretary (ICAR) with 112th foundation course of FOCARS

6.5.3 DG Interacts with 113th FOCARS Scientists at ICAR-NAARM

Dr. Himanshu Pathak, Secretary DARE and DG ICAR interacted with 113th FOCARS trainees on 29 July, 2023 at ICAR-NAARM, Hyderabad. This

113 FOCARS consists of 69 trainees from 32 ARS disciplines and represents 21 States / UTs of India.

Dr. Pathak, during the interactions, shared his thought-provoking experiences of life in the form of storytelling to scientist probationers in order to boost functional creativity, active participation and high motivation. He encouraged and



Figure 6.5.3: DG interacting with 113th FOCARS batch scientists

motivated young scientists for their dedicated services to the cause of the farming community in India. He also answered several questions and queries raised by the scientist trainees to provide insightful knowledge.

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM briefed the gathering about various innovations brought in 113 FOCARS viz., interactive sessions with women directors, young achievers, directors of ICAR institutes, startups and academy industry interface as per the changing needs of the young scientists of ICAR. He also highlighted the need for young scientists to sustain happiness and efficiency in any form of adversity.

6.5.4 DDG (Agri Extension) Addresses MDP Participants at ICAR-NAARM

ICAR-NAARM organized the VIII MDP for Newly Recruited Senior Scientists & Heads of Krishi Vigyan Kendras from 31 January to 14 February 2023. The valedictory session of the programme was held on 14 February 2023. Dr. Udham Singh Gautam, Deputy Director General (Agricultural Extension), ICAR was the Chief Guest and Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM presided over. Dr. Sheikh N. Meera, Director, ICAR-ATARI, Zone X, Hyderabad was the Guest of Honour.

Dr. Udham Singh Gautam in his valedictory address appreciated the capacity-building role of NAARM in imparting this much-needed training to Heads of KVKs. He highlighted the future plans of ICAR in strengthening the frontline extension system in general and KVKs in particular. In his presidential remarks, Dr. Ch. Srinivasa Rao highlighted the contributions of the Academy in strengthening the frontline extension role of ICAR, besides reassuring the capacity-building

support of the Academy to the KVK system. He called upon the Heads of KVKs to make the best use of various capacity-building and academic programs of the Academy to strengthen their human resource capacity. Dr. Nagulameera Shaik in his remarks lauded the contributions of KVKs in agricultural development in the country over the years. Fifteen newly recruited Heads of KVKs representing seven ATARI Zones participated in this program. Dr. Bharat S. Sontakki, Principal Scientist, and Dr. (Mrs.) Surya Rathore, Head (Acting), Extension Systems Management Division of the Academy was the Programme Director. The programme was jointly planned and implemented by the Agricultural Extension Division of ICAR, ICAR-NAARM, and ICAR-ATARIs exclusively for the newly recruited Senior Scientists, and Heads of KVKs in 3 phases viz., 1st Phase of 15 days at ICAR-NAARM, 2nd Phase of 10 days at Best KVK in other Zone and 3rd Phase of five days at respective ATARIs.



Figure 6.5.4: Visit of Dr. Udham Singh Gautam, Deputy Director General (Agricultural Extension) to ICAR NAARM

6.6 Other Major activities

6.6.1 भा.कृ.अनु.परिषद मुख्यालय द्वारा हैदराबाद स्थित स्थानीय संस्थानों के राजभाषा कार्यान्वयन का निरीक्षण

भा.कृ.अनु.प-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (नार्म) में दि. 16.02.2023 को परिषद द्वारा हैदराबाद स्थित स्थानीय संस्थानों में राजभाषा कार्यान्वयन संबंधी निरीक्षण किया गया। यह निरीक्षण, विशेषतः राजभाषा विभाग, गृह मंत्रालय के मानदण्डों के अनुरूप 'ग' क्षेत्र में हैदराबाद स्थित संस्थानों का किया गया एवं संजय बकोलिया (निदेशक जी ए सी एवं राजभाषा) इनके अध्यक्ष रहे। इसमें नौ स्थानीय संस्थानों के प्रशासनिक प्रमुख एवं राजभाषा अधिकारियों ने भाग लिया। समीक्षा सत्र में नार्म के निदेशक डॉ.सीएच. श्रीनिवास राव ने रेखांकित किया कि नार्म एक प्रशिक्षण का शीर्ष संस्थान हैं, अतः यहाँ हिंदी के साथ अन्य दक्षिण / क्षेत्रीय भाषाओं के प्रचार-प्रसार की व्यावहारिक प्रणाली अपनाई गई है। संस्थान की ओर से राजभाषा कार्यान्वयन की व्यावहारिक अनुपालनीयता पर डॉ.जे.रेणुका, संयुक्त निदेशक (रा.भा) ने प्रस्तुतीकरण दिया।



Figure 6.6.1: निरीक्षण के समापन में डॉ.सीएच.श्रीनिवास राव, निदेशक, नार्म सहभागियों को संबोधित करते हुए।

6.6.2 WIPO-IPO Roving Seminar on PCT organized at ICAR-NAARM

ICAR-NAARM hosted the ‘WIPO-IPO Roving Seminar on PCT’ organized by the Indian Patent Office in collaboration with the World Intellectual Property Organization on March 06, 2023. The objective of the event was to promote awareness in respect of protecting the innovations globally. The session was attended by more than 85 participants including personnel from various Govt. organizations, industry, IP professionals, and students. Ms. Christine, Director, PCT International Cooperation Division, Patents and Technology Sector, WIPO, in a special address mentioned about increasing trend of PCT filing from India and appreciated the role of IPO. Ms. Lisa Jorgenson, Deputy Director General, Patents and Technology Sector, WIPO, Geneva, in her video address, highlighted that the Asia region is emerging as a leader in technology advancements concerning PCT filings. During the inaugural session, Chief Guest, Dr. Anindya Sircar, Professor-IPR, NALSAR University of Law, Hyderabad, expressed his views about the evolution of the IPR regime in the country and mentioned the significant improvements in the IP filing process in the country in the last few years.



Figure 6.6.2a: Ms. Christine, Director, PCT International Cooperation Division, WIPO delivering her address



Figure 6.6.2b: Participants of the WIPO-IPO seminar

The event consisted of several sessions by numerous experts from WIPO, Geneva, and IPO, India to provide an overview of the PCT system including the benefits of filing PCT for innovators, the PCT filing procedure, the role of the Indian Patent Office as receiving office and live demonstration of ePCT filing, etc. Industry representatives also shared their experiences about the benefits of filing PCT. The program ended with a concluding remark by Mr. Sameer Swaroop, Deputy Controller, IPO.



Figure 6.6.2c: Delegates from WIPO-IPO with other dignitaries

6.6.3 Career Fairs Organized by NAHEP-Career Development Centers at SAUs

In addition to soft skills and personality development, another important key function of NAHEP component-2 sponsored “Career Development Center (CDC)” is to improve the knowledge canvas of students regarding innovative career opportunities. Dr RC Agrawal, National Director, NAHEP and DDG

(Education) asked the NAHEP-NAARM team to develop a plan of action for conducting “Student Career Counselling Fairs/ Expo, etc” through CDCs, which will help the knowledge enhancement regarding placements and professional opportunities. Responding to the

call, from January to March 2023, the CDCs have organized eight such events related to ‘Career Fairs and Career Expos’, ‘Campus Placement Drives’, etc. In these events, regional industries, MNCs, and NGOs participated including offers of placements to agri-graduates.

Table 6.1.1: The details of activities conducted by various CDC Centers

S.No	Name of the CDC	Event Name	Date
1.	IGKV, Raipur	Campus Outreach Programme	09 January 2023
2.	IGKV, Raipur	Seminar Series -2023 “ Career Counselling and Personality Development programme for Agricultural Students”	16-23 February 2023
3.	IGKV, Raipur	Campus Placements Drive	25 February 2023
4.	IGKV, Raipur	Campus Outreach Programme - Agzistence	01 March 2023
5.	IGKV, Raipur	Campus Outreach Programme – Bayer Crop Science	15 March 2023
6.	UBKV, Cooch Behar	UBKV - Career Expo (Job Fair, Career Guidance, Academia Industry Interface)	15 March 2023
7.	UBKV, Cooch Behar	Alumni Meet & Reunion Programme	15 March 2023
8.	CAU, Imphal	Job Fair 2023	24 March 2023

6.6.4 Special Lecture on Challenges in Extending Symbiotic Nitrogen Fixation Beyond Current Host Range

The NAAS Hyderabad Chapter in collaboration with ICAR- NAARM organized a special online lecture on “Challenges in Extending Symbiotic Nitrogen Fixation Beyond the Current Host Range”, delivered by Dr. Maitrayee DasGupta,

Professor, University of Kolkata on 10 August 2023. Dr. Maitrayee Das Gupta gave a detailed account of nitrogen fixation. She highlighted the various signalling pathways triggering nodule organogenesis and crack invasion in legumes. More than 100 participants from various ICAR institutions, SAUs, NAAS- Regional Chapter-Hyderabad fellows, and associates participated virtually.



Figure 6.6.3: A snapshot of the Lecture on Challenges in Extending Symbiotic Nitrogen Fixation Beyond the Current Host Range

6.6.5 Sensitization Programme for B.Sc. (Ag) Students

The NAAS Hyderabad chapter in association with ICAR- NAARM organized a Sensitization Programme for B.Sc. (Agriculture) Students from two private educational institutes on 11 - 12 August 2023 at Agricultural Research Station, (ANGRAU), Anantapuramu, Andhra Pradesh. The opportunities for higher education in the agriculture sector, nutrition sensitivity, and climate change issues were discussed with the students. Students were sensitized on aspects of value addition, food processing, etc. to reduce food losses due to extreme climatic events.



Figure 6.6.4: Sensitization Programme for B.Sc. (Agriculture) Students

6.6.6 ICAR-NAARM at G20 Technical Workshop on 'Climate Resilient Agriculture'

ICAR-NAARM participated in the G20 Technical Workshop on 'Climate Resilient Agriculture' held at Novotel Hotel, Shamshabad Airport, Hyderabad on 4-5 September 2023 and

exhibited posters depicting various activities and programs happening at the Academy for the visiting dignitaries from the G20 countries and other developed countries.



Figure 6.6.5a: ICAR- NAARM stall at G20 Technical Workshop



Figure 6.6.5b: G20 Technical Workshop on 'Climate Resilient Agriculture'

6.6.7 Training Workshop on "Fostering Academic Leadership in Agriculture"

A Training workshop on 'Fostering Academic Leadership in Agriculture' from 4 - 8 September 2023 in collaboration with the Faculty Development Centre, Govind Ballabh Pant University of Agriculture and Technology, Pantnagar under NAHEP Component 2. The major objective of this workshop is to enhance the efficiency of the Indian Agricultural Education System through education management and Academic leadership. Around 31 participants from 13 SAUs from 11 states attended this training program.



Figure 6.6.6: Participants of the Workshop on “Fostering Academic Leadership in Agriculture”

6.6.8 अकादमी में हिंदी पखवाड़े का आयोजन

अकादमी में हिंदी माह/ सैद्धांतिक रूप से 14 सितंबर – अक्तूबर, 2023 के दौरान उत्साह से मनाया गया। संस्थान सदैव राजभाषा के संकल्पशील अनुपालनीयता हेतु प्रतिबद्ध रहा है। अकादमी के अधिकारियों/कर्मचारियों के लिए विभिन्न हिंदी प्रतियोगिताएँ जैसे केवल एक मिनट, निबंध, गायन, आलेखन एवं टिप्पण एवं अंताक्षरी इत्यादि आयोजित किए गए। निजी एवं व्यावसायिक जीवन में तनाव प्रबंधन पर डॉ.ए.देबनाथ द्वारा महिला अधिकारी एवं कर्मचारियों के लिए विशेष व्याख्यान आयोजित किया गया। वैज्ञानिक समुदाय हेतु वैज्ञानिक लेख की हिंदी में सार प्रस्तुति संपन्न हुई। भावी फोर्कर्स प्रशिक्षु/प्रबंधन युवाओं के लिए वाद-विवाद एवं गायन प्रतियोगिताएँ आयोजित किए गये। चतुर्थ श्रेणी कर्मचारियों के लिए “मेरा कार्यालय – मेरा काम” विषय पर मिली जुली भाषा में प्रश्नोत्तरी साक्षात्कार सत्र चलाये गए।



Figure 6.6.7: हिंदी पखवाड़े के दौरान प्रतियोगिता की एक झलक

6.6.9 Workshops on “Sustainability of Career Development Centres”

ICAR-NAARM, in collaboration with CSK Himachal Pradesh Krishi Vishwavidyalaya (CSKHPKV), Palampur; Anand Agriculture University, Anand, Gujarat; Kerala Veterinary & Animal Sciences University, Wayanad, Kerala organized three one-day workshops titled “Sustainability of Career Development Centres” during October –November 2023. A total of 140 participants including Deans and Directors of 28 nearby SAUs, and 06 Private universities participated in this workshop. This 03 one-day workshop aimed to sensitize all agricultural universities to the importance of CDCs and encourage them to establish their own centers and to scale up and institutionalize these Career Development Centers across all Agricultural Universities.

Table 6.1.2: Details of the workshops on “Sustainability of Career Development Centres”

S. No	Place of the workshop	No of participants	Date of workshop
1.	CSK Himachal Pradesh Krishi Vishwavidyalaya (CSKHPKV), Palampur	56	04-10-2023
2.	Anand Agriculture University, Anand, Gujarat	34	10-10-2023
3.	Kerala Veterinary & Animal Sciences University, Wayanad, Kerala	50	07-11-2023
Total		140	



Figure 6.6.8: Workshop on Sustainability of Career Development Centres

6.6.10 ICAR - NAARM Cherishes Charity Work

The Academy has shared its big heart by distributing used bedsheets, towels, blankets, and pillow covers to orphanages/homes for the destitute. Like any other guest houses/hostels; the Halls of Residence, Scientists' Home, and International Guest House of the academy had replaced the old furnishings with the new ones. In this process, a local survey was conducted to find out the concerned orphanages/homes for the destitute who were in need and were ready to accept the items. Based on the survey, two takers were identified by the committee. A visit was made under the leadership of Dr. Surya Rathore with the support of the committee members comprising Dr. P Supriya and Mr. Sham Bahadur along with Mr. Shekhar Reddy on 20 April 2023, to Home for Aged, Shivarampalli and Cherish

Home for Girls, Kismathpur. These Orphanages accepted the items wholeheartedly. In all, 148 bed sheets, 80 pillow covers, 29 blankets, and 200 towels were distributed leading to a million dollar smiles among the habitants of these orphan's asylums.



Figure 6.6.9: ICAR – NAARM's Charity work

6.6.11 Demonstration-cum-Distribution of Agricultural Inputs and Farm Tool Kits under SCSP 2022-23

As a part of SC Sub-Plan implementation, ICAR-National Academy of Agricultural Research Management, Hyderabad organized a demonstration cum distribution of agricultural inputs and farm tool kits viz., tarpaulin covers, farm tools kits, Neema power powder, torch lights and bio-micro nutrient power to 346 and 326 scheduled caste farmers of the identified village, Cherukucherla of Midthur Mandal (Nandhyal District, Andhra Pradesh) on 27th June 2023 and Peddakadubur of Peddakadubur Mandal (Kurnool District, Andhra Pradesh) on 28th June 2023 respectively.

The demonstration-cum-distribution activity was taken up in the presence of Shri T. Mohan Rao, Joint Director of Agriculture (JDA), Nandyal District, Dr. N.C. Venkateshwarlu, ADR, RARS, Nandyal District, Shri V. Bharath Kumar Reddy, DAAB, Nandyal District, Mr. T. Ram Mohan Rao, DAO, Nandyal District and Mr. Peeru Naik, Mandal Agricultural Officer (MAO). Shri T. Mohan Rao, JDA, Nandyal District addressed the farmers and stressed on the utilization of bio-micronutrients, neema power powder, and different farm implements to improve

the livelihood of the beneficiaries. Dr. N.C. Venkateshwarlu, ADR, RARS, Nandyal District urged the beneficiaries to utilize the scheme effectively for their better livelihood. Shri T. Ram Mohan Rao, DAO, Nandyal District participated in the event and interacted with beneficiaries on various aspects under the SC sub-plan along with ICAR-NAARM officials.



Figure 6.6.10: ICAR-National Academy of Agricultural Research Management, Hyderabad organized a demonstration cum distribution of agricultural inputs and farm tool kits

6.7 Visits of Students/Trainees/Farmers

Sl No	Name of the College and details	Date of Visit	No. of students
1	Final B.Sc (Hons Agri) students from Imayam Institute of agriculture and technology, (affiliated to TNAU) Kannanur, Thuraiyur Taluk Tiruchirappalli District Tamilnadu	January 4, 2023	107
2	B Sc Ag Students from RG Deshmukh College of Agriculture Tiwasa	January 4, 2023	90
3	BSc (Hons) Horticulture students from College of Horticulture UHF Nauni Solan HP (Batch 1)	January 18, 2023	63
4	4 th year B.Sc (Hons Forestry) College of Horticulture and Forestry, (Dr Y.S. Parmar University of Horticulture and Forestry,) Batch – 2 Nauni Solan HP	January 23, 2023	67
5	Final year B.Sc (Ag Hons) students from College of Agriculture, Vellayani, Thiruvananthapuram, KAU, Kerala	January 24, 2023	124
6	Final year B.Sc (Hons) students from College of Cooperation, Banking and management, Kerala Agri University, Thrissur, Kerala	January 27, 2023	38
7	B Sc Agriculture Students J.K.K.Munirajah College of Agricultural Science, Palayam- 638 506 (Tamil Nadu)	January 30, 2023	120
8	Final year ABM students from Department of Agricultural and Rural Management Centre for Agricultural and Rural Development Studies Tamil Nadu Agricultural University, Coimbatore	January 30, 2023	41
9	4 th year B.Sc (Hons Horticulture) College of Horticulture and Forestry, (Dr Y.S. Parmar University of Horticulture and Forestry,) Neri Dt, HP	February 2, 2023	29
10	Final year B.Sc (Ag Hons) students from College of Agriculture, KLEF, Vaddeshwaram, Guntur - 522302	February 6, 2023	180
11	Final year B.Sc (Agri Hons) students from RVS agricultural college, Thanjavur, TN (Affiliated to TNAU)	February 8, 2023	117
12	3 rd year students from Biswanath College of Agriculture, Assam agril University, Biswanath Chariali, Assam	February 8, 2023	37
13	Final year B.Sc (Agri Hons) students from College of Agricultural Technology (Affiliated to TNAU), Theni, Tamilnadu	February 9, 2023	118
14	Final year B.Sc (Ag Hons) students from Dhanalakshmi agriculture College, (affiliated to TNAU) Perambalur, TN	February 10, 2023	111
15	Final year B.Sc (Ag Hons) students from College of Agriculture, Punjab Agril University, Ludhiana, Punjab	February 14, 2023	83
16	B.Sc (Biotechnology) students from College of Animal Biotechnology, Guru Angad Dev Veterinary and Animal Science University, (PAU) Ludhiana 141004, Punjab	February 16, 2023	22

Sl No	Name of the College and details	Date of Visit	No. of students
17	Final year B.Sc (Agri Hons) students from Don Bosco college of Agriculture (affiliated to TNAU), Sagayathottam, Tamilnadu	February 15, 2023	115
18	Final year B.Sc (Hons) Horticulture students from Horticultural college and Research Institute, Tamilnadu Agricultural University, Coimbatore	February 17, 2023	69
19	4 th year students of B.Sc. (Biotechnology) from College of Agriculture	February 17, 2023	48
20	BSc. (Hons.) Agriculture students of Usha Martin University, Ranchi	February 20 2023	84
21	Final year B.Sc. (Agri.) students of College of Agriculture, UAS, Raichur (Batch 1)	February 21, 2023	62
22	Final year B.Sc. (Agri.) students of College of Agriculture, UAS, Raichur (Batch 2)	February 23, 2023	63
23	Final year B.Sc. (Agri. Hons) students from MIT college of Agriculture and Technology, Vellalapatti, Musiri, Tamilnadu, (affiliated to TNAU)	February 23, 2023	109
24	Final year B.Sc (Ag Hons) students from Krishna College of Agriculture & Technology, Sreirapuram, Madurai, TN	February 24, 2023	86
25	Final year B.Sc (Ag Hons) students from Jaya Agricultural College (Affiliated to TNAU) Vyasapuram-631 210 Tamilnadu	February 27, 2023	61
26	Final year B.Sc (Ag Hons) students from College of Agriculture, Bheemaranagudi, UAS, Raichur, Karnataka	February 27, 2023	51
27	Final year B.Sc (Ag Hons) students from Kumaraguru Institute of Agriculture, (affiliated to TNAU), Erode, TN	February 27, 2023	118
28	Final year B.Sc (Ag Hons) students from Annamalai University, Chidambaram, Tamil Nadu Batch 1	February 27, 2023	98
29	Final year B.Sc (Ag Hons) students from Annamalai University, Thiruvannamalai, Tamil Nadu Batch 2	February 27, 2023	88
30	Final year B.Sc (Ag Hons) students from Nammazvar College of Agriculture and Technology, Kumuthy, Ramnad, Tamil Nadu	February 27, 2023	82
31	Final year B.Sc (Ag Hons) students from college of agriculture, UAS, Bheemaranagudi, Raichur (batch 2)	February 27, 2023	15
32	IV year B.Sc. (Hons) Agriculture students from Anbil Dharmalingam Agril. College and Research Institute, (TNAU) Navalurkuttapattu, Tiruchirappalli - 620 027, Tamil Nadu, India.	February 28, 2023	124
33	IV year B.Sc. (Horticulture) students Adhiparasakthi Horticultural College G.B.Nagar, Kalavai - 632 506 Ranipet District	March 1, 2023	117
34	B.Sc. (Hons.) Agriculture final students of PGP College of Agricultural Sciences, Palani Nagar, Namakkal - 637405	March 1, 2023	117

Sl No	Name of the College and details	Date of Visit	No. of students
35	Final year students of B.Sc. (Hons.) from Aravindhar Agricultural Institute of Technology Aravindhar Nagar, Kalasapakkam, 606 751 Thiruvannamalai District, Tamil Nadu	March 1, 2023	107
36	Final year students of B.Sc. (Hons.) from College of Agriculture and research institute, (Affiliated to TNAU) Thanjavur, TN	March 1, 2023	82
37	Final year B.Sc (Agri Hons) students from College of Agriculture, (University of agril sciences, Raichur), Kalaburagi, Karnataka	March 2, 2023	70
38	IV B.Sc (Agri) students from Adiparashakthi agricultural college (affiliated to TNAU)	March 2, 2023	116
39	IV B.Sc. (Hons.) Sericulture students from Forest College and Research Institute, TNAU, Mettupalayam,	March 2, 2023	24
40	IV B.Sc. Agri (Hons.) students from College of Agriculture and Research Institute, TNAU, Killikulam, Vallanad,	March 2, 2023	124
41	3 rd B.Tech (Biotechnology) students from TNAU, Coimbatore, TN	March 3, 2023	45
42	BSc (Agri) second year students from College of Agriculture, Dhule Maharashtra	March 3, 2023	140
43	IV B.Sc (Agri Hons) students from Palar Agricultural College, (Affiliated to TNAU) Kothamarikuppam (Village), Melpatti (Post), Vellore (Dist.) Tamil Nadu-635 805.	March 6, 2023	113
44	IV B.Sc. (Hons) Agri. Students from Faculty of Agriculture, Annamalai University, TN	March 7, 2023	100
45	IV B.Sc (Agri Hons) students from Faculty of Agriculture, Annamalai University, TN	March 9, 2023	99
46	IV year B. Sc. (Hons) Agriculture – School of Agriculture, Bharath Institute of Higher Education and Research, Tambaram, Chennai	March 10, 2023	49
47	SRM Institute of Science and Technology Vendhar Nagar, Baburayanpettai, Elapakkam Post, Maduranthagam Taluk, Chengalpattu District - 603 201.	March 14, 2023	180
48	College of Agriculture, University of Agril Sciences, Dharwad	March 20, 2023	43
49	IV B.Sc. (Hons) Agri. Students from Dept of Agri extension, Faculty of Agriculture, Annamalai University, TN	March 20, 2023	100
50	IV B.Sc. (Hons) Agri. Students from SRM College of Agricultural Sciences, SRM Institute of Sceicne and Technology, Chengalpattu, Tamilnadu	March 20, 2023	61
51	IV B.Tech (Hons) Agri.Engg Students from College of Agri. Engineering, University of Agricultural Sciences, Raichur	March 21, 2023	64
52	College of Agriculture, University of Agril Sciences, Dharwad	March23, 2023	58

Sl No	Name of the College and details	Date of Visit	No. of students
53	Students of VI sem (ABM) Mahatma Gandhi Krishi Vidyapeeth, Rahuri, Malegoan, Nashik Govt. College of Agri-Business Management, Kashti, Malegaon. Dist. Nashik (MS)	March 28, 2023	38
54	Final year B.Sc (agri engg) students from Agricultural Engineering College and Research Institute, TNAU, Coimbatore	March 31, 2023	36
55	Final year B.Tech. (Agri Engg) Agril engineering college and research institute, (TNAU) Kumalur, Trichy, TN	July 4, 2023	54
56	2 nd year B.Sc (Hons) students from College of Agriculture, Dhule (Mahatma Phule Krishi Vidyapeeth, Rahuri) Govt of Maharashtra	July 20, 2023	103
57	B.Sc. (Hons)(Community science), College of Community science, Vasathrao Naik Marathwada Krishi Vidyapeeth, Parbhani, Maharashtra	July 28, 2023	16+
58	2 nd year B.Sc. (Agri Hons) College of Agriculture, Golegaon, (VNMKV Parbhani, Maharashtra)	July 28, 2023	55
59	3 rd year students of B.Sc. (Agricultural Engineering), College of Agriculture, Vasathrao Naik Marathwada Krishi Vidyapeeth, Parbhani	August 11, 2023	113 students (Girls)
60	2 nd year B.Sc (Agri) students from School of Agriculture, Loyola Academy	August 11, 2023	100
61	Students of B.Sc. (Agricultural engineering), College of Agriculture, Vasathrao Naik Marathwada Krishi Vidyapeeth, Parbhani	August 14, 2023	89 (Boys)
62	1 st year B.Sc (Hons Agri) students from Dept of Agril and horticultural sciences, Vignan's Foundation for Science, Technology and Research, Vadlamudi, Guntur, Andhra Pradesh	August 21, 2023	63
63	2 nd year B.Sc (Agri) from College of Agriculture Nandurbar (Mahatma Phule Krishi Vidyapeeth, Rahuri) Maharashtra	August 22, 2023	47
64	Third year BVSc&AH students from Department of Veterinary Parasitology, CVAS., Mannuthy	August 28, 2023	94
65	3 rd year B.Sc (Agri) students from faculty of Agriculture, Annamalai University, Chidambaram, TN	August 28, 2023	89
66	VIth Semester (Final) Year students of Agriculture Business Management from College of Agricultural Business Management, Narayangaon (affiliated to Mahatma Phule Krishi Vidyapeeth (MPKV), Rahuri) Pune, Maharashtra, India	August 30, 2023	74
67	Final year students of B.Sc (Vet.) from college of Veterinary and animal sciences, Kerala Veterinary and animal sciences university, Wayanad, Kerala	August 30, 2023	73

Sl No	Name of the College and details	Date of Visit	No. of students
68	IV B.Sc. (Agri Hons) Faculty of Agriculture Annamalai University, Chidambaram, Tamilnadu (2 nd Batch)	September 5, 2023	120
69	IV B.Sc. (Agri Hons) Faculty of Agriculture Annamalai University, Chidambaram, Tamilnadu (3 rd Batch)	September 6, 2023	130
70	2 nd semester B.Sc (Agri. Hons) Punyashlok Ahilyadevi Holkar College of Agriculture, Halgaon, Ahmednagar, Maharashtra (affiliated to Mahatma Phule Krishi Vidyapeeth, Rahuri, Govt of Maharashtra)	September 6, 2023	64
71	Agri (UG) students of Acharya N G Ranga Agricultural University, Andhra Pradesh and Odisha University of Agriculture & Technology (OUAT) who are undergoing four-week training on Agro-based Entrepreneurship Development Programme at Ni-MSME, Hyd	September 6, 2023	68
72	IV B.Sc (Hons Agri) students from Dept. of Agriculture, Kalasalingam School of Agriculture and Horticulture, Kalasalingam Academy of Research and Education, Anand Nagar, Krishnankovil – 626126.TN	September 8, 2023	59
73	3 rd year B.Sc (Hons Agri) students from College of Agriculture, haradanahally farm, Chamarajanagara (affiliated to UAS, Bengaluru)	September 13, 2023	32
74	3 rd year B.Sc (Agri Hons) students from College of Agri., Dr. Punjabrao Deshmukh Krishi Vidyapeeth, Akola	September 25, 2023	124
75	Shriram College of Agriculture, Paniv, Solapur (affiliated to Mahatma Phule Krishi Vidyapeeth, Rahuri, Govt of Maharashtra)	September 26, 2023	110
76	Students from Odisha University of Agriculture and Technology (OUAT) who were trainees of one training prog.on School of Entrepreneurship Extension (SEE) National Institute for Micro, Small, and Medium Enterprises (ni-MSME), yousufguda, Hyd	September 27, 2023	20
77	M.Sc (Agri) students from Agricultural college, Bapatla, ANGRAU	September 27, 2023	6
78	Second-year B.Sc. (Hons.) Agriculture students of Dept. of Agricultural and Horticultural Sciences, Vignan University, Guntur, AP	October 9, 2023	84
79	3 rd Year B.Sc (Agri) students from Vasudhatai Deshmukh College of Agriculture, Bodna, Amaravati, Maharashtra (Affiliated to Dr PanjabRao Deshmukh Krishi Vidyapeeth, Akola)	October 9, 2023	54
80	3 rd year B.Sc (agri Hons), College of Agriculture, Late Ambadasarao Warpudkar, Warpud, TQ. dist. Parbhani (M.S)	October 12, 2023	31

Sl No	Name of the College and details	Date of Visit	No. of students
81	3 rd B.Sc (Hons) forestry from Forest college and research institute, Tamilnadu agricultural university, Mettupalayam, TN	October 25, 2023	54
82	3 rd year B.Sc (Ag) students from Ambedkar College of Agriculture (affiliated to VN marathwada Krishi Vidyapeeth), Gangalnagar, Beed, Maharashtra	November 1, 2023	90
83	Second B.Sc. (Hons.) Agriculture students from VIT School of Agricultural Innovations and Advanced Learning, Vellore Institute of Technology, Vellore, Tamil Nadu.	November 2, 2023	118
84	B.Tech (Engg) final students from MGM College of Agricultural Biotechnology, MGM Hills, Aurangabad	November 7, 2023	70
85	Final B.Sc. (Hons.) Horticulture students from College of Horticulture, Mudigere -577 132, Chikmagalur District, Karnataka (batch- 1)	November 20, 2023	38
86	IV final B.Sc. (Hons.) Horticulture students from College of Horticulture, Mudigere -577 132, Chikmagalur District, Karnataka (batch- 2)	November 22, 2023	38
87	U88G, PG and Ph.D Students from 2023-24 batch of IARI Meg89avarsity, Hyderabad hub, CRIDA, Hyderabad	November 23, 2023	40
88	Third-year B.Sc. (Hons.) Agriculture students from Amrita School of Agriculture, Coimbatore Campus, Amrita Vishwa Vidyapeetham, Arasampalyam, TN	December11, 2023	118
89	2 nd year B.F.Sc (Fisheries) students from College of Fisheries, OUAT, Rangilunda, Berhampur-760007 Ganjam, Odisha	December11, 2023	37
90	3 rd year B.Tech (Biotechnology) students from College of Biotechnology, Georai Tanda, Paithon Road, Aurangabad, Maharashtra (affiliated to VNMKV, Parbhani)	December 18, 2023	54
91	4 th Year B.Sc. (Horti) students from College of Horticulture & Forestry Thunag Distt, Mandi (HP) (Under Dr. Y. S. Parmar University of Horticulture and Forestry Nauni, Solan)	December 27, 2023	27

The Academy is situated within a tranquil and expansive campus spanning approximately 50 hectares. It encompasses a range of amenities, including a Faculty Center, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Children's Park, Guest Houses, Trainees Hostel, as well as Housing and Residential Quarters.

7.1 Foundation Stone Laying for New Facilities

Sri Narendra Singh Tomar, Hon'ble Union Minister of Agriculture & Farmers Welfare has laid the foundation stone for three new infrastructure facilities of the Academy Viz., New Faculty Building, Farmers Training Centre and Ladies/Girls Hostel on 17 June, 2023. The Hon'ble Minister of State for Agriculture & Farmers Welfare, Shri Kailash Choudhary; Dr. Himanshu Pathak, Secretary, DARE & Director General, ICAR and Dr. Ch. Srinivasa Rao, Director ICAR-NAARM) and staff of the Academy were present on this auspicious occasion. Dr. Himanshu Pathak, Secretary, DARE & Director General, ICAR welcomed the dignitaries and briefed them about the new infrastructure facilities that are being created at a cost of Rs. 27 crores to enable the Academy to pursue its vision and mandate of capacity building and think tank policy support. Addressing the gathering on this occasion, the Hon'ble Minister, Shri Narendra Singh Tomar appreciated the efforts and contributions of NAARM and hoped that the new facilities would

strengthen it further to render seamless support and services to Indian agriculture and farming community. Hon'ble MoS Shri Kailash Choudhary congratulated NAARM family and conveyed best wishes to all staff members of NAARM. Dr. Ch. Srinivasa Rao, Director ICAR-NAARM thanked the Hon'ble Ministers and Secretary & DG, ICAR for performing the Bhoomi Pooja and laying the foundation stones for the new infrastructure facilities.



Figure 7.1: Figure Union Minister performing bhoomi pooja

7.2 Library

The NAARM Library plays a pivotal role in delivering comprehensive information support services to the Institute's research and training endeavors. Its extensive collection comprises 31,458 books, 2 international journals, 14 national journals, 1 international online journal, and 1 online database/software, namely Indiastat.com. A significant portion of the library's publications is available in digital formats, including CD, VCD, and its proprietary digital repository, EPrints@NAARM.

Furthermore, the Academy is affiliated with CeRA and enjoys access to a plethora of journals and databases through reputable sources such as CSIRO, Springer, and Open J-Gate. The library has undergone digitization using the open-source software "Koha." Additionally, it boasts a substantial assortment of online patent search providers, GIS software, and various statistical tools.

The library services have been extended to students enrolled in programs supervised by institute scientists. These students receive guidance on utilizing reference resources, CeRA, online databases, and international journals. Photocopying services are also available on an official/payment basis, catering to the needs of Institute staff and other library patrons.



Figure 7.2: Library at ICAR-NAARM

7.3 Health Center

NAARM health centre provides comprehensive health services like preventive, curative and awareness regarding all health care services to our employees and their dependents, trainees who attend various training programs held at NAARM, PGDM-ABM students and all the pensioners along with their dependents from

all the neighbouring ICAR-institutions (approx. 2000 members) who have registered with our health centre. Health centre is efficiently taken care by Dr B.Ganesh Kumar, Officer-in charge and Dr T.Himabindu, Consultant Medical Officer (CMO). Mr Hussain and Mrs Sravanthi are providing excellent care regarding physiotherapy especially in managing pre-operative and post-operative cases. There are special efforts done by Mr Mahender (pharmacist) and Mrs Sunitha (staff nurse) Mr Ramulu (skilled supporting staff) in managing health centre and guiding patients according to their needs.

Apart from healthcare services, we also perform:

1. Education on health and life style management.
2. Regular yoga sessions done by Dr. (Maj) Debnath and few yoga professionals.
3. Diabetic camps (RBS/PPBS/FBS) done once in 15 days.
4. Cardiac camps – where cardiologists from nearby CGHS listed hospitals come and evaluates our patients conditions every month.
5. Eye camps done once in 3 months.
6. Orthopaedic camps done once in 2 months.

With above said camps and periodical yoga sessions our employees and pensioners are getting very much benefited by diagnosing their conditions in a very initial stages. Many of our pensioners, who are not able to attend health centre in any emergency conditions are also provided prescriptions and referral letters by having video consultation with our consultant medical officer 24/7. Hence, ICAR-NAARM health centre provides all the essential and

emergency medical facilities to the entire ICAR family located in Hyderabad with full devotion and dedication. Health centre conducts voluntary blood donation camps once in 3 months where many of our employees, trainees, students come forward with great enthusiasm and help people who requires blood, with a great moto of “A drop of blood can save a life “.



Figure 7.3: Blood donation camp

7.4 Sports Complex

The sports complex at NAARM features an outdoor stadium tailored for diverse activities, including cricket, football, and lawn tennis, among other sporting events. Additionally, the indoor complex houses facilities such as a badminton court, table tennis, billiards/snooker, and a well-equipped gymnasium. This facility is actively utilized by NAARM personnel and trainees. The NAARM sports contingent, leveraging the amenities provided, has achieved notable success by securing prizes in various levels of competition, including zonal, inter-zonal, and inter-institutional sports tournaments.



Figure 7.4: Sports Complex at NAARM

7.5 Agro-Technology Demonstration Blocks

The Agro-technological demonstrations block at the Academy covered with orchards, floriculture block (Jasmine & Rose), avenue plantation along with all main roads, Landscaping gardens, nursery, Herbal Garden, Agrobiodiversity park, children park etc. The total area under farm is around 30 has. The agro-technology park is developed to demonstrate various agro-technologies in different aspects of farm. These technologies related to agriculture and horticulture will act as show window developed by ICAR institutes as well as SAUs for the benefit of the students, farmers, researchers, enthusiasts and trainee participants of various training programmes.

7.6 Orchard Block

The orchard crop includes Mango (2.0 ha), Sweet lime (1.5 ha), Sapota (0.25 ha), Aonla (0.25 ha), Custard apple (1.0 ha), Phalsa (0.5 ha) and many



Figure 7.5: Clean cultural practices opted in Sweetlime cultivation

other miscellaneous fruit orchards were grown in an around 6.5 hectares. All the fruit orchards are pruned in time and fields were kept weed free as well YM application was taken up in time to harvest a good crop.

7.7 Floriculture Block

In floriculture hybrid HT roses were planted are timely pruned and are in full flowering presently. Jasmine and Golden rod (*Solidago nemoralis*) also planted and well grown now. These flowering plants / crops are cultivated in an area of 1.0 ha and available for the demonstration purpose for the benefit of the farmers and other visitors. The cut flower of roses, golden rod is used time to time for flower bouquets and flower vases arrangements for several occasions.



Figure 7.6a: Roses in full bloom at Rose Garden in floriculture Block

The floriculture block is also covered with the turf grass *St. Augustine* to stop the soil erosion as well as it also helps to absorb the atmospheric absorption of carbon into soil.



Figure 7.6b: Roses in full bloom and soil is covered with green turf grass to avoid water loss as well to stop soil erosion

7.8 Agro-biodiversity park

The agro-biodiversity park developed on 4.0 acres in the past consisting of various native tree species such as *Azadiracta indica* (Neem Tree), tamarind, *Ficus krishnae*, flame of forest, bamboo, Tree jasmine, *Delonix regia* (Gulmohor), Japanese fern tree, Kadamba, spathodia, red shower, pink shower etc. and several other ornamental plants such as acalipha, eranthimums, euphorbia, hibiscus etc. were planted. Few fruit plants such as mulberry, dragon fruit, Karonda are also planted so that birds can also be attracted in the campus. Around 2000 new ornamental/flowering bushes are planted at the agro-biodiversity park to create the aesthetic value and beautification of the area and also attract the butterflies. In this new plantation hibiscus, jatropa, acalipha, arelia, dreszenia and eranthimum plants were included.

On June 5th 2023 on the occasion of World Environmental Day a special campaign was taken

up and planted 600 ornamental plants in a single day. After that many more flowering bushes, flowering creepers, flowering plants, colorful foliage plants were planted by adding another 2500 plants to attract the butterflies, Bees, black beetle, birds and several species of insects. The



Figure 7.7a: New plantation at Agro-biodiversity park to attract butterflies, bees, birds and insects



Figure 7.7b: Theme based plantation at Biodiversity park at ICAR-NAARM



Figure 7.7c: Beautifully planted avenue trees at ICAR-NAARM Campus near Director's Office

plants such as *Euphorbia millii*, *Arelia*, *Sanchezia spesiosa*, *Brugmentia*, *Hibiscus*, *Plumeria*, *Almonda*, *Agave Americana*, *Aglonia*, *Eranthimum*, *Hamelia petans*, *Maranta* etc. were included in this area as new plantation so that this area can be converted into Butter Fly Park and to create the newness to bring the aesthetic value.



Figure 7.7d: *Tabebuia Argentea* "Yellow" flowering Avenue trees planted at ICAR-NAARM

7.9 New plantations at Nursery area to Guest Houses road

The road connecting from nursery to the guest houses is mostly used by all the trainee participants as well as VIPs visiting the campus. This area brought under the new plantations of flowering plants as well as ornamental plants. In this area we have taken up plantation of brugmentia, Ornamental flowering jatropa, nerium, techoma fFlowering tree, plumeria, arelia, acalypha, and dracaena.



Figure 7.8: New ornamental plants planted at Nursery to Guest Hose Road



7.10 Herbal Garden

Herbal garden is developed at the Academy for the benefit of visitors, students, farmers, researchers, trainee participants and for enthusiasts. Herbal garden is covered with 50 species including medicinal plants, herbal plants and aromatic plants. Self-explanatory boards are placed in each bed of the plants covering with scientific name, local name and use of plant parts and its health benefits.



Figure 7.9: Insulin plants at Herbal Garden

at several places like guest houses, biodiversity park, nursery area, lawn area and also new the new parking area near SBI. The propagated plants are hibiscus, dressenia, Arelia, brugmentia, ribbon grass, money plant creepers, flowering creepers, jatropha, golden rod, tube rose, jasmine plumeria, insulin, multivitamin, tulasi, ganesh creeper, yellow techoma, jasmine species etc. Due to plant propagation at nursery we are able to maintain our campus green and beautiful as well the expenses for purchasing plant material is avoided.



Figure 7.10a: *Sanchezia oblonga* plants propagated at Nursery

7.11 Self-sustainability by Plant propagation at Nursery

Self-sustainability is the key to create green environment in the campus by using the existing resources and a plant nursery is developed on 0.25 ha area which is very good source for us to propagate various ornamental and other flowering plants. In 2023 we have produced around 12000 plants and planted in the campus



Figure 7.10b: *Acalypha* plants propagated at Nursery

Recognitions and Publications

8

8.1 Awards and Recognitions

8.1.1 NAARM Foundation Day Awards

Award	Awardee
Annual Performance Awards (Period: 2022-23)	
Best Faculty Award	Dr BS Yashavanth, Scientist
Best Trainer Award	Dr S Senthil Vinayagam, Head ESM Division
Best Scientist Award	Dr A Dhandapani, Head ICM Division
Best Promising Young Scientist	Dr Sanjiv Kumar, Senior Scientist
Best Policy Paper	Strengthening Kisan Call Centers in India ICAR-NAARM Policy Paper Yashvanth B S, Sontakki B S, Soam S K, Venkateshwarlu G and Srinivasarao Ch
Best Popular Article	Doubling Farmers Income by 2022: Seeding the change Just Agriculture Magazine Venkatesan P, Sivaramane N & Srinivasarao Ch
Administrative Staff (Officer)	Sri Mukul Rajsingh, Senior Admin. Officer and Smt S. Shanthi, Private Secretary
Administrative Staff (Non-Officer)	Sri C Phani Raj, Assistant
Technical (T-1 to T-5 Grade)	Sri BK Venktram, Senior Tech. Assistant
Technical (T-6 to Above)	Smt G Aneeja, Chief Technical Officer
Skilled Support Staff (Male)	Sri K Daniel, SSS
Young Professionals (YPI&II)	G Trinadh YP-II MIS FMS M Vishwanath YP-II Cash & Bills (Admin.)
Project Staff	B Kalyani, Consultant, FARMER FIRST Project
Best PG Student	Ms Aishwarya GS & Mr Vinay Kumar
Institution Building	Dr P Vijender Reddy, Chief Technical Officer
Decadal Performance Awards (Period: 2012-23)	
Best Faculty	Dr SK Soam, Principal Scientist
Best Administrative Staff	Sri M Sridhar, Assistant
Best Technical Staff	Sri Sohail Ahmad Khan, Chief Tech Officer Sri Sham Bahadur, Asst. Chief Tech Officer
Best Skilled Support Staff	Sri M Ganesh Kumar, Skilled Support Staff

8.1.2 Other Recognitions Received by the Faculty of NAARM

Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member/Chairman of the following Committees:

International

- 1) Member, Executive Board of The International Dryland Development Commission (IDDC), Cairo, Egypt. (2016 – till date)
- 2) Member, Asian Long Term Experimental Network for Agriculture (ALTENA), Tsukuba, Japan (2017 – till date)

National

- 1) Member, Committee for Selection-cum-Standing Committee for Emeritus Scientist Scheme of ICAR.
- 2) Member (ICAR Representative), Board of Management, University of Agricultural Sciences, Raichur.
- 3) Member, Advisory Committee for M/s Agrinnovate India Limited (AgIn), New Delhi.
- 4) Member, ICAR - ICT Steering Committee
- 5) Member, ICAR - National Computing Facilities Management Committee (Level-3 Committee).
- 6) Guest Editor for the Open access and peer-reviewed Journal Agronomy Special Issue "Integrated Nutrient Management for Farming Sustainability".
- 7) Chairperson, Capacity Building Unit (CBU) for DARE/ICAR.
- 8) Member, Advisory Committee for M/s Agrinnovate India Limited (AgIn), New Delhi.

- 9) Member of the Council of the National Academy of Sciences, India (NASI) for the year 2023.
- 10) *Ex-officio* Member Board of Studies (BoS), IARI Hyderabad Hub constituted for UG, PG, Ph.D. programmes for the academic year 2023-24
- 11) Chairman, Committee for devising formats for output-outcome parameters, constituted by NCCT, Ministry of Cooperation, New Delhi.
- 12) Member, Steering Committee to prepare India's Adaptation Communication under UNFCCC, constituted by MoEF&CC, Govt. of India.
- 13) Chairman, Committee to recommend course curriculum for the Certificate Course on Agri-Logistics, constituted by NCCT, Ministry of Cooperation, Govt. of India.
- 14) Member, Food and Agriculture Division Council (FADC), Bureau of Indian Standards, Ministry of Consumer Affairs, Food and Public Distribution, Govt. of India.
- 15) Member, Management Committee of Extension Education Institute (Southern Region), Department of Agriculture and Cooperation & Farmers' Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.
- 16) Member, Executive Council, National Council for Cooperative Training (NCCT), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.
- 17) Member, Regional Advisory Group (RAG), NABARD

- 18) Member, Board of Management, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana (as ICAR Nominee)
- 19) Member, Board of Management, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh (as ICAR Nominee)
- 20) Member, Board of Management, University of Agricultural Sciences, Raichur, Karnataka (as ICAR Nominee).
- 21) Member, Academic Council, Professor Jayashankar Telangana State Agricultural University, (PJTSAU), Hyderabad, Telangana.
- 22) Member, Academic Council, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh.
- 23) Member, ICT Steering Committee of ICAR
- 24) Member, NAAS Sectional Committee in Natural Resource Management
- 25) Member, Section Committee on Life Sciences including Agricultural Sciences, Telangana Academy of Sciences (TAS), Hyderabad.
- 26) Member, ICAR Committee to revise and explore the possibility of filing APAR online.
- 27) Member of Chief Advisory Committee for Preparation of Integrated Horticulture Development Perspective Plan, Dept. of Horticulture and Sericulture, Govt. of Telangana.
- 28) Member, ICAR Steering committee to review and monitor the performance of National Agricultural Innovation Fund (NAIF) Scheme.
- 29) Member, Advisory Committee, World Bank funded ICAR-NAHEP Component-2 Project on Investment in ICAR Leadership for Agriculture Higher Education.
- 30) Member, ICAR Committee on Agricultural Research Service for a period of two years (2021-23).
- 31) Member, ICAR Committee to review Career Advancement Scheme (CAS) and Score Card for Scientists of ICAR.
- 32) Member, ICAR Committee on Development of Tech Dome for Central Vista Project.
- 33) Member, Performance Review Committee, NCCT, Ministry of Cooperation, Govt. of India
- 34) Member, Advisory Board, The Journal of Research - PJTSAU
- 35) Member, Advisory Board, The Journal of Research – ANGRAU, Andhra Pradesh from April 2018 onwards
- 36) Member, PGDM Committee for Post Graduate Diploma in Management - Agribusiness & Management of Vakunth Mehta National Institute of Cooperative Management (VAMNICOM), (Ministry of Agriculture & Farmers Welfare, Govt. of India), Pune, Maharashtra.
- 37) Member, ICAR Steering Committee for implementation of ICAR Research Data Management Guidelines.
- 38) Chairman, of Expert Committee on Agrobiodiversity, National Biodiversity Authority, (MoEF&CC, Govt. of India), Chennai, Tamil Nadu for a period of one year from October 2021.
- 39) Member, Executive Council of NAAS, for a period of one year from 1-1-2022.

40) Editor, Journal of Climate Change & Environmental Sustainability

41) Guest Editor, Journal Agronomy. Guest Editor for the Open access and peer-reviewed Journal Agronomy Special Issue "Integrated Nutrient Management for Farming Sustainability".

Dr. B. Ganesh Kumar

- Member, Research Advisory Committee (RAC) for ICAR-Central Institute of Fisheries Technology (ICAR-CIFT), Cochin since January, 2023.
- Member, Research Advisory Committee (RAC) for ICAR-National Institute on Foot and Mouth Disease (ICAR-NIFMD), Bhubaneswar since July, 2023.
- Member, Training Quality Improvement Measures Committee (TQIMC) of NIRD & PR, Hyderabad from February, 2016 till date.
- Member, Committee for Costing and Revision of Cost of Services being offered by ICAR-NIFMD, Bhubaneswar from November, 2023 till date.
- Panelist in the session on 'Knowledge Economy – Role of Academics in nurturing innovative entrepreneurs' in the National Conference on 'Building an Entrepreneurial Ecosystem in Universities- Opportunities and Challenges' organized by ASPIRE and DST-TEC at University of Hyderabad (UoH) on 3 February, 2023.
- Coordinated the Bilateral Meeting between Secretary-DARE & DG-ICAR and Vice Minister of Agriculture and Livestock of Brazil on the sidelines of G20 Agriculture Ministers' Meeting on 16 June, 2023.

- Chairman of the Technical Session on 'Entrepreneurship and Business Opportunities' in the International Seminar on 'Sensitizing the Millets Farming, Consumption and Nutritional Security - Challenges and Opportunities' held at Pandit Jawaharlal Nehru College of Agriculture & Research Institute (PAJANCOA & RI), Karaikal, Union Territory of Puducherry sponsored by Indian Council of Social Science Research (ICSSR), New Delhi during 9-10 November, 2023.

- Panelist in the session on 'Fisheries, Dairy and Animal Husbandry' in the Andhra Pradesh Gunvatta Sankalp organized by Quality Council of India under Department of Industrial Policy and Promotion, Ministry of Commerce and Industry, GoI at Hotel Novotel, Vijayawada on 19 December, 2023.

Dr M Ramesh Naik, Scientist, RSM Division nominated as a Member for Hyderabad Chapter, Indian Society of Agronomy (ISA-HYD).

Dr N Sivaramane got best popular article award on "Doubling Farmers Income by 2022: Seeding the change" *Just Agriculture Magazine*.

Dr Ranjit Kumar

- Member of Quinquennial Review Team (QRT) of ICAR- Indian institute of Oil Palm Research, Pedavegi.
- Nominated as the Eminent Scientist-Member in the Board of Studies for School of Agribusiness and Rural Management at RP Central Agricultural University, Pusa.

- Member, Quinquennial Review Team (QRT) of ICAR- Directorate of Floricultural Research (ICAR-DFR), Pune.
- Member, Institute Management Committee (IMC) of ICAR- Central Institute for Women in Agriculture (ICAR-CIWA), Bhubaneswar.
- Member, Editorial Board, Indian Journal of Agricultural Economics, Mumbai.
- External Examiner for evaluation of PhD thesis in the subject Home Science of University of Delhi, Delhi.
- Expert by Board of Studies (BOS) in Extension & Communication under the Faculty of Family & Community Sciences by the Syndicate, The Maharaja Sayajirao University of Baroda, Vadodara with effect from Sept. 1, 2023 for 3 years.
- Guest of Honour for valedictory function of the Training on Gender Equality and Challenges held during July 4-6, 2023 at VNMKV, Parbhani.

Dr P Supriya received Young Scientist Award from Society for Plant Research during National Conference on “Future of Agriculture and Agriculture for Future: Indian Perspective organised from 4-6 September 2023.”

Dr Surya Rathore

- Expert for Career Advancement Scheme (CAS) for the post of Professor (Home Science Extension) at Bihar Agricultural University, Sabour (Bihar)

8.1.3 ICAR-NAARM in Media



Investors and incubates meet at AgriUdaan

AgriUdaan, the central government's initiative for promoting agribusiness, organized a meeting between investors and incubates. The meeting was held at the AgriUdaan office in New Delhi. The meeting was attended by several investors and incubates. The meeting was a success and it was decided to launch the AgriUdaan initiative in the near future.

వ్యవసాయ ఉత్పాదక పరిశోధన పనిపై దృష్టి పెట్టాలి: డాక్టర్ బి.ఎన్.మాధుర్

వ్యవసాయ ఉత్పాదక పరిశోధన పనిపై దృష్టి పెట్టాలి. డాక్టర్ బి.ఎన్.మాధుర్. వ్యవసాయ ఉత్పాదక పరిశోధన పనిపై దృష్టి పెట్టాలి. డాక్టర్ బి.ఎన్.మాధుర్. వ్యవసాయ ఉత్పాదక పరిశోధన పనిపై దృష్టి పెట్టాలి. డాక్టర్ బి.ఎన్.మాధుర్.

నార్మ కృషి అభినందనీయం

నార్మ కృషి అభినందనీయం. నార్మ కృషి అభినందనీయం. నార్మ కృషి అభినందనీయం. నార్మ కృషి అభినందనీయం. నార్మ కృషి అభినందనీయం.

రాజేంద్రనగర్ సర్కిల్ లో ఘనంగా జాతీయ మహాత్మా గాంధీ మాజీ ప్రధాని లాల్ బహదూర్ శాస్త్రి జయంతి వేడుకలు

రాజేంద్రనగర్ సర్కిల్ లో ఘనంగా జాతీయ మహాత్మా గాంధీ మాజీ ప్రధాని లాల్ బహదూర్ శాస్త్రి జయంతి వేడుకలు. రాజేంద్రనగర్ సర్కిల్ లో ఘనంగా జాతీయ మహాత్మా గాంధీ మాజీ ప్రధాని లాల్ బహదూర్ శాస్త్రి జయంతి వేడుకలు.

కేంద్రీయ కృషి మంత్రి నే భూమి పూజన కియా

కేంద్రీయ కృషి మంత్రి నే భూమి పూజన కియా. కేంద్రీయ కృషి మంత్రి నే భూమి పూజన కియా. కేంద్రీయ కృషి మంత్రి నే భూమి పూజన కియా. కేంద్రీయ కృషి మంత్రి నే భూమి పూజన కియా.

నే హిందీ మిలాప కి ౧౯౮౭

నే హిందీ మిలాప కి ౧౯౮౭. నే హిందీ మిలాప కి ౧౯౮౭. నే హిందీ మిలాప కి ౧౯౮౭. నే హిందీ మిలాప కి ౧౯౮౭.

నార్మ మే గ్లోబల్ బావో ఇండియా రోడ్ షో

నార్మ మే గ్లోబల్ బావో ఇండియా రోడ్ షో. నార్మ మే గ్లోబల్ బావో ఇండియా రోడ్ షో. నార్మ మే గ్లోబల్ బావో ఇండియా రోడ్ షో. నార్మ మే గ్లోబల్ బావో ఇండియా రోడ్ షో.

Graduation Ceremony of NAARM's a-IDEA Agri-Start-ups

Graduation Ceremony of NAARM's a-IDEA Agri-Start-ups. Graduation Ceremony of NAARM's a-IDEA Agri-Start-ups. Graduation Ceremony of NAARM's a-IDEA Agri-Start-ups. Graduation Ceremony of NAARM's a-IDEA Agri-Start-ups.

సార్వజనిక జాతీయ సైన్స్ డే వేడుకలు

సార్వజనిక జాతీయ సైన్స్ డే వేడుకలు. సార్వజనిక జాతీయ సైన్స్ డే వేడుకలు. సార్వజనిక జాతీయ సైన్స్ డే వేడుకలు. సార్వజనిక జాతీయ సైన్స్ డే వేడుకలు.

సారంతి మేగా రక్తదాన శిబిరం

సారంతి మేగా రక్తదాన శిబిరం. సారంతి మేగా రక్తదాన శిబిరం. సారంతి మేగా రక్తదాన శిబిరం. సారంతి మేగా రక్తదాన శిబిరం.

8.1.4 Invited Lectures Delivered by the Faculty and Staff

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr Ch Srinivasa Rao	Nutrient Management Strategies for Climate Smart Agriculture	3-day Residential Programme on Fertilizers and Food Security: Challenges and Way Forward	The Fertiliser Association of India (18-20 January, 2023)	Hotel Aditya Park, Ameerpet, Hyderabad	January 01, 2023
	Food Security and Sustainable Agriculture for Achieving SDGs in the	Winter School Multi-disciplinary Refresher Course on Environmental Challenges and Sustainable Development for College and University Teachers	UGC-Human Resource Development Centre, BHU, Varanasi (February 6-20, 2023).	Virtual mode	February 02, 2023
	Climate Change Mitigation: Role of Science, Technology, Economic and Policy	5-day Training Programme on Climate Change: Challenges and Response for Women Scientists & Technologists working in the Government Sector (March 13-17, 2023)	National Academy of Administration, Mussoorie	Virtual mode	March 17, 2023
	Policy and Programs for Climate Resilient Agriculture	G20 Technical Workshop on Climate Resilient Agriculture during its G20 Presidency in 2023, (September 4 – 6, 2023)	ICAR & CRIDA	Hotel Novotel, Hyderabad Airport	September 05, 2023
	Technology, Policy, Programs and Institutions for Resilient Dryland Systems.	Webinar on: PRE - COP28 entitled: Impact of Climate Change on the Dry Areas	International Dryland Development Commission (IDDC); Regional Action on Climate Change (RACC), Japan; Nizami Ganjavi International Center (NGIC) the African League of Young Masters (ALYM), and hosted by Alexandria University	Virtual mode	September 11, 2023.

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr Ch Srinivasa Rao	Indias Action Plan on Climate Change, Key Policies and Programmes for Mitigating the Impact of Climate Change	Five Days Training Progrmame on Climate Risk Management: Policy and Governance for Scientists and technologists working in the Government sector (11 th to 15 th September, 2023)	Lal Bahadur Shastri National Academy of Administration (LBSNAA)	LBSNAA Mussorie	September 15, 2023
	Potential of Carbon Sequestration in Agriculture	Brainstorming Session on Green Carbon Credits	NAAS, New Delhi	Virtual mode	September 24, 2023
	Carbon Farming Opportunities and Action Plan towards Achieving SDGs in India	XVI Agricultural Science Congress	NAAS jointly with ICAR - CMFRI	ICAR-CMFRI, Kochi, Kerala	October 10-13, 2023
	Positive Thinking	Annual Group Meeting on Agroforestry	PJTSAU, Hyderabad	PJTSAU, Hyderabad	October 16, 2023
	Climate Change and SDGs	11th Refresher Course in Environmental Studies [Multi-disciplinary]	UGC-Human Resource Development Centre, BHU, Varanasi	Virtual mode	November 02, 2023
	Nature based soil management	10 days short course on Organic Farming/ Natural Farming – Principles and Practices	ANGRAU Regional Agricultural Research Station, Anakapalle	Virtual mode	November 17, 2023
	Why soil management critical for climate change mitigation?	Dr B. Ramamoorthy 5th Memorial Lecture	PJTSAU, Hyderabad	PJTSAU, Hyderabad	November 21, 2023
	Opportunities and challenges in agricultural and horticultural sectors	NAHEP sponsored National Training on Entrepreneurial Skill Development for B.Sc (Agri/Hort) students of RVSKVV, Gwalior	NAARM	NAARM	December 06, 2023
	Agricultural scenario in the context of Education	Off-Campus training on Enhancement of Teaching Competence for the Faculty of UAS, Raichur	NAARM	Virtual mode	December 06, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr Tavva Srinivas	Crop & Livestock Insurance and Crop Damage Assessment Methods and For Efficient Mitigation of Human-wildlife Conflict (HWC)	Holistic Approach to Human-Wildlife Conflict Mitigation and One Health		Online	May 15-16, 2023
Dr Ranjit Kumar	Understanding Market Opportunities for New Business Ideas		AgHub, PJTSAU, Hyderabad	Online	April 26, 2023
	Business Plan for long term growth of FPOs		ICAR- Directorate of Onion and Garlic Research, Pune	Online	May 04, 2023
	Natural Farming for Sustainable Agriculture: Farm level evidences		Bihar Agriculture Management & Extension Training Institute (BAMETI), Patna	Online	June 26, 2023
	From Farm to Fork: Role of FPOs and Agri-Startups	XVI Agricultural Science Congress & Expo		Kochi, Kerala	October 10-13, 2023
	Competitive Advantage through Brand Differentiation		SKUAST-Kashmir, Srinagar	Online	December 08, 2023
	Business planning and marketing strategies for young entrepreneur in dairy sector		ICAR-NDRI Southern Regional Station, Bengaluru	Online	December 14, 2023
	Repositioning Indian Agri-business to meet Contemporary Challenges		Administrative Staff College of India (ASCI), Hyderabad	Online	December 23, 2023
	Adoption of Natural Farming and farmers livelihood	International Agriculture Conference on Natural & Organic Farming: In context to Bhartiya Agriculture	Gujarat Natural Farming Science University	Anand (Online)	December 24, 2023
Dr Sanjiv Kumar	National Agriculture Market	Training Programme on Innovative Strategies to Facilitate Integration of Farmers with Market	MANAGE, Hyderabad	Hyderabad	September 23, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr Sanjiv Kumar	Business Plan for Starting Dairy Business	Training Program on Dairy Business Management	Anand Agricultural University, Anand	Anand	October 05, 2023
	Entrepreneurial opportunities in Agricultural Sector	Training on Role of Agribusiness in Self Reliance of India	CCSHAU, Hisar	Hisar	November 08, 2023
	Digital Marketing: Emerging areas in Agribusiness marketing	Training on Role of Agribusiness in Self Reliance of India	CCSHAU, Hisar	Hisar	November 08, 2023
	Concept of Supply Chain Management – traditional vs modern	Training program on Supply Chain Management in Agriculture	MANAGE, Hyderabad	Hyderabad	November 29, 2023
	Elements in agriculture supply chain management	Training program on Supply Chain Management in Agriculture	MANAGE, Hyderabad	Hyderabad	November 29, 2023
Dr S Senthil Vinayagam	Nurturing the Startups – Incubation Centers Experiences	Startup Conclave_India International Science Festival (IISF) organized by BIRAC, DBT	Department of Biotechnology	Bhopal	January 23, 2023
	a-Idea for Incubation and Business Development	Entrepreneurship Orientation Program on Medicinal and Aromatic Plants-VI (EOPMAP-VI)	ICAR-DMAPR with its Medi-Hub Technology Business Incubator (TBI)	Online	February 12, 2023
	Concept of Incubation Centres	TOT on Accounting & Financial Management under Non- Farm Livelihood Enterprises	NIRDPR	Hyderabad	February 23, 2023
	Communication Skills for Effective Teaching	Training Program on Digital Teaching Techniques	Kamdhenu University	Amreli, Gujarat	March 14, 2023
	Role of FPOs in Promotion of Startups	Millets Conference 2023: Future Super Food for The World	The Associated Chambers of Commerce and Industry of India (ASSOCHAM)	Novotel, HICC, Hitech City, Hyderabad	March 24, 2023
	Innovations for Sustainable Food	Startup20X	Atal Incubation Centre-Sri Krishnadevaraya University	Ananthapur	April 13, 2023
	Entrepreneurial Potential for GRIs and Agripreneurs	Orientation Workshop for Participating Institutes	National Institution of Technology	Warangal	May 04, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr S Senthil Vinayagam	Scope of Women Agripreneurs	training program on DST sponsored Women Entrepreneurship Development Programme	NGI TBI- Supported by NSTEDB and DST, Govt. of India	T M Palayam. Coimbatore	June 22, 2023
	Entrepreneurship and Innovation	National Seminar on Agripreneurship: A Key Driver in Agriculture 4.0	VIT	Vellore	July 29, 2023
	Business Development for the Entrepreneurship	Orientation Program on Medicinal and Aromatic Plants-VII (EOPMAP-VII)	ICAR-DMAPR with its Medi-Hub Technology Business Incubator (TBI)	Online	July 06, 2023
	Foundation Day Lecture on Agri-entrepreneurship opportunities: Idea to startup route for rural empowerment	Web series Theme Talk on Fasal se Safal	FloraFauna Science Foundation (FFSF)	Lucknow, UP	July 10, 2023
	Start-up culture among rural youth	Training program on Extension core competencies and skills for effective transfer for technology in agri and allied sector	Extension Education Institute	Hyderabad	August 30, 2023
	Extension strategies for enhancing market access	ITEC Program on Innovative Strategies to Facilitate Integration of Farmers with Market	MANAGE	Hyderabad	September 11, 2023
	Entrepreneurial skills for rural youth	Training on Sensitization Program for Extension Functionaries on Entrepreneurship Development through Geographical Indication in Agriculture	MANAGE, Hyd and KAU, Thrissur	Online	October 05, 2023
	Generation incubation and commercialization of ideas and innovations	Training Programme on Agripreneurship for Atma Nirbhar Bharat	Anand Agricultural University	Anand, Gujarat	October 07, 2023
	Agri -Startup Validation and Certification	National Online Collaborative Training on Agri Startups	MANAGE, Hyderabad and UHS, Bagalkot	Online	October 19, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr S Senthil Vinayagam	Student - Centered Learning and Digital Teaching Competencies	Training on Enhancement of Teaching Competence	UAS	Raichur	December 06, 2023
	Becoming a successful entrepreneur	Training Programme on Clean Meat Production and Value added Products Preparation	ICAR -NRC MEAT	Hyderabad	December 08, 2023
Dr D Thammi Raju	Curriculum Development	Enhancing Pedagogical Competencies for Agricultural Education	NAAS, New Delhi	New Delhi	August 4, 2023
	Teaching Management	Faculty Development Programme for GBPUAT	GBPUAT, Pantnagar	Online	August 10, 2023
Dr GRK Murthy	Teaching with Technology	Faculty Development Programme	MANUU	Hyderabad	November 18, 2023
	Integration of Technology in Teaching	11 th Education Council	OUAT	Bhubaneswar	December 14, 2023
Dr B Ganesh Kumar	Intensification of Livestock System in Middle and Low-Income Countries	International Training Programme on Natural Resource Management and Climate Change Adaptation	NIRD & PR, Hyderabad	Hyderabad	January 23, 2023
	Livestock as Alternative Livelihoods for Enhancing Households Income in the Watershed Areas	Workshop on Enhancing Livelihoods and Diversification under Watersheds under WDC-PMKSY 2.0	NIRD & PR, Hyderabad	Hyderabad	February 07, 2023
	Livestock based Livelihoods and Value-added Cow based products	ToT Training on Promotion of MGNREGA Convergence with Animal Husbandry Department	NIRD & PR, Hyderabad	Hyderabad	February 22, 2023
	Livestock Management and Adaptation to Climate Change	Training program on Climate Change Adaptation and Mitigation Strategies under WDC-PMKSY 2.0	NIRD & PR, Hyderabad	Hyderabad	March 16, 2023
	Composite pisciculture for the livelihood security in rural areas	Training program on Convergence of Pradhan mantra Matsya Sampad Yojana (PMMSY) with MGNREGS	NIRD & PR, Hyderabad	Hyderabad	March 21, 2023
	Integrating Livestock Farmers with Market	Training Programme on Promotion of FPO in Livestock Sector: Opportunities and Challenges	MANAGE, Hyderabad	Hyderabad	March 28, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr B Ganesh Kumar	Livestock based Food Production and its Role in Nutritional Security	Training Programme on Sustainable Food Systems	EEI, Hyderabad	Hyderabad	April 28, 2023
	Opportunities and Challenges in Export of Livestock Products	Training Programme on Export Management in Agri and Allied Products	EEI, Hyderabad	Hyderabad	June 06, 2023
	Marketing Strategies for Livestock Products	Training Programme on Branding, Marketing and Extension Methods for Promotion of Agri and Allied Products	EEI, Hyderabad	Hyderabad	July 05, 2023
	Entrepreneurship in Livestock Sector: Learning Opportunities	Faculty Development Programme for mid-career faculty members of TANUVAS	TANUVAS, Chennai under NAHEP-IDP	Chennai	August 21, 2023
	Farmer and Market Linkage in Fishery Sector	Training programme on Development of Managerial Skill, Leadership Development, Extension and Accounting for the Fisheries officers, Government of Odisha	NIRD & PR, Hyderabad	Hyderabad	September 22, 2023 October 13, 2023 October 31, 2023 December 19, 2023
	Linking Farmers to Market: Issues and Challenges	Training Programme on Agripreneurship Development for the Senior Officials of Dept. of Agriculture, Government of Odisha	MANAGE, Hyderabad	Hyderabad	September 29, 2023
	(i) Economics of dairy farming in different agro-climatic situation (ii) Marketing strategies for value added milk and milk products	International Training Programme under Indian Technical and Economic Cooperation (ITEC) on Recent Advances in dairy farming, Processing and Marketing Strategies	MANAGE, Hyderabad	Hyderabad	October 12, 2023
Dr K Srinivas	Innovation ecosystem in agriculture	Integrating Sciences & Institutions for Resilient Production System	SKUAST-K, Shalimar Campus	Shalimar Srinagar, (J&K)	March 17, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr K Srinivas	Repurposing Agriculture for Achieving SDGs: Issues and Strategies	XVI Agricultural Science Congress	National Academy of Agricultural Sciences (NAAS), ICAR-CMFRI,	Kochi, Kerala.	October 10-13, 2023
	Intellectual Property Rights, innovation & Agriculture	Agricultural Innovation & Entrepreneurship conclave	Navasari Agricultural University, Navsari, Gujarat		December 04, 2023
	Intellectual Property Rights, innovation & Entrepreneurship in agriculture and allied sector.	National workshop on Sensitizing the Faculties to Inspire the Youths in Bio-entrepreneurship	University of Agricultural Sciences, Dharwad		December 19, 2023
Dr R Venkattakumar	Improved Production Technologies on Chilli		TNAU, Coimbatore	Hyderabad	March 30, 2023
	Innovative Extension Models	Faculty Development Programme	TNAU, Coimbatore & MANAGE, Hyderabad	Hyderabad	June 14, 2023
	Communication through Mainstream Media	Leadership Skills and Management Techniques	TNAU, Coimbatore & MANAGE, Hyderabad	Hyderabad	August 24, 2023
	Content Development for Mainstream Media for Mobile Journalism	Collaborative Training Programme on Mobile Journalism	ICAR-IIHR, Bengaluru and MANAGE, Hyderabad	Hyderabad	October 28, 2023
	Team Building and Organizational Structure	Online lecture Series of Kalpa Agri-Business Incubator	ICAR-CPCRI, Kasaragod	Hyderabad	December 1, 2023
	Role of FPOs in Tapping the Value Addition Opportunities in Horticulture	Refresher Training Programme on Extension of Horticultural Technologies	ICAR-IIHR, Bengaluru and MANAGE, Hyderabad	Hyderabad	December 19, 2023
Dr Surya Rathore	Hands on Training on Experiential Learning Methods & Management games	Experiential Learning Methods & Management games from Jan 3-7, 2023.	Extension Education Institute (Southern region)	Hyderabad (Telangana)	January 05, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr Surya Rathore	Social Safeguards and GRM	NAHEP programme on Social Safeguards	Acharya NG Ranga Agricultural University, Guntur	Online	March 03, 2023
	Strengthening Extension delivery mechanism through KVK and ATMA	Workshop on Strengthening Extension delivery mechanism at district level through KVK and ATMA held during April 24-25, 2023	MANAGE, Hyderabad	Hyderabad (Telangana)	April 24, 2023
	Women Led development in Science	Webinar on Women Led Development in Science	ICAR- Central Soil Salinity Research Institute, Karnal	Online	May 02, 2023
	Introduction to Blogs and Writing Educational Blogs	Use of ICT in teaching for faculty of Dairy Science College	Kamdhenu University, Gandhinagar (Gujarat)	Gandhinagar (Gujarat)	July 4, 2023
	Flipped Classrooms	Use of ICT in teaching for faculty of Dairy Science College	Kamdhenu University, Gandhinagar (Gujarat)	Gandhinagar (Gujarat)	July 5, 2023
	Gender Mainstreaming in Agriculture	Training on Gender Equality and Challenges during July 4-6, 2023.	Vasanthrao Naik Marathwada Kri-shi Vishvavidhyalaya, Parbhani	Parbhani (Maharashtra)	July 6, 2023
	Vocational Scope of Agricultural & Community Sciences	Agri-Education Fair under IDP, NAHEP	Maharana Pratap University of Agriculture & Technology, Udaipur	Udaipur (Rajasthan)	September 16, 2023
Dr P Venkatesan	Qualitative Research: An Introduction to Thematic Analysis	Online Training on Advances in Agricultural Extension Research	MANAGE, Hyderabad and KAU	Regional Agricultural Research Station, Ambalavayal, KAU	May 04, 2023
	Farmer-Led Extension Approaches	MANAGE and TNAU Collaborative online Training on Faculty Development Programme for Extension	TNAU	Coimbatore	June 13, 2023
	Effective Communication Skill	On-line Training Programme on Leadership Skills and Management Techniques	TNAU	Coimbatore	August 21, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr NA Vijay Avinashilingam	Invited Speech on Entrepreneurship, Startups and the process of Incubation	Two days Entrepreneurship Development Program	DUVASU, Mathura	Mathura	February 24 – 25, 2023
	Invited Speech on Preparation of Business Plan	Two days Entrepreneurship Development Program	DUVASU, Mathura	Mathura	February 24 – 25, 2023
	Keynote speech on Science for Human well being	National Science Day- 2023	Department of S & H Sengunthar Engineering College (Autonomous)	Online	February 28, 2023
	Chairman and Session Moderator for Credit Linkages & Funding Support in Agriculture	Geo-Smart India Conference 2023	Geo Spatial World, Hyderabad International Convention Centre (HICC), Hyderabad, India	Physical	October 17-19, 2023
Dr Ramasubramanian V	Two lectures viz., (i) Classification and Regression Trees (CART) with Hands-on session (ii) Self-Organizing Maps (SOM) with Hands-on session	Online Training programme on Analysis of Agricultural Data using Statistical and Data Mining Techniques organized during July 11-20, 2023	ICAR-IASRI	New Delhi	July 18, 2023
Dr D Damodar Reddy	Rock-phosphate as a Source of Phosphorus in Organic Farming	ICAR Short Course on Recent Developments in Organic Production of Oilseeds for Ushering Productivity, Soil Health and Exports organized during 21.02.2023 - 02.03.2023	ICAR-IIOR	Hyderabad	March 01, 2023
Dr A Dhandapani	Data Analysis in IPM	Winter School: Integrated Pest Management: A Way Forward for Food Safety, Security & Sustainability	ICAR-NCIPM, New Delhi	New Delhi	January 16, 2023
	Data Analysis related to plant protection	ICT in Plant Protection (Online)	ICAR-NCIPM, New Delhi	New Delhi	November 09, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr SK Soam	Awareness about Career Development Center (CDC) by NAARM under National Agricultural Higher Education Project (NAHEP)	Faculty Development Program (FDP) centered on proposal/ grant writing has been planned during 12th June to 17th June 2023.	G H Raisonni University (GHRU)- Saikheda	Saikheda	June 15, 2023
	Funding opportunities at National and International levels	Faculty Development Program (FDP) centered on proposal/ grant writing has been planned during 12th June to 17th June 2023.	G H Raisonni University (GHRU)- Saikheda	Saikheda	June 15, 2023
	Science Communication	21 Days Refresher Course on Agroforestry during 17 July to 06 August 2023. Organized by Indian Society of Agroforestry and ICAR-Central Agroforestry Research Institute	ICAR-Central Agroforestry Research Institute	Jhansi	August 02, 2023
	Green economy, Food-Energy-Water Nexus (FEW), Climate Change Education CCE	International Innovation and Creativity Conclave (I2C2) Version - 2.0	World Forum for Education in association with Bharatiya Engineering Science and Technology Innovation University (BESTIU)	Vishakhapatnam	May 28-30, 2023
Dr M Balakrishnan	Bioinformatics tools for genome editing	In the ICAR winter school on Development, Evaluation and Biosafety Crops: Hands-on Training from 20 January to 9 February 2023	ICAR-IIRR	Hyderabad	January 21, 2023
	Keynote address on Bioinformatics Applications in Agriculture and Other sectors: Challenges and Opportunities	In the TECHEON-2023 as part of SUDHEE-2023	Department of MCA, CBIT (A)	Hyderabad	January 31, 2023

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr M Bal-akrishnan	Awareness on cyber-crime and security measures	Celebrating Vigilance Awareness Week as well as the PIDPI Awareness Campaign under from October 30th to November 15th, 2023	ICAR-CIARI	Port Blair, Andaman & Nicobar	December 02, 2023
	AI tools for analyzing misuse of pesticides on in the.	Faculty Development Programme on Artificial intelligence for agriculture innovation from 11-16th December 2023	BVRIT College of Engineering for Women,	Hyderabad	December 16, 2023
Dr PD Sreekanth	Use of QGIS for climate change experiments	Winter school on Implications and Impact Assessment of Climate Change on Pests Natural Enemies and Pollinators in Horticulture Ecosystem	ICAR-IIHR, Bengaluru	Bengaluru	February 7-27, 2023.
	Use of GIS in Dairy Science	Training programme on Use of Information and Communication Technology in Teaching for Faculty of Dairy Science College	Kamadhenu University, Amreli		July 6-7, 2023
Dr N Srinivasa Rao	Open source online tools for education	Training on digital teaching techniques	Kamdhenu University	Amreli	March 15, 2023
	Open source web tools for digital content management and its hands on training exercises	Development of mobile apps and its hands on training exercises	UBKV	Cooch Behar	March 30, 2023
	Career Development Centres for Students Excellence	Workshop on Fostering Academic Leadership in Agriculture	GBPUAT	Pantnagar	September 4-8, 2023
	CDCs and its monitoring system	Workshop on Sustainability of Career Development Centers	CSKHPKV	Palampur	October 03, 2023
	CDCs and its monitoring system	Workshop on Sustainability of Career Development Centers	AAU	Anand	October 09, 2023
	CDCs and its monitoring system	Workshop on Sustainability of Career Development Centers	KVASU	Wayanad	November 06, 2023
Dr BS Yashavanth	Data Visualization through ggplot2; Spatial Data Visualization in R	Multivariate Statistical Machine Learning Methods for Modelling Agricultural Data	ICAR-IIRR	Hyderabad	July 25, 23

Name of the Staff	Lecture	Programme	Organized by	Place	Date
Dr BS Yashavanth	Data Visualization using R	Data Analysis with R	Centre for e-Learning, Kerala Agriculture University	Online	August 10, 23
	Statistical backing in biodiversity data collection and analysis	Methods for documentation of biodiversity	ICFRE-Institute of Forest Biodiversity	Hyderabad	August 06, 23
Dr P Supriya	MicroRNA analysis	Tools and Techniques of Bioinformatics for Agriculture Research	Agricultural Knowledge Management Unit (AKMU) ICAR - Indian Agricultural Research Institute	Online	September 13, 2023
G Aneja, CTO	Communication skills and Gender Sensitization, and Soft skills, Gender balance	Different programmes organized for the various levels Staff members	MSTU, Income Tax dept, Hyderabad	Hyderabad	September 26, 2023 to December 21, 2023
Dr Laxman M Ahire	Biodiversity Conservation for Green Environment	International Conference on Decarbonizing Agriculture	Organized by Krishi Tantra at TMA Pai International Convention Center, Mangalore, Karnataka	International Conference on Decarbonizing Agriculture	November 25-27, 2023

8.1.5 Other Roles

Faculty	Role
A Dhandapani	- Member in Institute Management Committee (IMC) of ICAR-National Centre for Integrated Pest Management, New Delhi - Member in Institute Management Committee (IMC) of ICAR-Indian Agricultural Statistics Research Institute, New Delhi - Expert in State Level Technical Committee for reviewing Research Proposals at ANGRAU, Guntur - Expert in State Level Technical Committee for reviewing Research Proposals at PJTSAU, Hyderabad
M.Balakrishnan	- External Examiner for Ph.D. Thesis Evaluation at Anna University, TN; - NIT, Trichy, TN; IARI, New Delhi - Academic Council Member for the Academic Activities at Mall Reddy University, Hyderabad, Telangana - Member in Institute Management Committee (IMC) of Directorate of Poultry Research, Rajendranagar, Hyderabad-500030 - Technical Expert at Center for DNA Fingerprinting and Diagnostics, Inner Ring Road, Uppal, Hyderabad 500039.
M. Ramesh Naik	Nominated as a Member for Hyderabad Chapter, Indian Society of Agronomy (ISA-HYD).
N. Srinivasa Rao	- Member in KRISHI megh operation & management committee of IASRI, New Delhi by DDG (Edu), ICAR - Member in Analysis of requirement, scope, specifications, preparation & evaluation of Techno refreshment of ICAR Data Center of IASRI committee. - Member, bidding document committee on AMC and Techno Refreshment of ICAR Data Centre by IASRI, New Delhi
S K Soam	- Member of Organising Committee in International Conference on Blended Learning Ecosystem for Higher Education in Agriculture organized by ICAR - expert reviewer for the business proposals submitted to 20th call of Biotechnology Ignition Grant (BIG) under Biotechnology Industry Research Assistance Council (BIRAC) scheme organized by Department of Biotechnology, Govt of India. - Member of ITMC at ICAR-Indian Institute of Millet Research, Hyderabad. - Guest of Honour in the inaugural session of one-day awareness workshop on Intellectual Property protection for Biotechnological Research organized by ICAR- National Institute of Biotic Stress Management, Raipur on 15-03-2023.

V. Ramasubramanian	• As an expert, evaluated the research project proposal entitled "Evaluation of suitable model for Muga and Eri based Integrated Farming System" of Central Silk Board, Central Muga Eri Research & Training Institute, Central Silk Board, Jorhat and sent comments on 04.10.2023 • As expert member, review of research proposals of M.Sc. (Agricultural Statistics) students of PJTSAU, Hyderabad was done in online mode on 30.10.2023 • As Resource person for Professional Attachment Training (PAT) of Mr. Rakesh, N., Scientist, MGIFRI, Motihari, East Champaran on the topic "A Principal Component Analysis Outlook on Contribution of India's GDP Constituents" during 24.08.2023 to 23.11.2023
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8.2 Publications

8.2.1 Research Papers in Referred Journals

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8.2.2 Books & Monographs

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8.2.3 Book Chapters

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8.2.4 Popular and Newspaper Articles

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Sanjiv Kumar (2023). Role of Agri-Startups in Agricultural Value Chain in India. Paper presented in National Seminar on Sustainable Emerging Approach in Agri-Business Development at Nagaland University, Nagaland.

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8.2.5 Training Manuals /Technical bulletins

8.2.5.1 Training Manuals

Chandra Sekhara Rao C, Viswanatha Reddy K, Hema B, Damodar Reddy D, Poorna Bindu J & Sheshu Madhav M (2023). Central Tobacco Research Institute @75. ICAR-Central Tobacco Research Institute, Rajahmundry. Tech. Bull 1/2023, pp 1-18.

Dhandapani, A. (2023). Analysis of Experimental Data Using R. Training Manual, ICAR-NAARM.

Krishna Murthy A, Dhanalakshmi G, Balakrishnan M & Srinivasa Rao Ch (2023); Profitable backyard poultry in Telugu (Booklet), p30.

Ramesh Naik M, Srinivasa Rao Ch, Kiran S, Sunitha B, Bhagyalakshmi Y, Neelaveni S, Balkrishna Ch, V. Harikumar, S. Anusha, B. Mounika, G, Shilpa and DLN murthy 2023. Millets-Our Health. Pp46.

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Ramesh P, Debnath A, Hima Bindu T, Venkateshwarlu G & Srinivasarao Ch (2022). A Manual on Stress Management and Well-Being in the Workplace. ICAR-National Academy of Agricultural Research Management, Hyderabad, India, p48.

Senthil Vinayagam S, Sumanth Kumar VV, Yashavanth BS., Umesh Hudedamani, Vijay Avinashilingam NA, Venkateshwarlu G (2023). Startup Panorama. ICAR-NAARM, Hyderabad, 50.

Soam SK, Singh SB, Sarangthem I, Raju DT, Rao NS, Kumar A & Haldhar SM (2022). Socio-Economic Development of North Eastern India through Agri-graduates: Success Stories from Career Development Center at CAU Imphal. ICAR- National Academy of Agricultural Research Management, Hyderabad-500030.

Sontakki BS & Rathore S (2023). VIII Management Development Programme for Newly Recruited Senior Scientists and Heads of Krishi Vigyan Kendras Programme Directors 31 January – 14 February 2023 Training Manual. ICAR – National Academy of Agricultural Research Management, Hyderabad, India.

Sontakki BS, Venkatesan P, Rathore S, Vinayagam SS, Raju DT, Kumar A, Srikanth PD & Ahire LM (2023). Field Experience Training guidelines' 112th Foundation Course for Agricultural Research Service (112th FOCARS), April 2023. ICAR – National Academy of Agricultural Research Management, Hyderabad.

Umesh Hudedamani, Supriya P, Balakrishnan M, Satendra Magrauthia, Sundaram RM & Srinivasa Rao Ch (2023). Gene Editing and Technology Management. ICAR-NAARM, Hyderabad. Pp. 73.

8.2.5.2 Technical Bulletins

Alok Kumar, Srinivasarao, N., & Soam, S. (2020). National Workshops on Establishment of Career Development Centre in Agricultural Universities, Proceedings & Recommendations. ICAR-National Academy of Agricultural Research Management, Hyderabad, 500 030. Pp. 1-80.

Ramesh Naik M & Umesh Hudedamani (2023). Organic Farming in India: Production Constraints and Policy Recommendations. Climate Smart Agronomy for Resilient Production Systems and Livelihood Security. In: Extended Summaries of XXII Biennial National Symposium. The Indian Society of Agronomy, New Delhi, pp 41-42.

Saravanan Raj, Balasubramani, N., Surya Rathore, Bharat S Sontakki, Venkatesan P & Vijay Avinashilingam NA (2023). Report on “Strengthening Extension Delivery Mechanism at District Level through KVK & ATMA” Jointly prepared by National Institute of Agricultural Extension Management (MANAGE) and ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad, 36.

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National Academy of Agricultural Research Management, Hyderabad-500030. Pp 1-52. ISBN: 978-93-5779-988-1.

Srinivasarao Ch, Sreekanth PD & Murthy GRK (2023). Status and Scope of Policy Interventions in Digital Agriculture in India. In Souvenir – National Symposium on Digital Farming: The Future of Indian Agriculture, 2-3 February 2023, ICAR-IISS, Bhopal.

8.2.6 Papers presented in Conferences / Seminars

Aketi Gayatri J & Dastagiri MB (2023). *Impact of Globalization and Multinational Corporation on Farmers Suicides in India: An Over View, Effects, Strategies and Policies*. Texas A&M International University, 27th Annual Western Hemispheric Trade Conference Proceedings, held on April 12-14, 2023 in Laredo, Texas, USA. 198-207.

Dastagiri MB, Shailaja Thota & Ramesh Naik M (2023). *Global Agriculture First to benefit SDGs to People, Prosperity, Peace and Planet: Geo-Social, Economic, Political Analysis*. Texas A&M International University, 27th Annual Western Hemispheric Trade Conference Proceedings, held on April 12-14, 2023 in Laredo, Texas, USA. 69-91.

Dastagiri MB, Vijaya Kumari R, Shailaja T, Bharath Kumar M & Munni Manta (2023). *WTO Free Trade Agreements Impacts on Globalized World GDP, FDI, Trade and Poverty: Geo-Political, Trade and Economic Analysis*. Texas A&M International University, 27th Annual Western Hemispheric Trade Conference Proceedings, held on April 12-14, 2023 in Laredo, Texas, USA. 168-190.

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Sontakki, B. S. and Rathore, S. (2023). *VIII Management Development Programme for Newly Recruited Senior Scientists and Heads of Krishi Vigyan Kendras Programme Directors 31 January – 14 February 2023 Training Manual*. ICAR – National Academy of Agricultural Research Management, Hyderabad, India.

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Venkattakumar R & Vasanthi C (October, 2023). *Role of FPOs in Production and Marketing of Under-Exploited Vegetables*. In: International Seminar on Exotic and Underutilized Horticultural Crops: Priorities & Emerging Trends.

Venkattakumar R, Venkatesan P & Srinivasrao Ch (2023). *Stakeholder Engagement through Capacity Building for Resilient Agri-Eco Systems*. Lead Paper presented at the National Conference on “NexGen Extension for Evolving Resilient Agri-Ecosystems (NEERAE 2023)” jointly organized by EEI, GoI; PJTSAU, Hyderabad and Society of Extension Professionals, Hyderabad at EEI, Hyderabad during 25 - 27 September, 2023.

8.2.7 Policy Briefs

Raj S, Balasubramani N, Rathore S, Sontakki BS, Venkatesan P & Avinashalingam NA (2023). Report on Strengthening Extension Delivery Mechanism at District Level through KVK and ATMA, jointly prepared by National Institute of Agricultural Extension Management (MANAGE) & ICAR- National Academy of Agricultural Research Management (ICAR-NAARM), Hyderabad: 38p.

9.1 Budget Allocation and Expenditure during Jan-Dec 2023 (Rs. in lakhs)

S.No.	Head of Account	Budget	Expenditure
A. Grant in Aid - Capital			
1	Works	891.77	891.77
2	Equipment	127.04	23.22
3	Information Technology	6.14	6.14
4	Library Books & Journals	1.41	1.41
5	Vehicles & Vessels	0.00	0.00
6	Furniture & Fixtures	10.64	10.64
Total Capital		1037.00	933.18
Grants in Aid - Salaries (REVENUE)			
1	Establishment Expenses		
2	i. Establishment Charges	2846.66	2846.66
3	ii. Wages	0.84	0.84
4	iii. OTA	0.06	0.05
Total Salaries		2847.56	2847.55
B. Grant in Aid - General (REVENUE)			
1	Pension & Other Retirement Benefits	5376.39	5376.39
2	Domestic TA / Transfer TA	144.46	144.16
3	Research & Operational Expenses	350.00	324.18
4	Administrative Expenses	1699.01	1699.01
5	HRD	16.58	12.72
6	Miscellaneous Expenses	190.58	142.79
Total Grants in Aid - General		7777.02	7699.25
Grand Total (Capital+Salaries+Revenue)		11661.58	11479.98

9.2 Resource Generation

9.2.1 Sponsored Projects Budget (Rs. in lakhs)

Sl. No	Name of the Project	Fund received	Expenditure
1	ICAR-NAARM-MANAGE Project "Agri-Warehousing Management"	0.90	0.90
2	ICAR-NAARM-PJTSAU Digital Content for MOOCS	0.89	0.89
3	Skill Development in Bioinformatics Infrastructure Facility on Agril.Bio Technology under DBT	2.86	2.86
4	ICAR-NAARM-NDRI, Karnal for developing MOOC Courses on Commercial Dairy farming & Milk processing & Value Edition	1.22	1.22
5	International Consultancy Project -ICAR-NAARM-GIZ "Developing and Pilot testing Specialized Knowledge Products on Human Wildlife Conflict (HWC) Mitigation for Agriculture and Veterinary Sector in India	5.31	4.7
6	Identification & Characterization of superior haplotypes for agronomically important traits in Rice under DST-SERB	8.47	0.44
7	RUTAG 2.0	45.87	26.96
	Total	65.52	37.97

9.2.2 Off-Campus and Sponsored Programmes from Jan-Dec. 2023 (Rs. in lakhs)

Sl. No	Name of the Programme	Total Budget received
1	Foundation Course for Asst.Prof.of OUAT, Bhubaneswar 24.01.2023 to 13.02.2023	12.75
2	M/s.Coromandal International Ltd., CIL-NAARM Collaborative Training Programme for 13-17 Feb 2023 & 22-24 Feb 2023	9.62
3	M/s.Coromandal International Ltd NEAT phase 2 batch 3 March 13-17 2023 & MORE Batch 1 March 20-24 2023	6.18
4	M/s.Coromandal International Ltd NEAT phase 1 batch 1 Dec 12-16 2022 & NEAT phase 1 batch 2 19-23 Dec 2022	6.06
5	NDRI, Karnal for Developing MOOC Content	7.08
6	Indian patent office New Delhi Charges for using facilities for conducting PCT Roving seminar by 6.3.2023	1.76
7	UBKV, Koochbehar Off Campus Programme fee for training on Digital content creation & mgmt for faculty 28-30 March 2023	4.12

8	Faculty user charges for organizing millet conclave by NABARD	0.71
9	M/s.Coromandal International Ltd NEAT phase 1 batch 2 June 12-16 2023 & batch 1 June 5 to 9 2023	10.53
10	M/s.Coromandal International Ltd MORE training 4-9 Sep 2023 & NEAT phase 2 11-15 Sep 203 & oct 9-12 2023	22.73
11	M/s.Coromandal International Ltd NEAT phase 2 batch 2	5.05
12	OUAT Bhubaneswar Programme fee for the Faculty for FDP from 24.07.2023 to 03.08.2023	6.66
13	NDRI, Karnal Skill Development Training Programme for 4th year B.Tech Students	13.19
14	NDRI, Karnal for Developing MOOC Content	18.3
15	RVSKVY, Gwalior for Training of Students	15.18
16	UAS, Raichur for Off Campus Programme fee for "Enhancement of Teaching Competence" for Faculty	2.76
17	UHF, NAUNI, Solan, HP Programme Fee for training of Principal Farm Managers & Section officers	3.81
18	MAFSU, Mumbai Programme for for training of Assistant Professors & equivalent of Nagpur College	11.6
	Total	158.09

9.2.3 Resource Generation from Educational Programmes (Rs. in lakhs)

Sl. No	Educational Programmes	Receipts
1	PGDM (ABM) 2021-23	17.56
2	PGDM (ABM) 2022-24	233.01
3	PGDM (ABM) 2023-25	303.41
	Total	553.975

9.2.4 Resource Generation from other Activities (Rs. in lakhs)

Sl. No	Net Resource Generation from other activities	Receipts
1	Receipts from Farm Produce	1.47
2	Receipts from Guest House and Quarters	84.59
3	Receipts from Institute Training Programmes	23.65
4	Miscellaneous Receipts	27.43
	Total	137.14

9.2.5 Net receipts from Sponsored Research Projects from Jan to Dec 2023 (Rs. in lakhs)

Sl. No	Sponsored Research Projects	
1	Profiling the Resources Devoted in Agril.R&D Dev.of National framework for Evidence-based S&T Governance in India Scheme of DST	2.02
2	Development of MOOC Course, NDRI Karnal	6.00
3	Net Receipts from Educational Programmes	291.95
4	Net Receipts from IRGS Programmes/Schemes	11.16
	Total	311.13

9.3 Plan Schemes (Rs. in lakhs)

Sl. No.	Name of the Scheme	Fund / Budget	Expenditure
1	National Agricultural Innovation Fund (Component-I)-ZTMC	8.25	7.91
2	National Agricultural Innovation Fund (Component-II)-ABI	18.85	8.41
3	XII Plan EFC Main Scheme KRISHI Project	12.78	12.24
4	Management & Impact Assessment of Farmer First Project	10.63	9.83
5	NASE, New Delhi - Entrepreneurship Development through Farmer Led Innovations	12.45	1.52
6	Emeritus Scientist Scheme	8.75	4.91
7	NAHEP	288.57	163.70
	Total	360.28	208.52

Personnel

10

Human Resources at the Academy (as on 31.12.2023)

Category	Sanctioned Strength	In position	Abolished	Vacant
Scientific – RMP	2	2	00	00
Scientific - Faculty	60	29	00	31
Technical	42	19	00	23
Administrative (including Legal)	44	28	00	16
Multi-Tasking Staff (Regular posts)	33	25	00	10
Multi-Tasking Staff (Additionally Sanctioned posts)	18	15	03	- Only 16 CLTS were available for Regularisation - The post gets abolished on exit by the incumbent MTS
Total	199	118	03	78

Research Management Positions

Dr. Ch.Srinivasa Rao, Director

Dr. G.Venkateshwarlu, Joint Director

Scientific Staff

Agribusiness Management Division	
1	Dr.Tavva Srinivas, Principal Scientist and Head
2	Dr.Ranjit Kumar, Principal Scientist
3	Dr.Prem Chand Meena, Principal Scientist
4	Dr.N.Sivaramane, Principal Scientist
5	Dr.Sanjiv Kumar, Sr. Scientist
Education Systems Management Division	
1	Dr.S.Senthil Vinayagam, Principal Scientist, Head
2	Dr.Dhumantarao Thammi Raju, Principal Scientist
3	Dr.G.R.Ramakrishna Murthy, Principal Scientist
4	Dr.V.V.Sumanth Kumar, Principal Scientist
Human Resource Management Division	
1	Dr.B.Ganesh Kumar, Principal Scientist, Head
2	Dr.P.Ramesh, Principal Scientist
3	Dr.K.Srinivas, Principal Scientist
Extension Systems Management Division	
1	Dr.Venkattakumar, Principal Scientist, Head
2	Dr.(Mrs.)Surya Rathore, Principal Scientist

3	Dr.P.Venkatesan, Principal Scientist
4	Dr.Vijay Avinashilingam N.A., Principal Scientist
Research Systems Management Division	
1	Dr.Ramasubramanian V., Principal Scientist and Head
2	Dr. D.Damodar Reddy, Principal Scientist (Since 18.01.2023)
3	Dr.M.B.Dastagiri, Principal Scientist
4	Dr.P.Krishnan, Principal Scientist (On Deputation Since 29.12.2021)
5	Dr.Umesh Hudedamani, Scientist
6	Dr.M.Ramesh Naik, Scientist
Information and Communication Management Division	
1	Dr.A.Dhandapani and Head
2	Dr.S.K.Soam, Principal Scientist
3	Dr.M.Balakrishnan, Principal Scientist
4	Dr.P.D.Sreekanth, Principal Scientist
5	Dr.N.Srinivasa Rao, Principal Scientist
6	Dr.Yashavanth, B.S., Scientist
7	Dr.(Mrs.)Purru Supriya, Scientist

Administrative Staff

Chief Administrative Officer (Sr. Grade) Shri Vivek Purwar
Comptroller Shri Z.H.Khilji
Joint Director (OL) Dr. J. Renuka
Senior Administrative Officer Shri Mukul Rajsingh
Principal Private Secretary Shri K. Sanath Kumar
Principal Private Secretary Shri B.P.R.Vithal
Principal Private Secretary Smt.A.Mercy
Finance & Accounts Officer Shri Mariseti Naga Venkateswara Rao
Assistant Administrative Officers
1. Smt.K.K.Rukmani Ammal,
2. Dr.K.R.Ghanshyam
3. Shri C.J.Samuel
4. Shri T.Srinivas
Assistant Finance & Accounts Officer Smt.N.Vijayalakshmi
Private Secretaries
1. Smt.T.Vanisri,
2. Smt.S.Shanthi
3. Smt.Y.Anuradha
4. Smt.S.Sesha Sai
Assistants
1. Shri G.Raj Reddy
2. Smt.B.Padma Saroja
3. Shri R.Chandra Babu
4. Shri M.Sridhar
5. Smt.Y.Gayathri
6. Shri K.Suryanarayana
7. Shri M.K.Samson
Personal Assistant Smt.V.Shailaja
Upper Division Clerk
1. Smt.Rajashri S.Bokde
2. Shri P.Swamy
3. Shri M.Narsing Rao

Technical Category

Grade T-9 (Category-III)
1. Shri P.Namdev, Chief Technical Officer (Artist)
2. Dr.P.Vijender Reddy, Chief Technical Officer (Research Assistant)
3. Shri Sohail Ahmad Khan, Chief Technical Officer (Junior Engineer-Civil)
4. Smt.G.Aneeja, Chief Technical Officer (Assistant Editor)
5. Sh.M.Shekhar Reddy, Chief Technical Officer (Dark Room Assistant)
Grade T (7-8) (Category-III)
6. Dr.Ahire Laxman Maharu, Assistant Chief Technical Officer (Horticultural Technical Assistant)
7. Shri M.Ravi, Assistant Chief Technical Officer (Photographer-cum-Artist)
8. Shri P.Mohan Singh, Assistant Chief Technical Officer (Computer Assistant)
9. Shri Sham Bahadur, Assistant Chief Technical Officer (Catering in-charge)
10. Mrs.Savithri Murali, Assistant Chief Technical Officer (Catering in-charge)
Grade T-6 (Category-III)
11. Shri Kirttiranjan Baral, Sr. Technical Officer (Farm Superintend)
Grade T-5 (Category-II)
12. Shri K.Shivaiah, Technical Officer (Technician)
13. Shri S.Rajukumar, Technical Officer (Electronic Computer Operator)
14. Shri N.Prabhakar Rao, Technical Officer (Plumber)
15. Shri M.Srinivasa Rao, Technical Officer (Pump Driver)
Grade T-4 (Category-II)
16. Shri Pitla Srinivas, Senior Technical Assistant (Proof Reader)
17. Shri B.K.Venkatram, Senior Technical Assistant (Pump Driver)
18. Shri R.Siva Prasad, Senior Technical Assistant (Driver)
Grade T-3 (Category-II)
19. Sh.Danam Murahari, Technical Assistant (A.V.Operator)

Multi-Tasking Staff

1. Smt.S.Shakuntala
2. Shri K.Pentaiah
3. Shri P.Yadaiah
4. Shri G.Pentaiah
5. Shri R.Sattaiah
6. Shri L.Satyanarayana
7. Smt.C.Kausalya
8. Shri M.Ganesh Kumar
9. Shri Chilumula Venkatesham
10. Shri Kyasam Satyanarayana
11. Shri B.Premdas
12. Smt.V.Saroja
13. Shri S.Nayab Rasool
14. Shri B.Ashok
15. Shri K.Daniel
16. Shri M.Satyanarayana
17. Shri D.Srisailam
18. Shri B.Ratnaiah
19. Shri D.Babaiah
20. Shri K.Samson
21. Shri.K.Satnarayana/C
22. Shri K.Narasimha
23. Smt.K.Andalu
24. Shri N.Krishna
25. Smt.P.Amrutha
26. Smt.K.Lakshmi
27. Smt.L.Chandramma
28. Shri B.Raju
29. Shri MD.Sadath
30. Shri T.Srinivas
31. Smt.E.Yadamma
32. Smt.S.Lakshamma
33. Shri P.Venkatesh
34. Shri G.Ramulu S/o.G.Laxmaiah
35. Shri P.Yadaiah/K
36. Smt.R.Pramila
37. Smt.M.Laxmamma
38. Shri G.Ramulu S/o.G.Danaiah
39. Smt.M.Sharada
40. Shri N. Yadagiri

Retirements:

1. Dr.R.V.Satyanarayana Rao, Principal Scientist retired (on superannuation) w.e.f. 28.02.2023
2. Shri Kumba Satyanarayana, SSS retired (on superannuation) w.e.f. 28.02.2023
3. Shri Venkatesham S., SSS retired (on superannuation) w.e.f. 30.04.2023
4. Shri Satyanarayana G., SSS retired (on superannuation) w.e.f. 30.04.2023
5. Dr.Kareemulla K., Principal Scientist retired (on superannuation) w.e.f. 31.05.2023
6. Dr.Ravichandran S., Principal Scientist retired (on superannuation) w.e.f. 31.05.2023
7. Shri Srinivasu P., Assistant retired (on superannuation) w.e.f. 31.05.2023
8. Shri Dinesh M., Assistant Admn.Officer retired (on superannuation) w.e.f. 30.06.2023
9. Shri Laxman T., T-5 Technical Officer (Driver) retired (on superannuation) w.e.f. 30.06.2023
10. Shri. Ravi M.S., SSS retired (on superannuation) w.e.f. 30.06.2023
11. Shri M. Ashok, UDC retired (on superannuation) w.e.f. 31.07.2023
12. Shri C.Phani Raj, Assistant retired (on superannuation) w.e.f. 31.10.2023
13. Shri K.Sanath Kumar, PPS retired (on superannuation) w.e.f. 31.12.2023
14. Smt.P.Amrutha, MTS retired (on superannuation) w.e.f. 31.12.2023
15. Shri M.Satyanarayana, MTS retired (on superannuation) w.e.f. 31.12.2023

Transfers from ICAR-NAARM:

1. Dr.Alok Kumar, Pr. Scientist transferred and joined as Head, Division of Social Science at ICAR-CPRI, Shimla w.e.f. 11.07.2023
2. Shri T.V.Ramadas, Personal Assistant Promoted as Private Secretary & relieved from NAARM on 01.12.2023 (FN) to join in DPR.

Transfers to ICAR-NAARM:

1. Dr.D.Damodhar Reddy, Principal Scientist (Soil Science) joined the Academy w.e.f. 18.01.2023(FN) on transfer from ICAR-CTRI, Rajahmundry.
2. Dr.K.Srinivas, Principal Scientist(Agricultural Economics) joined the Academy w.e.f. 14.02.2023 (FN) on transfer from ICAR Headquarters.
3. Shri B.P.R.Vithal joined the Academy w.e.f. 26.12.2023 (FN) on transfer from ICAR-IIHR, Bengaluru

Appointments / New Joinings:

1. Dr.Ramasubramanian, V, Joined as Head, RSM Division w.e.f. 13.07.2023(FN)
2. Dr.S.Senthil Vinayagam Joined as Head, ESM Division w.e.f. 17.07.2023(FN)
3. Dr.A.Dhandapani Joined as Head, ICM Division w.e.f. 17.07.2023(FN)
4. Dr.B.Ganesh Kumar Joined as Head, HRM Division w.e.f. 17.07.2023(FN)
5. Dr. Tavva Srinivas Joined as Head, ABM Division w.e.f. 17.07.2023(AN)
6. Dr. R. Venkattakumar, Joined as Head, XSM Division w.e.f. 20.07.2023(FN)

Promotions:

1. Shri Z.H.Khilji, Chief Finance & Accounts Officer has been promoted to the post of Comptroller w.e.f. 02.01.2023 (FN).
2. Smt.G.Aneeta, T(7-8) Assistant Chief Technical Officer has been promoted to the grade of T-9 Chief Technical Officer w.e.f. 10.12.2020 (FN).
3. Shri M. Shekhar Reddy, T(7-8) Assistant Chief Technical Officer has been promoted to the grade of T-9 Chief Technical Officer w.e.f. 01.01.2022 (FN).
4. Shri Telu Srinivas, Assistant has been promoted to the post of Asst. Admn. Officer w.e.f. 03.07.2023 (FN)
5. Smt.Y.Anuradha, Personal Assistant has been promoted to the post of Private Secretary w.e.f. 10.11.2023 (FN).
6. Smt.S.Sesha Sai, Personal Assistant has been promoted to the post of Private Secretary w.e.f. 10.11.2023 (FN).
7. Smt.A.Mercy, Private Secretary has been promoted to the post of Principal Private Secretary w.e.f. 13.12.2023 (FN).

Financial Upgradations under MACPS:-

S.No.	Name	Designation	1 st /2 nd /3 rd Financial Upgradation with GP/ Level of Pay	Effective Date
1.	Smt. N. Vijaya Lakshmi	Asst. Finance & Accounts Officer	3 rd Financial Upgradation with Level #8 (GP : Rs.4800/-)	14.07.2023

Clearance of Probation Period and Confirmation

S.No.	Name	Post	Date of Clearance of Probation	Date of Confirmation
1.	Shri K.Satyanarayana	MTS	28.01.2023	29.01.2023
2.	Shri K.Narasimha	MTS	28.01.2023	29.01.2023
3.	Smt.K.Andalu	MTS	29.01.2023	30.01.2023
4.	Shri M.S.Ravi	MTS	29.01.2023	30.01.2023
5.	Smt.P.Amruha	MTS	08.08.2023	09.08.2023

Obituary: 1. Dr. Bharat S.Sontakki, Pr.Scientist breathed his last on 11.11.2023

Others:

- The following Principal Scientists taken over the charge of HOD in Acting Capacity w.e.f. 01.01.2023.
 - Dr.P.Ramesh as Head, HRM Division.
 - Dr.A.Dhandapani as Head, ICM Division.
 - Dr.Tavva Srinivas as Head, ABM Division.
 - Dr.(Mrs.) Surya Rathore as Head, XSM Division.
- Dr.R.V.S.Rao**, Principal Scientist (Retd.), ICAR-NAARM – engaged as **Adjunct Faculty** in the domain of Human Resource Management **w.e.f. 12.04.2023**
- Dr. D. Thammi Raju**, Principal Scientist – appointed as **Vigilance Officer w.e.f. 13.07.2023**
- Dr.T.Hima Bindu** joined as **Medical Doctor** on contractual basis **w.e.f. 26.10.2023**

List of Committees

1) Higher Education Committee

1.	Joint Director & Dean	Chairman
2.	Heads of all Divisions	Members
3.	CAO (SG)	Member
4.	Comptroller	Member
5.	Principal Course Coordinator, PGDM – ABM	Member
6.	Coordinators, PGDM-ABM	Member
7.	Coordinators, DTMA	Member
8.	Coordinators, DETM	Member
9.	Controller of Examinations	Member
10.	OIC, Hostels	Member
11.	OIC, PGS Unit	Member Secretary

2) Board of Studies

1.	Joint Director	Chairman
2.	Heads of all six Divisions	Members
3.	Head of Administration and Registrar	Member
4.	Comptroller	Member
5.	Principal Course Coordinator, PGD-ABM	Member
6.	Academic / Course Coordinators of PGD-ABM batches, DTMA and DETM	Member
7.	Coordinator for Degree Programmes	Member
8.	Controller of Examinations	Member
9.	Placement Coordinator	Member
10.	Officer In-charge, P.G.Studies Unit	Member-Secretary

3) Works Committee

1.	Dr.B.S.Sontakki, Pr.Scientist (Till 11.11.2023)	Chairman
2.	Dr.V.Ramasubramanian, HoD, RSM Division (From 17.11.2023)	Chairman
3.	Dr.G.R.Ramakrishna Murthy, Pr.Scientist	Member
4.	Dr.M.Balakrishnan, Pr.Scientist	Member
5.	Dr.P.D.Sreekanth, Pr.Scientist	Member
6.	Joint Director (Admn.) & Registrar	Member
7.	Comptroller	Member
8.	Dr. P. Vijender Reddy, CTO	Member
9.	Shri P. Mohan Singh, ACTO	Member

10.	Shri Sohail Ahmad Khan, CTO	Member Secretary
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4) Anti-Ragging Committee

1.	Joint Director / Dean	Chairman
3.	CAO(SG)	Member
4.	Academic Coordinators, PGDM (A) Programme	Member
5.	Dr.P.C.Meena, Pr.Scientist & OIC, P.G.S Unit	Member Secretary

5) Anti-Ragging Squad

1.	Dr.(Mrs.) Surya Rathore, Principal Scientist	Chairman
2.	Dr.B.Ganesh Kumar, Head, HRM Division	Member
3.	Dr.Umesh Hudedamani, Scientist	Member
4.	Comptroller	Member
5.	Officer Incharge, Hostel	Member Secretary

6) Grievance Committee

1.	Joint Director	Chairman
2.	Dr.G.R.K. Murthy, Principal Scientist	Member
3.	Shri Vivek Purwar, CAO (SG)	Member
4.	Shri Z.H. Khilji, Comptroller	Member
5.	Shri Mukul Rajsingh, SAO	Member
6.	Dr. Surya Rathore, Principal Scientist	Elected Member – Scientific Category
7.	Shri M. Ravi, ACTO	Elected Member – Technical Category
8.	Shri C. Julius Samuel, AAO	Elected Member – Admn. Category
9.	Shri Kyasam Satyanarayana, SSS	Elected Member – Skilled Supporting Category
10.	Mrs.K K Rukmani, AAO	Member Secretary

7) Farm Advisory Committee

1.	Dr.Mude Ramesh Naik, Scientist(SS)	Chairman
2.	Dr.B.S.Yashavanth, Scientist	Member
3.	Dr.P.Vijender Reddy, CTO	Member

4.	Dr.L.M.Ahire, ACTO	Member
5.	Dr.P.Mohan Singh, ACTO	Member
6.	Shri.M.Shekhar Reddy, ACTO	Member
7.	Shri.Danam Murahari, TA	Member
8.	Dr.P.Vijender Reddy, CTO	Member Secretary

8) Library Advisory Committee

1.	Dr.D.Damodar Reddy, Pr.Scientist	Chairman
2.	Heads of all Divisions	Members
3.	Dr.M.Balakrishnan, Pr.Scientist	Member
4.	Dr.V.V.Sumanth Kumar, Pr.Scientist	Member
5.	Head of Administration and Registrar	Member
6.	Comptroller	Member
7.	Dr.N.Srinivasa Rao, Pr.Scientist and OIC, Library	Member-Secretary

9) Newsletter Committee

1.	Dr.D.Damodar Reddy, Pr. Scientist	Chairman
2.	Dr.Sanjiv Kumar, Sr.Scientist	Member
3.	Dr. Umesh Hudedamani, Scientist	Member
4.	Dr.(Mrs.) Purru Supriya, Scientist	Member secretary

Dr.B.S. Yashavanth, Scientist will help the committee for the first issue and committee is responsible for April-June, 2023 Newsletter onwards.

10) Annual Report Committee (Till 05.12.2023)

1.	Joint Director	Chairman
2.	Head, ICM Divn.	Member
3.	Head, RSM Divn.	Member
4.	Head, HRM Divn.	Member
5.	Head, ESM Divn.	Member
6.	Head, ABM Divn.	Member
7.	Head, XSM Divn.	Member
8.	Dr.Tavva Srinivas, Pr.Scientist and Head, PME Cell	Member
9.	Dr.Sanjiv Kumar, Sr.Scientist, ABM Division	Member
10.	Dr.Yashavanth, B.S., Scientist, ICM Division	Member
11.	Dr.(Mrs.)P.Supriya, Scientist, ICM Division	Member
12.	Dr.Alok Kumar, Pr.Scientist, HRM Division.	Member-Secretary

10A) Annual Report Committee

(from 06.12.2023)

1.	Joint Director	Chairman
2.	Head, ICM Divn.	Member
3.	Head, RSM Divn.	Member
4.	Head, HRM Divn.	Member
5.	Head, ESM Divn.	Member
6.	Head, ABM Divn.	Member
7.	Head, XSM Divn.	Member
8.	Head of Admin. & Registrar	Member
9.	Comptroller	Member
11.	Dr.(Mrs.)P.Supriya, Scientist, ICM Division	Member-Secretary

11) Campus Residents Welfare Committee

1.	Dr.S.K.Soam, Principal Scientist	Chairman
2.	Dr.Umesh Hudedamani, Scientist	Member
3.	Dr.Mude Ramesh Naik, Scientist (SS)	Member
4.	Shri.Z.H.Khilji, Comptroller	Member
5.	Dr.B.S.Yashavanth, Scientist	Member Secretary

12) Examination Unit

1.	Dr.P.Ramesh, Pr.Scientist & COE	Chairman
3.	Sr. Admin. Officer	Member
4.	Shri Pitla Srinivas, Sr.Technical Assistant	Member
5.	Mrs Y.Anuradha, Private Secretary	Member

13) Hostel Management Committee

1.	Shri Vivek Purwar, CAO(SG)	Chairman
2.	Shri Z. H. Khilji, Comptroller	Member
3.	Shri Mukul Rajsingh, SAO	Member
4.	Dr. P. Vijender Reddy, CTO	Member
5.	Shri Sohail Ahmad Khan, CTO	Member
6.	Shri P. Mohan Singh, ACTO	Member
7.	Smt. Savithri Murali, ACTO	Member
8.	Shri K. Shivaiah, TO	Member
9.	Shri M. Sridhar, Assistant	Member
10.	Shri Sham Bahadur, ACTO	Member-Secretary
11.	Dr. M. BalaKrishnan, Pr. Scientist	Special invitee
12.	Dr.M.Ramesh Naik, Scientist (SS)	Special invitee
13.	Dr. (Mrs.) P. Supriya, Scientist	Special invitee

14) Institute Technology Management Committee

1.	Dr.Ch.Srinivasa Rao, Director	Chairman
2.	Joint Director	Member
3.	Dr.S.Senthil Vinayagam, Head, ESM Division	Member
4.	Dr.GRK Murthy, Pr.Scientist	Member
5.	Dr.V.V.Sumanth Kumar, Pr.Scientist	Member
6.	Joint Director (Admin) Registrar	Member
7.	Comptroller	Member
8.	Dr.RM Sundaram, Director, ICAR - IIRR.Hyd	Member
9.	Dr.Umesh Hudedemani, Scientist	Member Secretary

15) Sports Complex Maintenance Committee

1.	Dr. M. Ramesh Naik, Scientist(SS)	Chairman
2.	Dr.P.Vijender Reddy, CTO	Member
3.	Sh.Sohail Ahmad Khan, ACTO	Member
4.	Dr.Laxman M Ahire, ACTO	Member
5.	Sh.M.Shekar Reddy, ACTO	Member
6.	Sh.P.Mohan Singh, ACTO	Member
7.	Sh.Sham Bahadur, ACTO	Member
8.	Sh.M.K.Samson, UDC	Member
09.	Smt. K.K.Rukmani Ammal, AAO	Member
10.	Dr. (Mrs.) P. Supriya, Scientist	Member Secretary

16) Canteen Committee

1.	Dr. G. R. Ramakrishna Murthy, Pr. Scientist	Chairman
2.	Dr. P. Vijender Reddy, CTO	Member
3.	Shri S. Rajukumar, TO	Member
4.	Shri M. Sridhar, Asst. And Member, IJSC & CJSC	Member Secretary

17) Internal Complaints Committee

1.	Dr.(Mrs.) Surya Rathore , Pr. Scientist	Chairman
2.	Shri MukulRajsingh, SAO	Member
3.	Smt.G.Aneeja, Asst.Chief Technical officer	Member
4.	Smt.N.Vijayalaxmi, AF&AO	Member
5.	Smt.T.Vanisri, PS	Member

6.	Smt. P. Padmavathi, Kasturba Gandhi National Memorial Trust as 3 rd Party Representative (representing either NGO or other body who is familiar with the issues of sexual harassment)	Member
7.	Smt.KK.Rukmani Ammal, AAO	Member Secretary

18) Academic Council

1.	Dr.Ch.Srinivasa Rao, Director	Chairman
2.	All Faculty	Members
3.	Dr.A.Dhandapani, Head, ICM Divn.	Member Secretary -1

19) Purchase Advisory Committee (Stores)

1.	Dr.A.Dhandapani, Head, ICM Divn.	Chairman
2.	Dr.P.Ramesh, Principal Scientist	Member
3.	Dr.Umesh Hudedamani, Scientist	Member
4.	Head of Admin. & Registrar	Member
5.	Comptroller	Member
6.	Dr.P.Vijender Reddy, CTO	Member
7.	Shri.M.Sridhar, Assistant	Member
8.	Shri Mukul Rajsingh, SAO	Member Secretary

20) Technical Advisory Committee (Stores)

1.	Dr.R.Venkattakumar, Head, XSM Divn.	Chairman
2.	Dr.G.R.K.Murthy, Pr.Scientist	Member
3.	Dr.N.S.Rao. Pr.Scientist	Member
4.	Dr.V.V.Sumanth Kumar, Pr. Scientist	Member
5.	Dr. Laxman Ahire, ACTO	Member
6.	Shri Rajukumar, TO	Member Secretary

21) Admission Committee for PGDM-ABM (2024-26 Batch)

1.	Dr. B.Ganesh Kumar, Head, HRM Divn.	Chairman
2.	Head of Division, RSM	Members
3.	Head of Division, ESM	Member
4.	Head of Division, ICM	Member
5.	Head of Division, ABM	Member
6.	Head of Division, XSM	Member
7.	Dr.Sivaramane N, Principal Scientist & Academic Coordinator	Member

8.	Head of Administration & Registrar	Member
9.	Dr.Sanjiv Kumar, Sr. Scientist & Academic Coordinator	Member
10.	Dr.P.C.Meena, Principal Scientist & OIC-PGS Unit	Member Secretary

22) Applications Screening Committee for PGDM-ABM (2024-26 Batch)

1.	Dr. N. Sivaramane, Principal Scientist	Chairman
2.	Dr. P. C. Meena, Principal Scientist & OIC-PGS Unit	Member
3.	Dr. Sanjiv Kumar, Senior Scientist & Academic Coordinator	Member
4.	Shri Pitla Srinivas, Sr. Technical Assistant PGS Unit	Member Secretary

23) Course Coordinators for Academic Programmes

1.	Dr.B.Ganesh Kumar, HoD, HRM Division	Principal Course Coordinator, PGDM-ABM
2.	Dr.N.Sivaramane, Principal Scientist	Academic Coordinator, PGDM-ABM
3.	Dr.Sanjiv Kumar, Senior Scientist	Academic Coordinator, PGDM-ABM
4.	Dr.Umesh Hudedamani, Scientist	Course Coordinator, DTMA
5.	Dr.V.V.Sumanth Kumar, Principal Scientist	Course Coordinator, DETM
6.	Dr.(Mrs.) P.Supriya, Scientist	Co-Coordinator, DETM

24) RTI Functionaries designated at ICAR-NAARM under the provisions of RTI Act, 2005

#	Name of the Officer and Designation	Function under RTI Act, 2005
1.	Dr.G.Venkateshwarlu Joint Director	Appellate Authority
2.	Shri Vivek Purwar Chief Admn.Officer (Sr.Grade)	Transparency Officer
3.	Dr.M.Balakrishnan Principal Scientist	Central Public Information Officer (CPIO) for Scientific Matters
4.	Dr.P.D.Sreekanth Principal Scientist	Central Assistant Public Information Officer (CAPIO) for Scientific matters
5.	Dr.P.C.Meena Principal Scientist and OIC, P.G.Studies Unit (w.e.f. 31.08.2023)	Central Public Information Officer (CPIO) for PGS Unit matters
6.	Shri Mukul Rajsingh Senior Admn. Officer	Central Public Information Officer (CPIO) for Admin. Matters
	Dr.P.Vijender Reddy Chief Technical Officer	Central Public Information Officer (CPIO) for Technical Matters
7.	Dr.N.Sivaramane Principal Scientist (w.e.f. 31.08.2023)	Central Assistant Public Information Officer (CAPIO) for PGS Unit Matters
8.	Shri M.Dinesh Assistant Admn.Officer (Till 30.06.2023) Smt.K.K.Rukmani Ammal Assistant Admn.Officer (w.e.f. 31.08.2023)	Central Assistant Public Information Officer (CAPIO) for Technical and Admin. matters
9.	Smt.K.K.Rukmani Ammal Assistant Admn.Officer	Nodal Officer

25) Institute Deputation Committee (IDC)

1.	Head, ESM Division	Chairman
2.	Dr.A.Dhandapani, Pr.Scientist	Member
3.	OIC, Training Unit	Member
7.	Asst.Admin.Officer, Admin.I Section	Member-Secretary

26) House Allotment Committee

1.	Dr.V.Ramasubramanian, Head, RSM Divn.	Chairman
2.	Dr.M.Balakrishnan, Pr.Scientist	Member
3.	CAO(SG)	Member
4.	Comptroller	Member
5.	Dr.M.Ramesh Naik, Scientist (SS)	Member
6.	Shri Sohail Ahmad Khan, CTO and OIC, Works Unit (Civil)	Member
7.	Asst.Admin.Officer (Admin.IV)	Member-Secretary

27) Institute Corpus Fund Committee (ICFC)

1.	Director	Chairman
2.	Joint Director	Member
3.	Head of Division, Information & Communication Systems Management	Member
4.	Head PME	Member
5.	Chairman Works Committee	Member
6.	Chief Administrative Officer (Sr. Grade)	Member
7.	Comptroller	Member
8.	Senior Administrative Officer	Member-Secretary

28) Committee for Utilization of budget allocated under SC Sub-Plan.

1.	Dr. M. Balakrishnan, Principal Scientist	Chairman
2.	Shri Z. H. Khilji, Comptroller	Member
3.	SAO	Member
4.	Dr. P. Vijender Reddy, CTO	Member
5.	Dr. L. M. Ahire, ACTO	Member
6.	Dr. M. Ramesh Naik, Scientist (SS)	Member-Secretary

29) Committee for Software Renewal

1.	Dr. A.Dhandapani, Principal Scientist	Chairman
2.	Dr. N. Srinivasa Rao, Principal Scientist	Members

3.	Dr. V.V.Sumanth Kumar, Principal Scientist	Member
4.	Shri S.Rajukumar, Technical Officer	Member-Secretary

30) Committee to monitor the timely credit of NPS contribution

1.	Director
2.	Head of Admin.
3.	DDO
4.	Head of Finance

31) Committee for organizing the meeting of retired scientists with Hon'ble Minister of Agriculture and farmers Welfare

1.	Joint Director	Chairman
2.	Dr. Bharat S. Sontakki, Head, XSM Divn.	Member
3.	Dr. Alok Kumar, Pr. Scientist	Member
4.	Dr. P.D.Sreekanth, Pr. Scientist	Member
5.	Shri Vivek Purwar, CAO(SG)	Member
6.	Shri Z.H.Khilji, Comptroller	Member
7.	Dr.A.Dhandapani, Head, ICM Divn.	Member-Secretary

32) Committee for smooth conduct of 48th Foundation Day of ICAR-NAARM on 1st September 2023

1.	Shri Vivek Purwar, CAO (SG)	Chairman
2.	Dr. Umesh Hudedamani, Scientist	Members
3.	Shri Mukul Rajsingh, SAO	Member
4.	Dr.P.Vijender Reddy, CTO	Member
5.	Shri Sohail Ahmad Khan, CTO	Member
6.	Shri Sham Bahadur, ACTO	Member
7.	Shri P.Mohan Singh, ACTO	Member
8.	Shri C.J.Samuel, AAO	Member
9.	Smt K.K.Rukmani Ammal, AAO	Member-Secretary

33) Visitors and Editorial Services Committee

1.	Dr.B.Ganesh Kumar, HoD, HRM Divn.	Chairman
2.	Dr.M.Ramesh Naik, Scientist (SS)	Member
3.	Dr.(Mrs.) Purru Supriya, Scientist	Member
4.	Smt.Savithri Murali, ACTO	Member
5.	Smt.G.Aneja, CTO	Member-Secretary

34) Student's Welfare Committee

1.	Dr G Venkateshwarlu, JD&Dean	Chairman
2.	Dr.Tavva Srinivas, Head and ABM Divn	Member
3.	Dr. B Ganesh Kumar, Principal Coordinator (PGDM-ABM)	Member
4.	Dr Sivaramane N, Academic Coordinator (PGDM-ABM)	Member
5.	Dr PC Meena, OIC, PGS Unit	Member
6.	Dr M Balakrishnan, Warden	Member
7.	Dr P Supriya, Warden	Member
8.	Dr Mohan Singh, OIC, Electrical Section	Member
9.	Mr Sohail Khan, OIC, Maintenance Section	Member
10.	Dr Vijender Reddy, OIC, Training Unit	Member
11.	Dr Ramesh Naik, Chairman, Sports Committee	Member
12.	Mr Rajukumar, OIC, IT in-charge	Member
13.	Students' representatives (of both the batches)	Members
14.	Dr Sanjiv Kumar, Academic Coordinator (PGDM-ABM)	Member Secretary

35) Committee to expedite the process of getting MBA Equivalence for the PGD-ABM programme

1.	Joint Director	Chairman
2.	Head, ABM Division	Member
3.	Dr.B.Ganesh Kumar, Head, HRM Divison and Principal Coordinator (PGDM-ABM)	Member
4.	Dr.S.K.Soam, Pr.Scientist	Member
5.	Dr.N.Sivaramane, Pr.Scientist	Member
6.	Dr.P.C.Meena, Pr.Scientist and OIC, P.G.Studies Unit	Member-Secretary

36) Committee to monitor the progress of capital expenditure under NAHEP

1.	Dr.S.K.Soam, Pr.Scientist & CPI-1, NAHEP	Chairman
2.	Head of Administration	Member
3.	Comptroller	Member
4.	Senior Admin.Officer	Member
5.	Dr.N.Srinivasa Rao, Pr.Scientist	Member
6.	Finance & Accounts Officer	Member-Secretary

37) Committee to update the booklet "NAARM at a Glance"

1.	Dr.V.V.Sumanth Kumar, Pr.Scientist	Chairman
2.	Dr.Yashavanth,B.S., Scientist	Member
3.	Dr.(Mrs.)P.Supriya, Scientist	Member
4.	Dr.P.Vijender Reddy, CTO & OIC, Training Unit	Member-Secretary



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

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ICAR-National Academy of Agricultural Research Management

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